Prime Number Theorem And ...

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Chapter 1

The project

The project github page is `https://github.com/AlexKontorovich/PrimeNumberTheoremAnd`.

The project docs page is `https://alexkontorovich.github.io/PrimeNumberTheoremAnd/docs`.

The first main goal is to prove the Prime Number Theorem in Lean. (This remains one of the outstanding problems on Wiedijk's list of 100 theorems to formalize.) Note that PNT has been formalized before, first by Avigad et al in Isabelle, `https://arxiv.org/abs/cs/0509025` following the Selberg / Erdos method, then by Harrison in HOL Light `https://www.cl.cam.ac.uk/$\sim$jrh13/papers/mikefest.html` via Newman's proof. Carniero gave another formalization in Metamath of the Selberg / Erdos method: `https://arxiv.org/abs/1608.02029`, and Eberl-Paulson gave a formalization of Newman's proof in Isabelle: `https://www.isa-afp.org/entries/Prime_Number_Theorem.html`

Continuations of this project aim to extend this work to primes in progressions (Dirichlet's theorem), Chebotarev's density theorem, etc etc.

There are (at least) three approaches to PNT that we may want to pursue simultaneously. The quickest, at this stage, is likely to follow the "Euler Products" project by Michael Stoll, which has a proof of PNT missing only the Wiener-Ikehara Tauberian theorem.

The second develops some complex analysis (residue calculus on rectangles, argument principle, Mellin transforms), to pull contours and derive a PNT with an error term which is stronger than any power of log savings.

The third approach, which will be the most general of the three, is to: (1) develop the residue calculus et al, as above, (2) add the Hadamard factorization theorem, (3) use it to prove the zero-free region for zeta via Hoffstein-Lockhart+Goldfeld-Hoffstein-Liemann (which generalizes to higher degree L-functions), and (4) use this to prove the prime number theorem with exp-root-log savings.

A word about the expected "rate-limiting-steps" in each of the approaches.

(*) In approach (1), I think it will be the fact that the Fourier transform is a bijection on the Schwartz class. There is a recent PR (`https://github.com/leanprover-community/mathlib4/pull/9773`) with David Loeffler and Heather Macbeth making the first steps in that direction, just computing the (Frechet) derivative of the Fourier transform. One will need to iterate on that to get arbitrary derivatives, to conclude that the transform of a Schwartz function is Schwartz. Then to get the bijection, we need an inversion formula. We can derive Fourier inversion *from* Mellin inversion! So it seems that the most important thing to start is Perron's formula.

(*) In approach (2), there are two rate-limiting-steps, neither too serious (in my estimation). The first is how to handle meromorphic functions on rectangles. Perhaps in this project, it should not be done in any generality, but on a case by case basis. There are two simple poles whose residues need to be computed in the proof of the Perron formula, and one simple pole in the log-derivative of zeta, nothing too complicated, and maybe we shouldn't get bogged down in the general case. The other is the fact that the ϵ-smoothed Chebyshev function differs from the unsmoothed by ϵX (and not $\epsilon X \log X$, as follows from a trivial bound). This needs a Brun-Titchmarsh type theorem, perhaps can be done even more easily in this case with a Selberg sieve, on which there is (partial?) progress in Mathlib.

(*) In approach (3), it's obviously the Hadamard factorization, which needs quite a lot of nontrivial mathematics. (But after that, the math is not hard, on top of things in approach (2) – and if we're getting the strong error term, we can afford to lose $\log X$ in the Chebyshev discussion above...).

Chapter 2

Basic definitions and notation

Let p_n denote the n^{th} prime.

Definition 2.0.1 (First prime gap). $P(g)$ is the first prime p_n for which the prime gap $p_{n+1} - p_n$ is equal to g, or 0 if no such gap exists.

Definition 2.0.2 (pi). $\pi(x)$ is the number of primes less than or equal to x.

Definition 2.0.3 (li and Li). $\mathrm{li}(x) = \int_0^x \frac{dt}{\log t}$ (in the principal value sense) and $\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}$.

Definition 2.0.4 (Equation (2) of FKS2). $E_\psi(x) = |\psi(x) - x|/x$

Definition 2.0.5 (Definitions 1, 5, FKS2). We say that E_ψ satisfies a *classical bound* with parameters A, B, C, R, x_0 if for all $x \geq x_0$ we have

$$E_\psi(x) \leq A \left(\frac{\log x}{R}\right)^B \exp\left(-C \left(\frac{\log x}{R}\right)^{1/2}\right).$$

We say that it obeys a *numerical bound* with parameter $\varepsilon(x_0)$ if for all $x \geq x_0$ we have

$$E_\psi(x) \leq \varepsilon(x_0).$$

Definition 2.0.6 (Equation (1) of FKS2). $E_\pi(x) = |\pi(x) - \mathrm{Li}(x)|/\mathrm{Li}(x)$.

Definition 2.0.7 (Equation (2) of FKS2). $E_\theta(x) = |\theta(x) - x|/x$

Definition 2.0.8 (Definitions 1, 5, FKS2). We say that E_θ satisfies a *classical bound* with parameters A, B, C, R, x_0 if for all $x \geq x_0$ we have

$$E_\theta(x) \leq A \left(\frac{\log x}{R}\right)^B \exp\left(-C \left(\frac{\log x}{R}\right)^{1/2}\right).$$

We say that it obeys a *numerical bound* with parameter $\varepsilon(x_0)$ if for all $x \geq x_0$ we have

$$E_\theta(x) \leq \varepsilon(x_0).$$

Definition 2.0.9 (Definitions 1, 5, FKS2)**.** We say that E_π satisfies a *classical bound* with parameters A, B, C, R, x_0 if for all $x \geq x_0$ we have

$$E_\pi(x) \leq A \left(\frac{\log x}{R}\right)^B \exp\left(-C \left(\frac{\log x}{R}\right)^{1/2}\right).$$

We say that it obeys a *numerical bound* with parameter $\varepsilon(x_0)$ if for all $x \geq x_0$ we have

$$E_\pi(x) \leq \varepsilon(x_0).$$

Lemma 2.0.1 (Admissible bound decreasing for large x)**.** If $A, B, C, R > 0$ then the classical bound is monotone decreasing for $x \geq \exp(R(2B/C)^2)$.

Proof. Differentiate the bound and check the sign. ☐

Lemma 2.0.2 (Classic bound implies numerical bound)**.** A classical bound for $x \geq x_0$ implies a numerical bound for $x \geq \max(x_0, \exp(R(2B/C)^2))$.

Proof. Immediate from previous lemma ☐

Chapter 3

First approach: Wiener-Ikehara Tauberian theorem

3.1 A Fourier-analytic proof of the Wiener-Ikehara theorem

The Fourier transform of an absolutely integrable function $\psi : \mathbb{R} \to \mathbb{C}$ is defined by the formula

$$\hat{\psi}(u) := \int_{\mathbb{R}} e(-tu)\psi(t)\ dt$$

where $e(\theta) := e^{2\pi i\theta}$.

Let $f : \mathbb{N} \to \mathbb{C}$ be an arithmetic function such that $\sum_{n=1}^{\infty} \frac{|f(n)|}{n^\sigma} < \infty$ for all $\sigma > 1$. Then the Dirichlet series

$$F(s) := \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

is absolutely convergent for $\sigma > 1$.

Lemma 3.1.1 (first-fourier)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is integrable and $x > 0$, then for any $\sigma > 1$

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^\sigma} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) = \int_{\mathbb{R}} F(\sigma + it)\psi(t)x^{it}\ dt.$$

Proof. By the definition of the Fourier transform, the left-hand side expands as

$$\sum_{n=1}^{\infty} \int_{\mathbb{R}} \frac{f(n)}{n^\sigma} \psi(t) e(-\frac{1}{2\pi} t \log \frac{n}{x})\ dt$$

while the right-hand side expands as

$$\int_{\mathbb{R}} \sum_{n=1}^{\infty} \frac{f(n)}{n^{\sigma+it}} \psi(t)x^{it}\ dt.$$

Since

$$\frac{f(n)}{n^\sigma} \psi(t) e(-\frac{1}{2\pi} t \log \frac{n}{x}) = \frac{f(n)}{n^{\sigma+it}} \psi(t)x^{it}$$

the claim then follows from Fubini's theorem. □

Lemma 3.1.2 (second-fourier)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is absolutely integrable and $x > 0$, then for any $\sigma > 1$

$$\int_{-\log x}^{\infty} e^{-u(\sigma-1)} \hat{\psi}(\frac{u}{2\pi})\ du = x^{\sigma-1} \int_{\mathbb{R}} \frac{1}{\sigma+it-1} \psi(t) x^{it}\ dt.$$

Proof. The left-hand side expands as

$$\int_{-\log x}^{\infty} \int_{\mathbb{R}} e^{-u(\sigma-1)} \psi(t) e(-\frac{tu}{2\pi})\ dt\ du$$

so by Fubini's theorem it suffices to verify the identity

$$\begin{aligned} \int_{-\log x}^{\infty} e^{-u(\sigma-1)} e(-\frac{tu}{2\pi})\ du &= \int_{-\log x}^{\infty} e^{(it-\sigma+1)u}\ du \\ &= \frac{1}{it-\sigma+1} e^{(it-\sigma+1)u}\ \Big|_{-\log x}^{\infty} \\ &= x^{\sigma-1} \frac{1}{\sigma+it-1} x^{it} \end{aligned}$$

□

Now let $A \in \mathbb{C}$, and suppose that there is a continuous function $G(s)$ defined on $\mathrm{Re} s \geq 1$ such that $G(s) = F(s) - \frac{A}{s-1}$ whenever $\mathrm{Re} s > 1$. We also make the Chebyshev-type hypothesis

$$\sum_{n \leq x} |f(n)| \ll x \tag{3.1}$$

for all $x \geq 1$ (this hypothesis is not strictly necessary, but simplifies the arguments and can be obtained fairly easily in applications).

Lemma 3.1.3 (Preliminary decay bound I)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is absolutely integrable then

$$|\hat{\psi}(u)| \leq \|\psi\|_1$$

for all $u \in \mathbb{R}$. where C is an absolute constant.

Proof. Immediate from the triangle inequality. □

Lemma 3.1.4 (Preliminary decay bound II)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is absolutely integrable and of bounded variation, then

$$|\hat{\psi}(u)| \leq \|\psi\|_{TV}/2\pi|u|$$

for all non-zero $u \in \mathbb{R}$.

Proof. By Lebesgue–Stiejtes integration by parts we have

$$2\pi i u \hat{\psi}(u) = \int_{\mathbb{R}} e(-tu) d\psi(t)$$

and the claim then follows from the triangle inequality. □

Lemma 3.1.5 (Preliminary decay bound III)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is absolutely integrable, absolutely continuous, and ψ' is of bounded variation, then

$$|\hat{\psi}(u)| \leq \|\psi'\|_{TV}/(2\pi|u|)^2$$

for all non-zero $u \in \mathbb{R}$.

Proof. Should follow from previous lemma. □

Lemma 3.1.6 (Decay bound, alternate form)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is absolutely integrable, absolutely continuous, and ψ' is of bounded variation, then

$$|\hat{\psi}(u)| \leq (\|\psi\|_1 + \|\psi'\|_{TV}/(2\pi)^2)/(1+|u|^2)$$

for all $u \in \mathbb{R}$.

Proof. Should follow from previous lemmas. □

Lemma 3.1.7 (Decay bounds)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is C^2 and obeys the bounds

$$|\psi(t)|, |\psi''(t)| \leq A/(1+|t|^2)$$

for all $t \in \mathbb{R}$, then

$$|\hat{\psi}(u)| \leq CA/(1+|u|^2)$$

for all $u \in \mathbb{R}$, where C is an absolute constant.

Proof. From two integration by parts we obtain the identity

$$(1+u^2)\hat{\psi}(u) = \int_{\mathbf{R}} (\psi(t) - \frac{u}{4\pi^2}\psi''(t))e(-tu)\ dt.$$

Now apply the triangle inequality and the identity $\int_{\mathbf{R}} \frac{dt}{1+t^2}\ dt = \pi$ to obtain the claim with $C = \pi + 1/4\pi$. □

Lemma 3.1.8 (Limiting Fourier identity)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is C^2 and compactly supported and $x \geq 1$, then

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) - A \int_{-\log x}^{\infty} \hat{\psi}(\frac{u}{2\pi})\ du = \int_{\mathbb{R}} G(1+it)\psi(t)x^{it}\ dt.$$

Proof. By Lemma 3.1.1 and Lemma 3.1.2, we know that for any $\sigma > 1$, we have

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^\sigma} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) - Ax^{1-\sigma} \int_{-\log x}^{\infty} e^{-u(\sigma-1)}\hat{\psi}(\frac{u}{2\pi})\ du = \int_{\mathbb{R}} G(\sigma+it)\psi(t)x^{it}\ dt.$$

Now take limits as $\sigma \to 1$ using dominated convergence together with (3.1) and Lemma 3.1.7 to obtain the result. □

Corollary 3.1.1 (Corollary of limiting identity)**.** With the hypotheses as above, we have

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) = A \int_{-\infty}^{\infty} \hat{\psi}(\frac{u}{2\pi})\ du + o(1)$$

as $x \to \infty$.

Proof. Immediate from the Riemann-Lebesgue lemma, and also noting that $\int_{-\infty}^{-\log x} \hat{\psi}(\frac{u}{2\pi})\ du = o(1)$. □

Lemma 3.1.9 (Smooth Urysohn lemma)**.** If I is a closed interval contained in an open interval J, then there exists a smooth function $\Psi : \mathbb{R} \to \mathbb{R}$ with $1_I \leq \Psi \leq 1_J$.

Proof. A standard analysis lemma, which can be proven by convolving 1_K with a smooth approximation to the identity for some interval K between I and J. Note that we have "SmoothBumpFunction"s on smooth manifolds in Mathlib, so this shouldn't be too hard... □

Lemma 3.1.10 (Limiting identity for Schwartz functions)**.** The previous corollary also holds for functions ψ that are assumed to be in the Schwartz class, as opposed to being C^2 and compactly supported.

Proof. For any $R > 1$, one can use a smooth cutoff function (provided by Lemma 3.1.9 to write $\psi = \psi_{\leq R} + \psi_{>R}$, where $\psi_{\leq R}$ is C^2 (in fact smooth) and compactly supported (on $[-R, R]$), and $\psi_{>R}$ obeys bounds of the form

$$|\psi_{>R}(t)|, |\psi''_{>R}(t)| \ll R^{-1}/(1 + |t|^2)$$

where the implied constants depend on ψ. By Lemma 3.1.7 we then have

$$\hat{\psi}_{>R}(u) \ll R^{-1}/(1 + |u|^2).$$

Using this and (3.1) one can show that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}_{>R}(\frac{1}{2\pi} \log \frac{n}{x}), A \int_{-\infty}^{\infty} \hat{\psi}_{>R}(\frac{u}{2\pi})\ du \ll R^{-1}$$

(with implied constants also depending on A), while from Lemma 3.1.1 one has

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}_{\leq R}(\frac{1}{2\pi} \log \frac{n}{x}) = A \int_{-\infty}^{\infty} \hat{\psi}_{\leq R}(\frac{u}{2\pi})\ du + o(1).$$

Combining the two estimates and letting R be large, we obtain the claim. □

Lemma 3.1.11 (Bijectivity of Fourier transform)**.** The Fourier transform is a bijection on the Schwartz class. [Note: only surjectivity is actually used.]

Proof. This is a standard result in Fourier analysis. It can be proved here by appealing to Mellin inversion, Theorem **??**. In particular, given f in the Schwartz class, let $F : \mathbb{R}_+ \to \mathbb{C} : x \mapsto f(\log x)$ be a function in the "Mellin space"; then the Mellin transform of F on the imaginary axis $s = it$ is the Fourier transform of f. The Mellin inversion theorem gives Fourier inversion. □

Corollary 3.1.2 (Smoothed Wiener-Ikehara)**.** If $\Psi : (0, \infty) \to \mathbb{C}$ is smooth and compactly supported away from the origin, then,

$$\sum_{n=1}^{\infty} f(n)\Psi(\frac{n}{x}) = Ax \int_0^{\infty} \Psi(y)\ dy + o(x)$$

as $x \to \infty$.

Proof. By Lemma 3.1.11, we can write

$$y\Psi(y) = \hat{\psi}(\frac{1}{2\pi} \log y)$$

for all $y > 0$ and some Schwartz function ψ. Making this substitution, the claim is then equivalent after standard manipulations to

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) = A \int_{-\infty}^{\infty} \hat{\psi}(\frac{u}{2\pi})\ du + o(1)$$

and the claim follows from Lemma 3.1.10. ☐

Now we add the hypothesis that $f(n) \geq 0$ for all n.

Proposition 3.1.1 (Wiener-Ikehara in an interval)**.** For any closed interval $I \subset (0, +\infty)$, we have

$$\sum_{n=1}^{\infty} f(n) 1_I(\frac{n}{x}) = Ax|I| + o(x).$$

Proof. Use Lemma 3.1.9 to bound 1_I above and below by smooth compactly supported functions whose integral is close to the measure of $|I|$, and use the non-negativity of f. ☐

Corollary 3.1.3 (Wiener-Ikehara Theorem (1))**.** We have

$$\sum_{n \leq x} f(n) = Ax + o(x).$$

Proof. Apply the preceding proposition with $I = [\varepsilon, 1]$ and then send ε to zero (using (3.1) to control the error). ☐

3.2 Weak PNT

Theorem 3.2.1 (WeakPNT)**.** We have

$$\sum_{n \leq x} \Lambda(n) = x + o(x).$$

Proof. Already done by Stoll, assuming Wiener-Ikehara. ☐

3.3 Removing the Chebyshev hypothesis

In this section we do *not* assume the bound (3.1), but instead derive it from the other hypotheses.

Lemma 3.3.1 (limiting-fourier-variant)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is C^2 and compactly supported with f and $\hat{\psi}$ non-negative, and $0 < x$, then

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x}) - A \int_{-\log x}^{\infty} \hat{\psi}(\frac{u}{2\pi})\ du = \int_{\mathbb{R}} G(1 + it) \psi(t) x^{it}\ dt.$$

Proof. Repeat the proof of Lemma 3.3.1, but use monotone convergence instead of dominated convergence. (The proof should be simpler, as one no longer needs to establish domination for the sum.) ☐

Corollary 3.3.1 (crude-upper-bound)**.** If $\psi : \mathbb{R} \to \mathbb{C}$ is C^2 and compactly supported with f and $\hat{\psi}$ non-negative, then there exists a constant B such that

$$|\sum_{n=1}^{\infty} \frac{f(n)}{n} \hat{\psi}(\frac{1}{2\pi} \log \frac{n}{x})| \leq B$$

for all $x > 0$.

Proof. This readily follows from the previous lemma and the triangle inequality. $\square$

Corollary 3.3.2 (auto-cheby)**.** One has

$$\sum_{n \leq x} f(n) = O(x)$$

for all $x \geq 1$.

Proof. By applying Corollary 3.3.1 for a specific compactly supported function ψ, one can obtain a bound of the form $\sum_{(1-\varepsilon)x<n\leq x} f(n) = O(x)$ for all x and some absolute constant ε (which can be made explicit).

If C is a sufficiently large constant, the claim $|\sum_{n\leq x} f(n)| \leq Cx$ can now be proven by strong induction on x, as the claim for $(1-\varepsilon)x$ implies the claim for x by the triangle inequality (and the claim is trivial for $x < 1$). $\square$

Theorem 3.3.1 (Wiener-Ikehara Theorem (2))**.** We have

$$\sum_{n \leq x} f(n) = Ax + o(x).$$

Proof. Use Corollary 3.3.2 to remove the Chebyshev hypothesis in Theorem 3.1.3. $\square$

3.4 The prime number theorem in arithmetic progressions

Lemma 3.4.1 (WeakPNT-character)**.** If $q \geq 1$ and a is coprime to q, and $\mathrm{Re}s > 1$, we have

$$\sum_{n:n=a\ (q)} \frac{\Lambda(n)}{n^s} = -\frac{1}{\varphi(q)} \sum_{\chi\ (q)} \overline{\chi(a)} \frac{L'(s,\chi)}{L(s,\chi)}.$$

Proof. From the Fourier inversion formula on the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^\times$, we have

$$1_{n=a\ (q)} = \frac{\varphi(q)}{q} \sum_{\chi\ (q)} \overline{\chi(a)}\chi(n).$$

On the other hand, from standard facts about L-series we have for each character χ that

$$\sum_n \frac{\Lambda(n)\chi(n)}{n^s} = -\frac{L'(s,\chi)}{L(s,\chi)}.$$

Combining these two facts, we obtain the claim. $\square$

Proposition 3.4.1 (WeakPNT-AP-prelim)**.** If $q \geq 1$ and a is coprime to q, the Dirichlet series $\sum_{n \leq x : n = a\ (q)} \frac{\Lambda(n)}{n^s}$ converges for $\mathrm{Re}(s) > 1$ to $\frac{1}{\varphi(q)} \frac{1}{s-1} + G(s)$ where G has a continuous extension to $\mathrm{Re}(s) = 1$.

Proof. We expand out the left-hand side using Lemma 3.4.1. The contribution of the non-principal characters χ extend continuously to $\mathrm{Re}(s) = 1$ thanks to the non-vanishing of $L(s, \chi)$ on this line (which should follow from another component of this project), so it suffices to show that for the principal character χ_0, that

$$-\frac{L'(s, \chi_0)}{L(s, \chi_0)} - \frac{1}{s-1}$$

also extends continuously here. But we already know that

$$-\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1}$$

extends, and from Euler product machinery one has the identity

$$\frac{L'(s, \chi_0)}{L(s, \chi_0)} = \frac{\zeta'(s)}{\zeta(s)} + \sum_{p|q} \frac{\log p}{p^s - 1}.$$

Since there are only finitely many primes dividing q, and each summand $\frac{\log p}{p^s-1}$ extends continuously, the claim follows. ☐

Theorem 3.4.1 (WeakPNT-AP)**.** If $q \geq 1$ and a is coprime to q, we have

$$\sum_{n \leq x : n = a\ (q)} \Lambda(n) = \frac{x}{\varphi(q)} + o(x).$$

Proof. Apply Theorem 3.1.3 (or Theorem 3.3.1 to avoid checking the Chebyshev condition) using Proposition 3.4.1. ☐

3.5 The Chebotarev density theorem: the case of cyclotomic extensions

In this section, K is a number field, $L = K(\mu_m)$ for some natural number m, and $G = Gal(K/L)$.

The goal here is to prove the Chebotarev density theorem for the case of cyclotomic extensions.

Lemma 3.5.1 (Dedekind-factor)**.** We have

$$\zeta_L(s) = \prod_{\chi} L(\chi, s)$$

for $\Re(s) > 1$, where χ runs over homomorphisms from G to $\mathbb{C}^\times$ and L is the Artin L-function.

Proof. See Propositions 7.1.16, 7.1.19 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`. ☐

Lemma 3.5.2 (Simple pole)**.** ζ_L has a simple pole at $s = 1$.

Proof. See Theorem 7.1.12 of `https://www.math.ucla.edu/~sharifi/algnum.pdf`. □

Lemma 3.5.3 (Dedekind-nonvanishing)**.** For any non-principal character χ of $Gal(K/L)$, $L(\chi, s)$ does not vanish for $\Re(s) = 1$.

Proof. For $s = 1$, this will follow from Lemmas 3.5.1, 3.5.2. For the rest of the line, one should be able to adapt the arguments for the Dirichlet L-function. □

3.6 The Chebotarev density theorem: the case of abelian extensions

(Use the arguments in Theorem 7.2.2 of `https://www.math.ucla.edu/~sharifi/algnum.pdf` to extend the previous results to abelian extensions (actually just cyclic extensions would suffice))

3.7 The Chebotarev density theorem: the general case

(Use the arguments in Theorem 7.2.2 of `https://www.math.ucla.edu/~sharifi/algnum.pdf` to extend the previous results to arbitrary extensions

Lemma 3.7.1 (PNT for one character)**.** For any non-principal character χ of $Gal(K/L)$,

$$\sum_{N\mathfrak{p} \leq x} \chi(\mathfrak{p}) \log N\mathfrak{p} = o(x).$$

Proof. This should follow from Lemma 3.5.3 and the arguments for the Dirichlet L-function. (It may be more convenient to work with a von Mangoldt type function instead of $\log N\mathfrak{p}$). □

Chapter 4

Second approach

4.1 Residue calculus on rectangles

This files gathers definitions and basic properties about rectangles.

The border of a rectangle is the union of its four sides.

Definition 4.1.1 (RectangleBorder)**.** A Rectangle's border, given corners z and w is the union of the four sides.

Definition 4.1.2 (RectangleIntegral)**.** A RectangleIntegral of a function f is one over a rectangle determined by z and w in $\mathbb{C}$. We will sometimes denote it by $\int_z^w f$. (There is also a primed version, which is $1/(2\pi i)$ times the original.)

Definition 4.1.3 (UpperUIntegral)**.** An UpperUIntegral of a function f comes from $\sigma + i\infty$ down to $\sigma + iT$, over to $\sigma' + iT$, and back up to $\sigma' + i\infty$.

Definition 4.1.4 (LowerUIntegral)**.** A LowerUIntegral of a function f comes from $\sigma - i\infty$ up to $\sigma - iT$, over to $\sigma' - iT$, and back down to $\sigma' - i\infty$.

It is very convenient to define integrals along vertical lines in the complex plane, as follows.

Definition 4.1.5 (VerticalIntegral)**.** Let f be a function from $\mathbb{C}$ to $\mathbb{C}$, and let σ be a real number. Then we define

$$\int_{(\sigma)} f(s)ds = \int_{\sigma - i\infty}^{\sigma + i\infty} f(s)ds.$$

We also have a version with a factor of $1/(2\pi i)$.

Lemma 4.1.1 (DiffVertRect-eq-UpperLowerUs)**.** The difference of two vertical integrals and a rectangle is the difference of an upper and a lower U integrals.

Proof. Follows directly from the definitions. □

Theorem 4.1.1 (existsDifferentiableOn-of-bddAbove)**.** If f is differentiable on a set s except at $c \in s$, and f is bounded above on $s \setminus \{c\}$, then there exists a differentiable function g on s such that f and g agree on $s \setminus \{c\}$.

Proof. This is the Riemann Removable Singularity Theorem, slightly rephrased from what's in Mathlib. (We don't care what the function g is, just that it's holomorphic.) □

Theorem 4.1.2 (HolomorphicOn.vanishesOnRectangle)**.** *If f is holomorphic on a rectangle z and w, then the integral of f over the rectangle with corners z and w is 0.*

Proof. This is in a Mathlib PR. □

The next lemma allows to zoom a big rectangle down to a small square, centered at a pole.

Lemma 4.1.2 (RectanglePullToNhdOfPole)**.** *If f is holomorphic on a rectangle z and w except at a point p, then the integral of f over the rectangle with corners z and w is the same as the integral of f over a small square centered at p.*

Proof. Chop the big rectangle with two vertical cuts and two horizontal cuts into smaller rectangles, the middle one being the desired square. The integral over each of the outer rectangles vanishes, since f is holomorphic there. (The constant c being "small enough" here just means that the inner square is strictly contained in the big rectangle.) □

Lemma 4.1.3 (ResidueTheoremAtOrigin)**.** *The rectangle (square) integral of $f(s) = 1/s$ with corners $-1 - i$ and $1 + i$ is equal to $2\pi i$.*

Proof. This is a special case of the more general result above. □

Lemma 4.1.4 (ResidueTheoremOnRectangleWithSimplePole)**.** *Suppose that f is a holomorphic function on a rectangle, except for a simple pole at p. By the latter, we mean that there is a function g holomorphic on the rectangle such that, $f = g + A/(s - p)$ for some $A \in \mathbb{C}$. Then the integral of f over the rectangle is A.*

Proof. Replace f with $g + A/(s - p)$ in the integral. The integral of g vanishes by Lemma 4.1.2. To evaluate the integral of $1/(s - p)$, pull everything to a square about the origin using Lemma 4.1.2, and rescale by c; what remains is handled by Lemma 4.1.3. □

4.2 Perron Formula

In this section, we prove the Perron formula, which plays a key role in our proof of Mellin inversion.

The following is preparatory material used in the proof of the Perron formula, see Lemma 4.2.16.

Lemma 4.2.1 (zeroTendstoDiff)**.** *If the limit of 0 is $L_1 - L_2$, then $L_1 = L_2$.*

Proof. Obvious. □

Lemma 4.2.2 (RectangleIntegral-tendsTo-VerticalIntegral)**.** *Let $\sigma, \sigma' \in \mathbb{R}$, and $f : \mathbb{C} \to \mathbb{C}$ such that the vertical integrals $\int_{(\sigma)} f(s)ds$ and $\int_{(\sigma')} f(s)ds$ exist and the horizontal integral $\int_{(\sigma)}^{\sigma'} f(x + yi)dx$ vanishes as $y \to \pm\infty$. Then the limit of rectangle integrals*

$$\lim_{T\to\infty} \int_{\sigma - iT}^{\sigma' + iT} f(s)ds = \int_{(\sigma')} f(s)ds - \int_{(\sigma)} f(s)ds.$$

Proof. Almost by definition. □

Lemma 4.2.3 (RectangleIntegral-tendsTo-UpperU)**.** Let $\sigma, \sigma' \in \mathbb{R}$, and $f : \mathbb{C} \to \mathbb{C}$ such that the vertical integrals $\int_{(\sigma)} f(s)ds$ and $\int_{(\sigma')} f(s)ds$ exist and the horizontal integral $\int_{(\sigma)}^{\sigma'} f(x+yi)dx$ vanishes as $y \to \pm\infty$. Then the limit of rectangle integrals

$$\int_{\sigma+iT}^{\sigma'+iU} f(s)ds$$

as $U \to \infty$ is the "UpperUIntegral" of f.

Proof. Almost by definition. □

Lemma 4.2.4 (RectangleIntegral-tendsTo-LowerU)**.** Let $\sigma, \sigma' \in \mathbb{R}$, and $f : \mathbb{C} \to \mathbb{C}$ such that the vertical integrals $\int_{(\sigma)} f(s)ds$ and $\int_{(\sigma')} f(s)ds$ exist and the horizontal integral $\int_{(\sigma)}^{\sigma'} f(x+yi)dx$ vanishes as $y \to -\infty$. Then the limit of rectangle integrals

$$\int_{\sigma-iU}^{\sigma'-iT} f(s)ds$$

as $U \to \infty$ is the "LowerUIntegral" of f.

Proof. Almost by definition. □

TODO : Move to general section

Lemma 4.2.5 (limitOfConstant)**.** Let $a : \mathbb{R} \to \mathbb{C}$ be a function, and let $\sigma > 0$ be a real number. Suppose that, for all $\sigma, \sigma' > 0$, we have $a(\sigma') = a(\sigma)$, and that $\lim_{\sigma\to\infty} a(\sigma) = 0$. Then $a(\sigma) = 0$.

Proof.

$$\begin{aligned} \lim_{\sigma'\to\infty} a(\sigma) &= \lim_{\sigma'\to\infty} a(\sigma') \\ &= 0 \end{aligned}$$

□

Lemma 4.2.6 (limitOfConstantLeft)**.** Let $a : \mathbb{R} \to \mathbb{C}$ be a function, and let $\sigma < -3/2$ be a real number. Suppose that, for all $\sigma, \sigma' > 0$, we have $a(\sigma') = a(\sigma)$, and that $\lim_{\sigma\to-\infty} a(\sigma) = 0$. Then $a(\sigma) = 0$.

Proof.

$$\begin{aligned} \lim_{\sigma'\to-\infty} a(\sigma) &= \lim_{\sigma'\to-\infty} a(\sigma') \\ &= 0 \end{aligned}$$

□

Lemma 4.2.7 (tendsto-rpow-atTop-nhds-zero-of-norm-lt-one)**.** Let $x > 0$ and $x < 1$. Then

$$\lim_{\sigma\to\infty} x^\sigma = 0.$$

Proof. Standard. □

Lemma 4.2.8 (tendsto-rpow-atTop-nhds-zero-of-norm-gt-one)**.** Let $x > 1$. Then

$$\lim_{\sigma \to -\infty} x^\sigma = 0.$$

Proof. Standard. ☐

Lemma 4.2.9 (isHolomorphicOn)**.** Let $x > 0$. Then the function $f(s) = x^s/(s(s+1))$ is holomorphic on the half-plane $\{s \in \mathbb{C} : \Re(s) > 0\}$.

Proof. Composition of differentiabilities. ☐

Lemma 4.2.10 (integralPosAux)**.** The integral

$$\int_{\mathbb{R}} \frac{1}{|(1+t^2)(2+t^2)|^{1/2}} dt$$

is positive (and hence convergent - since a divergent integral is zero in Lean, by definition).

Proof. This integral is between $\frac{1}{2}$ and 1 of the integral of $\frac{1}{1+t^2}$, which is π. ☐

Lemma 4.2.11 (vertIntBound)**.** Let $x > 0$ and $\sigma > 1$. Then

$$\left| \int_{(\sigma)} \frac{x^s}{s(s+1)} ds \right| \leq x^\sigma \int_{\mathbb{R}} \frac{1}{|(1+t^2)(2+t^2)|^{1/2}} dt.$$

Proof. Triangle inequality and pointwise estimate. ☐

Lemma 4.2.12 (vertIntBoundLeft)**.** Let $x > 1$ and $\sigma < -3/2$. Then

$$\left| \int_{(\sigma)} \frac{x^s}{s(s+1)} ds \right| \leq x^\sigma \int_{\mathbb{R}} \frac{1}{|(1/4+t^2)(2+t^2)|^{1/2}} dt.$$

Proof. Triangle inequality and pointwise estimate. ☐

Lemma 4.2.13 (isIntegrable)**.** Let $x > 0$ and $\sigma \in \mathbb{R}$. Then

$$\int_{\mathbb{R}} \frac{x^{\sigma+it}}{(\sigma+it)(1+\sigma+it)} dt$$

is integrable.

Proof. By 4.2.9, f is continuous, so it is integrable on any interval.
Also, $|f(x)| = \Theta(x^{-2})$ as $x \to \infty$,
and $|f(-x)| = \Theta(x^{-2})$ as $x \to \infty$.
Since $g(x) = x^{-2}$ is integrable on $[a, \infty)$ for any $a > 0$, we conclude. ☐

Lemma 4.2.14 (tendsto-zero-Lower)**.** Let $x > 0$ and $\sigma', \sigma'' \in \mathbb{R}$. Then

$$\int_{\sigma'}^{\sigma''} \frac{x^{\sigma+it}}{(\sigma+it)(1+\sigma+it)} d\sigma$$

goes to 0 as $t \to -\infty$.

Proof. The numerator is bounded and the denominator tends to infinity. ☐

Lemma 4.2.15 (tendsto-zero-Upper)**.** Let $x > 0$ and $\sigma', \sigma'' \in \mathbb{R}$. Then

$$\int_{\sigma'}^{\sigma''} \frac{x^{\sigma+it}}{(\sigma+it)(1+\sigma+it)} d\sigma$$

goes to 0 as $t \to \infty$.

Proof. The numerator is bounded and the denominator tends to infinity. □

We are ready for the first case of the Perron formula, namely when $x < 1$:

Lemma 4.2.16 (formulaLtOne)**.** For $x > 0$, $\sigma > 0$, and $x < 1$, we have

$$\frac{1}{2\pi i} \int_{(\sigma)} \frac{x^s}{s(s+1)} ds = 0.$$

Proof. Let $f(s) = x^s/(s(s+1))$. Then f is holomorphic on the half-plane $\{s \in \mathbb{C} : \Re(s) > 0\}$. The rectangle integral of f with corners $\sigma - iT$ and $\sigma + iT$ is zero. The limit of this rectangle integral as $T \to \infty$ is $\int_{(\sigma')} - \int_{(\sigma)}$. Therefore, $\int_{(\sigma')} = \int_{(\sigma)}$.

But we also have the bound $\int_{(\sigma')} \leq x^{\sigma'} * C$, where $C = \int_{\mathbb{R}} \frac{1}{|(1+t)(1+t+1)|} dt$.

Therefore $\int_{(\sigma')} \to 0$ as $\sigma' \to \infty$.

So pulling contours gives $\int_{(\sigma)} = 0$. □

The second case is when $x > 1$. Here are some auxiliary lemmata for the second case. TODO: Move to more general section

Lemma 4.2.17 (keyIdentity)**.** Let $x \in \mathbb{R}$ and $s \neq 0, -1$. Then

$$\frac{x^\sigma}{s(1+s)} = \frac{x^\sigma}{s} - \frac{x^\sigma}{1+s}$$

Proof. By ring. □

Lemma 4.2.18 (diffBddAtZero)**.** Let $x > 0$. Then for $0 < c < 1/2$, we have that the function

$$s \mapsto \frac{x^s}{s(s+1)} - \frac{1}{s}$$

is bounded above on the rectangle with corners at $-c - i * c$ and $c + i * c$ (except at $s = 0$).

Proof. Applying Lemma 4.2.17, the function $s \mapsto x^s/s(s+1) - 1/s = x^s/s - x^0/s - x^s/(1+s)$. The last term is bounded for s away from -1. The first two terms are the difference quotient of the function $s \mapsto x^s$ at 0; since it's differentiable, the difference remains bounded as $s \to 0$. □

Lemma 4.2.19 (diffBddAtNegOne)**.** Let $x > 0$. Then for $0 < c < 1/2$, we have that the function

$$s \mapsto \frac{x^s}{s(s+1)} - \frac{-x^{-1}}{s+1}$$

is bounded above on the rectangle with corners at $-1 - c - i * c$ and $-1 + c + i * c$ (except at $s = -1$).

Proof. Applying Lemma 4.2.17, the function $s \mapsto x^s/s(s+1) - x^{-1}/(s+1) = x^s/s - x^s/(s+1) - (-x^{-1})/(s+1)$. The first term is bounded for s away from 0. The last two terms are the difference quotient of the function $s \mapsto x^s$ at -1; since it's differentiable, the difference remains bounded as $s \to -1$. ☐

Lemma 4.2.20 (residueAtZero)**.** Let $x > 0$. Then for all sufficiently small $c > 0$, we have that

$$\frac{1}{2\pi i}\int_{-c-i*c}^{c+i*c} \frac{x^s}{s(s+1)}ds = 1.$$

Proof. For $c > 0$ sufficiently small,

$x^s/(s(s+1))$ is equal to $1/s$ plus a function, g, say, holomorphic in the whole rectangle (by Lemma 4.2.18).

Now apply Lemma 4.1.4. ☐

Lemma 4.2.21 (residueAtNegOne)**.** Let $x > 0$. Then for all sufficiently small $c > 0$, we have that

$$\frac{1}{2\pi i}\int_{-c-i*c-1}^{c+i*c-1} \frac{x^s}{s(s+1)}ds = -\frac{1}{x}.$$

Proof. Compute the integral. ☐

Lemma 4.2.22 (residuePull1)**.** For $x > 1$ (of course $x > 0$ would suffice) and $\sigma > 0$, we have

$$\frac{1}{2\pi i}\int_{(\sigma)} \frac{x^s}{s(s+1)}ds = 1 + \frac{1}{2\pi i}\int_{(-1/2)} \frac{x^s}{s(s+1)}ds.$$

Proof. We pull to a square with corners at $-c - i*c$ and $c + i*c$ for $c > 0$ sufficiently small. By Lemma 4.2.20, the integral over this square is equal to 1. ☐

Lemma 4.2.23 (residuePull2)**.** For $x > 1$, we have

$$\frac{1}{2\pi i}\int_{(-1/2)} \frac{x^s}{s(s+1)}ds = -1/x + \frac{1}{2\pi i}\int_{(-3/2)} \frac{x^s}{s(s+1)}ds.$$

Proof. Pull contour from $(-1/2)$ to $(-3/2)$. ☐

Lemma 4.2.24 (contourPull3)**.** For $x > 1$ and $\sigma < -3/2$, we have

$$\frac{1}{2\pi i}\int_{(-3/2)} \frac{x^s}{s(s+1)}ds = \frac{1}{2\pi i}\int_{(\sigma)} \frac{x^s}{s(s+1)}ds.$$

Proof. Pull contour from $(-3/2)$ to (σ). ☐

Lemma 4.2.25 (formulaGtOne)**.** For $x > 1$ and $\sigma > 0$, we have

$$\frac{1}{2\pi i}\int_{(\sigma)} \frac{x^s}{s(s+1)}ds = 1 - 1/x.$$

Proof. Let $f(s) = x^s/(s(s+1))$. Then f is holomorphic on $\mathbb{C} \setminus 0, -1$.

First pull the contour from (σ) to $(-1/2)$, picking up a residue 1 at $s = 0$.

Next pull the contour from $(-1/2)$ to $(-3/2)$, picking up a residue $-1/x$ at $s = -1$.

Then pull the contour all the way to (σ') with $\sigma' < -3/2$.

For $\sigma' < -3/2$, the integral is bounded by $x^{\sigma'} \int_{\mathbb{R}} \frac{1}{|(1+t^2)(2+t^2)|^{1/2}} dt$.

Therefore $\int_{(\sigma')} \to 0$ as $\sigma' \to \infty$.

So pulling contours gives $\int_{(-3/2)} = 0$. $\square$

The two together give the Perron formula. (Which doesn't need to be a separate lemma.)

For $x > 0$ and $\sigma > 0$, we have

$$\frac{1}{2\pi i} \int_{(\sigma)} \frac{x^s}{s(s+1)} ds = \begin{cases} 1 - \frac{1}{x} & \text{if } x > 1 \\ 0 & \text{if } x < 1 \end{cases}.$$

4.3 Mellin transforms

Lemma 4.3.1 (PartialIntegration)**.** Let f, g be once differentiable functions from $\mathbb{R}_{>0}$ to $\mathbb{C}$ so that fg' and $f'g$ are both integrable, and $f \cdot g(x) \to 0$ as $x \to 0^+, \infty$. Then

$$\int_0^\infty f(x)g'(x)dx = -\int_0^\infty f'(x)g(x)dx.$$

Proof. Partial integration. $\square$

In this section, we define the Mellin transform (already in Mathlib, thanks to David Loeffler), prove its inversion formula, and derive a number of important properties of some special functions and bumpfunctions.

Def: (Already in Mathlib) Let f be a function from $\mathbb{R}_{>0}$ to $\mathbb{C}$. We define the Mellin transform of f to be the function $\mathcal{M}(f)$ from $\mathbb{C}$ to $\mathbb{C}$ defined by

$$\mathcal{M}(f)(s) = \int_0^\infty f(x)x^{s-1}dx.$$

[Note: My preferred way to think about this is that we are integrating over the multiplicative group $\mathbb{R}_{>0}$, multiplying by a (not necessarily unitary!) character $|\cdot|^s$, and integrating with respect to the invariant Haar measure dx/x. This is very useful in the kinds of calculations carried out below. But may be more difficult to formalize as things now stand. So we might have clunkier calculations, which "magically" turn out just right - of course they're explained by the aforementioned structure...]

Finally, we need Mellin Convolutions and properties thereof.

Definition 4.3.1 (MellinConvolution)**.** Let f and g be functions from $\mathbb{R}_{>0}$ to $\mathbb{C}$. Then we define the Mellin convolution of f and g to be the function $f * g$ from $\mathbb{R}_{>0}$ to $\mathbb{C}$ defined by

$$(f * g)(x) = \int_0^\infty f(y)g(x/y)\frac{dy}{y}.$$

Let us start with a simple property of the Mellin convolution.

Lemma 4.3.2 (MellinConvolutionSymmetric)**.** Let f and g be functions from $\mathbb{R}_{>0}$ to $\mathbb{R}$ or $\mathbb{C}$, for $x \neq 0$,

$$(f * g)(x) = (g * f)(x).$$

Proof. By Definition 4.3.1,

$$(f * g)(x) = \int_0^\infty f(y)g(x/y)\frac{dy}{y}$$

in which we change variables to $z = x/y$:

$$(f * g)(x) = \int_0^\infty f(x/z)g(z)\frac{dz}{z} = (g * f)(x).$$

$\square$

The Mellin transform of a convolution is the product of the Mellin transforms.

Theorem 4.3.1 (MellinConvolutionTransform)**.** Let f and g be functions from $\mathbb{R}_{>0}$ to $\mathbb{C}$ such that

$$(x, y) \mapsto f(y)\frac{g(x/y)}{y}x^{s-1} \tag{4.1}$$

is absolutely integrable on $[0, \infty)^2$. Then

$$\mathcal{M}(f * g)(s) = \mathcal{M}(f)(s)\mathcal{M}(g)(s).$$

Proof. By Definitions **??** and 4.3.1

$$\mathcal{M}(f * g)(s) = \int_0^\infty \int_0^\infty f(y)g(x/y)x^{s-1}\frac{dy}{y}dx$$

By (4.1) and Fubini's theorem,

$$\mathcal{M}(f * g)(s) = \int_0^\infty \int_0^\infty f(y)g(x/y)x^{s-1}dx\frac{dy}{y}$$

in which we change variables from x to $z = x/y$:

$$\mathcal{M}(f * g)(s) = \int_0^\infty \int_0^\infty f(y)g(z)y^{s-1}z^{s-1}dzdy$$

which, by Definition **??**, is

$$\mathcal{M}(f * g)(s) = \mathcal{M}(f)(s)\mathcal{M}(g)(s).$$

$\square$

The ν function has Mellin transform $\mathcal{M}(\nu)(s)$ which is entire and decays (at least) like $1/|s|$.

[Of course it decays faster than any power of $|s|$, but it turns out that we will just need one power.]

Theorem 4.3.2 (MellinOfPsi)**.** The Mellin transform of ν is

$$\mathcal{M}(\nu)(s) = O\left(\frac{1}{|s|}\right),$$

as $|s| \to \infty$ with $\sigma_1 \le \Re(s) \le 2$.

Proof. Integrate by parts:

$$\left| \int_0^\infty \nu(x) x^s \frac{dx}{x} \right| = \left| - \int_0^\infty \nu'(x) \frac{x^s}{s} dx \right|$$

$$\leq \frac{1}{|s|} \int_{1/2}^2 |\nu'(x)| x^{\Re(s)} dx.$$

Since $\Re(s)$ is bounded, the right-hand side is bounded by a constant times $1/|s|$. $\square$

We can make a delta spike out of this bumpfunction, as follows.

Definition 4.3.2 (DeltaSpike)**.** Let ν be a bumpfunction supported in $[1/2, 2]$. Then for any $\epsilon > 0$, we define the delta spike ν_ϵ to be the function from $\mathbb{R}_{>0}$ to $\mathbb{C}$ defined by

$$\nu_\epsilon(x) = \frac{1}{\epsilon} \nu \left(x^{\frac{1}{\epsilon}} \right).$$

This spike still has mass one:

Lemma 4.3.3 (DeltaSpikeMass)**.** For any $\epsilon > 0$, we have

$$\int_0^\infty \nu_\epsilon(x) \frac{dx}{x} = 1.$$

Proof. Substitute $y = x^{1/\epsilon}$, and use the fact that ν has mass one, and that dx/x is Haar measure. $\square$

The Mellin transform of the delta spike is easy to compute.

Theorem 4.3.3 (MellinOfDeltaSpike)**.** For any $\epsilon > 0$, the Mellin transform of ν_ϵ is

$$\mathcal{M}(\nu_\epsilon)(s) = \mathcal{M}(\nu)(\epsilon s).$$

Proof. Substitute $y = x^{1/\epsilon}$, use Haar measure; direct calculation. $\square$

In particular, for $s = 1$, we have that the Mellin transform of ν_ϵ is $1 + O(\epsilon)$.

Corollary 4.3.1 (MellinOfDeltaSpikeAt1)**.** For any $\epsilon > 0$, we have

$$\mathcal{M}(\nu_\epsilon)(1) = \mathcal{M}(\nu)(\epsilon).$$

Proof. This is immediate from the above theorem. $\square$

Lemma 4.3.4 (MellinOfDeltaSpikeAt1-asymp)**.** As $\epsilon \to 0$, we have

$$\mathcal{M}(\nu_\epsilon)(1) = 1 + O(\epsilon).$$

Proof. By Lemma 4.3.1,

$$\mathcal{M}(\nu_\epsilon)(1) = \mathcal{M}(\nu)(\epsilon)$$

which by Definition **??** is

$$\mathcal{M}(\nu)(\epsilon) = \int_0^\infty \nu(x) x^{\epsilon - 1} dx.$$

Since $\nu(x)x^{\epsilon-1}$ is integrable (because ν is continuous and compactly supported),

$$\mathcal{M}(\nu)(\epsilon) - \int_0^\infty \nu(x)\frac{dx}{x} = \int_0^\infty \nu(x)(x^{\epsilon-1} - x^{-1})dx.$$

By Taylor's theorem,

$$x^{\epsilon-1} - x^{-1} = O(\epsilon)$$

so, since ν is absolutely integrable,

$$\mathcal{M}(\nu)(\epsilon) - \int_0^\infty \nu(x)\frac{dx}{x} = O(\epsilon).$$

We conclude the proof using Theorem 4.3.5. □

Let $1_{(0,1]}$ be the function from $\mathbb{R}_{>0}$ to $\mathbb{C}$ defined by

$$1_{(0,1]}(x) = \begin{cases} 1 & \text{if } x \leq 1 \\ 0 & \text{if } x > 1 \end{cases}.$$

This has Mellin transform: [Note: this already exists in mathlib]

Theorem 4.3.4 (MellinOf1)**.** The Mellin transform of $1_{(0,1]}$ is

$$\mathcal{M}(1_{(0,1]})(s) = \frac{1}{s}.$$

Proof. This is a straightforward calculation. □

What will be essential for us is properties of the smooth version of $1_{(0,1]}$, obtained as the Mellin convolution of $1_{(0,1]}$ with ν_ϵ.

Definition 4.3.3 (Smooth1)**.** Let $\epsilon > 0$. Then we define the smooth function $\widetilde{1_\epsilon}$ from $\mathbb{R}_{>0}$ to $\mathbb{C}$ by

$$\widetilde{1_\epsilon} = 1_{(0,1]} * \nu_\epsilon.$$

Proof. Let $c := 2^\epsilon > 1$, in terms of which we wish to prove

$$-1 < c \log c - c.$$

Letting $f(x) := x \log x - x$, we can rewrite this as $f(1) < f(c)$. Since

$$\frac{d}{dx} f(x) = \log x > 0,$$

f is monotone increasing on $[1, \infty)$, and we are done. □

In particular, we have the following two properties.

Lemma 4.3.5 (Smooth1Properties-below)**.** Fix $\epsilon > 0$. There is an absolute constant $c > 0$ so that: If $0 < x \leq (1 - c\epsilon)$, then

$$\widetilde{1_\epsilon}(x) = 1.$$

Proof. Opening the definition, we have that the Mellin convolution of $1_{(0,1]}$ with ν_ϵ is

$$\int_0^\infty 1_{(0,1]}(y)\nu_\epsilon(x/y)\frac{dy}{y} = \int_0^1 \nu_\epsilon(x/y)\frac{dy}{y}.$$

The support of ν_ϵ is contained in $[1/2^\epsilon, 2^\epsilon]$, so it suffices to consider $y \in [1/2^\epsilon x, 2^\epsilon x]$ for nonzero contributions. If $x < 2^{-\epsilon}$, then the integral is the same as that over $(0, \infty)$:

$$\int_0^1 \nu_\epsilon(x/y)\frac{dy}{y} = \int_0^\infty \nu_\epsilon(x/y)\frac{dy}{y},$$

in which we change variables to $z = x/y$ (using $x > 0$):

$$\int_0^\infty \nu_\epsilon(x/y)\frac{dy}{y} = \int_0^\infty \nu_\epsilon(z)\frac{dz}{z},$$

which is equal to one by Lemma 4.3.3. We then choose

$$c := \log 2,$$

which satisfies

$$c > \frac{1 - 2^{-\epsilon}}{\epsilon}$$

by Lemma **??**, so

$$1 - c\epsilon < 2^{-\epsilon}.$$

□

Lemma 4.3.6 (Smooth1Properties-above)**.** Fix $0 < \epsilon < 1$. There is an absolute constant $c > 0$ so that: if $x \geq (1 + c\epsilon)$, then

$$\widetilde{1_\epsilon}(x) = 0.$$

Proof. Again the Mellin convolution is

$$\int_0^1 \nu_\epsilon(x/y)\frac{dy}{y},$$

but now if $x > 2^\epsilon$, then the support of ν_ϵ is disjoint from the region of integration, and hence the integral is zero. We choose

$$c := 2 \log 2.$$

By Lemma **??**,

$$c > 2\frac{1 - 2^{-\epsilon}}{\epsilon} > 2^\epsilon\frac{1 - 2^{-\epsilon}}{\epsilon} = \frac{2^\epsilon - 1}{\epsilon},$$

so

$$1 + c\epsilon > 2^\epsilon.$$

□

Lemma 4.3.7 (Smooth1Nonneg)**.** If ν is nonnegative, then $\widetilde{1_\epsilon}(x)$ is nonnegative.

Proof. By Definitions 4.3.3, 4.3.1 and 4.3.2

$$\widetilde{1_\epsilon}(x) = \int_0^\infty 1_{(0,1]}(y) \frac{1}{\epsilon} \nu((x/y)^{\frac{1}{\epsilon}}) \frac{dy}{y}$$

and all the factors in the integrand are nonnegative. □

Lemma 4.3.8 (Smooth1LeOne)**.** If ν is nonnegative and has mass one, then $\widetilde{1_\epsilon}(x) \leq 1$, $\forall x > 0$.

Proof. By Definitions 4.3.3, 4.3.1 and 4.3.2

$$\widetilde{1_\epsilon}(x) = \int_0^\infty 1_{(0,1]}(y) \frac{1}{\epsilon} \nu((x/y)^{\frac{1}{\epsilon}}) \frac{dy}{y}$$

and since $1_{(0,1]}(y) \leq 1$, and all the factors in the integrand are nonnegative,

$$\widetilde{1_\epsilon}(x) \leq \int_0^\infty \frac{1}{\epsilon} \nu((x/y)^{\frac{1}{\epsilon}}) \frac{dy}{y}$$

(because in mathlib the integral of a non-integrable function is 0, for the inequality above to be true, we must prove that $\nu((x/y)^{\frac{1}{\epsilon}})/y$ is integrable; this follows from the computation below). We then change variables to $z = (x/y)^{\frac{1}{\epsilon}}$:

$$\widetilde{1_\epsilon}(x) \leq \int_0^\infty \nu(z) \frac{dz}{z}$$

which by Theorem 4.3.5 is 1. □

Combining the above, we have the following three Main Lemmata of this section on the Mellin transform of $\widetilde{1_\epsilon}$.

Lemma 4.3.9 (MellinOfSmooth1a)**.** Fix $\epsilon > 0$. Then the Mellin transform of $\widetilde{1_\epsilon}$ is

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = \frac{1}{s} \left(\mathcal{M}(\nu) (\epsilon s) \right).$$

Proof. By Definition 4.3.3,

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = \mathcal{M}(1_{(0,1]} * \nu_\epsilon)(s).$$

We wish to apply Theorem 4.3.1. To do so, we must prove that

$$(x, y) \mapsto 1_{(0,1]}(y) \nu_\epsilon(x/y)/y$$

is integrable on $[0, \infty)^2$. It is actually easier to do this for the convolution: $\nu_\epsilon * 1_{(0,1]}$, so we use Lemma 4.3.2: for $x \neq 0$,

$$1_{(0,1]} * \nu_\epsilon(x) = \nu_\epsilon * 1_{(0,1]}(x).$$

Now, for $x = 0$, both sides of the equation are 0, so the equation also holds for $x = 0$. Therefore,

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = \mathcal{M}(\nu_\epsilon * 1_{(0,1]})(s).$$

Now,

$$(x, y) \mapsto \nu_\epsilon(y) 1_{(0,1]}(x/y) \frac{x^{s-1}}{y}$$

has compact support that is bounded away from $y = 0$ (specifically $y \in [2^{-\epsilon}, 2^\epsilon]$ and $x \in (0, y]$), so it is integrable. We can thus apply Theorem 4.3.1 and find

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = \mathcal{M}(\nu_\epsilon)(s)\mathcal{M}(1_{(0,1]})(s).$$

By Lemmas 4.3.4 and 4.3.3,

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = \frac{1}{s}\mathcal{M}(\nu)(\epsilon s).$$

□

Lemma 4.3.10 (MellinOfSmooth1b). Given $0 < \sigma_1 \leq \sigma_2$, for any s such that $\sigma_1 \leq \mathcal{Re}(s) \leq \sigma_2$, we have

$$\mathcal{M}(\widetilde{1_\epsilon})(s) = O\left(\frac{1}{\epsilon |s|^2}\right).$$

Proof. Use Lemma 4.3.9 and the bound in Lemma 4.3.2. □

Lemma 4.3.11 (MellinOfSmooth1c). At $s = 1$, we have

$$\mathcal{M}(\widetilde{1_\epsilon})(1) = 1 + O(\epsilon)).$$

Proof. Follows from Lemmas 4.3.9, 4.3.1 and 4.3.4. □

Lemma 4.3.12 (Smooth1ContinuousAt). Fix a nonnegative, continuously differentiable function F on $\mathbb{R}$ with support in $[1/2, 2]$. Then for any $\epsilon > 0$, the function $x \mapsto \int_{(0,\infty)} x^{1+it}\widetilde{1_\epsilon}(x)dx$ is continuous at any $y > 0$.

Proof. Use Lemma **??** to write $\widetilde{1_\epsilon}(x)$ as an integral over an integral near 1, in particular avoiding the singularity at 0. The integrand may be bounded by $2^\epsilon \nu_\epsilon(t)$ which is independent of x and we can use dominated convergence to prove continuity. □

Let ν be a bumpfunction.

Theorem 4.3.5 (SmoothExistence). There exists a smooth (once differentiable would be enough), nonnegative "bumpfunction" ν, supported in $[1/2, 2]$ with total mass one:

$$\int_0^\infty \nu(x)\frac{dx}{x} = 1.$$

Proof. Same idea as Urysohn-type argument. □

4.4 Zeta Bounds

Already on Mathlib (with a shortened proof):

Theorem 4.4.1 (hasDerivAt-conj-conj). Let $f : \mathbb{C} \to \mathbb{C}$ be a complex differentiable function at $p \in \mathbb{C}$ with derivative a. Then the function $g(z) = \overline{f(\overline{z})}$ is complex differentiable at $\overline{p}$ with derivative $\overline{a}$.

Proof. We expand the definition of the derivative and compute. ☐

Submitted to Mathlib:

Theorem 4.4.2 (deriv-conj-conj)**.** Let $f : \mathbb{C} \to \mathbb{C}$ be a function at $p \in \mathbb{C}$ with derivative a. Then the derivative of the function $g(z) = \overline{f(\overline{z})}$ at $\overline{p}$ is $\overline{a}$.

Proof. We proceed by case analysis on whether f is differentiable at p. If f is differentiable at p, then we can apply the previous theorem. If f is not differentiable at p, then neither is g, and both derivatives have the default value of zero. ☐

Theorem 4.4.3 (conj-riemannZeta-conj-aux1)**.** Conjugation symmetry of the Riemann zeta function in the half-plane of convergence. Let $s \in \mathbb{C}$ with $\Re(s) > 1$. Then $\overline{\zeta(\overline{s})} = \zeta(s)$.

Proof. We expand the definition of the Riemann zeta function as a series and find that the two sides are equal term by term. ☐

Theorem 4.4.4 (conj-riemannZeta-conj)**.** Conjugation symmetry of the Riemann zeta function. Let $s \in \mathbb{C}$. Then

$$\overline{\zeta(\overline{s})} = \zeta(s).$$

Proof. By the previous lemma, the two sides are equal on the half-plane $\{s \in \mathbb{C} : \Re(s) > 1\}$. Then, by analytic continuation, they are equal on the whole complex plane. ☐

Theorem 4.4.5 (riemannZeta-conj)**.** Conjugation symmetry of the Riemann zeta function. Let $s \in \mathbb{C}$. Then

$$\zeta(\overline{s}) = \overline{\zeta(s)}.$$

Proof. This follows as an immediate corollary of Theorem 4.4.4. ☐

Theorem 4.4.6 (deriv-riemannZeta-conj)**.** Conjugation symmetry of the derivative of the Riemann zeta function. Let $s \in \mathbb{C}$. Then

$$\zeta'(\overline{s}) = \overline{\zeta'(s)}.$$

Proof. We apply the derivative conjugation symmetry to the Riemann zeta function and use the conjugation symmetry of the Riemann zeta function itself. ☐

Theorem 4.4.7 (intervalIntegral-conj)**.** The conjugation symmetry of the interval integral. Let $f : \mathbb{R} \to \mathbb{C}$ be a measurable function, and let $a, b \in \mathbb{R}$. Then

$$\int_a^b \overline{f(x)}\, dx = \overline{\int_a^b f(x)\, dx}.$$

Proof. We unfold the interval integral into an integral over a uIoc and use the conjugation property of integrals. ☐

We record here some prelimiaries about the zeta function and general holomorphic functions.

Theorem 4.4.8 (ResidueOfTendsTo)**.** If a function f is holomorphic in a neighborhood of p and $\lim_{s \to p}(s - p)f(s) = A$, then $f(s) = \frac{A}{s-p} + O(1)$ near p.

Proof. The function $(s - p) \cdot f(s)$ bounded, so by Theorem 4.1.1, there is a holomorphic function, g, say, so that $(s - p)f(s) = g(s)$ in a neighborhood of $s = p$, and $g(p) = A$. Now because g is holomorphic, near $s = p$, we have $g(s) = A + O(s - p)$. Then when you divide by $(s - p)$, you get $f(s) = A/(s - p) + O(1)$. □

Theorem 4.4.9 (riemannZetaResidue)**.** The Riemann zeta function $\zeta(s)$ has a simple pole at $s = 1$ with residue 1. In particular, the function $\zeta(s) - \frac{1}{s-1}$ is bounded in a neighborhood of $s = 1$.

Proof. From `riemannZeta_residue_one` (in Mathlib), we know that $(s-1)\zeta(s)$ goes to 1 as $s \to 1$. Now apply Theorem 4.4.8. (This can also be done using ζ_0 below, which is expressed as $1/(s - 1)$ plus things that are holomorphic for $\Re(s) > 0$...) □

Theorem 4.4.10 (nonZeroOfBddAbove)**.** If a function f has a simple pole at a point p with residue $A \neq 0$, then f is nonzero in a punctured neighborhood of p.

Proof. We know that $f(s) = \frac{A}{s-p} + O(1)$ near p, so we can write

$$f(s) = \left(f(s) - \frac{A}{s - p}\right) + \frac{A}{s - p}.$$

The first term is bounded, say by M, and the second term goes to ∞ as $s \to p$. Therefore, there exists a neighborhood V of p such that for all $s \in V \setminus \{p\}$, we have $f(s) \neq 0$. □

Theorem 4.4.11 (logDerivResidue)**.** If f is holomorphic in a neighborhood of p, and there is a simple pole at p, then f'/f has a simple pole at p with residue -1:

$$\frac{f'(s)}{f(s)} = \frac{-1}{s - p} + O(1).$$

Proof. Using Theorem 4.1.1, there is a function g holomorphic near p, for which $f(s) = A/(s - p) + g(s) = h(s)/(s - p)$. Here $h(s) := A + g(s)(s - p)$ which is nonzero in a neighborhood of p (since h goes to A which is nonzero). Then $f'(s) = (h'(s)(s - p) - h(s))/(s - p)^2$, and we can compute the quotient:

$$\frac{f'(s)}{f(s)} + 1/(s - p) = \frac{h'(s)(s - p) - h(s)}{h(s)} \cdot \frac{1}{(s - p)} + 1/(s - p) = \frac{h'(s)}{h(s)}.$$

Since h is nonvanishing near p, this remains bounded in a neighborhood of p. □

Theorem 4.4.12 (BddAbove-to-IsBigO)**.** If f is bounded above in a punctured neighborhood of p, then f is $O(1)$ in that neighborhood.

Proof. Elementary. □

Let's also record that if a function f has a simple pole at p with residue A, and g is holomorphic near p, then the residue of $f \cdot g$ is $A \cdot g(p)$.

Theorem 4.4.13 (ResidueMult)**.** If f has a simple pole at p with residue A, and g is holomorphic near p, then the residue of $f \cdot g$ at p is $A \cdot g(p)$. That is, we assume that

$$f(s) = \frac{A}{s - p} + O(1)$$

near p, and that g is holomorphic near p. Then

$$f(s) \cdot g(s) = \frac{A \cdot g(p)}{s - p} + O(1).$$

Proof. Elementary calculation.

$$f(s) * g(s) - \frac{A * g(p)}{s-p} = \left(f(s) * g(s) - \frac{A * g(s)}{s-p}\right) + \left(\frac{A * g(s) - A * g(p)}{s-p}\right).$$

The first term is $g(s)(f(s) - \frac{A}{s-p})$, which is bounded near p by the assumption on f and the fact that g is holomorphic near p. The second term is A times the log derivative of g at p, which is bounded by the assumption that g is holomorphic. □

As a corollary, the log derivative of the Riemann zeta function has a simple pole at $s = 1$:

Theorem 4.4.14 (riemannZetaLogDerivResidue)**.** The log derivative of the Riemann zeta function $\zeta(s)$ has a simple pole at $s = 1$ with residue -1: $-\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1} = O(1)$.

Proof. This follows from Theorem 4.4.11 and Theorem 4.4.9. □

Definition 4.4.1 (riemannZeta0)**.** For any natural $N \geq 1$, we define

$$\zeta_0(N, s) := \sum_{1 \leq n \leq N} \frac{1}{n^s} + \frac{-N^{1-s}}{1-s} + \frac{-N^{-s}}{2} + s \int_N^\infty \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx$$

Lemma 4.4.1 (sum-eq-int-deriv)**.** Let $a < b$, and let ϕ be continuously differentiable on $[a, b]$. Then

$$\sum_{a < n \leq b} \phi(n) = \int_a^b \phi(x) \, dx + \left(\lfloor b \rfloor + \frac{1}{2} - b\right) \phi(b) - \left(\lfloor a \rfloor + \frac{1}{2} - a\right) \phi(a) - \int_a^b \left(\lfloor x \rfloor + \frac{1}{2} - x\right) \phi'(x) \, dx.$$

Proof. Specialize Abel summation from Mathlib to the trivial arithmetic function and then manipulate integrals. □

Lemma 4.4.2 (ZetaSum-aux1)**.** Let $0 < a < b$ be natural numbers and $s \in \mathbb{C}$ with $s \neq 1$ and $s \neq 0$. Then

$$\sum_{a < n \leq b} \frac{1}{n^s} = \frac{b^{1-s} - a^{1-s}}{1-s} + \frac{b^{-s} - a^{-s}}{2} + s \int_a^b \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx.$$

Proof. Apply Lemma 4.4.1 to the function $x \mapsto x^{-s}$. □

Lemma 4.4.3 (ZetaBnd-aux1a)**.** For any $0 < a < b$ and $s \in \mathbb{C}$ with $\sigma = \Re(s) > 0$,

$$\int_a^b \left| \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx \right| \leq \frac{a^{-\sigma} - b^{-\sigma}}{\sigma}.$$

Proof. Apply the triangle inequality

$$\left| \int_a^b \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx \right| \leq \int_a^b \frac{1}{x^{\sigma+1}} \, dx,$$

and evaluate the integral. □

Lemma 4.4.4 (ZetaSum-aux2)**.** Let N be a natural number and $s \in \mathbb{C}$, $\Re(s) > 1$. Then

$$\sum_{N < n} \frac{1}{n^s} = \frac{-N^{1-s}}{1-s} + \frac{-N^{-s}}{2} + s \int_N^\infty \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx.$$

Proof. Apply Lemma 4.4.2 with $a = N$ and $b \to \infty$. $\square$

Lemma 4.4.5 (ZetaBnd-aux1b)**.** For any $N \geq 1$ and $s = \sigma + tI \in \mathbb{C}$, $\sigma > 0$,

$$\left| \int_N^\infty \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx \right| \leq \frac{N^{-\sigma}}{\sigma}.$$

Proof. Apply Lemma 4.4.3 with $a = N$ and $b \to \infty$. $\square$

Lemma 4.4.6 (ZetaBnd-aux1)**.** For any $N \geq 1$ and $s = \sigma + tI \in \mathbb{C}$, $\sigma =\in (0, 2], 2 < |t|$,

$$\left| s \int_N^\infty \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx \right| \leq 2|t| \frac{N^{-\sigma}}{\sigma}.$$

Proof. Apply Lemma 4.4.5 and estimate $|s| \ll |t|$. $\square$

Big-Oh version of Lemma 4.4.6.

Lemma 4.4.7 (ZetaBnd-aux1p)**.** For any $N \geq 1$ and $s = \sigma + tI \in \mathbb{C}$, $\sigma =\in (0, 2], 2 < |t|$,

$$\left| s \int_N^\infty \frac{\lfloor x \rfloor + 1/2 - x}{x^{s+1}} \, dx \right| \ll |t| \frac{N^{-\sigma}}{\sigma}.$$

Proof. Apply Lemma 4.4.5 and estimate $|s| \ll |t|$. $\square$

Theorem 4.4.15 (HolomorphicOn-riemannZeta0)**.** For any $N \geq 1$, the function $\zeta_0(N, s)$ is holomorphic on $\{s \in \mathbb{C} \mid \Re(s) > 0 \land s \neq 1\}$.

Proof. The function $\zeta_0(N, s)$ is a finite sum of entire functions, plus an integral that's absolutely convergent on $\{s \in \mathbb{C} \mid \Re(s) > 0 \land s \neq 1\}$ by Lemma 4.4.5. $\square$

Lemma 4.4.8 (isPathConnected-aux)**.** The set $\{s \in \mathbb{C} \mid \Re(s) > 0 \land s \neq 1\}$ is path-connected.

Proof. Construct explicit paths from 2 to any point, either a line segment or two joined ones. $\square$

Lemma 4.4.9 (Zeta0EqZeta)**.** For $\Re(s) > 0$, $s \neq 1$, and for any N,

$$\zeta_0(N, s) = \zeta(s).$$

Proof. Use Lemma 4.4.4 and the Definition 4.4.1. $\square$

Lemma 4.4.10 (ZetaBnd-aux2)**.** Given $n \leq t$ and σ with $1 - A/\log t \leq \sigma$, we have that

$$|n^{-s}| \leq n^{-1} e^A.$$

Proof. Use $|n^{-s}| = n^{-\sigma} = e^{-\sigma \log n} \leq \exp(-\left(1 - \frac{A}{\log t}\right) \log n) \leq n^{-1} e^A$, since $n \leq t$. $\square$

Lemma 4.4.11 (ZetaUpperBnd)**.** For any $s = \sigma + tI \in \mathbb{C}$, $1/2 \leq \sigma \leq 2, 3 < |t|$ and any $0 < A < 1$ sufficiently small, and $1 - A/\log|t| \leq \sigma$, we have

$$|\zeta(s)| \ll \log t.$$

Proof. First replace $\zeta(s)$ by $\zeta_0(N,s)$ for $N = \lfloor |t| \rfloor$. We estimate:

$$|\zeta_0(N,s)| \ll \sum_{1\le n\le |t|} |n^{-s}| + \frac{-|t|^{1-\sigma}}{|1-s|} + \frac{-|t|^{-\sigma}}{2} + |t|\cdot|t|^{-\sigma}/\sigma$$

$$\ll e^A \sum_{1\le n<|t|} n^{-1} + |t|^{1-\sigma}$$

, where we used Lemma 4.4.10 and Lemma 4.4.6. The first term is $\ll \log|t|$. For the second term, estimate

$$|t|^{1-\sigma} \le |t|^{1-(1-A/\log|t|)} = |t|^{A/\log|t|} \ll 1.$$

□

Lemma 4.4.12 (DerivUpperBnd-aux7)**.** For any $s = \sigma + tI \in \mathbb{C}$, $1/2 \le \sigma \le 2, 3 < |t|$, and any $0 < A < 1$ sufficiently small, and $1 - A/\log|t| \le \sigma$, we have

$$\left\| s \cdot \int_N^\infty \left(\lfloor x\rfloor + \frac{1}{2} - x\right)\cdot x^{-s-1}\cdot(-\log x)\right\| \le 2\cdot|t|\cdot N^{-\sigma}/\sigma\cdot\log|t|.$$

Proof. Estimate $|s| = |\sigma + tI|$ by $|s| \le 2+|t| \le 2|t|$ (since $|t| > 3$). Estimating $|\lfloor x\rfloor + 1/2 - x|$ by 1, and using $|x^{-s-1}| = x^{-\sigma-1}$, we have

$$\left\| s \cdot \int_N^\infty \left(\lfloor x\rfloor + \frac{1}{2} - x\right)\cdot x^{-s-1}\cdot(-\log x)\right\| \le 2\cdot|t|\int_N^\infty x^{-\sigma}\cdot(\log x).$$

For the last integral, integrate by parts, getting:

$$\int_N^\infty x^{-\sigma-1}\cdot(\log x) = \frac{1}{\sigma}N^{-\sigma}\cdot\log N + \frac{1}{\sigma^2}\cdot N^{-\sigma}.$$

Now use $\log N \le \log|t|$ to get the result. □

Lemma 4.4.13 (ZetaDerivUpperBnd)**.** For any $s = \sigma + tI \in \mathbb{C}$, $1/2 \le \sigma \le 2, 3 < |t|$, there is an $A > 0$ so that for $1 - A/\log t \le \sigma$, we have

$$|\zeta'(s)| \ll \log^2 t.$$

Proof. First replace $\zeta(s)$ by $\zeta_0(N,s)$ for $N = \lfloor |t| \rfloor$. Differentiating term by term, we get:

$$\zeta'(s) = -\sum_{1\le n<N} n^{-s}\log n + \frac{N^{1-s}}{(1-s)^2} + \frac{N^{1-s}\log N}{1-s} + \frac{N^{-s}\log N}{2} + \int_N^\infty \frac{\lfloor x\rfloor + 1/2 - x}{x^{s+1}}\,dx - s\int_N^\infty \log x\frac{\lfloor x\rfloor + 1/2 - x}{x^{s+1}}\,dx$$

Estimate as before, with an extra factor of $\log|t|$. □

Lemma 4.4.14 (ZetaNear1BndFilter)**.** As $\sigma \to 1^+$,

$$|\zeta(\sigma)| \ll 1/(\sigma - 1).$$

Proof. Zeta has a simple pole at $s = 1$. Equivalently, $\zeta(s)(s-1)$ remains bounded near 1. Lots of ways to prove this. Probably the easiest one: use the expression for $\zeta_0(N,s)$ with $N = 1$ (the term $N^{1-s}/(1-s)$ being the only unbounded one). □

Lemma 4.4.15 (ZetaNear1BndExact)**.** There exists a $c > 0$ such that for all $1 < \sigma \leq 2$,

$$|\zeta(\sigma)| \leq c/(\sigma - 1).$$

Proof. Split into two cases, use Lemma 4.4.14 for σ sufficiently small and continuity on a compact interval otherwise. □

Lemma 4.4.16 (ZetaLowerBound3)**.** There exists a $c > 0$ such that for all $1 < \sigma <= 2$ and $3 < |t|$,

$$c\frac{(\sigma-1)^{3/4}}{(\log|t|)^{1/4}} \leq |\zeta(\sigma + tI)|.$$

Proof. Combine Lemma **??** with upper bounds for $|\zeta(\sigma)|$ (from Lemma 4.4.15) and $|\zeta(\sigma + 2it)|$ (from Lemma 4.4.11). □

Lemma 4.4.17 (ZetaInvBound1)**.** For all $\sigma > 1$,

$$1/|\zeta(\sigma + it)| \leq |\zeta(\sigma)|^{3/4}|\zeta(\sigma + 2it)|^{1/4}$$

Proof. The identity

$$1 \leq |\zeta(\sigma)|^3|\zeta(\sigma + it)|^4|\zeta(\sigma + 2it)|$$

for $\sigma > 1$ is already proved by Michael Stoll in the EulerProducts PNT file. □

Lemma 4.4.18 (ZetaInvBound2)**.** For $\sigma > 1$ (and $\sigma \leq 2$),

$$1/|\zeta(\sigma + it)| \ll (\sigma - 1)^{-3/4}(\log|t|)^{1/4},$$

as $|t| \to \infty$.

Proof. Combine Lemma 4.4.17 with the bounds in Lemmata 4.4.15 and 4.4.11. □

Lemma 4.4.19 (Zeta-eq-int-derivZeta)**.** For any $t \neq 0$ (so we don't pass through the pole), and $\sigma_1 < \sigma_2$,

$$\int_{\sigma_1}^{\sigma_2} \zeta'(\sigma + it)dt = \zeta(\sigma_2 + it) - \zeta(\sigma_1 + it).$$

Proof. This is the fundamental theorem of calculus. □

Lemma 4.4.20 (Zeta-diff-Bnd)**.** For any $A > 0$ sufficiently small, there is a constant $C > 0$ so that whenever $1 - A/\log t \leq \sigma_1 < \sigma_2 \leq 2$ and $3 < |t|$, we have that:

$$|\zeta(\sigma_2 + it) - \zeta(\sigma_1 + it)| \leq C(\log|t|)^2(\sigma_2 - \sigma_1).$$

Proof. Use Lemma 4.4.19 and estimate trivially using Lemma 4.4.13. □

Lemma 4.4.21 (ZetaInvBnd)**.** For any $A > 0$ sufficiently small, there is a constant $C > 0$ so that whenever $1 - A/\log^9|t| \leq \sigma < 1 + A/\log^9|t|$ and $3 < |t|$, we have that:

$$1/|\zeta(\sigma + it)| \leq C\log^7|t|.$$

Proof. Let σ be given in the prescribed range, and set $\sigma' := 1 + A/\log^9|t|$. Then

$$|\zeta(\sigma+it)| \geq |\zeta(\sigma'+it)| - |\zeta(\sigma+it) - \zeta(\sigma'+it)| \geq C(\sigma'-1)^{3/4}\log|t|^{-1/4} - C\log^2|t|(\sigma'-\sigma)$$

$$\geq CA^{3/4}\log|t|^{-7} - C\log^2|t|(2A/\log^9|t|),$$

where we used Lemma 4.4.18 and Lemma 4.4.20. Now by making A sufficiently small (in particular, something like $A = 1/16$ should work), we can guarantee that

$$|\zeta(\sigma+it)| \geq \frac{C}{2}(\log|t|)^{-7},$$

as desired. □

Annoyingly, it is not immediate from this that ζ doesn't vanish there! That's because $1/0 = 0$ in Lean. So we give a second proof of the same fact (refactor this later), with a lower bound on ζ instead of upper bound on $1/\zeta$.

Lemma 4.4.22 (ZetaLowerBnd)**.** For any $A > 0$ sufficiently small, there is a constant $C > 0$ so that whenever $1 - A/\log^9|t| \leq \sigma < 1$ and $3 < |t|$, we have that:

$$|\zeta(\sigma+it)| \geq C\log^7|t|.$$

Proof. Follow same argument. □

Now we get a zero free region.

Lemma 4.4.23 (ZetaZeroFree)**.** There is an $A > 0$ so that for $1 - A/\log^9|t| \leq \sigma < 1$ and $3 < |t|$,

$$\zeta(\sigma+it) \neq 0.$$

Proof. Apply Lemma 4.4.22. □

Lemma 4.4.24 (LogDerivZetaBnd)**.** There is an $A > 0$ so that for $1 - A/\log^9|t| \leq \sigma < 1 + A/\log^9|t|$ and $3 < |t|$,

$$|\frac{\zeta'}{\zeta}(\sigma+it)| \ll \log^9|t|.$$

Proof. Combine the bound on $|\zeta'|$ from Lemma 4.4.13 with the bound on $1/|\zeta|$ from Lemma 4.4.21. □

Theorem 4.4.16 (ZetaNoZerosOn1Line)**.** The zeta function does not vanish on the 1-line.

Proof. This fact is already proved in Stoll's work. □

Then, since ζ doesn't vanish on the 1-line, there is a $\sigma < 1$ (depending on T), so that the box $[\sigma, 1] \times_{\mathbb{C}} [-T, T]$ is free of zeros of ζ.

Lemma 4.4.25 (ZetaNoZerosInBox)**.** For any $T > 0$, there is a constant $\sigma < 1$ so that

$$\zeta(\sigma'+it) \neq 0$$

for all $|t| \leq T$ and $\sigma' \geq \sigma$.

Proof. Assume not. Then there is a sequence $|t_n| \le T$ and $\sigma_n \to 1$ so that $\zeta(\sigma_n + it_n) = 0$. By compactness, there is a subsequence $t_{n_k} \to t_0$ along which $\zeta(\sigma_{n_k} + it_{n_k}) = 0$. If $t_0 \neq 0$, use the continuity of ζ to get that $\zeta(1 + it_0) = 0$; this is a contradiction. If $t_0 = 0$, ζ blows up near 1, so can't be zero nearby. □

We now prove that there's an absolute constant σ_0 so that ζ'/ζ is holomorphic on a rectangle $[\sigma_2, 2] \times_{\mathbb{C}} [-3, 3] \setminus \{1\}$.

Lemma 4.4.26 (LogDerivZetaHolcSmallT)**.** There is a $\sigma_2 < 1$ so that the function

$$\frac{\zeta'}{\zeta}(s)$$

is holomorphic on $\{\sigma_2 \le \Re s \le 2, |\Im s| \le 3\} \setminus \{1\}$.

Proof. The derivative of ζ is holomorphic away from $s = 1$; the denominator $\zeta(s)$ is nonzero in this range by Lemma 4.4.25. □

Lemma 4.4.27 (LogDerivZetaHolcLargeT)**.** There is an $A > 0$ so that for all $T > 3$, the function $\frac{\zeta'}{\zeta}(s)$ is holomorphic on $\{1 - A/\log^9 T \le \Re s \le 2, |\Im s| \le T\} \setminus \{1\}$.

Proof. The derivative of ζ is holomorphic away from $s = 1$; the denominator $\zeta(s)$ is nonzero in this range by Lemma 4.4.23. □

Lemma 4.4.28 (LogDerivZetaBndUnif)**.** There exist $A, C > 0$ such that

$$|\frac{\zeta'}{\zeta}(\sigma + it)| \le C \log |t|^9$$

whenever $|t| > 3$ and $\sigma > 1 - A/\log |t|^9$.

Proof. For σ close to 1 use Lemma 4.4.24, otherwise estimate trivially. □

4.5 Proof of Medium PNT

The approach here is completely standard. We follow the use of $\mathcal{M}(\widetilde{1_\epsilon})$ as in [Kontorovich 2015].

Definition 4.5.1 (ChebyshevPsi)**.** The (second) Chebyshev Psi function is defined as

$$\psi(x) := \sum_{n \le x} \Lambda(n),$$

where $\Lambda(n)$ is the von Mangoldt function.

It has already been established that zeta doesn't vanish on the 1 line, and has a pole at $s = 1$ of order 1. We also have the following.

Theorem 4.5.1 (LogDerivativeDirichlet)**.** We have that, for $\Re(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

Proof. Already in Mathlib. □

The main object of study is the following inverse Mellin-type transform, which will turn out to be a smoothed Chebyshev function.

Definition 4.5.2 (SmoothedChebyshev)**.** Fix $\epsilon > 0$, and a bumpfunction supported in $[1/2, 2]$. Then we define the smoothed Chebyshev function ψ_ϵ from $\mathbb{R}_{>0}$ to $\mathbb{C}$ by

$$\psi_\epsilon(X) = \frac{1}{2\pi i}\int_{(\sigma)} \frac{-\zeta'(s)}{\zeta(s)} \mathcal{M}(\widetilde{1_\epsilon})(s)X^s ds,$$

where we'll take $\sigma = 1 + 1/\log X$.

Lemma 4.5.1 (SmoothedChebyshevDirichlet-aux-integrable)**.** Fix a nonnegative, continuously differentiable function F on $\mathbb{R}$ with support in $[1/2, 2]$, and total mass one, $\int_{(0,\infty)} F(x)/xdx = 1$. Then for any $\epsilon > 0$, and $\sigma \in (1, 2]$, the function

$$x \mapsto \mathcal{M}(\widetilde{1_\epsilon})(\sigma + ix)$$

is integrable on $\mathbb{R}$.

Proof. By Lemma 4.3.10 the integrand is $O(1/t^2)$ as $t \to \infty$ and hence the function is integrable. □

Lemma 4.5.2 (SmoothedChebyshevDirichlet-aux-tsum-integral)**.** Fix a nonnegative, continuously differentiable function F on $\mathbb{R}$ with support in $[1/2, 2]$, and total mass one, $\int_{(0,\infty)} F(x)/xdx = 1$. Then for any $\epsilon > 0$ and $\sigma \in (1, 2]$, the function $x \mapsto \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma+it}} \mathcal{M}(\widetilde{1_\epsilon})(\sigma + it)x^{\sigma+it}$ is equal to $\sum_{n=1}^{\infty} \int_{(0,\infty)} \frac{\Lambda(n)}{n^{\sigma+it}} \mathcal{M}(\widetilde{1_\epsilon})(\sigma + it)x^{\sigma+it}$.

Proof. Interchange of summation and integration. □

Theorem 4.5.2 (SmoothedChebyshevDirichlet)**.** We have that

$$\psi_\epsilon(X) = \sum_{n=1}^{\infty} \Lambda(n)\widetilde{1_\epsilon}(n/X).$$

Proof. We have that

$$\psi_\epsilon(X) = \frac{1}{2\pi i}\int_{(2)} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} \mathcal{M}(\widetilde{1_\epsilon})(s)X^s ds.$$

We have enough decay (thanks to quadratic decay of $\mathcal{M}(\widetilde{1_\epsilon})$) to justify the interchange of summation and integration. We then get

$$\psi_\epsilon(X) = \sum_{n=1}^{\infty} \Lambda(n)\frac{1}{2\pi i}\int_{(2)} \mathcal{M}(\widetilde{1_\epsilon})(s)(n/X)^{-s} ds$$

and apply the Mellin inversion formula. □

The smoothed Chebyshev function is close to the actual Chebyshev function.

Theorem 4.5.3 (SmoothedChebyshevClose)**.** We have that

$$\psi_\epsilon(X) = \psi(X) + O(\epsilon X \log X).$$

Proof. Take the difference. By Lemma 4.3.6 and 4.3.5, the sums agree except when $1-c\epsilon \leq n/X \leq 1+c\epsilon$. This is an interval of length $\ll \epsilon X$, and the summands are bounded by $\Lambda(n) \ll \log X$.

□

Returning to the definition of ψ_ϵ, fix a large T to be chosen later, and set $\sigma_0 = 1+1/logX$, $\sigma_1 = 1-A/\log T^9$, and $\sigma_2 < \sigma_1$ a constant. Pull contours (via rectangles!) to go from $\sigma_0 - i\infty$ up to $\sigma_0 - iT$, then over to $\sigma_1 - iT$, up to $\sigma_1 - 3i$, over to $\sigma_2 - 3i$, up to $\sigma_2 + 3i$, back over to $\sigma_1 + 3i$, up to $\sigma_1 + iT$, over to $\sigma_0 + iT$, and finally up to $\sigma_0 + i\infty$.

In the process, we will pick up the residue at $s = 1$. We will do this in several stages. Here the interval integrals are defined as follows:

Definition 4.5.3 (I).

$$I_1(\nu, \epsilon, X, T) := \frac{1}{2\pi i} \int_{-\infty}^{-T} \left(\frac{-\zeta'}{\zeta}(\sigma_0 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_0 + ti) X^{\sigma_0 + ti} \; i \; dt$$

Definition 4.5.4 (I).

$$I_2(\nu, \epsilon, X, T, \sigma_1) := \frac{1}{2\pi i} \int_{\sigma_1}^{\sigma_0} \left(\frac{-\zeta'}{\zeta}(\sigma - iT) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma - iT) X^{\sigma - iT} \; d\sigma$$

Definition 4.5.5 (I).

$$I_{37}(\nu, \epsilon, X, T, \sigma_1) := \frac{1}{2\pi i} \int_{-T}^{T} \left(\frac{-\zeta'}{\zeta}(\sigma_1 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_1 + ti) X^{\sigma_1 + ti} \; i \; dt$$

Definition 4.5.6 (I).

$$I_8(\nu, \epsilon, X, T, \sigma_1) := \frac{1}{2\pi i} \int_{\sigma_1}^{\sigma_0} \left(\frac{-\zeta'}{\zeta}(\sigma + Ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma + Ti) X^{\sigma + Ti} \; d\sigma$$

Definition 4.5.7 (I).

$$I_9(\nu, \epsilon, X, T) := \frac{1}{2\pi i} \int_{T}^{\infty} \left(\frac{-\zeta'}{\zeta}(\sigma_0 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_0 + ti) X^{\sigma_0 + ti} \; i \; dt$$

Definition 4.5.8 (I).

$$I_3(\nu, \epsilon, X, T, \sigma_1) := \frac{1}{2\pi i} \int_{-T}^{-3} \left(\frac{-\zeta'}{\zeta}(\sigma_1 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_1 + ti) X^{\sigma_1 + ti} \; i \; dt$$

Definition 4.5.9 (I).

$$I_7(\nu, \epsilon, X, T, \sigma_1) := \frac{1}{2\pi i} \int_{3}^{T} \left(\frac{-\zeta'}{\zeta}(\sigma_1 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_1 + ti) X^{\sigma_1 + ti} \; i \; dt$$

Definition 4.5.10 (I).

$$I_4(\nu, \epsilon, X, \sigma_1, \sigma_2) := \frac{1}{2\pi i} \int_{\sigma_2}^{\sigma_1} \left(\frac{-\zeta'}{\zeta}(\sigma - 3i) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma - 3i) X^{\sigma - 3i} \; d\sigma$$

Definition 4.5.11 (I).

$$I_6(\nu, \epsilon, X, \sigma_1, \sigma_2) := \frac{1}{2\pi i} \int_{\sigma_2}^{\sigma_1} \left(\frac{-\zeta'}{\zeta}(\sigma + 3i) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma + 3i) X^{\sigma+3i}\, d\sigma$$

Definition 4.5.12 (I).

$$I_5(\nu, \epsilon, X, \sigma_2) := \frac{1}{2\pi i} \int_{-3}^{3} \left(\frac{-\zeta'}{\zeta}(\sigma_2 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_2 + ti) X^{\sigma_2+ti}\, i\, dt$$

Lemma 4.5.3 (dlog-riemannZeta-bdd-on-vertical-lines)**.** For $\sigma_0 > 1$, there exists a constant $C > 0$ such that

$$\forall t \in \mathbb{R}, \quad \left\| \frac{\zeta'(\sigma_0 + ti)}{\zeta(\sigma_0 + ti)} \right\| \leq C.$$

Proof. Write as Dirichlet series and estimate trivially using Theorem 4.5.1. □

Lemma 4.5.4 (SmoothedChebyshevPull1-aux-integrable)**.** The integrand

$$\zeta'(s)/\zeta(s) \mathcal{M}(\widetilde{1_\epsilon})(s) X^s$$

is integrable on the contour $\sigma_0 + ti$ for $t \in \mathbb{R}$ and $\sigma_0 > 1$.

Proof. The $\zeta'(s)/\zeta(s)$ term is bounded, as is X^s, and the smoothing function $\mathcal{M}(\widetilde{1_\epsilon})(s)$ decays like $1/|s|^2$ by Theorem 4.3.10. Actually, we already know that $\mathcal{M}(\widetilde{1_\epsilon})(s)$ is integrable from Theorem 4.5.1, so we should just need to bound the rest. □

Lemma 4.5.5 (BddAboveOnRect)**.** Let $g : \mathbb{C} \to \mathbb{C}$ be a holomorphic function on a rectangle, then g is bounded above on the rectangle.

Proof. Use the compactness of the rectangle and the fact that holomorphic functions are continuous. □

Theorem 4.5.4 (SmoothedChebyshevPull1)**.** We have that

$$\psi_\epsilon(X) = \mathcal{M}(\widetilde{1_\epsilon})(1) X^1 + I_1 - I_2 + I_{37} + I_8 + I_9.$$

Proof. Pull rectangle contours and evaluate the pole at $s = 1$. □

Next pull contours to another box.

Lemma 4.5.6 (SmoothedChebyshevPull2)**.** We have that

$$I_{37} = I_3 - I_4 + I_5 + I_6 + I_7.$$

Proof. Mimic the proof of Lemma 4.5.4. □

We insert this information in ψ_ϵ. We add and subtract the integral over the box $[1 - \delta, 2] \times_{\mathbb{C}} [-T, T]$, which we evaluate as follows

Theorem 4.5.5 (ZetaBoxEval)**.** For all $\epsilon > 0$ sufficiently close to 0, the rectangle integral over $[1 - \delta, 2] \times_{\mathbb{C}} [-T, T]$ of the integrand in ψ_ϵ is

$$\frac{X^1}{1} \mathcal{M}(\widetilde{1_\epsilon})(1) = X(1 + O(\epsilon)),$$

where the implicit constant is independent of X.

Proof. Unfold the definitions and apply Lemma 4.3.11. □

It remains to estimate all of the integrals.

This auxiliary lemma is useful for what follows.

Lemma 4.5.7 (IBound-aux1)**.** Given a natural number k and a real number $X_0 > 0$, there exists $C \geq 1$ so that for all $X \geq X_0$,

$$\log^k X \leq C \cdot X.$$

Proof. We use the fact that $\log^k X/X$ goes to 0 as $X \to \infty$. Then we use the extreme value theorem to find a constant C that works for all $X \geq X_0$. □

Lemma 4.5.8 (I1Bound)**.** We have that

$$|I_1(\nu, \epsilon, X, T)\ | \ll \frac{X}{\epsilon T}.$$

Same with I_9.

Proof. Unfold the definitions and apply the triangle inequality.

$$|I_1(\nu, \epsilon, X, T)| = \left| \frac{1}{2\pi i} \int_{-\infty}^{-T} \left(\frac{-\zeta'}{\zeta}(\sigma_0 + ti) \right) \mathcal{M}(\widetilde{1}_\epsilon)(\sigma_0 + ti) X^{\sigma_0 + ti}\ i\ dt \right|$$

By Theorem 4.5.3 (once fixed!!), $\zeta'/\zeta(\sigma_0 + ti)$ is bounded by $\zeta'/\zeta(\sigma_0)$, and Theorem 4.4.14 gives $\ll 1/(\sigma_0 - 1)$ for the latter. This gives:

$$\leq \frac{1}{2\pi} \left| \int_{-\infty}^{-T} C \log X \cdot \frac{C'}{\epsilon|\sigma_0 + ti|^2} X^{\sigma_0}\ dt \right|,$$

where we used Theorem 4.3.10. Continuing the calculation, we have

$$\leq \log X \cdot C'' \frac{X^{\sigma_0}}{\epsilon} \int_{-\infty}^{-T} \frac{1}{t^2}\ dt\ \ \leq\ \ C''' \frac{X \log X}{\epsilon T},$$

where we used that $\sigma_0 = 1 + 1/\log X$, and $X^{\sigma_0} = X \cdot X^{1/\log X} = e \cdot X$. □

Lemma 4.5.9 (I2Bound)**.** Assuming a bound of the form of Lemma 4.4.28 we have that

$$|I_2(\nu, \epsilon, X, T)| \ll \frac{X}{\epsilon T}.$$

Proof. Unfold the definitions and apply the triangle inequality.

$$|I_2(\nu, \epsilon, X, T, \sigma_1)| = \left| \frac{1}{2\pi i} \int_{\sigma_1}^{\sigma_0} \left(\frac{-\zeta'}{\zeta}(\sigma - Ti) \right) \cdot \mathcal{M}(\widetilde{1}_\epsilon)(\sigma - Ti) \cdot X^{\sigma - Ti}\ d\sigma \right|$$

$$\leq \frac{1}{2\pi} \int_{\sigma_1}^{\sigma_0} C \cdot \log T^9 \frac{C'}{\epsilon|\sigma - Ti|^2} X^{\sigma_0}\ d\sigma \leq C'' \cdot \frac{X \log T^9}{\epsilon T^2},$$

where we used Theorems 4.3.10, the hypothesised bound on zeta and the fact that $X^\sigma \leq X^{\sigma_0} = X \cdot X^{1/\log X} = e \cdot X$. Since $T > 3$, we have $\log T^9 \leq C''' T$. □

Lemma 4.5.10 (I8I2)**.** Symmetry between I_2 and I_8:

$$I_8(\nu, \epsilon, X, T) = -\overline{I_2(\nu, \epsilon, X, T)}.$$

Proof. This is a direct consequence of the definitions of I_2 and I_8. □

Lemma 4.5.11 (I8Bound)**.** We have that

$$|I_8(\nu, \epsilon, X, T)| \ll \frac{X}{\epsilon T}.$$

Proof. We deduce this from the corresponding bound for I_2, using the symmetry between I_2 and I_8. □

Lemma 4.5.12 (log-pow-over-xsq-integral-bounded)**.** For every n there is some absolute constant $C > 0$ such that

$$\int_3^T \frac{(\log x)^9}{x^2} dx < C$$

Proof. Induct on n and just integrate by parts. □

Lemma 4.5.13 (I3Bound)**.** Assuming a bound of the form of Lemma 4.4.28 we have that

$$|I_3(\nu, \epsilon, X, T)| \ll \frac{X}{\epsilon} X^{-\frac{A}{(\log T)^9}}.$$

Same with I_7.

Proof. Unfold the definitions and apply the triangle inequality.

$$\begin{aligned}|I_3(\nu, \epsilon, X, T, \sigma_1)| &= \left| \frac{1}{2\pi i} \int_{-T}^{3} \left(\frac{-\zeta'}{\zeta}(\sigma_1 + ti) \right) \mathcal{M}(\tilde{1}_\epsilon)(\sigma_1 + ti) X^{\sigma_1 + ti}\ i\ dt \right| \\ &\leq \frac{1}{2\pi} \int_{-T}^{3} C \cdot \log t^9 \frac{C'}{\epsilon |\sigma_1 + ti|^2} X^{\sigma_1}\ dt,\end{aligned}$$

where we used Theorems 4.3.10 and the hypothesised bound on zeta. Now we estimate $X^{\sigma_1} = X \cdot X^{-A/\log T^9}$, and the integral is absolutely bounded. □

Lemma 4.5.14 (I4Bound)**.** We have that

$$|I_4(\nu, \epsilon, X, \sigma_1, \sigma_2)| \ll \frac{X}{\epsilon} X^{-\frac{A}{(\log T)^9}}.$$

Same with I_6.

Proof. The analysis of I_4 is similar to that of I_2, (in Lemma 4.5.9) but even easier. Let C be the sup of $-\zeta'/\zeta$ on the curve $\sigma_2 + 3i$ to $1 + 3i$ (this curve is compact, and away from the pole at $s = 1$). Apply Theorem 4.3.10 to get the bound $1/(\epsilon |s|^2)$, which is bounded by C'/ϵ. And X^s is bounded by $X^{\sigma_1} = X \cdot X^{-A/\log T^9}$. Putting these together gives the result. □

Lemma 4.5.15 (I5Bound)**.** We have that

$$|I_5(\nu, \epsilon, X, \sigma_2)| \ll \frac{X^{\sigma_2}}{\epsilon}.$$

Proof. Here ζ'/ζ is absolutely bounded on the compact interval $\sigma_2 + i[-3, 3]$, and X^s is bounded by X^{σ_2}. Using Theorem 4.3.10 gives the bound $1/(\epsilon |s|^2)$, which is bounded by C'/ϵ. Putting these together gives the result. □

4.6 MediumPNT

Theorem 4.6.1 (MediumPNT)**.** We have

$$\sum_{n \leq x} \Lambda(n) = x + O(x \exp(-c(\log x)^{1/10})).$$

Proof. Evaluate the integrals. □

Chapter 5

Third Approach

5.1 Hadamard factorization

In this file, we prove the Hadamard Factorization theorem for functions of finite order, and prove that the zeta function is such.

5.2 Hoffstein-Lockhart

In this file, we use the Hoffstein-Lockhart construction to prove a zero-free region for zeta.

Hoffstein-Lockhart + Goldfeld-Hoffstein-Liemann

Instead of the "slick" identity $3+4\cos\theta+\cos 2\theta = 2(\cos\theta+1)^2 \geq 0$, we use the following more robust identity.

Theorem 5.2.1. For any $p > 0$ and $t \in \mathbb{R}$,

$$3 + p^{2it} + p^{-2it} + 2p^{it} + 2p^{-it} \geq 0.$$

Proof. This follows immediately from the identity

$$|1 + p^{it} + p^{-it}|^2 = 1 + p^{2it} + p^{-2it} + 2p^{it} + 2p^{-it} + 2.$$

□

[Note: identities of this type will work in much greater generality, especially for higher degree L-functions.]

This means that, for fixed t, we define the following alternate function.

Definition 5.2.1. For $\sigma > 1$ and $t \in \mathbb{R}$, define

$$F(\sigma) := \zeta^3(\sigma)\zeta^2(\sigma+it)\zeta^2(\sigma-it)\zeta(\sigma+2it)\zeta(\sigma-2it).$$

Theorem 5.2.2. Then F is real-valued, and whence $F(\sigma) \geq 1$ there.

Proof. That $\log F(\sigma) \geq 0$ for $\sigma > 1$ follows from Theorem 5.2.1. □

[Note: I often prefer to avoid taking logs of functions that, even if real-valued, have to be justified as being such. Instead, I like to start with "logF" as a convergent Dirichlet series, show that it is real-valued and non-negative, and then exponentiate...]

From this and Hadamard factorization, we deduce the following.

Theorem 5.2.3. There is a constant $c > 0$, so that $\zeta(s)$ does not vanish in the region $\sigma > 1 - \frac{c}{\log t}$, and moreover,

$$-\frac{\zeta'}{\zeta}(\sigma + it) \ll (\log t)^2$$

there.

Proof. Use Theorem 5.2.2 and Hadamard factorization. □

This allows us to quantify precisely the relationship between T and δ in Theorem 4.4.25....

5.3 Strong PNT

Definition 5.3.1. Given a complex function f, we define the function

$$g(z) := \begin{cases} \frac{f(z)}{z}, & z \neq 0; \\ f'(0), & z = 0. \end{cases}$$

Lemma 5.3.1. Let f be a complex function and let $z \neq 0$. Then, with g defined as in Definition 5.3.1,

$$g(z) = \frac{f(z)}{z}.$$

Proof. This follows directly from the definition of g. □

Lemma 5.3.2. Let f be a complex function analytic on an open set s containing 0 such that $f(0) = 0$. Then, with g defined as in Definition 5.3.1, g is analytic on s.

Proof. We need to show that g is complex differentiable at every point in s. For $z \neq 0$, this follows directly from the definition of g and the fact that f is analytic on s. For $z = 0$, we use the definition of the derivative and the fact that $f(0) = 0$:

$$\lim_{z \to 0} \frac{g(z) - g(0)}{z - 0} = \lim_{z \to 0} \frac{\frac{f(z)}{z} - f'(0)}{z} = \lim_{z \to 0} \frac{f(z) - f'(0)z}{z^2} = \lim_{z \to 0} \frac{f(z) - f(0) - f'(0)(z - 0)}{(z - 0)^2} = 0,$$

where the last equality follows from the definition of the derivative of f at 0. Thus, g is complex differentiable at 0 with derivative 0, completing the proof. □

Lemma 5.3.3. Let f be a complex function analytic on the closed ball $|z| \leq R$ such that $f(0) = 0$. Then, with g defined as in Definition 5.3.1, g is analytic on $|z| \leq R$.

Proof. The proof is similar to that of Lemma 5.3.2, but we need to consider two cases: when x is on the boundary of the closed ball and when it is in the interior. In the first case, we take a small open ball around x that lies entirely within the closed ball, and apply Lemma 5.3.2 on this smaller ball. In the second case, we can take the entire open ball centered at 0 with radius R, and again apply Lemma 5.3.2. In both cases, we use the fact that $f(0) = 0$ to ensure that the removable singularity at 0 is handled correctly. □

Definition 5.3.2. Given a complex function f and a real number M, we define the function

$$f_M(z) := \frac{g(z)}{2M - f(z)},$$

where g is defined as in Definition 5.3.1.

Lemma 5.3.4. Let $M > 0$. Let f be analytic on the closed ball $|z| \leq R$ such that $f(0) = 0$ and suppose that $2M - f(z) \neq 0$ for all $|z| \leq R$. Then, with f_M defined as in Definition 5.3.2, f_M is analytic on $|z| \leq R$.

Proof. This follows directly from Lemma 5.3.3 and the fact that the difference of two analytic functions is analytic. □

Lemma 5.3.5. Let $M > 0$ and let x be a complex number such that $\Re x \leq M$. Then, $|x| \leq |2M - x|$.

Proof. We square both sides and simplify to obtain the equivalent inequality

$$0 \leq 4M^2 - 4M\Re x,$$

which follows directly from the assumption $\Re x \leq M$ and the positivity of M. □

Theorem 5.3.1 (borelCaratheodory-closedBall)**.** Let R, $M > 0$. Let f be analytic on $|z| \leq R$ such that $f(0) = 0$ and suppose $\Re f(z) \leq M$ for all $|z| \leq R$. Then for any $0 < r < R$,

$$\sup_{|z| \leq r} |f(z)| \leq \frac{2Mr}{R - r}.$$

Proof. Let

$$f_M(z) = \frac{f(z)/z}{2M - f(z)}.$$

Note that $2M - f(z) \neq 0$ because $\Re(2M - f(z)) = 2M - \Re f(z) \geq M > 0$. Additionally, since $f(z)$ has a zero at 0, we know that $f(z)/z$ is analytic on $|z| \leq R$. Likewise, $f_M(z)$ is analytic on $|z| \leq R$.

Now note that $|f(z)| \leq |2M - f(z)|$ since $\Re f(z) \leq M$. Thus we have that

$$|f_M(z)| = \frac{|f(z)|/|z|}{|2M - f(z)|} \leq \frac{1}{|z|}.$$

Now by the maximum modulus principle, we know the maximum of $|f_M|$ must occur on the boundary where $|z| = R$. Thus, $|f_M(z)| \leq 1/R$ for all $|z| \leq R$. So for $|z| = r$ we have

$$|f_M(z)| = \frac{|f(z)|/r}{|2M - f(z)|} \leq \frac{1}{R} \implies R\,|f(z)| \leq r\,|2M - f(z)| \leq 2Mr + r\,|f(z)|.$$

Which by algebraic manipulation gives

$$|f(z)| \leq \frac{2Mr}{R - r}.$$

Once more, by the maximum modulus principle, we know the maximum of $|f|$ must occur on the boundary where $|z| = r$. Thus, the desired result immediately follows □

Lemma 5.3.6 (DerivativeBound)**.** Let R, $M > 0$ and $0 < r < r' < R$. Let f be analytic on $|z| \leq R$ such that $f(0) = 0$ and suppose $\Re f(z) \leq M$ for all $|z| \leq R$. Then we have that

$$|f'(z)| \leq \frac{2M(r')^2}{(R - r')(r' - r)^2}$$

for all $|z| \leq r$.

Proof. By Lemma 5.3.7 we know that

$$f'(z) = \frac{1}{2\pi i} \oint_{|w|=r'} \frac{f(w)}{(w-z)^2}\, dw = \frac{1}{2\pi} \int_0^{2\pi} \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2}\, dt.$$

Thus,

$$|f'(z)| = \left| \frac{1}{2\pi} \int_0^{2\pi} \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2}\, dt \right| \le \frac{1}{2\pi} \int_0^{2\pi} \left| \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2} \right| dt. \tag{5.1}$$

Now applying Theorem 5.3.1, and noting that $r' - r \le |r'e^{it} - z|$, we have that

$$\left| \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2} \right| \le \frac{2M(r')^2}{(R-r')(r'-r)^2}.$$

Substituting this into Equation (5.2) and evaluating the integral completes the proof. ☐

This upstreamed from https://github.com/math-inc/strongpnt/tree/main

Lemma 5.3.7 (cauchy-formula-deriv)**.** Let f be analytic on $|z| \le R$. For any z with $|z| \le r$ and any r' with $0 < r < r' < R$ we have

$$f'(z) = \frac{1}{2\pi i} \oint_{|w|=r'} \frac{f(w)}{(w-z)^2}\, dw = \frac{1}{2\pi} \int_0^{2\pi} \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2}\, dt.$$

Proof. This is just Cauchy's integral formula for derivatives. ☐

Lemma 5.3.8 (DerivativeBound)**.** Let R, $M > 0$ and $0 < r < r' < R$. Let f be analytic on $|z| \le R$ such that $f(0) = 0$ and suppose $\Re f(z) \le M$ for all $|z| \le R$. Then we have that

$$|f'(z)| \le \frac{2M(r')^2}{(R-r')(r'-r)^2}$$

for all $|z| \le r$.

Proof. By Lemma 5.3.7 we know that

$$f'(z) = \frac{1}{2\pi i} \oint_{|w|=r'} \frac{f(w)}{(w-z)^2}\, dw = \frac{1}{2\pi} \int_0^{2\pi} \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2}\, dt.$$

Thus,

$$|f'(z)| = \left| \frac{1}{2\pi} \int_0^{2\pi} \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2}\, dt \right| \le \frac{1}{2\pi} \int_0^{2\pi} \left| \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2} \right| dt. \tag{5.2}$$

Now applying Theorem 5.3.1, and noting that $r' - r \le |r'e^{it} - z|$, we have that

$$\left| \frac{r'e^{it}\, f(r'e^{it})}{(r'e^{it}-z)^2} \right| \le \frac{2M(r')^2}{(R-r')(r'-r)^2}.$$

Substituting this into Equation (5.2) and evaluating the integral completes the proof. ☐

Theorem 5.3.2 (BorelCaratheodoryDeriv)**.** Let R, $M > 0$. Let f be analytic on $|z| \leq R$ such that $f(0) = 0$ and suppose $\Re f(z) \leq M$ for all $|z| \leq R$. Then for any $0 < r < R$,

$$|f'(z)| \leq \frac{16MR^2}{(R-r)^3}$$

for all $|z| \leq r$.

Proof. Using Lemma 5.3.8 with $r' = (R+r)/2$, and noting that $r < R$, we have that

$$|f'(z)| \leq \frac{4M(R+r)^2}{(R-r)^3} \leq \frac{16MR^2}{(R-r)^3}.$$

□

Definition 5.3.3 (PathIntegral)**.** Let $f : \mathbb{C} \to \mathbb{C}$. Define the functon $I_f : \mathbb{C} \to \mathbb{C}$ by

$$I_f(z) = z \int_0^1 f(tz)\, dt.$$

Theorem 5.3.3 (LogOfAnalyticFunction)**.** Let $0 < r < R$. Let $B : \overline{\mathbb{D}_R} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_R}$ with $B(z) \neq 0$ for all $z \in \overline{\mathbb{D}_R}$. Then there exists $J_B : \overline{\mathbb{D}_r} \to \mathbb{C}$ that is analytic on neighborhoods of points in $\overline{\mathbb{D}_r}$ such that

- $J_B(0) = 0$
- $J_B'(z) = B'(z)/B(z)$
- $\log|B(z)| - \log|B(0)| = \Re J_B(z)$

for all $z \in \overline{\mathbb{D}_r}$.

Proof. We let $J_B(z) = I_{B'/B}(z)$. Then clearly, $J_B(0) = 0$. Now note that

$$I_{B'/B}(z) = z \int_0^1 (B'/B)(tz)\, dt = \int_0^z (B'/B)(u)\, du.$$

Thus by the fundamental theorem of calculus we have that $J_B'(z) = B'(z)/B(z)$. Now let $H(z) = \exp(J_B(z))/B(z)$ and note that

$$H'(z) = (B(z)\, J_B'(z) - B'(z)) \left(\frac{\exp(J_B(z))}{(B(z))^2} \right).$$

Thus, H is constant since we know that $B(z)\, J_B'(z) - B(z) = 0$ from $J_B'(z) = B'(z)/B(z)$. So since $H(0) = \exp(J_B(0))/B(0) = 1/B(0)$ we know $H(z) = 1/B(0)$ for all z. So we have,

$$\frac{1}{B(0)} = \frac{\exp(J_B(z))}{B(z)} \implies \left|\frac{B(z)}{B(0)}\right| = \exp(\Re J_B(z)).$$

Taking the logarithm of both sides completes the proof. □

Definition 5.3.4 (SetOfZeros)**.** Let $R > 0$ and $f : \overline{\mathbb{D}_R} \to \mathbb{C}$. Define the set of zeros $\mathcal{K}_f(R) = \{\rho \in \mathbb{C} : |\rho| \leq R,\ f(\rho) = 0\}$.

Definition 5.3.5 (ZeroOrder)**.** Let $0 < R < 1$ and $f : \mathbb{C} \to \mathbb{C}$ be analtyic on neighborhoods of points in $\overline{\mathbb{D}_1}$. For any zero $\rho \in \mathcal{K}_f(R)$, we define $m_f(\rho)$ as the order of the zero ρ w.r.t f.

Lemma 5.3.9 (ZeroFactorization)**.** Let $f : \overline{\mathbb{D}_1} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) \neq 0$. For all $\rho \in \mathcal{K}_f(1)$ there exists $h_\rho(z)$ that is analytic at ρ, $h_\rho(\rho) \neq 0$, and $f(z) = (z - \rho)^{m_f(\rho)}\, h_\rho(z)$.

Proof. Since f is analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ we know that there exists a series expansion about ρ:

$$f(z) = \sum_{0 \leq n} a_n\, (z - \rho)^n.$$

Now if we let m be the smallest number such that $a_m \neq 0$, then

$$f(z) = \sum_{0 \leq n} a_n\, (z - \rho)^n = \sum_{m \leq n} a_n\, (z - \rho)^n = (z - \rho)^m \sum_{m \leq n} a_n\, (z - \rho)^{n-m} = (z - \rho)^m\, h_\rho(z).$$

Trivially, $h_\rho(z)$ is analytic at ρ (we have written down the series expansion); now note that

$$h_\rho(\rho) = \sum_{m \leq n} a_n (\rho - \rho)^{n-m} = \sum_{m \leq n} a_n 0^{n-m} = a_m \neq 0.$$

□

Definition 5.3.6 (CFunction)**.** Let $0 < r < R < 1$, and $f : \overline{\mathbb{D}_1} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) \neq 0$. We define a function $C_f : \overline{\mathbb{D}_R} \to \mathbb{C}$ as follows. This function is constructed by dividing $f(z)$ by a polynomial whose roots are the zeros of f inside $\overline{\mathbb{D}_r}$.

$$C_f(z) = \begin{cases} \dfrac{f(z)}{\prod_{\rho \in \mathcal{K}_f(r)} (z - \rho)^{m_f(\rho)}} & \text{for } z \notin \mathcal{K}_f(r) \\ \dfrac{h_z(z)}{\prod_{\rho \in \mathcal{K}_f(r) \setminus \{z\}} (z - \rho)^{m_f(\rho)}} & \text{for } z \in \mathcal{K}_f(r) \end{cases}$$

where $h_z(z)$ comes from Lemma 5.3.9.

Definition 5.3.7 (BlaschkeB)**.** Let $0 < r < R < 1$, and $f : \overline{\mathbb{D}_1} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) \neq 0$. We define a function $B_f : \overline{\mathbb{D}_R} \to \mathbb{C}$ as follows.

$$B_f(z) = C_f(z) \prod_{\rho \in \mathcal{K}_f(r)} \left(R - \frac{z\overline{\rho}}{R}\right)^{m_f(\rho)}$$

Lemma 5.3.10 (BlaschkeOfZero)**.** Let $0 < r < R < 1$, and $f : \overline{\mathbb{D}_1} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) \neq 0$. Then

$$|B_f(0)| = |f(0)| \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{|\rho|}\right)^{m_f(\rho)}.$$

Proof. Since $f(0) \neq 0$, we know that $0 \notin \mathcal{K}_f(r)$. Thus,

$$C_f(0) = \frac{f(0)}{\prod_{\rho \in \mathcal{K}_f(r)} (-\rho)^{m_f(\rho)}}.$$

Thus, substituting this into Definition 5.3.7,

$$|B_f(0)| = |C_f(0)| \prod_{\rho \in \mathcal{K}_f(r)} R^{m_f(\rho)} = |f(0)| \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{|\rho|}\right)^{m_f(\rho)}.$$

□

Lemma 5.3.11 (DiskBound)**.** Let $B > 1$ and $0 < R < 1$. If $f : \mathbb{C} \to \mathbb{C}$ is a function analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $|f(z)| \leq B$ for $|z| \leq R$, then $|B_f(z)| \leq B$ for $|z| \leq R$ also.

Proof. For $|z| = R$, we know that $z \notin \mathcal{K}_f(r)$. Thus,

$$C_f(z) = \frac{f(z)}{\prod_{\rho \in \mathcal{K}_f(r)} (z-\rho)^{m_f(\rho)}}.$$

Thus, substituting this into Definition 5.3.7,

$$|B_f(z)| = |f(z)| \prod_{\rho \in \mathcal{K}_f(r)} \left|\frac{R - z\overline{\rho}/R}{z-\rho}\right|^{m_f(\rho)}.$$

But note that

$$\left|\frac{R - z\overline{\rho}/R}{z-\rho}\right| = \frac{|R^2 - z\overline{\rho}|/R}{|z-\rho|} = \frac{|z| \cdot |\overline{z-\rho}|/R}{|z-\rho|} = 1.$$

So we have that $|B_f(z)| = |f(z)| \leq B$ when $|z| = R$. Now by the maximum modulus principle, we know that the maximum of $|B_f|$ must occur on the boundary where $|z| = R$. Thus $|B_f(z)| \leq B$ for all $|z| \leq R$. □

Theorem 5.3.4 (ZerosBound)**.** Let $B > 1$ and $0 < r < R < 1$. If $f : \mathbb{C} \to \mathbb{C}$ is a function analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) = 1$ and $|f(z)| \leq B$ for $|z| \leq R$, then

$$\sum_{\rho \in \mathcal{K}_f(r)} m_f(\rho) \leq \frac{\log B}{\log(R/r)}.$$

Proof. Since $f(0) = 1$, by Lemma 5.3.10 we know that

$$|B_f(0)| = |f(0)| \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{|\rho|}\right)^{m_f(\rho)} = \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{|\rho|}\right)^{m_f(\rho)}.$$

Thus, substituting this into Definition 5.3.7,

$$(R/r)^{\sum_{\rho \in \mathcal{K}_f(r)} m_f(\rho)} = \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{r}\right)^{m_f(\rho)} \leq \prod_{\rho \in \mathcal{K}_f(r)} \left(\frac{R}{|\rho|}\right)^{m_f(\rho)} = |B_f(0)| \leq B$$

whereby Lemma 5.3.11 we know that $|B_f(z)| \leq B$ for all $|z| \leq R$. Taking the logarithm of both sides and rearranging gives the desired result. □

Definition 5.3.8 (JBlaschke)**.** Let $B > 1$ and $0 < R < 1$. If $f : \mathbb{C} \to \mathbb{C}$ is a function analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) = 1$, define $L_f(z) = J_{B_f}(z)$ where J is from Theorem 5.3.3 and B_f is from Definition 5.3.7.

Lemma 5.3.12 (BlaschkeNonZero)**.** Let $0 < r < R < 1$ and $f : \overline{\mathbb{D}_1} \to \mathbb{C}$ be analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$. Then $B_f(z) \neq 0$ for all $z \in \overline{\mathbb{D}_r}$.

Proof. Suppose that $z \in \mathcal{K}_f(r)$. Then we have that

$$C_f(z) = \frac{h_z(z)}{\displaystyle\prod_{\rho \in \mathcal{K}_f(r)\setminus\{z\}} (z-\rho)^{m_f(\rho)}}.$$

where $h_z(z) \neq 0$ according to Lemma 5.3.9. Thus, substituting this into Definition 5.3.7,

$$|B_f(z)| = |h_z(z)| \cdot \left| R - \frac{|z|^2}{R} \right|^{m_f(z)} \prod_{\rho \in \mathcal{K}_f(r)\setminus\{z\}} \left| \frac{R - z\overline{\rho}/R}{z - \rho} \right|^{m_f(\rho)}. \tag{5.3}$$

Trivially, $|h_z(z)| \neq 0$. Now note that

$$\left| R - \frac{|z|^2}{R} \right| = 0 \implies |z| = R.$$

However, this is a contradiction because $z \in \overline{\mathbb{D}_r}$ tells us that $|z| \le r < R$. Similarly, note that

$$\left| \frac{R - z\overline{\rho}/R}{z - \rho} \right| = 0 \implies |z| = \frac{R^2}{|\overline{\rho}|}.$$

However, this is also a contradiction because $\rho \in \mathcal{K}_f(r)$ tells us that $R < R^2/|\overline{\rho}| = |z|$, but $z \in \overline{\mathbb{D}_r}$ tells us that $|z| \le r < R$. So, we know that

$$\left| R - \frac{|z|^2}{R} \right| \neq 0 \qquad \text{and} \qquad \left| \frac{R - z\overline{\rho}/R}{z - \rho} \right| \neq 0 \quad \text{for all} \quad \rho \in \mathcal{K}_f(r) \setminus \{z\}.$$

Applying this to Equation (5.3) we have that $|B_f(z)| \neq 0$. So, $B_f(z) \neq 0$.

Now suppose that $z \notin \mathcal{K}_f(r)$. Then we have that

$$C_f(z) = \frac{f(z)}{\displaystyle\prod_{\rho \in \mathcal{K}_f(r)} (z-\rho)^{m_f(\rho)}}.$$

Thus, substituting this into Definition 5.3.7,

$$|B_f(z)| = |f(z)| \prod_{\rho \in \mathcal{K}_f(r)} \left| \frac{R - z\overline{\rho}/R}{z - \rho} \right|^{m_f(\rho)}. \tag{5.4}$$

We know that $|f(z)| \neq 0$ since $z \notin \mathcal{K}_f(r)$. Now note that

$$\left| \frac{R - z\overline{\rho}/R}{z - \rho} \right| = 0 \implies |z| = \frac{R^2}{|\overline{\rho}|}.$$

However, this is a contradiction because $\rho \in \mathcal{K}_f(r)$ tells us that $R < R^2/|\overline{\rho}| = |z|$, but $z \in \overline{\mathbb{D}_r}$ tells us that $|z| \leq r < R$. So, we know that

$$\left|\frac{R - z\overline{\rho}/R}{z - \rho}\right| \neq 0 \quad \text{for all} \quad \rho \in \mathcal{K}_f(r).$$

Applying this to Equation (5.4) we have that $|B_f(z)| \neq 0$. So, $B_f(z) \neq 0$.

We have shown that $B_f(z) \neq 0$ for both $z \in \mathcal{K}_f(r)$ and $z \notin \mathcal{K}_f(r)$, so the result follows. □

Theorem 5.3.5 (JBlaschkeDerivBound)**.** Let $B > 1$ and $0 < r' < r < R < 1$. If $f : \mathbb{C} \to \mathbb{C}$ is a function analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) = 1$ and $|f(z)| \leq B$ for all $|z| \leq R$, then for all $|z| \leq r'$

$$|L_f'(z)| \leq \frac{16 \log(B)\, r^2}{(r - r')^3}$$

Proof. By Lemma 5.3.11 we immediately know that $|B_f(z)| \leq B$ for all $|z| \leq R$. Now since $L_f = J_{B_f}$ by Definition 5.3.8, by Theorem 5.3.3 we know that

$$L_f(0) = 0 \qquad \text{and} \qquad \Re L_f(z) = \log|B_f(z)| - \log|B_f(0)| \leq \log|B_f(z)| \leq \log B$$

for all $|z| \leq r$. So by Theorem 5.3.2, it follows that

$$|L_f'(z)| \leq \frac{16 \log(B)\, r^2}{(r - r')^3}$$

for all $|z| \leq r'$. □

Theorem 5.3.6 (FinalBound)**.** Let $B > 1$ and $0 < r' < r < R' < R < 1$. If $f : \mathbb{C} \to \mathbb{C}$ is a function analytic on neighborhoods of points in $\overline{\mathbb{D}_1}$ with $f(0) = 1$ and $|f(z)| \leq B$ for all $|z| \leq R$, then for all $z \in \overline{\mathbb{D}_{r'}} \setminus \mathcal{K}_f(R')$ we have

$$\left|\frac{f'}{f}(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho}\right| \leq \left(\frac{16r^2}{(r - r')^3} + \frac{1}{(R^2/R' - R')\,\log(R/R')}\right) \log B.$$

Proof. Since $z \in \overline{\mathbb{D}_{r'}} \setminus \mathcal{K}_f(R')$ we know that $z \notin \mathcal{K}_f(R')$; thus, by Definition 5.3.6 we know that

$$C_f(z) = \frac{f(z)}{\prod_{\rho \in \mathcal{K}_f(R')} (z - \rho)^{m_f(\rho)}}.$$

Substituting this into Definition 5.3.7 we have that

$$B_f(z) = f(z) \prod_{\rho \in \mathcal{K}_f(R')} \left(\frac{R - z\overline{\rho}/R}{z - \rho}\right)^{m_f(\rho)}.$$

Taking the complex logarithm of both sides we have that

$$\operatorname{Log} B_f(z) = \operatorname{Log} f(z) + \sum_{\rho \in \mathcal{K}_f(R')} m_f(\rho) \operatorname{Log}(R - z\overline{\rho}/R) - \sum_{\rho \in \mathcal{K}_f(R')} m_f(\rho) \operatorname{Log}(z - \rho).$$

Taking the derivative of both sides we have that

$$\frac{B_f'}{B_f}(z) = \frac{f'}{f}(z) + \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - R^2/\overline{\rho}} - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho}.$$

By Definition 5.3.8 and Theorem 5.3.3, since $L_f(z) = J_{B_f}(z)$ we have $L_f'(z) = J_{B_f}'(z) = (B_f'/B_f)(z)$. Thus,

$$\frac{f'}{f}(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho} = L_f'(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - R^2/\overline{\rho}}.$$

Now since $z \in \overline{\mathbb{D}_{R'}}$ and $\rho \in \mathcal{K}_f(R')$, we know that $R^2/R' - R' \leq |z - R^2/\overline{\rho}|$. Thus by the triangle inequality we have

$$\left| \frac{f'}{f}(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho} \right| \leq |L_f'(z)| + \left(\frac{1}{R^2/R' - R'} \right) \sum_{\rho \in \mathcal{K}_f(R')} m_f(\rho).$$

Now by Theorem 5.3.4 and 5.3.5 we get our desired result with a little algebraic manipulation. □

Theorem 5.3.7 (ZetaFixedLowerBound)**.** For all $t \in \mathbb{R}$ one has

$$|\zeta(3/2 + it)| \geq \frac{\zeta(3)}{\zeta(3/2)}.$$

Proof. From the Euler product expansion of ζ, we have that for $\Re s > 1$

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Thus, we have that

$$\frac{\zeta(2s)}{\zeta(s)} = \prod_p \frac{1 - p^{-s}}{1 - p^{-2s}} = \prod_p \frac{1}{1 + p^{-s}}.$$

Now note that $|1 - p^{-(3/2+it)}| \leq 1 + |p^{-(3/2+it)}| = 1 + p^{-3/2}$. Thus,

$$|\zeta(3/2 + it)| = \prod_p \frac{1}{|1 - p^{-(3/2+it)}|} \geq \prod_p \frac{1}{1 + p^{-3/2}} = \frac{\zeta(3)}{\zeta(3/2)}$$

for all $t \in \mathbb{R}$ as desired. □

Lemma 5.3.13 (ZetaAltFormula)**.** Let

$$\zeta_0(s) = 1 + \frac{1}{s-1} - s \int_1^\infty \{x\}\, x^{-s} \, \frac{dx}{x}.$$

We have that $\zeta(s) = \zeta_0(s)$ for $\sigma > 1$.

Proof. Note that for $\sigma > 1$ we have

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \sum_{n=1}^{\infty} \frac{n}{n^s} - \sum_{n=1}^{\infty} \frac{n-1}{n^s} = \sum_{n=1}^{\infty} \frac{n}{n^s} - \sum_{n=0}^{\infty} \frac{n}{(n+1)^s} = \sum_{n=1}^{\infty} \frac{n}{n^s} - \sum_{n=1}^{\infty} \frac{n}{(n+1)^s}.$$

Thus

$$\zeta(s) = \sum_{n=1}^{\infty} n\,(n^{-s} - (n+1)^{-s}).$$

Now we note that

$$s \int_n^{n+1} x^{-s}\,\frac{dx}{x} = s\left(-\frac{1}{s}\,x^{-s}\right)_n^{n+1} = n^{-s} - (n+1)^{-s}.$$

So, substituting this we have

$$\zeta(s) = \sum_{n=1}^{\infty} n\,(n^{-s} - (n+1)^{-s}) = s\sum_{n=1}^{\infty} n \int_n^{n+1} x^{-s}\,\frac{dx}{x} = s\int_1^{\infty} \lfloor x \rfloor\, x^{-s}\,\frac{dx}{x}.$$

But noting that $\lfloor x \rfloor = x - \{x\}$ we have that

$$\zeta(s) = s\int_1^{\infty} \lfloor x \rfloor\, x^{-s}\,\frac{dx}{x} = s\int_1^{\infty} x^{-s}\,dx - s\int_1^{\infty} \{x\}\, x^{-s}\,\frac{dx}{x}.$$

Evaluating the first integral completes the result. □

Lemma 5.3.14 (ZetaAltFormulaAnalytic)**.** We have that $\zeta_0(s)$ is analytic for all $s \in S$ where $S = \{s \in \mathbb{C} : \Re s > 0,\ s \neq 1\}$.

Proof. Note that we have

$$\left|\int_1^{\infty} \{x\}\, x^{-s}\,\frac{dx}{x}\right| \leq \int_1^{\infty} |\{x\}\, x^{-s-1}|\,dx \leq \int_1^{\infty} x^{-\sigma-1}\,dx = \frac{1}{\sigma}.$$

So this integral converges uniformly on compact subsets of S, which tells us that it is analytic on S. So it immediately follows that $\zeta_0(s)$ is analytic on S as well, since S avoids the pole at $s = 1$ coming from the $(s-1)^{-1}$ term. □

Lemma 5.3.15 (ZetaExtend)**.** We have that

$$\zeta(s) = 1 + \frac{1}{s-1} - s\int_1^{\infty} \{x\}\, x^{-s}\,\frac{dx}{x}$$

for all $s \in S$.

Proof. This is an immediate consequence of the identity theorem. □

Theorem 5.3.8 (GlobalBound)**.** For all $s \in \mathbb{C}$ with $|s| \leq 1$ and $t \in \mathbb{R}$ with $|t| \geq 2$, we have that

$$|\zeta(s + 3/2 + it)| \leq 7 + 2\,|t|.$$

Proof. For the sake of clearer proof writing let $z = s + 3/2 + it$. Since $|s| \le 1$ we know that $1/2 \le \Re z$; additionally, as $|t| \ge 2$, we know $1 \le |\Im z|$. So, $z \in S$. Thus, from Lemma 5.3.15 we know that

$$|\zeta(z)| \le 1 + \frac{1}{|z-1|} + |z| \cdot \left| \int_1^\infty \{x\}\, x^{-z} \,\frac{dx}{x} \right|$$

by applying the triangle inequality. Now note that $|z-1| \ge 1$. Likewise,

$$|z| \cdot \left| \int_1^\infty \{x\}\, x^{-z} \,\frac{dx}{x} \right| \le |z| \int_1^\infty |\{x\}\, x^{-z-1}|\, dx \le |z| \int_1^\infty x^{-\Re z - 1}\, dx = \frac{|z|}{\Re z} \le 2\,|z|.$$

Thus we have that,

$$|\zeta(s+3/2+it)| = |\zeta(z)| \le 1 + 1 + 2\,|z| = 2 + 2\,|s+3/2+it| \le 2 + 2\,|s| + 3 + 2\,|it| \le 7 + 2\,|t|.$$

□

Theorem 5.3.9 (LogDerivZetaFinalBound)**.** Let $t \in \mathbb{R}$ with $|t| \ge 2$ and $0 < r' < r < R' < R < 1$. If $f(z) = \zeta(z + 3/2 + it)$, then for all $z \in \overline{\mathbb{D}_{r'}} \setminus \mathcal{K}_f(R')$ we have that

$$\left| \frac{f'}{f}(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho} \right| \ll \left(\frac{16 r^2}{(r - r')^3} + \frac{1}{(R^2/R' - R')\, \log(R/R')} \right) \log|t|.$$

Proof. Let $g(z) = \zeta(z + 3/2 + it)/\zeta(3/2 + it)$. Note that $g(0) = 1$ and for $|z| \le R$

$$|g(z)| = \frac{|\zeta(z + 3/2 + it)|}{|\zeta(3/2 + it)|} \le \frac{\zeta(3/2)}{\zeta(3)} \cdot (7 + 2\,|t|) \le \frac{13\,\zeta(3/2)}{3\,\zeta(3)}\, |t|$$

by Theorems 5.3.7 and 5.3.8. Thus by Theorem 5.3.6 we have that

$$\left| \frac{g'}{g}(z) - \sum_{\rho \in \mathcal{K}_g(R')} \frac{m_g(\rho)}{z - \rho} \right| \le \left(\frac{16 r^2}{(r - r')^3} + \frac{1}{(R^2/R' - R')\, \log(R/R')} \right) \left(\log|t| + \log\left(\frac{13\,\zeta(3/2)}{3\,\zeta(3)} \right) \right).$$

Now note that $f'/f = g'/g$, $\mathcal{K}_f(R') = \mathcal{K}_g(R')$, and $m_g(\rho) = m_f(\rho)$ for all $\rho \in \mathcal{K}_f(R')$. Thus we have that,

$$\left| \frac{f'}{f}(z) - \sum_{\rho \in \mathcal{K}_f(R')} \frac{m_f(\rho)}{z - \rho} \right| \ll \left(\frac{16 r^2}{(r - r')^3} + \frac{1}{(R^2/R' - R')\, \log(R/R')} \right) \log|t|$$

where the implied constant C is taken to be

$$C \ge 1 + \frac{\log((13\,\zeta(3/2))/(3\,\zeta(3)))}{\log 2}.$$

□

Definition 5.3.9 (ZeroWindows)**.** Let $\mathcal{Z}_t = \{\rho \in \mathbb{C} : \zeta(\rho) = 0,\ |\rho - (3/2 + it)| \le 5/6\}$.

Lemma 5.3.16 (SumBoundI)**.** For all $\delta \in (0,1)$ and $t \in \mathbb{R}$ with $|t| \ge 2$ we have

$$\left| \frac{\zeta'}{\zeta}(1 + \delta + it) - \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{1 + \delta + it - \rho} \right| \ll \log|t|.$$

Proof. We apply Theorem 5.3.9 where $r' = 2/3$, $r = 3/4$, $R' = 5/6$, and $R = 8/9$. Thus, for all $z \in \overline{\mathbb{D}_{2/3}} \setminus \mathcal{K}_f(5/6)$ we have that

$$\left| \frac{\zeta'}{\zeta}(z + 3/2 + it) - \sum_{\rho \in \mathcal{K}_f(5/6)} \frac{m_f(\rho)}{z - \rho} \right| \ll \log|t|$$

where $f(z) = \zeta(z + 3/2 + it)$ for $t \in \mathbb{R}$ with $|t| \geq 3$. Now if we let $z = -1/2 + \delta$, then $z \in (-1/2, 1/2) \subseteq \overline{\mathbb{D}_{2/3}}$. Additionally, $f(z) = \zeta(1 + \delta + it)$, where $1 + \delta + it$ lies in the zero-free region where $\sigma > 1$. Thus, $z \notin \mathcal{K}_f(5/6)$. So,

$$\left| \frac{\zeta'}{\zeta}(1 + \delta + it) - \sum_{\rho \in \mathcal{K}_f(5/6)} \frac{m_f(\rho)}{-1/2 + \delta - \rho} \right| \ll \log|t|.$$

But now note that if $\rho \in \mathcal{K}_f(5/6)$, then $\zeta(\rho + 3/2 + it) = 0$ and $|\rho| \leq 5/6$. Thus, $\rho + 3/2 + it \in \mathcal{Z}_t$. Additionally, note that $m_f(\rho) = m_\zeta(\rho + 3/2 + it)$. So changing variables using these facts gives us that

$$\left| \frac{\zeta'}{\zeta}(1 + \delta + it) - \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{1 + \delta + it - \rho} \right| \ll \log|t|.$$

$\square$

Lemma 5.3.17 (ShiftTwoBound)**.** For all $\delta \in (0, 1)$ and $t \in \mathbb{R}$ with $|t| \geq 2$ we have

$$-\Re\left(\frac{\zeta'}{\zeta}(1 + \delta + 2it)\right) \ll \log|t|.$$

Proof. Note that, for $\rho \in \mathcal{Z}_{2t}$

$$\begin{aligned} \Re\left(\frac{1}{1 + \delta + 2it - \rho}\right) &= \Re\left(\frac{1 + \delta - 2it - \overline{\rho}}{(1 + \delta + 2it - \rho)(1 + \delta - 2it - \overline{\rho})}\right) \\ &= \frac{\Re(1 + \delta - 2it - \overline{\rho})}{|1 + \delta + 2it - \rho|^2} = \frac{1 + \delta - \Re\rho}{(1 + \delta - \Re\rho)^2 + (2t - \Im\rho)^2}. \end{aligned}$$

Now since $\rho \in \mathcal{Z}_{2t}$, we have that $|\rho - (3/2 + 2it)| \leq 5/6$. So, we have $\Re\rho \in (2/3, 7/3)$ and $\Im\rho \in (2t - 5/6, 2t + 5/6)$. Thus, we have that

$$1/3 < 1 + \delta - \Re\rho \qquad \text{and} \qquad (1 + \delta - \Re\rho)^2 + (2t - \Im\rho)^2 < 16/9 + 25/36 = 89/36.$$

Which implies that

$$0 \leq \frac{12}{89} < \frac{1 + \delta - \Re\rho}{(1 + \delta - \Re\rho)^2 + (2t - \Im\rho)^2} = \Re\left(\frac{1}{1 + \delta + 2it - \rho}\right). \tag{5.5}$$

Note that, from Lemma 5.3.16, we have

$$\sum_{\rho \in \mathcal{Z}_{2t}} m_\zeta(\rho) \Re\left(\frac{1}{1 + \delta + 2it - \rho}\right) - \Re\left(\frac{\zeta'}{\zeta}(1 + \delta + 2it)\right) \leq \left| \frac{\zeta'}{\zeta}(1 + \delta + 2it) - \sum_{\rho \in \mathcal{Z}_{2t}} \frac{m_\zeta(\rho)}{1 + \delta + 2it - \rho} \right| \ll \log|2t|.$$

Since $m_\zeta(\rho) \geq 0$ for all $\rho \in \mathcal{Z}_{2t}$, the inequality from Equation (5.5) tells us that by subtracting the sum from both sides we have

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta+2it)\right) \ll \log|2t|.$$

Noting that $\log|2t| = \log(2) + \log|t| \leq 2\log|t|$ completes the proof. □

Lemma 5.3.18 (ShiftOneBound)**.** There exists $C > 0$ such that for all $\delta \in (0,1)$ and $t \in \mathbb{R}$ with $|t| \geq 3$; if $\zeta(\rho) = 0$ with $\rho = \sigma + it$, then

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta+it)\right) \leq -\frac{1}{1+\delta-\sigma} + C\log|t|.$$

Proof. Note that for $\rho' \in \mathcal{Z}_t$

$$\begin{aligned}\Re\left(\frac{1}{1+\delta+it-\rho'}\right) &= \Re\left(\frac{1+\delta-it-\overline{\rho'}}{(1+\delta+it-\rho')(1+\delta-it-\overline{\rho'})}\right)\\ &= \frac{\Re(1+\delta-it-\overline{\rho'})}{|1+\delta+it-\rho'|^2} = \frac{1+\delta-\Re\rho'}{(1+\delta-\Re\rho')^2+(t-\Im\rho')^2}.\end{aligned}$$

Now since $\rho' \in \mathcal{Z}_t$, we have that $|\rho - (3/2+it)| \leq 5/6$. So, we have $\Re\rho' \in (2/3, 7/3)$ and $\Im\rho' \in (t-5/6, t+5/6)$. Thus we have that

$$1/3 < 1+\delta-\Re\rho' \qquad \text{and} \qquad (1+\delta-\Re\rho')^2+(t-\Im\rho')^2 < 16/9+25/36 = 89/36.$$

Which implies that

$$0 \leq \frac{12}{89} < \frac{1+\delta-\Re\rho'}{(1+\delta-\Re\rho')^2+(t-\Im\rho')^2} = \Re\left(\frac{1}{1+\delta+it-\rho'}\right). \tag{5.6}$$

Note that, from Lemma 5.3.16, we have

$$\sum_{\rho\in\mathcal{Z}_t} m_\zeta(\rho)\,\Re\left(\frac{1}{1+\delta+it-\rho}\right) - \Re\left(\frac{\zeta'}{\zeta}(1+\delta+it)\right) \leq \left|\frac{\zeta'}{\zeta}(1+\delta+it) - \sum_{\rho\in\mathcal{Z}_t}\frac{m_\zeta(\rho)}{1+\delta+it-\rho}\right| \ll \log|t|.$$

Since $m_\zeta(\rho) \geq 0$ for all $\rho' \in \mathcal{Z}_t$, the inequality from Equation (5.6) tells us that by subtracting the sum over all $\rho' \in \mathcal{Z}_t \setminus \{\rho\}$ from both sides we have

$$\frac{m_\zeta(\rho)}{\Re(1+\delta+it-\rho)} - \Re\left(\frac{\zeta'}{\zeta}(1+\delta+it)\right) \ll \log|t|.$$

But of course we have that $\Re(1+\delta+it-\rho) = 1+\delta-\sigma$. So subtracting this term from both sides and recalling the implied constant we have

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta+it)\right) \leq -\frac{m_\zeta(\rho)}{1+\delta-\sigma} + C\log|t|.$$

We have that $\sigma \leq 1$ since ζ is zero free on the right half plane $\sigma > 1$. Thus $0 < 1+\delta-\sigma$. Noting this in combination with the fact that $1 \leq m_\zeta(\rho)$ completes the proof. □

Lemma 5.3.19 (ShiftZeroBound)**.** For all $\delta \in (0,1)$ we have

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta)\right) \leq \frac{1}{\delta} + O(1).$$

Proof. From Theorem 4.4.14 we know that

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{s-1} + O(1).$$

Changing variables $s \mapsto 1+\delta$ and applying the triangle inequality we have that

$$-\Re\left(\frac{\zeta'}{\zeta}(1+\delta)\right) \leq \left|-\frac{\zeta'}{\zeta}(1+\delta)\right| \leq \frac{1}{\delta} + O(1).$$

□

Lemma 5.3.20 (ThreeFourOneTrigIdentity)**.** We have that

$$0 \leq 3 + 4\cos\theta + \cos 2\theta$$

for all $\theta \in \mathbb{R}$.

Proof. We know that $\cos(2\theta) = 2\cos^2\theta - 1$, thus

$$3 + 4\cos\theta + \cos 2\theta = 2 + 4\cos\theta + 2\cos^2\theta = 2\,(1+\cos\theta)^2.$$

Noting that $0 \leq 1 + \cos\theta$ completes the proof. □

Theorem 5.3.10 (ZeroInequality)**.** There exists a constant $0 < E < 1$ such that for all $\rho = \sigma + it$ with $\zeta(\rho) = 0$ and $|t| \geq 2$, one has

$$\sigma \leq 1 - \frac{E}{\log|t|}.$$

Proof. From Theorem 4.5.1 when $\Re s > 1$ we have

$$-\frac{\zeta'}{\zeta}(s) = \sum_{1\leq n} \frac{\Lambda(n)}{n^s}.$$

Thus,

$$-3\,\frac{\zeta'}{\zeta}(1+\delta) - 4\,\frac{\zeta'}{\zeta}(1+\delta+it) - \frac{\zeta'}{\zeta}(1+\delta+2it) = \sum_{1\leq n} \Lambda(n)\, n^{-(1+\delta)}\left(3 + 4n^{-it} + n^{-2it}\right).$$

Now applying Euler's identity

$$\begin{aligned}-3\,\Re\left(\frac{\zeta'}{\zeta}(1+\delta)\right) - 4\,\Re\left(\frac{\zeta'}{\zeta}(1+\delta+it)\right) - \Re\left(\frac{\zeta'}{\zeta}(1+\delta+2it)\right)&\\ = \sum_{1\leq n} \Lambda(n)\, n^{-(1+\delta)}\,(3 + 4\cos(-it\log n) + \cos(-2it\log n))&\end{aligned}$$

By Lemma 5.3.20 we know that the series on the right hand side is bounded below by 0, and by Lemmas 5.3.17, 5.3.18, and 5.3.19 we have an upper bound on the left hand side. So,

$$0 \leq \frac{3}{\delta} + 3A - \frac{4}{1+\delta-\sigma} + 4B\log|t| + C\log|t|$$

where A, B, and C are the implied constants coming from Lemmas 5.3.19, 5.3.18, and 5.3.17 respectively. By choosing $D \geq 3A/\log 2 + 4B + C$ we have

$$\frac{4}{1+\delta-\sigma} \leq \frac{3}{\delta} + D\log|t|$$

by some manipulation. Now if we choose $\delta = (2D\log|t|)^{-1}$ then we have

$$\frac{4}{1-\sigma+1/(2D\log|t|)} \leq 7D\log|t|.$$

So with some manipulation we have that

$$\sigma \leq 1 - \frac{1}{14D\log|t|}.$$

This is exactly the desired result with the constant $E = (14D)^{-1}$ □

Definition 5.3.10 (DeltaT)**.** Let $\delta_t = E/\log|t|$ where E is the constant coming from Theorem 5.3.10.

Lemma 5.3.21 (DeltaRange)**.** For all $t \in \mathbb{R}$ with $|t| \geq 2$ we have that

$$\delta_t < 1/14.$$

Proof. Note that $\delta_t = E/\log|t|$ where E is the implied constant from Lemma 5.3.10. But we know that $E = (14D)^{-1}$ where $D \geq 3A/\log 2 + 4B + C$ where A, B, and C are the constants coming from Lemmas 5.3.19, 5.3.18, and 5.3.17 respectively. Thus,

$$E \leq \frac{1}{14\,(3A/\log 2 + 4B + C)}.$$

But note that $A \geq 0$ and $B \geq 0$ by Lemmas 5.3.19 and 5.3.18 respectively. However, we have that

$$C \geq 2 + \frac{2\log((13\,\zeta(3/2))/(3\,\zeta(3)))}{\log 2}$$

by Theorem 5.3.9 with Lemmas 5.3.16 and 5.3.17. So, by a very lazy estimate we have $C \geq 2$ and $E \leq 1/28$. Thus,

$$\delta_t = \frac{E}{\log|t|} \leq \frac{1}{28\,\log 2} < \frac{1}{14}.$$

□

Lemma 5.3.22 (SumBoundII)**.** For all $t \in \mathbb{R}$ with $|t| \geq 2$ and $z = \sigma + it$ where $1 - \delta_t/3 \leq \sigma \leq 3/2$, we have that

$$\left|\frac{\zeta'}{\zeta}(z) - \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{z-\rho}\right| \ll \log|t|.$$

Proof. By Lemma 5.3.21 we have that

$$-11/21 < -1/2 - \delta_t/3 \leq \sigma - 3/2 \leq 0.$$

We apply Theorem 5.3.9 where $r' = 2/3$, $r = 3/4$, $R' = 5/6$, and $R = 8/9$. Thus for all $z \in \overline{\mathbb{D}_{2/3}} \setminus \mathcal{K}_f(5/6)$ we have that

$$\left| \frac{\zeta'}{\zeta}(z + 3/2 + it) - \sum_{\rho \in \mathcal{K}_f(5/6)} \frac{m_f(\rho)}{z - \rho} \right| \ll \log |t|$$

where $f(z) = \zeta(z + 3/2 + it)$ for $t \in \mathbb{R}$ with $|t| \geq 3$. Now if we let $z = \sigma - 3/2$, then $z \in (-11/21, 0) \subseteq \overline{\mathbb{D}_{2/3}}$. Additionally, $f(z) = \zeta(\sigma + it)$, where $\sigma + it$ lies in the zero free region given by Lemma 5.3.10 since $\sigma \geq 1 - \delta_t/3 \geq 1 - \delta_t$. Thus, $z \notin \mathcal{K}_f(5/6)$. So,

$$\left| \frac{\zeta'}{\zeta}(\sigma + it) - \sum_{\rho \in \mathcal{K}_f(5/6)} \frac{m_f(\rho)}{\sigma - 3/2 - \rho} \right| \ll \log |t|.$$

But now note that if $\rho \in \mathcal{K}_f(5/6)$, then $\zeta(\rho + 3/2 + it) = 0$ and $|\rho| \leq 5/6$. Additionally, note that $m_f(\rho) = m_\zeta(\rho + 3/2 + it)$. So changing variables using these facts gives us that

$$\left| \frac{\zeta'}{\zeta}(\sigma + it) - \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{\sigma + it - \rho} \right| \ll \log |t|.$$

□

Lemma 5.3.23 (GapSize)**.** Let $t \in \mathbb{R}$ with $|t| \geq 3$ and $z = \sigma + it$ where $1 - \delta_t/3 \leq \sigma \leq 3/2$. Additionally, let $\rho \in \mathcal{Z}_t$. Then we have that

$$|z - \rho| \geq \delta_t/6.$$

Proof. Let $\rho = \sigma' + it'$ and note that since $\rho \in \mathcal{Z}_t$, we have $t' \in (t - 5/6, t + 5/6)$. Thus, if $t > 1$ we have

$$\log |t'| \leq \log |t + 5/6| \leq \log |2t| = \log 2 + \log |t| \leq 2 \log |t|.$$

And otherwise if $t < -1$ we have

$$\log |t'| \leq \log |t - 5/6| \leq \log |2t| = \log 2 + \log |t| \leq 2 \log |t|.$$

So by taking reciprocals and multiplying through by a constant we have that $\delta_t \leq 2\delta_{t'}$. Now note that since $\rho \in \mathcal{Z}_t$ we know that $\sigma' \leq 1 - \delta_{t'}$ by Theorem 5.3.10 (here we use the fact that $|t| \geq 3$ to give us that $|t'| \geq 2$). Thus,

$$\delta_t/6 \leq \delta_{t'} - \delta_t/3 = 1 - \delta_t/3 - (1 - \delta_{t'}) \leq \sigma - \sigma' \leq |z - \rho|.$$

□

Lemma 5.3.24 (LogDerivZetaUniformLogSquaredBoundStrip)**.** There exists a constant $F \in (0, 1/2)$ such that for all $t \in \mathbb{R}$ with $|t| \geq 3$ one has

$$1 - \frac{F}{\log |t|} \leq \sigma \leq 3/2 \implies \left| \frac{\zeta'}{\zeta}(\sigma + it) \right| \ll \log^2 |t|$$

where the implied constant is uniform in σ.

Proof. Take $F = E/3$ where E comes from Theorem 5.3.10. Then we have that $\sigma \geq 1 - \delta_t/3$. So, we apply Lemma 5.3.22, which gives us that

$$\left| \frac{\zeta'}{\zeta}(z) - \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{z - \rho} \right| \ll \log |t|.$$

Using the reverse triangle inequality and rearranging, we have that

$$\left| \frac{\zeta'}{\zeta}(z) \right| \leq \sum_{\rho \in \mathcal{Z}_t} \frac{m_\zeta(\rho)}{|z - \rho|} + C \log |t|$$

where C is the implied constant in Lemma 5.3.22. Now applying Lemma 5.3.23 we have that

$$\left| \frac{\zeta'}{\zeta}(z) \right| \leq \frac{6}{\delta_t} \sum_{\rho \in \mathcal{Z}_t} m_\zeta(\rho) + C \log |t|.$$

Now let $f(z) = \zeta(z + 3/2 + it)/\zeta(3/2 + it)$ with $\rho = \rho' + 3/2 + it$. Then if $\rho \in \mathcal{Z}_t$ we have that

$$0 = \zeta(\rho) = \zeta(\rho' + 3/2 + it) = f(\rho')$$

with the same multiplicity of zero, that is $m_\zeta(\rho) = m_f(\rho')$. And also if $\rho \in \mathcal{Z}_t$ then

$$5/6 \geq |\rho - (3/2 + it)| = |\rho'|.$$

Thus we change variables to have that

$$\left| \frac{\zeta'}{\zeta}(z) \right| \leq \frac{6}{\delta_t} \sum_{\rho' \in \mathcal{K}_f(5/6)} m_f(\rho') + C \log |t|.$$

Now note that $f(0) = 1$ and for $|z| \leq 8/9$ we have

$$|f(z)| = \frac{|\zeta(z + 3/2 + it)|}{|\zeta(3/2 + it)|} \leq \frac{\zeta(3/2)}{\zeta(3)} \cdot (7 + 2|t|) \leq \frac{13\,\zeta(3/2)}{3\,\zeta(3)} |t|$$

by Theorems 5.3.7 and 5.3.8. Thus by Theorem 5.3.4 we have that

$$\sum_{\rho' \in \mathcal{K}_f(5/6)} m_f(\rho') \leq \frac{\log |t| + \log(13\,\zeta(3/2)/(3\,\zeta(3)))}{\log((8/9)/(5/6))} \leq D \log |t|$$

where D is taken to be sufficiently large. Recall, by definition that, $\delta_t = E/\log |t|$ with E coming from Theorem 5.3.10. By using this fact and the above, we have that

$$\left| \frac{\zeta'}{\zeta}(z) \right| \ll \log^2 |t| + \log |t|$$

where the implied constant is taken to be bigger than $\max(6D/E, C)$. We know that the RHS is bounded above by $\ll \log^2 |t|$; so the result follows. $\square$

Theorem 5.3.11 (LogDerivZetaUniformLogSquaredBound)**.** There exists a constant $F \in (0, 1/2)$ such that for all $t \in \mathbb{R}$ with $|t| \geq 3$ one has

$$1 - \frac{F}{\log |t|} \leq \sigma \implies \left| \frac{\zeta'}{\zeta}(\sigma + it) \right| \ll \log^2 |t|$$

where the implied constant is uniform in σ.

Proof. Note that

$$\left|\frac{\zeta'}{\zeta}(\sigma+it)\right| = \sum_{1\le n}\frac{\Lambda(n)}{|n^{\sigma+it}|} = \sum_{1\le n}\frac{\Lambda(n)}{n^{\sigma}} = -\frac{\zeta'}{\zeta}(\sigma) \le \left|\frac{\zeta'}{\zeta}(\sigma)\right|.$$

From Theorem 4.4.14, and applying the triangle inequality we know that

$$\left|\frac{\zeta'}{\zeta}(s)\right| \le \frac{1}{|s-1|} + C.$$

where $C > 0$ is some constant. Thus, for $\sigma \ge 3/2$ we have that

$$\left|\frac{\zeta'}{\zeta}(\sigma+it)\right| \le \left|\frac{\zeta'}{\zeta}(\sigma)\right| \le \frac{1}{\sigma-1} + C \le 2 + C \ll 1 \ll \log^2|t|.$$

Putting this together with Lemma 5.3.24 completes the proof. □

Theorem 5.3.12 (LogDerivZetaLogSquaredBoundSmallt)**.** For $T > 0$ and $\sigma' = 1-\delta_T/3 = 1 - F/\log T$, if $|t| \le T$ then we have that

$$\left|\frac{\zeta'}{\zeta}(\sigma'+it)\right| \ll \log^2(2+T).$$

Proof. Note that if $|t| \ge 3$ then from Theorem 5.3.11 we have that

$$\left|\frac{\zeta'}{\zeta}(\sigma'+it)\right| \ll \log^2|t| \le \log^2 T \le \log^2(2+T).$$

Otherwise, if $|t| \le 3$, then from Theorem 4.4.14 and applying the triangle inequality we know

$$\left|\frac{\zeta'}{\zeta}(\sigma'+it)\right| \le \frac{1}{|(\sigma'-1)+it|} + C \le \frac{\log T}{F} + C$$

where $C \ge 0$. Thus, we have that

$$\left|\frac{\zeta'}{\zeta}(\sigma'+it)\right| \le \left(\frac{\log T}{F\log 2} + \frac{C}{\log 2}\right)\log(2+|t|) \le \left(\frac{\log(2+T)}{F\log 2} + \frac{C}{\log 2}\right)\log(2+T) \ll \log^2(2+T).$$

□

From here out we closely follow our previous proof of the Medium PNT and we modify it using our new estimate in Theorem 5.3.11. Recall Definition 4.5.2; for fixed $\varepsilon > 0$ and a bump function ν supported on $[1/2, 2]$ we have

$$\psi_\varepsilon(X) = \frac{1}{2\pi i}\int_{(\sigma)}\left(-\frac{\zeta'}{\zeta}(s)\right)\mathcal{M}(\tilde{1}_\varepsilon)(s)\,X^s\,ds$$

where $\sigma = 1 + 1/\log X$. Let $T > 3$ be a large constant to be chosen later, and we take $\sigma' = 1-\delta_T/3 = 1-F/\log T$ with F coming from Theorem 5.3.11. We integrate along the σ vertical line, and we pull contours accumulating the pole at $s = 1$ when we integrate along the curves

- I_1: $\sigma - i\infty$ to $\sigma - iT$

- I_2: $\sigma' - iT$ to $\sigma - iT$
- I_3: $\sigma' - iT$ to $\sigma' + iT$
- I_4: $\sigma' + iT$ to $\sigma + iT$
- I_5: $\sigma + iT$ to $\sigma + i\infty$.

Definition 5.3.11 (I1New)**.** Let

$$I_1(\nu, \varepsilon, X, T) = \frac{1}{2\pi i} \int_{-\infty}^{-T} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right) \mathcal{M}(\tilde{1}_\varepsilon)(\sigma + it)\, X^{\sigma+it}\, dt.$$

Definition 5.3.12 (I5New)**.** Let

$$I_5(\nu, \varepsilon, X, T) = \frac{1}{2\pi i} \int_{T}^{\infty} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right) \mathcal{M}(\tilde{1}_\varepsilon)(\sigma + it)\, X^{\sigma+it}\, dt.$$

Lemma 5.3.25 (I1NewBound)**.** We have that

$$|I_1(\nu, \varepsilon, X, T)| \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Proof. Note that $|I_1(\nu, \varepsilon, X, T)| =$

$$\left| \frac{1}{2\pi i} \int_{-\infty}^{-T} \left(-\frac{\zeta'}{\zeta}(\sigma + it) \right) \mathcal{M}(\tilde{1}_\varepsilon)(\sigma + it)\, X^{\sigma+it}\, dt \right| \ll \int_{-\infty}^{-T} \left| \frac{\zeta'}{\zeta}(\sigma + it) \right| \cdot |\mathcal{M}(\tilde{1}_\varepsilon)(\sigma+it)| \cdot X^{\sigma}\, dt.$$

Applying Theorem 5.3.11 and Lemma 4.3.10, we have that

$$|I_1(\nu, \varepsilon, X, T)| \ll \int_{-\infty}^{-T} \log^2 |t| \cdot \frac{X^\sigma}{\varepsilon\, |\sigma + it|^2}\, dt \ll \frac{X}{\varepsilon} \int_{T}^{\infty} \frac{\sqrt{t}\, dt}{t^2} \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Here we are using the fact that $\log^2 t$ grows slower than $\sqrt{t}$, $|\sigma + it|^2 \geq t^2$, and $X^\sigma = X \cdot X^{1/\log X} = eX$. □

Lemma 5.3.26 (I5NewBound)**.** We have that

$$|I_5(\nu, \varepsilon, X, T)| \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Proof. By symmetry, note that

$$|I_1(\nu, \varepsilon, X, T)| = |\overline{I_5(\nu, \varepsilon, X, T)}| = |I_5(\nu, \varepsilon, X, T)|.$$

Applying Lemma 5.3.25 completes the proof. □

Definition 5.3.13 (I2New)**.** Let

$$I_2(\nu, \varepsilon, X, T) = \frac{1}{2\pi i} \int_{\sigma'}^{\sigma} \left(-\frac{\zeta'}{\zeta}(\sigma_0 - iT) \right) \mathcal{M}(\tilde{1}_\varepsilon)(\sigma_0 - iT)\, X^{\sigma_0 - iT}\, d\sigma_0.$$

Definition 5.3.14 (I4New)**.** Let

$$I_4(\nu, \varepsilon, X, T) = \frac{1}{2\pi i} \int_{\sigma'}^{\sigma} \left(-\frac{\zeta'}{\zeta}(\sigma_0 + iT) \right) \mathcal{M}(\tilde{1}_\varepsilon)(\sigma_0 + iT)\, X^{\sigma_0 + iT}\, d\sigma_0.$$

Lemma 5.3.27 (I2NewBound)**.** We have that

$$|I_2(\nu,\varepsilon,X,T)| \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Proof. Note that $|I_2(\nu,\varepsilon,X,T)| =$

$$\left|\frac{1}{2\pi i}\int_{\sigma'}^{\sigma}\left(-\frac{\zeta'}{\zeta}(\sigma_0 - iT)\right)\mathcal{M}(\tilde{1}_\varepsilon)(\sigma_0 - iT)\,X^{\sigma_0 - iT}\,d\sigma_0\right| \ll \int_{\sigma'}^{\sigma}\left|\frac{\zeta'}{\zeta}(\sigma_0 - iT)\right|\cdot|\mathcal{M}(\tilde{1}_\varepsilon)(\sigma_0 - iT)|\cdot X^{\sigma_0}\,d\sigma_0.$$

Applying Theorem 5.3.11 and Lemma 4.3.10, we have that

$$|I_2(\nu,\varepsilon,X,T)| \ll \int_{\sigma'}^{\sigma}\log^2 T\cdot\frac{X^{\sigma_0}}{\varepsilon\,|\sigma_0 - iT|^2}\,d\sigma_0 \ll \frac{X\,\log^2 T}{\varepsilon\,T^2}\int_{\sigma'}^{\sigma}d\,\sigma_0 = \frac{X\,\log^2 T}{\varepsilon\,T^2}\,(\sigma - \sigma').$$

Here we are using the fact that $X^{\sigma_0} \le X^{\sigma} = X\cdot X^{1/\log X} = eX$ and $|\sigma_0 - iT|^2 \ge T^2$. Now note that

$$|I_2(\nu,\varepsilon,X,T)| \ll \frac{X\,\log^2 T}{\varepsilon\,T^2}\,(\sigma - \sigma') = \frac{X\,\log^2 T}{\varepsilon\,T^2\,\log X} + \frac{FX\,\log T}{\varepsilon\,T^2} \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Here we are using the fact that $\log T \ll T^{3/2}$, $\log^2 T \ll T^{3/2}$, and $X/\log X \ll X$. $\square$

Lemma 5.3.28 (I4NewBound)**.** We have that

$$|I_4(\nu,\varepsilon,X,T)| \ll \frac{X}{\varepsilon\sqrt{T}}.$$

Proof. By symmetry, note that

$$|I_2(\nu,\varepsilon,X,T)| = |\overline{I_4(\nu,\varepsilon,X,T)}| = |I_4(\nu,\varepsilon,X,T)|.$$

Applying Lemma 5.3.27 completes the proof. $\square$

Definition 5.3.15 (I3New)**.** Let

$$I_3(\nu,\varepsilon,X,T) = \frac{1}{2\pi i}\int_{-T}^{T}\left(-\frac{\zeta'}{\zeta}(\sigma' + it)\right)\mathcal{M}(\tilde{1}_\varepsilon)(\sigma' + it)\,X^{\sigma' + it}\,dt.$$

Lemma 5.3.29 (I3NewBound)**.** We have that

$$|I_3(\nu,\varepsilon,X,T)| \ll \frac{X^{1-F/\log T}\sqrt{T}}{\varepsilon}.$$

Proof. Note that $|I_3(\nu,\varepsilon,X,T)| =$

$$\left|\frac{1}{2\pi i}\int_{-T}^{T}\left(-\frac{\zeta'}{\zeta}(\sigma' + it)\right)\mathcal{M}(\tilde{1}_\varepsilon)(\sigma' + it)\,X^{\sigma' + it}\,dt\right| \ll \int_{-T}^{T}\left|\frac{\zeta'}{\zeta}(\sigma' + it)\right|\cdot|\mathcal{M}(\tilde{1}_\varepsilon)(\sigma' + it)|\cdot X^{\sigma'}\,dt.$$

Applying Theorem 5.3.12 and Lemma 4.3.10, we have that

$$|I_3(\nu,\varepsilon,X,T)| \ll \int_{-T}^{T}\log^2(2+T)\cdot\frac{X^{\sigma'}}{\varepsilon\,|\sigma' + it|^2}\,dt \ll \frac{X^{1-F/\log T}\,\sqrt{T}}{\varepsilon}\int_0^T\frac{dt}{|\sigma' + it|^2}.$$

Here we are using the fact that this integrand is symmetric in t about 0 and that $\log^2(2+T) \ll \sqrt{T}$ for sufficiently large T. Now note that, by Lemma 5.3.21, we have

$$\frac{1}{|\sigma' + it|^2} = \frac{1}{(1-\delta_T/3)^2 + t^2} < \frac{1}{(41/42)^2 + t^2}.$$

Thus,

$$|I_3(\nu, \varepsilon, X, T)| \ll \frac{X^{1-F/\log T}\sqrt{T}}{\varepsilon} \int_0^T \frac{dt}{|\sigma' + it|^2} \leq \frac{X^{1-F/\log T}\sqrt{T}}{\varepsilon} \int_0^\infty \frac{dt}{(41/42)^2 + t^2}.$$

The integral on the right hand side evaluates to $21\pi/41$, which is just a constant, so the desired result follows. $\square$

Theorem 5.3.13 (SmoothedChebyshevPull3)**.** We have that

$$\psi_\varepsilon(X) = \mathcal{M}(\tilde{1}_\varepsilon)(1)\, X^1 + I_1 - I_2 + I_3 + I_4 + I_5.$$

Proof. Pull contours and accumulate the pole of ζ'/ζ at $s = 1$. $\square$

Theorem 5.3.14 (StrongPNT)**.** We have

$$\sum_{n \leq x} \Lambda(n) = x + O\left(x \exp(-c\sqrt{\log x})\right).$$

Proof. By Theorem 4.5.3 and 5.3.13 we have that

$$\mathcal{M}(\tilde{1}_\varepsilon)(1)\, x^1 + I_1 - I_2 + I_3 + I_4 + I_5 = \psi(x) + O(\varepsilon x \log x).$$

Applying Theorem 4.3.11 and Lemmas 5.3.25, 5.3.27, 5.3.29, 5.3.28, and 5.3.26 we have that

$$\psi(x) = x + O(\varepsilon x) + O(\varepsilon x \log x) + O\left(\frac{x}{\varepsilon\sqrt{T}}\right) + O\left(\frac{x^{1-F/\log T}\sqrt{T}}{\varepsilon}\right).$$

We absorb the $O(\varepsilon x)$ term into the $O(\varepsilon x \log x)$ term and balance the last two terms in T.

$$\frac{x}{\varepsilon\sqrt{T}} = \frac{x^{1-F/\log T}\sqrt{T}}{\varepsilon} \implies T = \exp(\sqrt{F \log x}).$$

Thus,

$$\psi(x) = x + O(\varepsilon x \log x) + O\left(\frac{x}{\varepsilon \exp((1/2) \cdot \sqrt{F \log x})}\right).$$

Now we balance the last two terms in ε.

$$\varepsilon x \log x = \frac{x}{\varepsilon \exp((1/2) \cdot \sqrt{F \log x})} \implies \varepsilon \log x = \frac{\sqrt{\log x}}{\exp((1/4) \cdot \sqrt{F \log x})}.$$

Thus,

$$\psi(x) = x + O\left(x \exp(-(\sqrt{F}/4) \cdot \sqrt{\log x})\sqrt{\log x}\right).$$

Absorbing the $\sqrt{\log x}$ into the $\exp(-(\sqrt{F}/4) \cdot \sqrt{\log x})$ completes the proof. $\square$

Chapter 6

Elementary Corollaries

Lemma 6.0.1 (finsum-range-eq-sum-range)**.** For any arithmetic function f and real number x, one has

$$\sum_{n \leq x} f(n) = \sum_{n \leq \lfloor x \rfloor_+} f(n)$$

and

$$\sum_{n < x} f(n) = \sum_{n < \lceil x \rceil_+} f(n).$$

Proof. Straightforward. □

Theorem 6.0.1 (chebyshev-asymptotic)**.** One has

$$\sum_{p \leq x} \log p = x + o(x).$$

Proof. From the prime number theorem we already have

$$\sum_{n \leq x} \Lambda(n) = x + o(x)$$

so it suffices to show that

$$\sum_{j \geq 2} \sum_{p^j \leq x} \log p = o(x).$$

Only the terms with $j \leq \log x / \log 2$ contribute, and each j contributes at most $\sqrt{x} \log x$ to the sum, so the left-hand side is $O(\sqrt{x} \log^2 x) = o(x)$ as required. □

Corollary 6.0.1 (primorial-bounds)**.** We have

$$\prod_{p \leq x} p = \exp(x + o(x))$$

Proof. Exponentiate Theorem **??**. □

Theorem 6.0.2 (pi-asymp)**.** There exists a function $c(x)$ such that $c(x) = o(1)$ as $x \to \infty$ and

$$\pi(x) = (1 + c(x)) \int_2^x \frac{dt}{\log t}$$

for all x large enough.

Proof. We have the identity

$$\pi(x) = \frac{1}{\log x} \sum_{p \le x} \log p + \int_2^x (\sum_{p \le t} \log p) \frac{dt}{t \log^2 t}$$

as can be proven by interchanging the sum and integral and using the fundamental theorem of calculus. For any ε, we know from Theorem **??** that there is x_ε such that $\sum_{p \le t} \log p = t + O(\varepsilon t)$ for $t \ge x_\varepsilon$, hence for $x \ge x_\varepsilon$

$$\pi(x) = \frac{1}{\log x}(x + O(\varepsilon x)) + \int_{x_\varepsilon}^x (t + O(\varepsilon t)) \frac{dt}{t \log^2 t} + O_\varepsilon(1)$$

where the $O_\varepsilon(1)$ term can depend on x_ε but is independent of x. One can evaluate this after an integration by parts as

$$\pi(x) = (1 + O(\varepsilon)) \int_{x_\varepsilon}^x \frac{dt}{\log t} + O_\varepsilon(1)$$

$$= (1 + O(\varepsilon)) \int_2^x \frac{dt}{\log t}$$

for x large enough, giving the claim. □

Corollary 6.0.2 (pi-alt)**.** One has

$$\pi(x) = (1 + o(1)) \frac{x}{\log x}$$

as $x \to \infty$.

Proof. An integration by parts gives

$$\int_2^x \frac{dt}{\log t} = \frac{x}{\log x} - \frac{2}{\log 2} + \int_2^x \frac{dt}{\log^2 t}.$$

We have the crude bounds

$$\int_2^{\sqrt{x}} \frac{dt}{\log^2 t} = O(\sqrt{x})$$

and

$$\int_{\sqrt{x}}^x \frac{dt}{\log^2 t} = O(\frac{x}{\log^2 x})$$

and combining all this we obtain

$$\int_2^x \frac{dt}{\log t} = \frac{x}{\log x} + O(\frac{x}{\log^2 x})$$

$$= (1 + o(1)) \frac{x}{\log x}$$

and the claim then follows from Theorem 6.0.2. □

Proposition 6.0.1 (pn-asymptotic)**.** One has

$$p_n = (1 + o(1))n \log n$$

as $n \to \infty$.

Proof. Use Corollary 6.0.2 to show that $n = \pi(p_n) \sim p_n / \log p_n$ Taking logs gives $\log n \sim \log p_n - \log\log p_n \sim \log p_n$. Multiplying these gives $p_n \sim n \log n$ from which the result follows. ☐

Corollary 6.0.3 (pn-pn-plus-one)**.** We have $p_{n+1} - p_n = o(p_n)$ as $n \to \infty$.

Proof. Easy consequence of preceding proposition. ☐

Corollary 6.0.4 (prime-between)**.** For every $\varepsilon > 0$, there is a prime between x and $(1+\varepsilon)x$ for all sufficiently large x.

Proof. Use Corollary 6.0.2 to show that $\pi((1 + \varepsilon)x) - \pi(x)$ goes to infinity as $x \to \infty$. ☐

Proposition 6.0.2. We have $|\sum_{n\le x} \frac{\mu(n)}{n}| \le 1$.

Proof. From Möbius inversion $1_{n=1} = \sum_{d|n} \mu(d)$ and summing we have

$$1 = \sum_{d\le x} \mu(d) \lfloor \frac{x}{d} \rfloor$$

for any $x \ge 1$. Since $\lfloor \frac{x}{d} \rfloor = \frac{x}{d} - \epsilon_d$ with $0 \le \epsilon_d < 1$ and $\epsilon_x = 0$, we conclude that

$$1 \ge x \sum_{d\le x} \frac{\mu(d)}{d} - (x - 1)$$

and the claim follows. ☐

Proposition 6.0.3 (Möbius form of prime number theorem)**.** We have $\sum_{n\le x} \mu(n) = o(x)$.

Proof. From the Dirichlet convolution identity

$$\mu(n) \log n = -\sum_{d|n} \mu(d)\Lambda(n/d)$$

and summing we obtain

$$\sum_{n\le x} \mu(n) \log n = -\sum_{d\le x} \mu(d) \sum_{m\le x/d} \Lambda(m).$$

For any $\varepsilon > 0$, we have from the prime number theorem that

$$\sum_{m\le x/d} \Lambda(m) = x/d + O(\varepsilon x/d) + O_\varepsilon(1)$$

(divide into cases depending on whether x/d is large or small compared to ε). We conclude that

$$\sum_{n\le x} \mu(n) \log n = -x \sum_{d\le x} \frac{\mu(d)}{d} + O(\varepsilon x \log x) + O_\varepsilon(x).$$

Applying (6.0.2) we conclude that

$$\sum_{n\leq x}\mu(n)\log n = O(\varepsilon x\log x) + O_\varepsilon(x).$$

and hence

$$\sum_{n\leq x}\mu(n)\log x = O(\varepsilon x\log x) + O_\varepsilon(x) + O(\sum_{n\leq x}(\log x - \log n)).$$

From Stirling's formula one has

$$\sum_{n\leq x}(\log x - \log n) = O(x)$$

thus

$$\sum_{n\leq x}\mu(n)\log x = O(\varepsilon x\log x) + O_\varepsilon(x)$$

and thus

$$\sum_{n\leq x}\mu(n) = O(\varepsilon x) + O_\varepsilon(\frac{x}{\log x}).$$

Sending $\varepsilon \to 0$ we obtain the claim. $\square$

Proposition 6.0.4. We have $\sum_{n\leq x}\lambda(n) = o(x)$.

Proof. From the identity

$$\lambda(n) = \sum_{d^2|n}\mu(n/d^2)$$

and summing, we have

$$\sum_{n\leq x}\lambda(n) = \sum_{d\leq\sqrt{x}}\sum_{n\leq x/d^2}\mu(n).$$

For any $\varepsilon > 0$, we have from Proposition 6.0.3 that

$$\sum_{n\leq x/d^2}\mu(n) = O(\varepsilon x/d^2) + O_\varepsilon(1)$$

and hence on summing in d

$$\sum_{n\leq x}\lambda(n) = O(\varepsilon x) + O_\varepsilon(x^{1/2}).$$

Sending $\varepsilon \to 0$ we obtain the claim. $\square$

Proposition 6.0.5 (Alternate Möbius form of prime number theorem)**.** We have $\sum_{n\leq x}\mu(n)/n = o(1)$.

Proof. As in the proof of Theorem 6.0.2, we have

$$1 = \sum_{d\leq x}\mu(d)\lfloor\frac{x}{d}\rfloor$$

$$= x\sum_{d\leq x}\frac{\mu(d)}{d} - \sum_{d\leq x}\mu(d)\{\frac{x}{d}\}$$

so it will suffice to show that

$$\sum_{d \le x} \mu(d)\{\frac{x}{d}\} = o(x).$$

Let N be a natural number. It suffices to show that

$$\sum_{d \le x} \mu(d)\{\frac{x}{d}\} = O(x/N).$$

if x is large enough depending on N. We can split the left-hand side as the sum of

$$\sum_{d \le x/N} \mu(d)\{\frac{x}{d}\}$$

and

$$\sum_{j=1}^{N-1} \sum_{x/(j+1) < d \le x/j} \mu(d)(x/d - j).$$

The first term is clearly $O(x/N)$. For the second term, we can use Theorem 6.0.3 and summation by parts (using the fact that $x/d - j$ is monotone and bounded) to find that

$$\sum_{x/(j+1) < d \le x/j} \mu(d)(x/d - j) = o(x)$$

for any given j, so in particular

$$\sum_{x/(j+1) < d \le x/j} \mu(d)(x/d - j) = O(x/N^2)$$

for all $j = 1, \dots, N-1$ if x is large enough depending on N. Summing all the bounds, we obtain the claim. □

6.1 Consequences of the PNT in arithmetic progressions

Theorem 6.1.1 (Prime number theorem in AP)**.** If $a\ (q)$ is a primitive residue class, then one has

$$\sum_{p \le x : p = a\ (q)} \log p = \frac{x}{\phi(q)} + o(x).$$

Proof. This is a routine modification of the proof of Theorem 6.0.1. □

Corollary 6.1.1 (Dirichlet's theorem)**.** Any primitive residue class contains an infinite number of primes.

Proof. If this were not the case, then the sum $\sum_{p \le x : p = a\ (q)} \log p$ would be bounded in x, contradicting Theorem 6.1.1. □

6.2 Consequences of the Chebotarev density theorem

Lemma 6.2.1 (Cyclotomic Chebotarev)**.** For any a coprime to m,

$$\sum_{N\mathfrak{p} \le x; N\mathfrak{p} = a\ (m)} \log N\mathfrak{p} = \frac{1}{|G|} \sum_{N\mathfrak{p} \le x} \log N\mathfrak{p}.$$

Proof. This should follow from Lemma 3.7.1 by a Fourier expansion. □

Chapter 7

Explicit estimates

We will try to systematically collect explicit estimates related to the prime number theorem from the literature, and formalize them in a modular fashion. We divide such estimates into four classes:

- *Zeta function* explicit estimates: bounds on the zeta function and its zeroes.
- *Primary* explicit estimates: those that are directly control $\psi(x)$ and $M(x)$, usually via information on the zeta function.
- *Secondary* explicit estimates: these are useful general-purpose estimates on functions relating to the primes, such as bounds on the n-th prime, or estimates for the prime counting function $\pi(x)$. These are generally derived from primary estimates and elementary arguments.
- *Tertiary* explicit estimates: these are bespoke applications to particular problems in analytic number theory or combinatorics that often require secondary estimates as input.

In this project we will state the best available zeta and primary estimates known in the literature, and try to formalize at least some of them; state the best available secondary estimates known in the literature, as well as various tools from passing from primary to secondary estimates, and formalize these; and then finally formalize some tertiary estimates as applications of the secondary ones.

Chapter 8

Zeta function estimates

8.1 Definitions

Definition 8.1.1. ρ is understood to lie in the set $\{s : \zeta(s) = 0\}$, counted with multiplicity. We will often restrict the zeroes ρ to a rectangle $\{\Re\rho \in I, \Im\rho \in J\}$, for instance through sums of the form $\sum_{\Re\rho\in I,\Im\rho\in J} f(\rho)$.

Definition 8.1.2. We say that the Riemann hypothesis has been verified up to height T if there are no zeroes in the rectangle $\{\Re\rho \in (0.5, 1), \Im\rho \in [0, T]\}$.

Definition 8.1.3 (Section 1.1, FKS2)**.** We say that one has a classical zero-free region with parameter R if $zeta(s)$ has no zeroes in the region $Re(s) \geq 1 - 1/R * \log|\Im s|$ for $\Im(s) > 3$.

Definition 8.1.4 (Zero counting function N(T))**.** The number of zeroes of imaginary part between 0 and T, counting multiplicity

Definition 8.1.5 (Riemann von Mangoldt estimate)**.** An estimate of the form $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}| \leq b_1 \log T + b_2 \log\log T + b_3$ for $T \geq 2$.

Definition 8.1.6 (Zero density bound)**.** An estimate of the form $N(\sigma, T) \leq c_1 T^p \log^q T + c_2 \log^2 T - \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}| \leq b_1 \log T + b_2 \log\log T + b_3$ for $T \geq 2$.

8.2 The estimates of Kadiri, Lumley, and Ng

In this section we establish the primary results of [20].

8.3 The zeta function bounds of Rosser and Schoenfeld

In this section we formalize the zeta function bounds of Rosser and Schoenfeld.

TODO: Add more results and proofs here, and reorganize the blueprint

Theorem 8.3.1 (Rosser–Schoenfeld Theorem 19)**.** One has a Riemann von Mangoldt estimate with parameters 0.137, 0.443, and 1.588.

Proof. □

8.4 Approximating the Riemann zeta function

We want a good explicit estimate on

$$\sum_{n\le a}\frac{1}{n^s}-\int_0^a\frac{du}{u^s},$$

for a a half-integer. As it turns out, this is the same problem as that of approximating $\zeta(s)$ by a sum $\sum_{n\le a} n^{-s}$. This is one of the two[1] main, standard ways of approximating $\zeta(s)$.

The non-explicit version of the result was first proved in [16, Lemmas 1 and 2]. The proof there uses first-order Euler-Maclaurin combined with a decomposition of $\lfloor x\rfloor - x + 1/2$ that turns out to be equivalent to Poisson summation. The exposition in [30, §4.7–4.11] uses first-order Euler-Maclaurin and van de Corput's Process B; the main idea of the latter is Poisson summation.

There are already several explicit versions of the result in the literature. In [5], [18] and [28], what we have is successively sharper explicit versions of Hardy and Littlewood's original proof. The proof in [10, Lemma 2.10] proceeds simply by a careful estimation of the terms in high-order Euler-Maclaurin; it does not use Poisson summation. Finally, [8] is an explicit version of [30, §4.7–4.11]; it gives a weaker bound than [28] or [10]. The strongest of these results is [28].

We will give another version here, in part because we wish to relax conditions – we will work with $|\Im s| < 2\pi a$ rather than $|\Im s| \le a$ – and in part to show that one can prove an asymptotically optimal result easily and concisely. We will use first-order Euler-Maclaurin and Poisson summation. We assume that a is a half-integer; if one inserts the same assumption into [10, Lemma 2.10], one can improve the result there, yielding an error term closer to the one here.

For additional context, see the Zulip discussion at `https://leanprover.zulipchat.com/#narrow/channel/423402-PrimeNumberTheorem.2B/topic/Let.20us.20formalize.20an.20appendix`

Definition 8.4.1 (e)**.** We recall that $e(\alpha) = e^{2\pi i\alpha}$.

8.4.1 The decay of a Fourier transform

Our first objective will be to estimate the Fourier transform of $t^{-s}\mathbb{1}_{[a,b]}$. In particular, we will show that, if a and b are half-integers, the Fourier cosine transform has quadratic decay *when evaluated at integers*. In general, for real arguments, the Fourier transform of a discontinuous function such as $t^{-s}\mathbb{1}_{[a,b]}$ does not have quadratic decay.

Lemma 8.4.1 (Fourier transform of a truncated power law)**.** Let $s = \sigma + i\tau$, $\sigma \ge 0$, $\tau \in \mathbb{R}$. Let $\nu \in \mathbb{R}\setminus\{0\}$, $b > a > \frac{|\tau|}{2\pi|\nu|}$. Then

$$\int_a^b t^{-s}e(\nu t)dt = \left.\frac{t^{-\sigma}e(\varphi_\nu(t))}{2\pi i\varphi_\nu'(t)}\right|_a^b + \sigma\int_a^b\frac{t^{-\sigma-1}}{2\pi i\varphi_\nu'(t)}e(\varphi_\nu(t))dt + \int_a^b\frac{t^{-\sigma}\varphi_\nu''(t)}{2\pi i(\varphi_\nu'(t))^2}e(\varphi_\nu(t))dt, \tag{8.1}$$

where $\varphi_\nu(t) = \nu t - \frac{\tau}{2\pi}\log t$.

[1] The other one is the approximate functional equation.

Proof. We write $t^{-s}e(\nu t) = t^{-\sigma}e(\varphi_\nu(t))$ and integrate by parts with $u = t^{-\sigma}/(2\pi i\varphi_\nu'(t))$, $v = e(\varphi_\nu(t))$. Here $\varphi_\nu'(t) = \nu - \tau/(2\pi t) \neq 0$ for $t \in [a,b]$ because $t \geq a > |\tau|/(2\pi|\nu|)$ implies $|\nu| > |\tau|/(2\pi t)$. Clearly

$$udv = \frac{t^{-\sigma}}{2\pi i\varphi_\nu'(t)} \cdot 2\pi i\varphi_\nu'(t)e(\varphi_\nu(t))dt = t^{-\sigma}e(\varphi_\nu(t))dt,$$

while

$$du = \left(\frac{-\sigma t^{-\sigma-1}}{2\pi i\varphi_\nu'(t)} - \frac{t^{-\sigma}\varphi_\nu''(t)}{2\pi i(\varphi_\nu'(t))^2}\right)dt.$$

□

Lemma 8.4.2 (Total variation of a function with monotone absolute value)**.** Let $g : [a,b] \to \mathbb{R}$ be continuous, with $|g(t)|$ non-increasing. Then g is monotone, and $\|g\|_{\mathrm{TV}} = |g(a)| - |g(b)|$.

Proof. Suppose g changed sign: $g(a') > 0 > g(b')$ or $g(a') < 0 < g(b')$ for some $a \leq a' < b' \leq b$. By IVT, there would be an $r \in [a', b']$ such that $g(r) = 0$. Since $|g|$ is non-increasing, $g(b') = 0$; contradiction. So, g does not change sign: either $g \leq 0$ or $g \geq 0$.

Thus, there is an $\varepsilon \in \{-1, 1\}$ such that $g(t) = \varepsilon|g(t)|$ for all $t \in [a,b]$. Hence, g is monotone. Then $\|g\|_{\mathrm{TV}} = |g(a) - g(b)|$. Since $|g(a)| \geq |g(b)|$ and $g(a)$, $g(b)$ are either both non-positive or non-negative, $|g(a) - g(b)| = |g(a)| - |g(b)|$. □

Lemma 8.4.3 (Non-stationary phase estimate)**.** Let $\varphi : [a,b] \to \mathbb{R}$ be C^1 with $\varphi'(t) \neq 0$ for all $t \in [a,b]$. Let $h : [a,b] \to \mathbb{R}$ be such that $g(t) = h(t)/\varphi'(t)$ is continuous and $|g(t)|$ is non-increasing. Then

$$\left|\int_a^b h(t)e(\varphi(t))dt\right| \leq \frac{|g(a)|}{\pi}.$$

Proof. Since φ is C^1, $e(\varphi(t))$ is C^1, and $h(t)e(\varphi(t)) = \frac{h(t)}{2\pi i\varphi'(t)}\frac{d}{dt}e(\varphi(t))$ everywhere. By Lemma 8.4.2, g is of bounded variation. Hence, we can integrate by parts:

$$\int_a^b h(t)e(\varphi(t))dt = \left.\frac{h(t)e(\varphi(t))}{2\pi i\varphi'(t)}\right|_a^b - \int_a^b e(\varphi(t))\, d\left(\frac{h(t)}{2\pi i\varphi'(t)}\right).$$

The first term on the right has absolute value $\leq \frac{|g(a)|+|g(b)|}{2\pi}$. Again by Lemma 8.4.2,

$$\left|\int_a^b e(\varphi(t))\, d\left(\frac{h(t)}{2\pi i\varphi'(t)}\right)\right| \leq \frac{1}{2\pi}\|g\|_{\mathrm{TV}} = \frac{|g(a)| - |g(b)|}{2\pi}.$$

We are done by $\frac{|g(a)|+|g(b)|}{2\pi} + \frac{|g(a)|-|g(b)|}{2\pi} = \frac{|g(a)|}{\pi}$. □

Lemma 8.4.4 (A decreasing function)**.** Let $\sigma \geq 0$, $\tau \in \mathbb{R}$, $\nu \in \mathbb{R} \setminus \{0\}$. Let $b > a > \frac{|\tau|}{2\pi|\nu|}$. Then, for any $k \geq 1$, $f(t) = t^{-\sigma-k}|2\pi\nu - \tau/t|^{-k-1}$ is decreasing on $[a,b]$.

Proof. Let $a \leq t \leq b$. Since $\left|\frac{\tau}{t\nu}\right| < 2\pi$, we see that $2\pi - \frac{\tau}{\nu t} > 0$, and so $|2\pi\nu - \tau/t|^{-k-1} = |\nu|^{-k-1}\left(2\pi - \frac{\tau}{t\nu}\right)^{-k-1}$. Now we take logarithmic derivatives:

$$t(\log f(t))' = -(\sigma + k) - (k+1)\frac{\tau/t}{2\pi\nu - \tau/t} = -\sigma - \frac{2\pi k + \frac{\tau}{t\nu}}{2\pi - \frac{\tau}{t\nu}} < -\sigma \leq 0,$$

since, again by $\frac{|\tau|}{t|\nu|} < 2\pi$ and $k \geq 1$, we have $2\pi k + \frac{\tau}{t\nu} > 0$, and, as we said, $2\pi - \frac{\tau}{t\nu} > 0$. □

Lemma 8.4.5 (Estimating an integral)**.** Let $s = \sigma + i\tau$, $\sigma \geq 0$, $\tau \in \mathbb{R}$. Let $\nu \in \mathbb{R} \setminus \{0\}$, $b > a > \frac{|\tau|}{2\pi|\nu|}$. Then

$$\int_a^b t^{-s} e(\nu t) dt = \left. \frac{t^{-\sigma} e(\varphi_\nu(t))}{2\pi i \varphi'_\nu(t)} \right|_a^b + \frac{a^{-\sigma-1}}{2\pi^2} O^*\left(\frac{\sigma}{(\nu - \vartheta)^2} + \frac{|\vartheta|}{|\nu - \vartheta|^3} \right),$$

where $\varphi_\nu(t) = \nu t - \frac{\tau}{2\pi} \log t$ and $\vartheta = \frac{\tau}{2\pi a}$.

Proof. Apply Lemma 8.4.1. Since $\varphi'_\nu(t) = \nu - \tau/(2\pi t)$, we know by Lemma 8.4.4 (with $k = 1$) that $g_1(t) = \frac{t^{-\sigma-1}}{(\varphi'_\nu(t))^2}$ is decreasing on $[a, b]$. We know that $\varphi'_\nu(t) \neq 0$ for $t \geq a$ by $a > \frac{|\tau|}{2\pi|\nu|}$, and so we also know that $g_1(t)$ is continuous for $t \geq a$. Hence, by Lemma 8.4.3,

$$\left| \int_a^b \frac{t^{-\sigma-1}}{2\pi i \varphi'_\nu(t)} e(\varphi_\nu(t)) dt \right| \leq \frac{1}{2\pi} \cdot \frac{|g_1(a)|}{\pi} = \frac{1}{2\pi^2} \frac{a^{-\sigma-1}}{|\nu - \vartheta|^2},$$

since $\varphi'_\nu(a) = \nu - \vartheta$. We remember to include the factor of σ in front of an integral in (8.1).

Since $\varphi'_\nu(t)$ is as above and $\varphi''_\nu(t) = \tau/(2\pi t^2)$, we know by Lemma 8.4.4 (with $k = 2$) that $g_2(t) = \frac{t^{-\sigma}|\varphi''_\nu(t)|}{|\varphi'_\nu(t)|^3} = \frac{|\tau|}{2\pi} \frac{t^{-\sigma-2}}{|\varphi'_\nu(t)|^3}$ is decreasing on $[a, b]$ we also know, as before, that $g_2(t)$ is continuous. Hence, again by Lemma 8.4.3,

$$\left| \int_a^b \frac{t^{-\sigma} \varphi''_\nu(t)}{2\pi i (\varphi'_\nu(t))^2} e(\varphi_\nu(t)) dt \right| \leq \frac{1}{2\pi} \frac{|g_2(a)|}{\pi} = \frac{1}{2\pi^2} \frac{a^{-\sigma-1}|\vartheta|}{|\nu - \vartheta|^3}.$$

$\square$

Lemma 8.4.6 (Estimating an sum)**.** Let $s = \sigma + i\tau$, $\sigma, \tau \in \mathbb{R}$. Let $n \in \mathbb{Z}_{>0}$. Let $a, b \in \mathbb{Z} + \frac{1}{2}$, $b > a > \frac{|\tau|}{2\pi n}$. Write $\varphi_\nu(t) = \nu t - \frac{\tau}{2\pi} \log t$. Then

$$\frac{1}{2} \sum_{\nu = \pm n} \left. \frac{t^{-\sigma} e(\varphi_\nu(t))}{2\pi i \varphi'_\nu(t)} \right|_a^b = \left. \frac{(-1)^n t^{-s} \cdot \frac{\tau}{2\pi t}}{2\pi i \left(n^2 - \left(\frac{\tau}{2\pi t} \right)^2 \right)} \right|_a^b.$$

Proof. Since $e(\varphi_\nu(t)) = e(\nu t) t^{-i\tau} = (-1)^\nu t^{-i\tau}$ for any half-integer t and any integer ν,

$$\left. \frac{t^{-\sigma} e(\varphi_\nu(t))}{2\pi i \varphi'_\nu(t)} \right|_a^b = \left. \frac{(-1)^\nu t^{-s}}{2\pi i \varphi'_\nu(t)} \right|_a^b$$

for $\nu = \pm n$. Clearly $(-1)^\nu = (-1)^n$. Since $\varphi'_\nu(t) = \nu - \alpha$ for $\alpha = \frac{\tau}{2\pi t}$,

$$\frac{1}{2} \sum_{\nu = \pm n} \frac{1}{\varphi'_\nu(t)} = \frac{1/2}{n - \alpha} + \frac{1/2}{-n - \alpha} = \frac{-\alpha}{\alpha^2 - n^2} = \frac{\alpha}{n^2 - \alpha^2}.$$

$\square$

It is this easy step that gives us quadratic decay on n. It is just as in the proof of van der Corput's Process B in, say, [29, I.6.3, Thm. 4].

Proposition 8.4.1 (Estimating a Fourier cosine integral)**.** Let $s = \sigma + i\tau$, $\sigma \geq 0$, $\tau \in \mathbb{R}$. Let $a, b \in \mathbb{Z} + \frac{1}{2}$, $b > a > \frac{|\tau|}{2\pi}$. Write $\vartheta = \frac{\tau}{2\pi a}$. Then, for any integer $n \geq 1$,

$$\int_a^b t^{-s} \cos 2\pi nt \, dt = \left(\frac{(-1)^n t^{-s}}{2\pi i} \cdot \frac{\frac{\tau}{2\pi t}}{n^2 - \left(\frac{\tau}{2\pi t}\right)^2} \right) \Bigg|_a^b$$
$$+ \frac{a^{-\sigma-1}}{4\pi^2} O^* \left(\frac{\sigma}{(n-\vartheta)^2} + \frac{\sigma}{(n+\vartheta)^2} + \frac{|\vartheta|}{|n-\vartheta|^3} + \frac{|\vartheta|}{|n+\vartheta|^3} \right).$$

Proof. Write $\cos 2\pi nt = \frac{1}{2}(e(nt) + e(-nt))$. Since $n \geq 1$ and $a > \frac{|\tau|}{2\pi}$, we know that $a > \frac{|\tau|}{2\pi n}$, and so we can apply Lemma 8.4.5 with $\nu = \pm n$. We then apply Lemma 8.4.6 to combine the boundary contributions $|_a^b$ for $\nu = \pm n$. □

8.4.2 Approximating zeta(s)

We start with an application of Euler-Maclaurin.

Lemma 8.4.7 (Identity for a partial sum of zeta(s) for integer b)**.** Let $b > 0$, $b \in \mathbb{Z}$. Then, for all $s \in \mathbb{C} \setminus \{1\}$ with $\Re s > 0$,

$$\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) + \frac{b^{1-s}}{1-s} + \frac{b^{-s}}{2} + s \int_b^\infty \left(\{y\} - \frac{1}{2} \right) \frac{dy}{y^{s+1}}. \tag{8.2}$$

Proof. Assume first that $\Re s > 1$. By first-order Euler-Maclaurin,

$$\sum_{n > b} \frac{1}{n^s} = \int_b^\infty \frac{dy}{y^s} + \int_b^\infty \left(\{y\} - \frac{1}{2} \right) d\left(\frac{1}{y^s} \right).$$

Here $\int_b^\infty \frac{dy}{y^s} = -\frac{b^{1-s}}{1-s}$ and $d\left(\frac{1}{y^s}\right) = -\frac{s}{y^{s+1}} dy$. Hence, by $\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) - \sum_{n > b} \frac{1}{n^s}$ for $\Re s > 1$,

$$\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) + \frac{b^{1-s}}{1-s} + \int_b^\infty \left(\{y\} - \frac{1}{2} \right) \frac{s}{y^{s+1}} dy.$$

Since the integral converges absolutely for $\Re s > 0$, both sides extend holomorphically to $\{s \in \mathbb{C} : \Re s > 0, s \neq 1\}$; thus, the equation holds throughout that region. □

Lemma 8.4.8 (Identity for a partial sum of zeta(s))**.** Let $b > 0$, $b \in \mathbb{Z} + \frac{1}{2}$. Then, for all $s \in \mathbb{C} \setminus \{1\}$ with $\Re s > 0$,

$$\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) + \frac{b^{1-s}}{1-s} + s \int_b^\infty \left(\{y\} - \frac{1}{2} \right) \frac{dy}{y^{s+1}}. \tag{8.3}$$

Proof. Assume first that $\Re s > 1$. By first-order Euler-Maclaurin and $b \in \mathbb{Z} + \frac{1}{2}$,

$$\sum_{n > b} \frac{1}{n^s} = \int_b^\infty \frac{dy}{y^s} + \int_b^\infty \left(\{y\} - \frac{1}{2} \right) d\left(\frac{1}{y^s} \right).$$

Here $\int_b^\infty \frac{dy}{y^s} = -\frac{b^{1-s}}{1-s}$ and $d\left(\frac{1}{y^s}\right) = -\frac{s}{y^{s+1}} dy$. Hence, by $\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) - \sum_{n > b} \frac{1}{n^s}$ for $\Re s > 1$,

$$\sum_{n \leq b} \frac{1}{n^s} = \zeta(s) + \frac{b^{1-s}}{1-s} + \int_b^\infty \left(\{y\} - \frac{1}{2} \right) \frac{s}{y^{s+1}} dy.$$

Since the integral converges absolutely for $\Re s > 0$, both sides extend holomorphically to $\{s \in \mathbb{C} : \Re s > 0, s \neq 1\}$; thus, the equation holds throughout that region. □

Lemma 8.4.9 (Estimate for a partial sum of $\zeta(s)$)**.** Let $b > a > 0$, $b \in \mathbb{Z} + \frac{1}{2}$. Then, for all $s \in \mathbb{C} \setminus \{1\}$ with $\sigma = \Re s > 0$,

$$\sum_{n \le a} \frac{1}{n^s} = -\sum_{a < n \le b} \frac{1}{n^s} + \zeta(s) + \frac{b^{1-s}}{1-s} + O^*\left(\frac{|s|}{2\sigma b^\sigma}\right).$$

Proof. By Lemma 8.4.8, $\sum_{n \le a} = \sum_{n \le b} - \sum_{a < n \le b}$, $\left|\{y\} - \frac{1}{2}\right| \le \frac{1}{2}$ and $\int_b^\infty \frac{dy}{|y^{s+1}|} = \frac{1}{\sigma b^\sigma}$. □

Lemma 8.4.10 (Poisson summation for a partial sum of $\zeta(s)$)**.** Let $a, b \in \mathbb{R} \setminus \mathbb{Z}$, $b > a > 0$. Let $s \in \mathbb{C} \setminus \{1\}$. Define $f : \mathbb{R} \to \mathbb{C}$ by $f(y) = 1_{[a,b]}(y)/y^s$. Then

$$\sum_{a < n \le b} \frac{1}{n^s} = \frac{b^{1-s} - a^{1-s}}{1-s} + \lim_{N \to \infty} \sum_{n=1}^{N} (\hat{f}(n) + \hat{f}(-n)).$$

Proof. Since $a \notin \mathbb{Z}$, $\sum_{a < n \le b} \frac{1}{n^s} = \sum_{n \in \mathbb{Z}} f(n)$. By Poisson summation (as in [21, Thm. D.3])

$$\sum_{n \in \mathbb{Z}} f(n) = \lim_{N \to \infty} \sum_{n=-N}^{N} \hat{f}(n) = \hat{f}(0) + \lim_{N \to \infty} \sum_{n=1}^{N} (\hat{f}(n) + \hat{f}(-n)),$$

where we use the facts that f is in L^1, of bounded variation, and (by $a, b \notin \mathbb{Z}$) continuous at every integer. Now

$$\hat{f}(0) = \int_{\mathbb{R}} f(y) dy = \int_a^b \frac{dy}{y^s} = \frac{b^{1-s} - a^{1-s}}{1-s}.$$

□

We could prove these equations starting from Euler's product for $\sin \pi z$.

Lemma 8.4.11 (Euler/Mittag-Leffler expansion for cosec)**.** Let $z \in \mathbb{C}$, $z \notin \mathbb{Z}$. Then

$$\frac{\pi}{\sin \pi z} = \frac{1}{z} + \sum_{n > 0} (-1)^n \left(\frac{1}{z-n} + \frac{1}{z+n}\right).$$

Proof. Let us start from the Mittag-Leffler expansion $\pi \cot \pi s = \frac{1}{s} + \sum_n \left(\frac{1}{s-n} + \frac{1}{s+n}\right)$.

Applying the trigonometric identity $\cot u - \cot\left(u + \frac{\pi}{2}\right) = \cot u + \tan u = \frac{2}{\sin 2u}$ with $u = \pi z/2$, and letting $s = z/2$, $s = (z+1)/2$, we see that

$$\begin{aligned}
\frac{\pi}{\sin \pi z} &= \frac{\pi}{2} \cot \frac{\pi z}{2} - \frac{\pi}{2} \cot \frac{\pi(z+1)}{2} \\
&= \frac{1/2}{z/2} + \sum_n \left(\frac{1/2}{\frac{z}{2} - n} + \frac{1/2}{\frac{z}{2} + n}\right) - \frac{1/2}{(z+1)/2} - \sum_n \left(\frac{1/2}{\frac{z+1}{2} - n} + \frac{1/2}{\frac{z+1}{2} + n}\right) \\
&= \frac{1}{z} + \sum_n \left(\frac{1}{z - 2n} + \frac{1}{z + 2n}\right) - \sum_n \left(\frac{1}{z - (2n-1)} + \frac{1}{z + (2n-1)}\right)
\end{aligned}$$

after reindexing the second sum. Regrouping terms again, we obtain our equation. □

Lemma 8.4.12 (Euler/Mittag-Leffler expansion for cosec squared)**.** Let $z \in \mathbb{C}$, $z \notin \mathbb{Z}$. Then

$$\frac{\pi^2}{\sin^2 \pi z} = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}.$$

Proof. Differentiate the expansion of $\pi\cot\pi z$ term-by-term because it converges uniformly on compact subsets of $\mathbb{C}\setminus\mathbb{Z}$. By $(\pi\cot\pi z)' = -\frac{\pi^2}{\sin^2\pi z}$ and $\left(\frac{1}{z\pm n}\right)' = -\frac{1}{(z\pm n)^2}$, we are done. □

Lemma 8.4.13 (Estimate for an inverse cubic series)**.** For $\vartheta\in\mathbb{R}$ with $0\le|\vartheta|<1$,

$$\sum_n\left(\frac{1}{(n-\vartheta)^3}+\frac{1}{(n+\vartheta)^3}\right)\le\frac{1}{(1-|\vartheta|)^3}+2\zeta(3)-1.$$

Proof. Since $\frac{1}{(n-\vartheta)^3}+\frac{1}{(n+\vartheta)^3}$ is even, we may replace ϑ by $|\vartheta|$. Then we rearrange the sum:

$$\sum_{n=1}^{\infty}\left(\frac{1}{(n-|\vartheta|)^3}+\frac{1}{(n+|\vartheta|)^3}\right)=\frac{1}{(1-|\vartheta|)^3}+\sum_{n=1}^{\infty}\left(\frac{1}{\left(n+1-|\vartheta|\right)^3}+\frac{1}{\left(n+|\vartheta|\right)^3}\right).$$

We may write $(n+1-|\vartheta|)^3$, $(n+|\vartheta|)^3$ as $(n+\frac{1}{2}-t)^3$, $(n+\frac{1}{2}+t)^3$ for $t=|\vartheta|-1/2$. Since $1/u^3$ is convex, $\frac{1}{(n+1/2-t)^3}+\frac{1}{(n+1/2+t)^3}$ reaches its maximum on $[-1/2,1/2]$ at the endpoints. Hence

$$\sum_{n=1}^{\infty}\left(\frac{1}{\left(n+1-|\vartheta|\right)^3}+\frac{1}{\left(n+|\vartheta|\right)^3}\right)\le\sum_{n=1}^{\infty}\left(\frac{1}{n^3}+\frac{1}{(n+1)^3}\right)=2\zeta(3)-1.$$

□

Lemma 8.4.14 (Estimate for a Fourier sum)**.** Let $s=\sigma+i\tau$, $\sigma\ge0$, $\tau\in\mathbb{R}$, with $s\ne1$. Let $b>a>0$, $a,b\in\mathbb{Z}+\frac{1}{2}$, with $a>\frac{|\tau|}{2\pi}$. Define $f:\mathbb{R}\to\mathbb{C}$ by $f(y)=1_{[a,b]}(y)/y^s$. Write $\vartheta=\frac{\tau}{2\pi a}$, $\vartheta_-=\frac{\tau}{2\pi b}$. Then

$$\sum_n(\hat f(n)+\hat f(-n))=\frac{a^{-s}g(\vartheta)}{2i}-\frac{b^{-s}g(\vartheta_-)}{2i}+O^*\left(\frac{C_{\sigma,\vartheta}}{a^{\sigma+1}}\right)$$

with absolute convergence, where $g(t)=\frac{1}{\sin\pi t}-\frac{1}{\pi t}$ for $t\ne0$, $g(0)=0$, and

$$C_{\sigma,\vartheta}=\begin{cases}\frac{\sigma}{2}\left(\frac{1}{\sin^2\pi\vartheta}-\frac{1}{(\pi\vartheta)^2}\right)+\frac{|\vartheta|}{2\pi^2}\left(\frac{1}{(1-|\vartheta|)^3}+2\zeta(3)-1\right) & \text{for } \vartheta\ne0,\\ \sigma/6 & \text{for } \vartheta=0.\end{cases}\tag{8.4}$$

Proof. By Proposition 8.4.1, multiplying by 2 (since $e(-nt)+e(nt)=2\cos2\pi nt$),

$$\begin{aligned}\hat f(n)+\hat f(-n)&=\frac{a^{-s}}{2\pi i}\frac{(-1)^{n+1}2\vartheta}{n^2-\vartheta^2}-\frac{b^{-s}}{2\pi i}\frac{(-1)^{n+1}2\vartheta_-}{n^2-\vartheta_-^2}\\&\quad+\frac{a^{-\sigma-1}}{2\pi^2}O^*\left(\frac{\sigma}{(n-\vartheta)^2}+\frac{\sigma}{(n+\vartheta)^2}+\frac{|\vartheta|}{(n-\vartheta)^3}+\frac{|\vartheta|}{(n+\vartheta)^3}\right),\end{aligned}\tag{8.5}$$

where $\vartheta_-=\tau/(2\pi b)$. Note $|\vartheta_-|\le|\vartheta|<1$. By the Lemma 8.4.11,

$$\sum_n\frac{(-1)^{n+1}2z}{n^2-z^2}=\frac{\pi}{\sin\pi z}-\frac{1}{z}$$

for $z\ne0$, while $\sum_n\frac{(-1)^{n+1}2z}{n^2-z^2}=\sum_n0=0$ for $z=0$. Moreover, by Lemmas 8.4.12 and 8.4.13, for $\vartheta\ne0$,

$$\sum_n\left(\frac{\sigma}{(n-\vartheta)^2}+\frac{\sigma}{(n+\vartheta)^2}\right)\le\sigma\cdot\left(\frac{\pi^2}{\sin^2\pi\vartheta}-\frac{1}{\vartheta^2}\right),$$

$$\sum_n \left(\frac{|\vartheta|}{(n-\vartheta)^3} + \frac{|\vartheta|}{(n+\vartheta)^3}\right) \le |\vartheta| \cdot \left(\frac{1}{(1-|\vartheta|)^3} + 2\zeta(3) - 1\right).$$

If $\vartheta = 0$, then $\sum_n \left(\frac{\sigma}{(n-\vartheta)^2} + \frac{\sigma}{(n+\vartheta)^2}\right) = 2\sigma \sum_{n=1}^{\infty} \frac{1}{n^2} = \sigma\frac{\pi^2}{3}$. □

Proposition 8.4.2 (Approximation of zeta(s) by a partial sum)**.** Let $s = \sigma + i\tau$, $\sigma \ge 0$, $\tau \in \mathbb{R}$, with $s \ne 1$. Let $a \in \mathbb{Z} + \frac{1}{2}$ with $a > \frac{|\tau|}{2\pi}$. Then

$$\zeta(s) = \sum_{n\le a} \frac{1}{n^s} - \frac{a^{1-s}}{1-s} + c_\vartheta a^{-s} + O^*\left(\frac{C_{\sigma,\vartheta}}{a^{\sigma+1}}\right), \tag{8.6}$$

where $\vartheta = \frac{\tau}{2\pi a}$, $c_\vartheta = \frac{i}{2}\left(\frac{1}{\sin \pi\vartheta} - \frac{1}{\pi\vartheta}\right)$ for $\vartheta \ne 0$, $c_0 = 0$, and $C_{\sigma,\vartheta}$ is as in (8.4).

Proof. Assume first that $\sigma > 0$. Let $b \in \mathbb{Z} + \frac{1}{2}$ with $b > a$, and define $f(y) = \frac{1_{[a,b]}(y)}{y^s}$. By Lemma 8.4.9 and Lemma 8.4.10,

$$\sum_{n\le a} \frac{1}{n^s} = \zeta(s) + \frac{a^{1-s}}{1-s} - \lim_{N\to\infty} \sum_{n=1}^{N} (\hat{f}(n) + \hat{f}(-n)) + O^*\left(\frac{2|s|}{\sigma b^\sigma}\right).$$

We apply Lemma 8.4.14 to estimate $\lim_{N\to\infty} \sum_{n=1}^{N} (\hat{f}(n) + \hat{f}(-n))$. We obtain

$$\sum_{n\le a} \frac{1}{n^s} = \zeta(s) + \frac{a^{1-s}}{1-s} - \frac{a^{-s} g(\vartheta)}{2i} + O^*\left(\frac{C_{\sigma,\vartheta}}{a^{\sigma+1}}\right) + \frac{b^{-s} g(\vartheta_-)}{2i} + O^*\left(\frac{2|s|}{\sigma b^\sigma}\right),$$

where $\vartheta_- = \frac{\tau}{2\pi b}$ and $g(t)$ is as in Lemma 8.4.14, and so $-\frac{g(\vartheta)}{2i} = c_\vartheta$. We let $b \to \infty$ through the half-integers, and obtain (8.6), since $b^{-\sigma} \to 0$, $\vartheta_- \to 0$ and $g(\vartheta_-) \to g(0) = 0$ as $b \to \infty$.

Finally, the case $\sigma = 0$ follows since all terms in (8.6) extend continuously to $\sigma = 0$. □

Remark 8.4.1. The term $c_\vartheta a^{-s}$ in (8.6) does not seem to have been worked out before in the literature; the factor of i in c_ϑ was a surprise. For the sake of comparison, let us note that, if $a \ge x$, then $|\vartheta| \le 1/2\pi$, and so $|c_\vartheta| \le |c_{\pm 1/2\pi}| = 0.04291\ldots$ and $|C_{\sigma,\vartheta}| \le |C_{\sigma,\pm 1/2\pi}| \le 0.176\sigma + 0.246$. While c_ϑ is optimal, $C_{\sigma,\vartheta}$ need not be – but then that is irrelevant for most applications.

Chapter 9

Primary explicit estimates

9.1 Definitions

In this section we define the basic types of primary estimates we will work with in the project.

9.2 A Lemma involving the Möbius Function

In this section we establish a lemma involving sums of the Möbius function.

Definition 9.2.1 (Q)**.** $Q(x)$ is the number of squarefree integers $\leq x$.

Definition 9.2.2 (R)**.** $R(x) = Q(x) - x/\zeta(2)$.

Definition 9.2.3 (M)**.** $M(x)$ is the summatory function of the Möbius function.

Sublemma 9.2.1 (Mobius Lemma 1, initial step)**.** For any $x > 0$,

$$Q(x) = \sum_{k \leq x} M\left(\sqrt{\frac{x}{k}}\right)$$

.

Proof. We compute

$$Q(x) = \sum_{n \leq x} \sum_{d : d^2 | n} \mu(d) = \sum_{k, d : kd^2 \leq x} \mu(d)$$

giving the claim. □

Lemma 9.2.1 (Mobius Lemma 1)**.** For any $x > 0$,

$$R(x) = \sum_{k \leq x} M\left(\sqrt{\frac{x}{k}}\right) - \int_0^x M\left(\sqrt{\frac{x}{u}}\right) du. \tag{9.1}$$

Proof. The equality is immediate from Theorem 9.2.1 and exchanging the order of $\sum$ and $\int$, as is justified by $\sum_n |\mu(n)| \int_0^{x/n^2} du \leq \sum_n x/n^2 < \infty$)

$$\int_0^x M\left(\sqrt{\frac{x}{u}}\right) du = \int_0^x \sum_{n \leq \sqrt{\frac{x}{u}}} \mu(n) du = \sum_n \mu(n) \int_0^{\frac{x}{n^2}} du = x \sum_n \frac{\mu(n)}{n^2} = \frac{x}{\zeta(2)}.$$

□

Since our sums start from 1, the sum $\sum_{k\leq K}$ is empty for $K=0$.

Sublemma 9.2.2 (Mobius Lemma 2 - first step)**.** For any $K \leq x$,

$$\sum_{k\leq x} M\left(\sqrt{\frac{x}{k}}\right) = \sum_{k\leq K} M\left(\sqrt{\frac{x}{k}}\right) + \sum_{K<k\leq x+1} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} M\left(\sqrt{\frac{x}{k}}\right) du.$$

Proof. This is just splitting the sum at K. ☐

Sublemma 9.2.3 (Mobius Lemma 2 - second step)**.** For any $K \leq x$, for $f(u) = M(\sqrt{x/u})$,

$$\sum_{K<k\leq x+1} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} f(u)du = \int_{K+\frac{1}{2}}^{\lfloor x\rfloor+\frac{3}{2}} f(u)du = \int_{K+\frac{1}{2}}^{x} f(u)du,$$

Proof. This is just splitting the integral at K, since $f(u) = M(\sqrt{x/u}) = 0$ for $x > u$. ☐

Lemma 9.2.2 (Mobius Lemma 2)**.** For any $x > 0$ and any integer $K \geq 0$,

$$\begin{aligned} R(x) = \sum_{k\leq K} M\left(\sqrt{\frac{x}{k}}\right) - \int_{0}^{K+\frac{1}{2}} M\left(\sqrt{\frac{x}{u}}\right) du \\ - \sum_{K<k\leq x+1} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} \left(M\left(\sqrt{\frac{x}{u}}\right) - M\left(\sqrt{\frac{x}{k}}\right)\right) du \end{aligned} \tag{9.2}$$

Proof. We split into two cases. If $K > x$, the second line of (9.2) is empty, and the first one equals (9.1), by $M(t) = 0$ for $t < 1$, so (9.2) holds.

Now suppose that $K \leq x$. Then we combine Sublemma 9.2.2 and Sublemma 9.2.3 with Lemma 9.2.1 to give the claim. ☐

9.3 The estimates of Fiori, Kadiri, and Swidinsky

In this section we establish the primary results of Fiori, Kadiri, and Swidinsky [14].

TODO: reorganize this blueprint and add proofs.

Theorem 9.3.1 (FKS Theorem 2.7)**.** Let H_0 denote a verification height for RH. Let $10^9/H_0 \leq k \leq 1$, $t > 0$, $H \in [1002, H_0)$, $\alpha > 0$, $\delta \geq 1$, $\eta_0 = 0.23622$, $1 + \eta_0 \leq \mu \leq 1 + \eta$, and $\eta \in (\eta_0, 1/2)$ be fixed. Let $\sigma > 1/2 + d/\log H_0$. Then for any $T \geq H_0$, one has

$$N(\sigma,T) \leq (T-H)\log T/(2\pi d) * \log(1+CC_1(\log(kT))^{2\sigma}(\log T)^{4(1-\sigma)}T^{8/3(1-\sigma)}/(T-H)) + CC_2 * \log^2 T/2\pi d$$

and

$$N(\sigma,T) \leq \frac{CC_1}{2\pi d}(\log kT)^{2\sigma}(\log T)^{5-4*\sigma}T^{8/3(1-\sigma)} + CC_2 * \log^2 T/2\pi d$$

.

Proof. ☐

Definition 9.3.1 (FKS Corollary 2.9)**.** For each $\sigma_1, \sigma_2, \tilde{c}_1, \tilde{c}_2$ given in Table 8, we have $N(\sigma,T) \leq \tilde{c}_1 T^{p(\sigma)} \log^{q(\sigma)} + \tilde{c}_2 \log^2 T$ for $\sigma_1 \leq \sigma \leq \sigma_2$ with $p(\sigma) = 8/3(1-\sigma)$ and $q(\sigma) = 5 - 2\sigma$.

Theorem 9.3.2 (FKS Lemma 2.1)**.** If $|N(T) - (T/2\pi \log(T/2\pi e) + 7/8)| \leq R(T)$ then $\sum_{U \leq \gamma < V} 1/\gamma \leq B_1(U, V)$.

Proof. □

Theorem 9.3.3 (FKS Corollary 2.3)**.** For each pair T_0, S_0 in Table 1 we have, for all $V > T_0$, $\sum_{0 < \gamma < V} 1/\gamma < S_0 + B_1(T_0, V)$.

Proof. □

Theorem 9.3.4 (FKS Lemma 2.5)**.** Let $T_0 \geq 2$ and $\gamma > 0$. Assume that there exist c_1, c_2, p, q, T_0 for which one has a zero density bound. Assume $\sigma \geq 5/8$ and $T_0 \leq U < V$. Then $s_0(\sigma, U, V) \leq B_0(\sigma, U, V)$.

Proof. □

Theorem 9.3.5 (FKS Remark 2-6-a)**.** $\Gamma(3, x) = (x^2 + 2(x + 1))e^{-x}$.

Proof. □

Theorem 9.3.6 (FKS Remark 2-6-b)**.** For $s > 1$, one has $\Gamma(s, x) \sim x^{s-1}e^{-x}$.

Proof. □

Theorem 9.3.7 (FKS Theorem 3.1)**.** Let $x > e^{50}$ and $50 < T < x$. Then $E_\psi(x) \leq \sum_{|\gamma| < T} |x^{\rho-1}/\rho| + 2\log^2 x/T$.

Proof. □

Theorem 9.3.8 (FKS Theorem 3.2)**.** For any $\alpha \in (0, 1/2]$ and $\omega \in [0, 1]$ there exist M, x_M such that for $\max(51, \log x) < T < (x^\alpha - 2)/5$ and some $T^* \in [T, 2.45T]$,

$$|\psi(x) - (x - \sum_{|\gamma| \leq T^*} x^\rho/\rho)| \leq Mx/T * log^{1-\omega} x$$

for all $x \geq x_M$.

Proof. □

Theorem 9.3.9 (FKS Proposition 3.4)**.** Let $x > e^{50}$ and $3\log x < T < \sqrt{x}/3$. Then $E_\psi(x) \leq \sum_{|\gamma| < T} |x^{\rho-1}/\rho| + 2\log^2 x/T$.

Proof. □

Theorem 9.3.10 (FKS Proposition 3.6)**.** Let $\sigma_1 \in (1/2, 1)$ and let (T_0, S_0) be taken from Table 1. Then $\Sigma_0^{\sigma_1} \leq 2x^{-1/2}(S_0 + B_1(T_0, T)) + (x_1^{\sigma_1-1} - x^{-1/2})B_1(H_0, T)$.

Proof. □

Theorem 9.3.11 (FKS equation (3.13))**.** $\Sigma_a^b = 2 * \sum_{H_a \leq \gamma < T; a \leq \beta < b} \frac{x^{\beta-1}}{\gamma}$.

Proof. □

Theorem 9.3.12 (FKS Remark 3.7)**.** If $\sigma < 1 - 1/R \log H_0$ then $H_\sigma = H_0$.

Proof. □

Theorem 9.3.13 (FKS Proposition 3.8)**.** Let $N \geq 2$ be an integer. If $5/8 \leq \sigma_1 < \sigma_2 \leq 1$, $T \geq H_0$, then $\Sigma_{\sigma_1}^{\sigma_2} \leq 2x^{-(1-\sigma_1)+(\sigma_2-\sigma_1/N)} B_0(\sigma_1, H_{\sigma_1}, T) + 2x^{(1-\sigma_1)}(1-x^{-(\sigma_2-\sigma_1)/N}) \sum_{n=1}^{N-1} B_0(\sigma^{(n)}, H^{(n)}, T) x^{(\sigma_2-\sigma_1)(n-}$

Proof. □

Theorem 9.3.14 (FKS Corollary 3.10)**.** If $\sigma_1 \geq 0.9$ then $\Sigma_{\sigma_1}^{\sigma_2} \leq 0.00125994 x^{\sigma_2 - 1}$.

Proof. □

Theorem 9.3.15 (FKS Proposition 3.11)**.** Let $5/8 < \sigma_2 \leq 1$, $t_0 = t_0(\sigma_2, x) = \max(H_{\sigma_2}, \exp(\sqrt{\log x}/R))$ and $T > 0$. Let $K \geq 2$ and consider a strictly increasing sequence $(t_k)_{k=0}^K$ such that $t_k = T$. Then $\Sigma_{\sigma_2}^1 \leq 2N(\sigma_2, T)x^{-1/R\log t_0}/t_0$ and $\Sigma_{\sigma_2}^1 \leq 2((\sum_{k=1}^{K-1} N(\sigma_2, t_k)(x^{-1/R\log t_{k-1}}/t_{k-1} - x^{-1/(R\log t_k)}/t_k)) + x^{-1/R\log t_{K-1}}/t_{K-1} N(\sigma_2, T))$.

Proof. □

Theorem 9.3.16 (FKS Corollary 3.12)**.** Let $5/8 < \sigma_2 \leq 1$, $t_0 = t_0(\sigma_2, x) = \max\left(H_{\sigma_2}, \exp\left(\sqrt{\frac{\log x}{R}}\right)\right)$, $T > t_0$. Let $K \geq 2$, $\lambda = (T/t_0)^{1/K}$, and consider $(t_k)_{k=0}^K$ the sequence given by $t_k = t_0 \lambda^k$. Then

$$\Sigma_{\sigma_2}^1 = 2 \sum_{\substack{0<\gamma<T \\ \sigma_2 \leq \beta < 1}} \frac{x^{\beta-1}}{\gamma} \leq \varepsilon_4(x, \sigma_2, K, T),$$

where

$$\varepsilon_4(x, \sigma_2, K, T) = 2 \sum_{k=1}^{K-1} \frac{x^{-\frac{1}{R\log t_k}}}{t_k} \left(\tilde{N}(\sigma_2, t_{k+1}) - \tilde{N}(\sigma_2, t_k)\right) + 2\tilde{N}(\sigma_2, t_1) \frac{x^{-\frac{1}{R(\log t_0)}}}{t_0},$$

and $\tilde{N}(\sigma, T)$ satisfy (ZDB) $N(\sigma, T) \leq \tilde{N}(\sigma, T)$.

Proof. □

Theorem 9.3.17 (FKS Proposition 3-14)**.** Fix $K \geq 2$ and $c > 1$, and set t_0, T, and σ_2 as functions of x defined by

$$t_0 = t_0(x) = \exp\left(\sqrt{\frac{\log x}{R}}\right), \quad T = t_0^c, \quad \text{and} \quad \sigma_2 = 1 - \frac{2}{R\log t_0}. \tag{9.3}$$

Then, with $\varepsilon_4(x, \sigma_2, K, T)$ as defined in (3.22), we have that as $x \to \infty$,

$$\varepsilon_4(x, \sigma_2, K, T) = (1 + o(1))C \frac{(\log t_0)^{3 + \frac{4}{R\log t_0}}}{t_0^2}, \quad \text{with } C = 2c_1 e^{\frac{16w_1}{3R}} w_1^3, \text{ and } w_1 = 1 + \frac{c-1}{K}, \tag{9.4}$$

where c_1 is an admissible value for (ZDB) on some interval $[\sigma_1, 1]$. Moreover, both $\varepsilon_4(x, \sigma_2, K, T)$ and $\frac{\varepsilon_4(x,\sigma_2,K,T)t_0^2}{(\log t_0)^3}$ are decreasing in x for $x > \exp(Re^2)$.

Proof. □

Theorem 9.3.18 (FKS Theorem 1.1)**.** For any x_0 with $\log x_0 > 1000$, and all $0.9 < \sigma_2 < 1$, $2 \leq c \leq 30$, and $N, K \geq 1$ the formula $\varepsilon(x_0) := \varepsilon(x_0, \sigma_2, c, N, K)$ as defined in (4.1) gives an effectively computable bound

$$E_\psi(x) \leq \varepsilon(x_0) \quad \text{for all } x \geq x_0.$$

Proof. □

Theorem 9.3.19 (FKS Theorem 1.1b)**.** Moreover, a collection of values, $\varepsilon(x_0)$ computed with well chosen parameters are provided in Table 5.

Proof. □

Theorem 9.3.20 (FKS Lemma 5.2)**.** For all $0 < \log x \leq 2100$ we have that

$$E_\psi(x) \leq 2(\log x)^{3/2} \exp\left(-0.8476836\sqrt{\log x}\right).$$

Proof. □

Theorem 9.3.21 (FKS Lemma 5.3)**.** For all $2100 < \log x \leq 200000$ we have that

$$E_\psi(x) \leq 9.22022(\log x)^{3/2} \exp\left(-0.8476836\sqrt{\log x}\right).$$

Proof. □

Theorem 9.3.22 (FKS Theorem 1.2b)**.** If $\log x_0 \geq 1000$ then we have an admissible bound for E_ψ with the indicated choice of $A(x_0)$, $B = 3/2$, $C = 2$, and $R = 5.5666305$.

Proof. □

Theorem 9.3.23 (FKS1 Corollary 1.3)**.** For all x > 2 we have $E_\psi(x) \leq 121.096(\log x/R)^{3/2} \exp(-2\sqrt{\log x/R})$ with $R = 5.5666305$.

Proof. □

Theorem 9.3.24 (FKS1 Corollary 1.4)**.** For all x > 2 we have $E_\psi(x) \leq 9.22022(\log x)^{3/2} \exp(-0.8476836\sqrt{\log x})$.

Proof. TODO. □

9.4 Numerical content of BKLNW Appendix A

Purely numerical calculations from Appendix A of [2]. This is kept in a separate file from the main file to avoid heavy recompilations. Because of this, this file should not import any other files from the PNT+ project, other than further numerical data files.

Sublemma 9.4.1 (BKLNW Table 8 vs expanded Table 8)**.** The value of $\varepsilon(b)$ arising from Table 8 of [2] is weaker than that from the expanded version of Table 8 available in the arXiv.

Proof. Routine computation. □

9.5 Appendix A of BKLNW

In this file we record the results from Appendix A of [2]. In this appendix, the authors derive explicit estimates on the error term in the prime number theorem for the Chebyshev function ψ assuming various inputs on the zeros of the Riemann zeta function, including a zero-density estimate, a classical zero-free region, and numerical verification of RH up to some height.

Sublemma 9.5.1 (Equation (A.7))**.** Let $x \geq e^{1000}$ and T satisfies $50 < T \leq x$. Then

$$\frac{\psi(x) - x}{x} = \sum_{|\gamma|<T} \frac{x^{\rho-1}}{\rho} + \mathcal{O}^*\left(\frac{2(\log x)^2}{T}\right)$$

where $A = \mathcal{O}^*(B)$ means $|A| \leq B$.

Proof. See [11, Theorem 1.3]. □

Definition 9.5.1 (Equation (A.8))**.** We denote

$$s_0(b, T) = \frac{2b^2}{T}.$$

Definition 9.5.2 (Definition of Sigma 1)**.** We denote

$$\Sigma_1 := \sum_{|\gamma| \leq T; \beta < 1-\delta} \frac{x^{\rho-1}}{\rho}$$

Definition 9.5.3 (Definition of Sigma 2)**.** We denote

$$\Sigma_2 := \sum_{|\gamma| \leq T; \beta \geq 1-\delta} \frac{x^{\rho-1}}{\rho}$$

Sublemma 9.5.2 (Equation (A.9))**.** We have

$$\sum_{|\gamma|<T} \frac{x^{\rho-1}}{\rho} = \Sigma_1 + \Sigma_2$$

Proof. Follows directly from the definitions of Σ and Σ . □

Sublemma 9.5.3 (Equation (A.10))**.** We have

$$|\Sigma_1| \leq x^{-\delta}\left(\frac{1}{2\pi}(\log(T/2\pi))^2 + 1.8642\right).$$

Proof. See [27, Lemma 2.10]. □

Definition 9.5.4 (Equation (A.11))**.** We denote

$$s_1(b, \delta, T) = e^{-\delta b}\left(\frac{1}{2\pi}(\log(T/2\pi))^2 + 1.8642\right).$$

Sublemma 9.5.4 (Equation (A.12))**.** We have

$$|\Sigma_2| \leq 2\sum_{k=0}^{K-1} \frac{\lambda^{k+1} x^{-\frac{1}{R\log(T/\lambda^k)}}}{T} N\left(1-\delta, \frac{T}{\lambda^k}\right).$$

Proof. An argument of Pintz [**?**, Pintz1980]s employed. The interval $[0, T]$ is split into subintervals $[T/\lambda^{k+1}, T/\lambda^k]$ where $\lambda > 1$, $0 \le k \le K-1$, and $K = \lfloor \frac{\log T/H}{\log \lambda} \rfloor + 1$. Then use the zero-free region to bound $\Re\rho$. □

Sublemma 9.5.5 (Equation (A.13))**.** We have

$$|\Sigma_2| \le \frac{2\lambda}{T} \sum_{k=0}^{K-1} \lambda^k x^{-\frac{1}{R\log(T/\lambda^k)}} \left(c_1 \left(\frac{T}{\lambda^k}\right)^{\frac{8\delta}{3}} (\log(T/\lambda^k))^{3+2\delta} + c_2(\log(T/\lambda^k))^2 \right).$$

Proof. Inserting (A.6) into the result of (A.12). □

Definition 9.5.5 (Equation (A.14))**.** We denote

$$\begin{aligned} s_2(b, \lambda, K, T) = \frac{2\lambda}{T} \sum_{k=0}^{K-1} \exp\left(k\log\lambda - \frac{b}{R(\log T - k\log\lambda)}\right) \\ \times \left(c_1 \left(\frac{T}{\lambda^k}\right)^{\frac{8\delta}{3}} (\log(T/\lambda^k))^{3+2\delta} + c_2(\log(T/\lambda^k))^2 \right). \end{aligned} \tag{9.5}$$

Theorem 9.5.1 (Theorem 13)**.** Let b_1, b_2 satisfy $1000 \le b_1 < b_2$. Let $0.001 \le \delta \le 0.025$, $\lambda > 1$, $H < T < e^{b_1}$, and $K = \left\lfloor \frac{\log \frac{T}{H}}{\log \lambda} \right\rfloor + 1$. Then for all $x \in [e^{b_1}, e^{b_2}]$

$$\left| \frac{\psi(x) - x}{x} \right| \le s_0(b_2, T) + s_1(b_1, \delta, T) + s_2(b_1, \delta, \lambda, K, T),$$

where s_0, s_1, s_2 are respectively defined in Definitions 9.5.1, 9.5.4, and 9.5.5

Proof. Follows from combining Sublemmas **??**, **??**, **??**, and **??**. □

Definition 9.5.6 (Equation (A.16))**.** We define

$$k(\sigma, x_0) = \left(\exp\left(\frac{10 - 16\sigma}{3} \left(\frac{\log x_0}{R}\right)^{1/2} \right) \left(\frac{\log x_0}{R}\right)^{5-2\sigma} \right)^{-1}.$$

Definition 9.5.7 (Equation (A.17))**.** We define

$$c_3(\sigma, x_0) = 2\exp\left(-2\left(\frac{\log x_0}{R}\right)^{1/2}\right)(\log x_0)^2 k(\sigma, x_0).$$

Definition 9.5.8 (Equation (A.18))**.** We define

$$c_4(\sigma, x_0) = x_0^{\sigma-1}\left(\frac{2\log x_0}{\pi R} + 1.8642\right)k(\sigma, x_0).$$

Definition 9.5.9 (Equation (A.19))**.** We define

$$c_5(\sigma, x_0) = 8.01 \cdot c_2(\sigma)\exp\left(-2\left(\frac{\log x_0}{R}\right)^{1/2}\right)\frac{\log x_0}{R}k(\sigma, x_0).$$

Definition 9.5.10 (Equation (A.20))**.** We define

$$A(\sigma, x_0) = 2.0025 \cdot 25^{-2\sigma} \cdot c_1(\sigma) + c_3(\sigma, x_0) + c_4(\sigma, x_0) + c_5(\sigma, x_0).$$

Definition 9.5.11 (Equation (A.21))**.** We define

$$B = 5/2 - \sigma,$$

and

$$C = 16\sigma/3 - \frac{10}{3}.$$

Theorem 9.5.2 (Theorem 14)**.** Let $x_0 \geq 1000$ and let $\sigma \in [0.75, 1)$. For all $x \geq e^{x_0}$,

$$\frac{|\psi(x) - x|}{x} \leq A \left(\frac{\log x}{R}\right)^B \exp\left(-C \left(\frac{\log x}{R}\right)^{1/2}\right)$$

where A, B, and C are defined in Definitions 9.5.10, 9.5.11.

Proof. This is proven by Platt and Trudgian [23] ☐

Theorem 9.5.3 (Equation (A.26))**.** For $100 \leq x \leq 10^{19}$, one has

$$|(x - \psi(x))/\sqrt{x}| \leq 0.94.$$

Proof. This follows from Theorem 10.6.1. TODO: create a primary Buthe section to place this result ☐

Lemma 9.5.1 (Lemma 15)**.** Let B_0, B, and c be positive constants such that

$$|(x - \psi(x))/\sqrt{x}| \leq c \text{ for all } B_0 < x \leq B \tag{A.27}$$

is known. Furthermore, assume for every $b_0 > 0$ there exists $\varepsilon(b_0) > 0$ such that

$$|\psi(x) - x| \leq \varepsilon(b_0)x \quad \text{for all } x \geq e^{b_0}. \tag{A.28}$$

Let b be positive such that $e^b \in (B_0, B]$. Then, for all $x \geq e^b$ we have

$$\left|\frac{\psi(x) - x}{x}\right| \leq \max\left(\frac{c}{e^{\frac{b}{2}}}, \varepsilon(\log B)\right). \tag{A.29}$$

Proof. Multiplying both sides of (**??**) by $\frac{1}{\sqrt{x}}$ gives

$$\left|\frac{\psi(x) - x}{x}\right| \leq \frac{c}{e^{\frac{b}{2}}} \quad \text{for all } e^b \leq x \leq B$$

as $\frac{1}{\sqrt{x}} \leq \frac{1}{e^{\frac{b}{2}}}$. Then, for $x \geq B$ we apply (**??**) with $b_0 = \log B$. Combining these bounds, we derive (**??**). ☐

Corollary 9.5.1 (Corollary 15.1)**.** Let b be a positive constant such that $\log 11 < b \leq 19\log(10)$. Then we have

$$\left|\frac{\psi(x) - x}{x}\right| \leq \max\left\{\frac{0.94}{e^{\frac{b}{2}}}, \varepsilon(19\log 10)\right\} \quad \text{for all } x \geq e^b.$$

Note that by Table 8, we have $\varepsilon(19\log 10) = 1.93378 \cdot 10^{-8}$.

Proof. By [4, (1.5)], (A.27) holds with $B_0 = 11$, $B = 10^{19}$, and $c = 0.94$. Thus we may apply Lemma 9.5.1 with $B_0 = 11$, $B = 10^{19}$, and $c = 0.94$ from [4, (1.5)] to obtain the claim. $\square$

Definition 9.5.12 (Logan's function)**.** We define Logan's function

$$\ell_{c,\varepsilon}(\xi) = \frac{c}{\sinh c} \frac{\sin(\sqrt{(\xi\varepsilon)^2 - c^2})}{\sqrt{(\xi\varepsilon)^2 - c^2}}.$$

Definition 9.5.13 (Fourier transform of Logan's function)**.** We define

$$\eta_{c,\varepsilon}(\xi) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\xi} \ell_{c,\varepsilon}(t)\, dt.$$

Definition 9.5.14 (Definition of Buthe's function mu)**.** We define the auxiliary functions

$$\mu_{c,\varepsilon}(t) = \begin{cases} -\int_t^\infty \eta_{c,\varepsilon}(\tau) d\tau & t < 0, \\ -\mu_{c,\varepsilon}(-t) & t > 0, \\ 0 & t = 0, \end{cases}$$

$$\nu_{c,\varepsilon}(t) = \int_{-\infty}^t \mu_{c,\varepsilon}(\tau) d\tau.$$

Theorem 9.5.4 (Theorem 16)**.** Let $0 < \varepsilon < 10^{-3}$, $c \geq 3$, $x_0 \geq 100$ and $\alpha \in [0, 1)$ such that the inequality

$$B_0 := \frac{\varepsilon e^{-\varepsilon} x_0 |\nu_c(\alpha)|}{2(\mu_c)_+(\alpha)} > 1$$

holds. We denote the zeros of the Riemann zeta function by $\rho = \beta + i\gamma$ with $\beta, \gamma \in \mathbb{R}$. Then, if $\beta = \frac{1}{2}$ holds for $0 < \gamma \leq \frac{c}{\varepsilon}$, the inequality

$$|\psi(x) - x| \leq x e^{\varepsilon\alpha} (\mathcal{E}_1 + \mathcal{E}_2 + \mathcal{E}_3)$$

holds for all $x \geq e^{\varepsilon\alpha} x_0$, where

$$\mathcal{E}_1 = e^{2\varepsilon} \log(e^\varepsilon x_0) \left[\frac{2\varepsilon|\nu_c(\alpha)|}{\log B_0} + \frac{2.01\varepsilon}{\sqrt{x_0}} + \frac{\log\log(2x_0^2)}{2x_0} \right] + e^{\varepsilon\alpha} - 1,$$

$$\mathcal{E}_2 = 0.16 \frac{1 + x_0^{-1}}{\sinh c} e^{0.71\sqrt{c\varepsilon}} \log\left(\frac{c}{\varepsilon}\right), \quad \text{and}$$

$$\mathcal{E}_3 = \frac{2}{\sqrt{x_0}} \sum_{0 < \gamma < \frac{c}{\varepsilon}} \frac{\ell_{c,\varepsilon}(\gamma)}{\gamma} + \frac{2}{x_0}.$$

The $\nu_c(\alpha) = \nu_{c,1}(\alpha)$ and $\mu_c(\alpha) = \mu_{c,1}(\alpha)$ where $\nu_{c,\varepsilon}(\alpha)$ and $\mu_{c,\varepsilon}(\alpha)$ are defined by [3, p. 2490].

Proof. This is [3, Theorem 1]. $\square$

Note: This thesis of Bhattacharjee [1] will be a good resource when formalizing this result.

Theorem 9.5.5 (Theorem 2)**.** If $b > 0$ then $|\psi(x) - x| \leq \varepsilon(b)x$ for all $x \geq \exp(b)$, where ε is as in [2, Table 8].

Proof. Values for $20 \le b \le 2000$ are computed using Theorem **??**, and values for $2500 \le b \le 25000$ are computed as using Theorem 9.5.1. For $b > 25000$ we use Theorem 9.5.2. ☐

Corollary 9.5.2 (Corollary 15.1, alternative version)**.** Let b be a positive constant such that $\log 11 < b \le 19\log(10)$. Then we have

$$\left|\frac{\psi(x)-x}{x}\right| \le \max\left\{\frac{0.94}{e^{\frac{b}{2}}}, 1.93378 \cdot 10^{-8}\right\} \quad \text{for all } x \ge e^b.$$

Proof. From Table 8 we have $\varepsilon(19\log 10) = 1.93378 \cdot 10^{-8}$. Now apply Corollary 9.5.1 and Theorem **??**. ☐

9.6 Chirre-Helfgott's estimates for sums of nonnegative arithmetic functions

We record some estimates from [6] for summing non-negative functions, with a particular interest in estimating ψ.

9.6.1 Fourier-analytic considerations

Some material from [6, Section 2], slightly rearranged to take advantage of existing results in the repository.

Sublemma 9.6.1 (CH2 Proposition 2.3, substep 1)**.** Let a_n be a sequence with $\sum_{n>1} \frac{|a_n|}{n\log^\beta n} < \infty$ for some $\beta > 1$. Write $G(s) = \sum_n a_n n^{-s} - \frac{1}{s-1}$ for $\mathrm{Re} s > 1$. Let φ be absolutely integrable, supported on $[-1,1]$, and has Fourier decay $\hat{\psi}(y) = O(1/|y|^\beta)$. Then for any $x > 0$ and $\sigma > 1$

$$\frac{1}{2\pi}\sum a_n \frac{x}{n^\sigma}\hat{\psi}(\frac{T}{2\pi}\log\frac{n}{x}) = \frac{1}{2\pi T}\int_{-T}^{T}\varphi(\frac{t}{T})G(\sigma+it)x^{it}\,dt + \int_{-T\log x/2\pi}^{\infty} e^{-y(\sigma-1)}\hat{\varphi}(y)\,dy)\frac{x^{2-\sigma}}{T}.$$

Proof. Use Lemma 3.1.1 and Lemma 3.1.2, similar to the proof of 'limiting_fourier_aux'. ☐

Proposition 9.6.1 (CH2 Proposition 2.3)**.** Let a_n be a sequence with $\sum_{n>1} \frac{|a_n|}{n\log^\beta n} < \infty$ for some $\beta > 1$. Assume that $\sum_n a_n n^{-s} - \frac{1}{s-1}$ extends continuously to a function G defined on $1 + i[-T,T]$. Let φ be absolutely integrable, supported on $[-1,1]$, and has Fourier decay $\hat{\varphi}(y) = O(1/|y|^\beta)$. Then for any $x > 0$,

$$\frac{1}{2\pi}\sum a_n \frac{x}{n}\hat{\varphi}(\frac{T}{2\pi}\log\frac{n}{x}) = \frac{1}{2\pi i T}\int_{1-iT}^{1+iT}\varphi(\frac{s-1}{iT})G(s)x^s\,ds + (\varphi(0) - \int_{-\infty}^{-T\log x/2\pi}\hat{\varphi}(y)\,dy)\frac{x}{T}.$$

Proof. Apply Sublemma 9.6.1 and take the limit as $\sigma \to 1^+$, using the continuity of G and the dominated convergence theorem, as well as the Fourier inversion formula. ☐

Definition 9.6.1 (CH2 Definition of S, (2.8))**.** $S_\sigma(x)$ is equal to $\sum_{n\le x} a_n/n^\sigma$ if $\sigma < 1$ and $\sum_{n\ge x} a_n/n^\sigma$ if $\sigma > 1$.

Definition 9.6.2 (CH2 Definition of I, (2.9))**.** $I_\lambda(u) = 1_{[0,\infty)}(\mathrm{sgn}(\lambda)u)e^{-\lambda u}$.

Sublemma 9.6.2 (CH2 Equation (2.10))**.** $S_\sigma(x) = x^{-\sigma}\sum_n a_n \frac{x}{n} I_\lambda(\frac{T}{2\pi}\log\frac{n}{x})$ where $\lambda = 2\pi(\sigma-1)/T$.

Proof. Routine manipulation. □

Proposition 9.6.2 (CH2 Proposition 2.4, upper bound)**.** Let a_n be a non-negative sequence with $\sum_{n>1}\frac{|a_n|}{n\log^\beta n} < \infty$ for some $\beta > 1$. Assume that $\sum_n a_n n^{-s} - \frac{1}{s-1}$ extends continuously to a function G defined on $1 + i[-T, T]$. Let φ_+ be absolutely integrable, supported on $[-1, 1]$, and has Fourier decay $\hat{\varphi}_+(y) = O(1/|y|^\beta)$. Let $\sigma \neq 1$ and write $\lambda = 2\pi(\sigma-1)/T$. Assume $I_\lambda(y) \leq \hat{\varphi}_+(y)$ for all y. Then for any $x \geq 1$,

$$S_\sigma(x) \leq \frac{2\pi x^{1-\sigma}}{T}\varphi_+(0) + \frac{x^{-\sigma}}{T}\int_{-T}^{T}\varphi_+(t/T)G(1+it)x^{1+it}\,dt - \frac{1_{(-\infty,1)}(\sigma)}{1-\sigma}.$$

Proof. By the nonnegativity of a_n we have

$$S_\sigma(x) \leq x^{-\sigma}\sum_n a_n \frac{x}{n}\hat{\varphi}_+(\frac{T}{2\pi}\log\frac{n}{x}).$$

By Proposition 9.6.1 we can express the right-hand side as

$$\frac{1}{2\pi i T}\int_{1-iT}^{1+iT}\varphi_+(\frac{s-1}{iT})G(s)x^s\,ds + (\varphi_+(0) - \int_{-\infty}^{-T\log x/2\pi}\hat{\varphi}_+(y)\,dy)\frac{x}{T}.$$

If $\lambda > 0$, then $I_\lambda(y) = 0$ for negative y, so

$$-\int_{-\infty}^{-T\log x/2\pi}\hat{\varphi}_+(y)\,dy \leq 0.$$

If $\lambda < 0$, then $I_\lambda(y) = e^{-\lambda y}$ for y negative, so

$$-\int_{-\infty}^{-T\log x/2\pi} I_\lambda(y)\,dy \leq e^{\lambda T\log x/2\pi}/(-\lambda) = x^{\sigma-1}/(-\lambda).$$

hence

$$-\int_{-\infty}^{-T\log x/2\pi}\hat{\varphi}_+(y)\,dy \leq -x^{\sigma-1}/(-\lambda).$$

Since $x^{-\sigma} * (2\pi x/T) * x^{\sigma-1}/(-\lambda) = 1/(1-\sigma)$, the result follows. □

Proposition 9.6.3 (CH2 Proposition 2.4, lower bound)**.** Let a_n be a non-negative sequence with $\sum_{n>1}\frac{|a_n|}{n\log^\beta n} < \infty$ for some $\beta > 1$. Assume that $\sum_n a_n n^{-s} - \frac{1}{s-1}$ extends continuously to a function G defined on $1 + i[-T, T]$. Let φ_- be absolutely integrable, supported on $[-1, 1]$, and has Fourier decay $\hat{\varphi}_-(y) = O(1/|y|^\beta)$. Let $\sigma \neq 1$ and write $\lambda = 2\pi(\sigma-1)/T$. Assume $\hat{\varphi}_-(y) \leq I_\lambda(y)$ for all y. Then for any $x \geq 1$ and $\sigma \neq 1$,

$$S_\sigma(x) \geq \frac{2\pi x^{1-\sigma}}{T}\varphi_-(0) + \frac{x^{-\sigma}}{T}\int_{-T}^{T}\varphi_-(t/T)G(1+it)x^{1+it}\,dt - \frac{1_{(-\infty,1)}(\sigma)}{1-\sigma}.$$

Proof. Similar to the proof of Proposition 9.6.2; see [6, Proposition 2.4] for details. □

9.6.2 Extremal approximants to the truncated exponential

In this section we construct extremal approximants to the truncated exponential function and establish their basic properties, following [6, Section 4], although we skip the proof of their extremality. As such, the material here is organized rather differently from that in the paper.

Definition 9.6.3 (Definition of Phi-circ)**.**

$$\Phi^{\pm,\circ}_{\nu}(z) := \frac{1}{2}(\coth\frac{w}{2} \pm 1)$$

where

$$w = -2\pi i z + \nu.$$

Theorem 9.6.1 (Phi-circ meromorphic)**.**

$$\Phi^{\pm,\circ}_{\nu}(z)$$

is meromorphic.

Proof. This follows from the definition of $\Phi^{\pm,\circ}_{\nu}$ and the properties of the coth function. ☐

Lemma 9.6.1 (Phi-circ poles)**.** The poles of

$$\Phi^{\pm,\circ}_{\nu}(z)$$

are of the form $n - i\nu/2\pi$ for $n \in \mathbb{Z}$.

Proof. This follows from the definition of $\Phi^{\pm,\circ}_{\nu}$ and the properties of the coth function. ☐

Lemma 9.6.2 (Phi-circ residues)**.** The residue of

$$\Phi^{\pm,\circ}_{\nu}(z)$$

at $n - i\nu/2\pi$ is $i/2\pi$.

Proof. This follows from the definition of $\Phi^{\pm,\circ}_{\nu}$ and the properties of the coth function. ☐

Lemma 9.6.3 (Phi-circ poles simple)**.** The poles of

$$\Phi^{\pm,\circ}_{\nu}(z)$$

are all simple.

Proof. This follows from the definition of $\Phi^{\pm,\circ}_{\nu}$ and the properties of the coth function. ☐

Definition 9.6.4 (Definition of B)**.** $B^{\pm}(s) = s/2(\coth(s/2) \pm 1)$ with the convention $B^{\pm}(0) = 1$.

Lemma 9.6.4 (Continuity of B at 0)**.** $B^{\pm}$ is continuous at 0.

Proof. L'Hôpital's rule can be applied to show the continuity at 0. ☐

Definition 9.6.5 (Definition of Phi-star)**.**

$$\Phi^{\pm,*}_{\nu}(z) := (B^{\pm}(w) - B^{\pm}(v))/(2\pi i)$$

where

$$w = -2\pi i z + \nu.$$

Theorem 9.6.2 (Phi-star at zero)**.**

$$\Phi_\nu^{\pm,*}(0) = 0.$$

Proof. This follows from the definition of $B^\pm$ and the fact that $B^\pm(0) = 1$. □

Theorem 9.6.3 (Phi-star meromorphic)**.**

$$\Phi_\nu^{\pm,*}(z)$$

is meromorphic.

Proof. This follows from the definition of $\Phi_\nu^{\pm,*}$ and the properties of the $B^\pm$ function. □

Lemma 9.6.5 (Phi-star poles)**.** The poles of

$$\Phi_\nu^{\pm,*}(z)$$

are of the form $n - i\nu/2\pi$ for $n \in \mathbb{Z}\backslash\{0\}$.

Proof. This follows from the definition of $\Phi_\nu^{\pm,*}$ and the properties of the $B^\pm$ function. □

Lemma 9.6.6 (Phi-star residues)**.** The residue of

$$\Phi_\nu^{\pm,*}(z)$$

at $n - i\nu/2\pi$ is $-in/2\pi$.

Proof. This follows from the definition of $\Phi_\nu^{\pm,*}$ and the properties of the $B^\pm$ function. □

Lemma 9.6.7 (Phi-star poles simple)**.** The poles of

$$\Phi_\nu^{\pm,*}(z)$$

are all simple.

Proof. This follows from the definition of $\Phi_\nu^{\pm,*}$ and the properties of the $B^\pm$ function. □

Lemma 9.6.8 (Phi pole cancellation)**.** $\Phi_\nu^{\sigma,\circ}(z) \pm \Phi_\nu^{\sigma,*}(z)$ is regular at $\pm 1 - i\nu/2\pi$.

Proof. The residues cancel out. □

Definition 9.6.6 (Definition of phi-pm)**.**

$$\varphi_\nu^\pm(t) := 1_{[-1,1]}(t)(\Phi_\nu^{\pm,\circ}(t) + \mathrm{sgn}(t)\Phi_\nu^{\pm,*}(t)).$$

Lemma 9.6.9 (phi is C2 on [-1,0)**.**] φ is C^2 on $[-1, 0]$.

Proof. Since $\Phi_\nu^{\pm,\circ}(z)$ and $\Phi_\nu^{\pm,\circ}(z)$ have no poles on $\mathbb{R}$, they have no poles on some open neighborhood of $[-1, 1]$. Hence they are C^2 on this interval. Since $w(0) =\not\ni u$, we see that $\Phi_\nu^{\pm,*}(0) = 0$, giving the claim. □

Lemma 9.6.10 (phi is C2 on [0,1)**.**] φ is C^2 on $[0, 1]$.

Proof. Since $\Phi_\nu^{\pm,\circ}(z)$ and $\Phi_\nu^{\pm,\circ}(z)$ have no poles on $\mathbb{R}$, they have no poles on some open neighborhood of $[-1, 1]$. Hence they are C^2 on this interval. Since $w(0) = \nu$, we see that $\Phi_\nu^{\pm,*}(0) = 0$, giving the claim. □

Lemma 9.6.11 (phi is continuous)**.** φ is continuous on $[0,1]$.

Proof. By the preceding lemmas it suffices to verify continuity at $0,-1,1$. Continuity at 0 is clear. For $t=-1,1$, by $\coth\frac{w(t)}{2}=\coth\frac{\nu}{2}$, we see that $B^{\pm}(w(t))=\left(\frac{\nu}{2}-\pi it\right)\left(\coth\frac{\nu}{2}\pm 1\right)$, and so

$$\Phi_\nu^{\pm,\star}(t)=-t\cdot\frac{1}{2}\left(\coth\frac{\nu}{2}\pm 1\right)=-t\,\Phi_\nu^{\pm,\circ}(t);$$

hence, by Definition 9.6.6, $\varphi_\nu^{\pm}(t)=0$. Thus, $\varphi_\nu^{\pm}$ is continuous at -1 and at 1. ☐

Lemma 9.6.12 (bound on phi-circ-right)**.** Let $0<\nu_0\le\nu_1$ and $c>-\nu_0/2\pi$, then there exists C such that for all $\nu\in[\nu_0,\nu_1]$, $\Im z\ge c$ one has $|\Phi_\nu^{\pm,\circ}(z)|\le C$.

Proof. The function $\coth w=1+\frac{2}{e^{2w}-1}$ is bounded away from the imaginary line $\Re w=0$, that is, it is bounded on $\Re w\ge\kappa$ and $\Re w\le-\kappa$ for any $\kappa>0$. The map $w(z)=\nu-2\pi iz$ sends the line $\Im z=-\frac{\nu}{2\pi}$ to the imaginary line, and the region $\Im z\ge c$ is sent to $\Re w\ge 2\pi c+\nu$. ☐

Lemma 9.6.13 (bound on phi-circ-left)**.** Let $0<\nu_0\le\nu_1$ and $c<-\nu_1/2\pi$, then there exists C such that for all $\nu\in[\nu_0,\nu_1]$, $\Im z\le c$ one has $|\Phi_\nu^{\pm,\circ}(z)|\le C$.

Proof. Similar to previous lemma. ☐

Lemma 9.6.14 (bound on phi-star-right)**.** Let $0<\nu_0\le\nu_1$ and $c>-\nu_0/2\pi$, then there exists C such that for all $\nu\in[\nu_0,\nu_1]$, $\Im z\ge c$ one has $|\Phi_\nu^{\pm,\star}(z)|\le C(|z|+1)$.

Proof. The bound on $\Phi_\nu^{\pm,\star}$ follows from the bound on $\Phi_\nu^{\pm,\circ}$ by $\Phi^{\pm,\star}(z)=\frac{1}{2\pi i}(w\,\Phi^{\pm,\circ}(w)-\nu\,\Phi^{\pm,\circ}(\nu))$. ☐

Lemma 9.6.15 (bound on phi-star-left)**.** Let $0<\nu_0\le\nu_1$ and $c<-\nu_1/2\pi$, then there exists C such that for all $\nu\in[\nu_0,\nu_1]$, $\Im z\le c$ one has $|\Phi_\nu^{\pm,\star}(z)|\le C(|z|+1)$.

Proof. Similar to previous lemma. ☐

Lemma 9.6.16 (B-plus is increasing)**.** For real t, $B^+(t)$ is increasing.

Proof. For all $t\neq 0$, by the identities $2\cosh\frac{t}{2}\sinh\frac{t}{2}=\sinh t$ and $2\sinh^2\frac{t}{2}=\cosh t-1$,

$$\frac{dB^{\pm}(t)}{dt}=\frac{\cosh\frac{t}{2}\sinh\frac{t}{2}-\frac{t}{2}\pm\sinh^2\frac{t}{2}}{2\sinh^2\frac{t}{2}}=\frac{\pm(e^{\pm t}-(1\pm t))}{4\sinh^2\frac{t}{2}}.$$

Since e^u is convex, $e^u\ge 1+u$ for all $u\in\mathbb{R}$. We apply this inequality with $u=t$ and $u=-t$ and obtain the conclusion for $t\neq 0$. Since $B^{\pm}(t)$ is continuous at $t=0$, we are done. . ☐

Lemma 9.6.17 (B-minus is decreasing)**.** For real t, $B^-(t)$ is decreasing.

Proof. Similar to previous. . ☐

Sublemma 9.6.3 (Fourier transform of varphi)**.**

$$\widehat{\varphi_\nu^{\pm}}(x)=\int_{-1}^{1}\varphi_\nu^{\pm}(t)\,e(-tx)\,dt=\int_{-1}^{0}(\Phi_\nu^{\pm,\circ}(t)-\Phi_\nu^{\pm,\star}(t))e(-tx)\,dt+\int_{0}^{1}(\Phi_\nu^{\pm,\circ}(t)+\Phi_\nu^{\pm,\star}(t))e(-tx)\,dt.$$

Proof. By the definition of the Fourier transform, and the fact that $\varphi_\nu^{\pm}$ is supported on $[-1,1]$. ☐

Sublemma 9.6.4 (Contour shifting upwards)**.** If $x < 0$, then

$$\widehat{\varphi_\nu^\pm}(x) = \int_{-1+i\infty}^{-1} (\Phi_\nu^{\pm,\circ}(z) - \Phi_\nu^{\pm,\star}(z)) e(-zx)\, dz + \int_1^{1+i\infty} (\Phi_\nu^{\pm,\circ}(z) + \Phi_\nu^{\pm,\star}(z)) e(-zx)\, dz + 2\int_0^{i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz. \tag{9.6}$$

Proof. Since $\Phi_\nu^{\pm,\circ}(z) \pm \Phi_\nu^{\pm,\star}(z)$ has no poles in the upper half plane, we can shift contours upwards, as we may: for $\Im z \to \infty$, $e(-zx) = e^{-2\pi i z x}$ decays exponentially on $\Im z$, while, by Lemma 1.3, $\Phi_\nu^{\pm,\circ}(z) \pm \Phi_\nu^{\pm,\star}(z)$ grows at most linearly, and so the contribution of a moving horizontal segment goes to 0 as $\Im z \to \infty$. □

Sublemma 9.6.5 (B affine periodic)**.** For any integer m,

$$B^\pm(w(z-m)) = B^\pm(w(z) + 2\pi i m) = B^\pm(w(z)) + 2\pi i m\, \Phi_\nu^{\pm,\circ}(z).$$

Proof. This follows from the πi-periodicity of coth. □

Sublemma 9.6.6 (Phi star affine periodic)**.** For any integer m,

$$\Phi_\nu^{\pm,\star}(z-m) = \Phi_\nu^{\pm,\star}(z) + m\, \Phi_\nu^{\pm,\circ}(z).$$

Proof. Follows from previous lemma. □

Sublemma 9.6.7 (Simplified formula for upward contour shift)**.** If $x < 0$, then $\widehat{\varphi_\nu^\pm}(x)$ equals

$$\frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu + y) - B^\pm(\nu))\, e^{xy}\, dy.$$

Proof. We have $\Phi_\nu^{\pm,\circ}(z) - \Phi_\nu^{\pm,\star}(z) = -\Phi_\nu^{\pm,\star}(z+1)$ and $\Phi_\nu^{\pm,\circ}(z) + \Phi_\nu^{\pm,\star}(z) = \Phi_\nu^{\pm,\star}(z-1)$, and so the formula in the previous lemma simplifies to

$$\begin{aligned} &2\int_0^{i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz - \int_0^{i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-(z-1)x)\, dz - \int_0^{i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-(z+1)x)\, dz \\ &= (2 - e(x) - e(-x)) \int_0^\infty \Phi_\nu^{\pm,\star}\left(\frac{iy}{2\pi}\right) e\left(\frac{xy}{2\pi}\right) dy = \frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu + y) - B^\pm(\nu))\, e^{xy}\, dy. \end{aligned}$$

□

Sublemma 9.6.8 (Contour shifting downwards)**.** If $x > 0$, then

$$\begin{aligned} \widehat{\varphi_\nu^\pm}(x) = &\left(\int_{-1}^{-1-i\infty} + \int_{-\frac{1}{2}-i\infty}^{-\frac{1}{2}}\right) (\Phi_\nu^{\pm,\circ}(z) - \Phi_\nu^{\pm,\star}(z)) e(-zx)\, dz \\ &+ \int_{-\frac{1}{2}}^{\frac{1}{2}} \Phi_\nu^{\pm,\circ}(z)\, e(-zx)\, dz - \int_{-\frac{1}{2}}^{0} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz + \int_0^{\frac{1}{2}} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz \\ &+ \left(\int_{\frac{1}{2}}^{\frac{1}{2}-i\infty} + \int_{1-i\infty}^{1}\right) (\Phi_\nu^{\pm,\circ}(z) + \Phi_\nu^{\pm,\star}(z)) e(-zx)\, dz. \end{aligned} \tag{9.7}$$

Proof. We would like to integrate along $\Re z = 0$, but $\Phi_\nu^{\pm,\circ}(z)$ has a pole at $z = -\frac{i\nu}{2\pi}$; when dealing with this issue, we have to take care not to introduce poles on the lines $\Re z = -1$ and $\Re z = 1$ by separating $\Phi_\nu^{\pm,\circ}$ and $\Phi_\nu^{\pm,\star}$ prematurely. As $\Im z \to -\infty$, $e(-zx) = e^{-2\pi i z x}$ decays exponentially on $\Im z$, while, by Lemma 1.3, $\Phi_\nu^{\pm,\circ}(z) \pm \Phi_\nu^{\pm,\star}(z)$ grows at most linearly. □

Sublemma 9.6.9 (First contour limit)**.**

$$\int_{-\frac{1}{2}-i\infty}^{-\frac{1}{2}} \Phi_\nu^{\pm,\circ}(z)\, e(-zx)\, dz + \int_{-\frac{1}{2}}^{\frac{1}{2}} \Phi_\nu^{\pm,\circ}(z)\, e(-zx)\, dz + \int_{\frac{1}{2}}^{\frac{1}{2}-i\infty} \Phi_\nu^{\pm,\circ}(z)\, e(-zx)\, dz = e\Big(-\Big(-\frac{i\nu}{2\pi}\Big)\, x\Big) = e^{-\nu x}$$

Proof. since the pole is at $-\frac{i\nu}{2\pi}$, the residue of $\Phi_\nu^{\pm,\circ}(z)$ at the pole is $\frac{i}{2\pi}$, and our path goes clockwise. . ☐

Sublemma 9.6.10 (Second contour limit)**.**

$$-\int_{-\frac{1}{2}-i\infty}^{-\frac{1}{2}} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz - \int_{-\frac{1}{2}}^{0} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz = \int_0^{-i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz.$$

Proof. Again by Cauchy's theorem and decay as $\Im z \to -\infty$ ☐

Sublemma 9.6.11 (Third contour limit)**.**

$$\int_0^{\frac{1}{2}} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz + \int_{\frac{1}{2}}^{\frac{1}{2}-i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz = -\int_{-i\infty}^{0} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz.$$

Proof. Similar to previous. ☐

Sublemma 9.6.12 (Simplified formula for downward contour shift)**.** If $x > 0$, then $\widehat{\varphi_\nu^{\pm}}(x) - e^{-\nu x}$ equals

$$-\frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu) - B^\pm(\nu - y))\, e^{-xy}\, dy.$$

Proof.

$$2\int_0^{-i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-zx)\, dz - \int_0^{-i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-(z-1)x)\, dz - \int_0^{-i\infty} \Phi_\nu^{\pm,\star}(z)\, e(-(z+1)x)\, dz$$
$$= (2 - e(x) - e(-x)) \int_0^\infty \Phi_\nu^{\pm,\star}\Big(-\frac{iy}{2\pi}\Big)\, e\Big(-\frac{yx}{2\pi i}\Big)\, d\Big(-\frac{iy}{2\pi}\Big)$$
$$= -\frac{2i}{\pi} \sin^2 \pi x \int_0^\infty \Phi_\nu^{\pm,\star}\Big(-\frac{iy}{2\pi}\Big)\, e^{-xy}\, dy = -\frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu - y) - B^\pm(\nu))\, e^{-xy}\, dy.$$

☐

Lemma 9.6.18 (Fourier formula for negative x)**.** Let $\nu > 0$, $x < 0$. Then

$$\widehat{\varphi_\nu^{\pm}}(x) - I_\nu(x) = \frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu + y) - B^\pm(\nu))\, e^{xy}\, dy.$$

Proof. This follows from the previous lemma. ☐

Lemma 9.6.19 (Fourier formula for positive x)**.** Let $\nu > 0$, $x > 0$. Then

$$\widehat{\varphi_\nu^{\pm}}(x) - e^{-\nu x} = -\frac{\sin^2 \pi x}{\pi^2} \int_0^\infty (B^\pm(\nu) - B^\pm(\nu - y))\, e^{-xy}\, dy.$$

Proof. This follows from the previous lemma. ☐

TODO: Lemmas 4.2, 4.3, 4.4

9.6.3 Contour shifting

TODO: incorporate material from [6, Section 5].

9.6.4 The main theorem

TODO: incorporate material from [6, Section 6].

9.6.5 Applications to psi

TODO: incorporate material from [6, Section 7] onwards.

Corollary 9.6.1 (Corollary 1.2, part a)**.** Assume the Riemann hypothesis holds up to height $T \geq 10^7$. For $x > \max(T, 10^9)$,

$$|\psi(x) - x \cdot \frac{\pi}{T} \coth(\frac{\pi}{T})| \leq \pi T^{-1} \cdot x + \frac{1}{2\pi} \log^2(T/(2\pi)) - \frac{1}{6\pi} \log(T/(2\pi))\sqrt{x},$$

Proof. TBD. □

Corollary 9.6.2 (Corollary 1.2, part b)**.** Assume the Riemann hypothesis holds up to height $T \geq 10^7$. For $x > \max(T, 10^9)$,

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} \leq \pi\sqrt{T}^{-1} + \frac{1}{2\pi} \log^2(T/(2\pi)) - \frac{1}{6\pi} \log(T/(2\pi))\frac{1}{x},$$

where $\gamma = 0.577215...$ is Euler's constant.

Proof. TBD. □

Corollary 9.6.3 (Corollary 1.3, part a)**.** For $x \geq 1$,

$$|\psi(x) - x| \leq \pi \cdot 3 \cdot 10^{-12} \cdot x + 113.67\sqrt{x},$$

where $\psi(x)$ is the Chebyshev function.

Proof. TBD. □

Corollary 9.6.4 (Corollary 1.3, part b)**.** For $x \geq 1$,

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x - \gamma + O^*(\pi \cdot \sqrt{3} \cdot 10^{-12} + 113.67/x).$$

Proof. TBD. □

9.7 Summary of results

In this section we list some papers that we plan to incorporate into this section in the future, and list some results that have not yet been moved into dedicated paper sections.

References to add:

None yet.

Chapter 10

Secondary explicit estimates

Theorem 10.0.1 (Lower bound on li(2))**.** $\mathrm{li}(2) \geq 1.039$

Proof. □

Theorem 10.0.2 (Upper bound on li(2))**.** $\mathrm{li}(2) \leq 1.06$

Proof. □

Theorem 10.0.3 (Bounds on li(2))**.** $1.039 \leq \mathrm{li}(2) \leq 1.06$

Proof. □

Theorem 10.0.4 (Symmetric form equals principal value)**.** $\int_0^1 \left(\frac{1}{\log(1+t)} + \frac{1}{\log(1-t)}\right) dt = \mathrm{li}(2)$

Proof. □

10.1 Definitions

In this section we define the basic types of secondary estimates we will work with in the project.

Sublemma 10.1.1 (Log upper bound)**.** For $t > -1$, one has $\log(1+t) \leq t$.

Proof. This follows from the mean value theorem. □

Sublemma 10.1.2 (First log lower bound)**.** For $t \geq 0$, one has $t - \frac{t^2}{2} \leq \log(1+t)$.

Proof. Use Taylor's theorem with remainder and the fact that the second derivative of $\log(1+t)$ is at most 1 for $t \geq 0$. □

Sublemma 10.1.3 (Second log lower bound)**.** For $0 \leq t \leq t_0 < 1$, one has $\frac{t}{t_0} \log(1 - t_0) \leq \log(1-t)$.

Proof. Use concavity of log. □

Sublemma 10.1.4 (Symmetrization of inverse log)**.** For $0 < t \leq 1/2$, one has $|\frac{1}{\log(1+t)} + \frac{1}{\log(1-t)}| \leq \frac{16\log(4/3)}{3}$.

Proof. The expression can be written as $\frac{|\log(1-t^2)|}{|\log(1-t)||\log(1+t)|}$. Now use the previous upper and lower bounds, noting that $t^2 \leq 1/4$. NOTE: this gives the slightly weaker bound of $16\log(4/3)/3$, but this can suffice for applications. The sharp bound would require showing that the left-hand side is monotone in t, which looks true but not so easy to verify. ☐

Remark 10.1.1 (li minus Li)**.** $\mathrm{li}(x) - \mathrm{Li}(x) = \mathrm{li}(2)$.

Proof. This follows from the previous estimate. ☐

Lemma 10.1.1 (Ramanujan-Soldner constant)**.** $\mathrm{li}(2) = 1.0451\ldots$.

Proof. Symmetrize the integral and use and some numerical integration. ☐

Sublemma 10.1.5 (Weak bounds on li(2))**.** $1.039 \leq \mathrm{li}(2) \leq 1.06$.

Proof. ☐

Sublemma 10.1.6 (Symmetric form equals principal value)**.** $\int_0^1 \left(\frac{1}{\log(1+t)} + \frac{1}{\log(1-t)}\right) dt = \mathrm{li}(2)$

Proof. ☐

10.2 Chebyshev's estimates

We record Chebyshev's estimates on ψ. The material here is adapted from the presentation of Diamond [9].

Definition 10.2.1 (The function T)**.** $T(x) := \sum_{n \leq x} \log n$.

Lemma 10.2.1 (Upper bound on T)**.** For $x \geq 1$, we have $T(x) \leq x\log x - x + 1 + \log x$.

Proof. Upper bound $\log n$ by $\int_n^{n+1} \log t\ dt$ for $n < x - 1$ and by $\log x$ for $x - 1 < n \leq x$ to bound

$$T(x) \leq \int_1^x \log t\ dt + \log x$$

giving the claim. ☐

Lemma 10.2.2 (Lower bound on T)**.** For $x \geq 1$, we have $T(x) \geq x\log x - x + 1 - \log x$.

Proof. Drop the $n = 1$ term. Lower bound $\log n$ by $\int_{n-1}^n \log t\ dt$ for $2 \leq n < x$ to bound

$$T(x) \geq \int_1^{\lfloor x \rfloor} \log t\ dt \geq \int_1^x \log t\ dt - \log x$$

giving the claim. ☐

Lemma 10.2.3 (Relating T and von Mangoldt)**.** For $x \geq 0$, we have $T(x) = \sum_{n \leq x} \Lambda(n)\lfloor x/n \rfloor$.

Proof. This follows from the identity $\log n = \sum_{d|n} \Lambda(d)$ and rearranging sums. ☐

Definition 10.2.2 (E function)**.** If $\nu : \mathbb{N} \to \mathbb{R}$, let $E : \mathbb{R} \to \mathbb{R}$ denote the function $E(x) := \sum_m \nu(m)\lfloor x/m \rfloor$.

Lemma 10.2.4 (Relating a weighted sum of T to an E-weighted sum of von Mangoldt)**.** If $\nu : \mathbb{N} \to \mathbb{R}$ is finitely supported, then

$$\sum_m \nu(m) T(x/m) = \sum_{n \le x} E(x/n) \Lambda(n).$$

Proof. □

Definition 10.2.3 (Chebyshev's weight u)**.** $\nu = e_1 - e_2 - e_3 - e_5 + e_{30}$, where e_n is the Kronecker delta at n.

Lemma 10.2.5 (Cancellation property of u)**.** One has $\sum_n \nu(n)/n = 0$

Proof. This follows from direct computation. □

Lemma 10.2.6 (E initially constant)**.** One has $E(x) = 1$ for $1 \le x < 6$.

Proof. This follows from direct computation. □

Lemma 10.2.7 (E is periodic)**.** One has $E(x + 30) = E(x)$.

Proof. This follows from direct computation. □

Lemma 10.2.8 (E lies between 0 and 1)**.** One has $0 \le E(x) \le 1$ for all $x \ge 0$.

Proof. This follows from direct computation for $0 \le x < 30$, and then by periodicity for larger x. □

Definition 10.2.4 (The U function)**.** We define $U(x) := \sum_m \nu(m) T(x/m)$.

Proposition 10.2.1 (Lower bounding ψ by a weighted sum of T)**.** For any $x > 0$, one has $\psi(x) \ge U(x)$.

Proof. Use Lemma 10.2.4 and Lemma 10.2.8. □

Proposition 10.2.2 (Upper bounding a difference of ψ by a weighted sum of T)**.** For any $x > 0$, one has $\psi(x) - \psi(x/6) \le U(x)$.

Proof. Use Lemma 10.2.4, Lemma 10.2.8, and Lemma 10.2.6. □

Definition 10.2.5 (The constant a)**.** We define $a := -\sum_m \nu(m) \log m/m$.

Lemma 10.2.9 (Numerical value of a)**.** We have $0.92129 \le a \le 0.92130$.

Proof. □

Lemma 10.2.10 (Bounds for U)**.** For $x \ge 30$, we have $|U(x) - ax| \le 5 \log x - 5$.

Proof. Use Lemma 10.2.1, Lemma 10.2.2, the definition of a, and the triangle inequality, also using that $\log(2) + \log(3) + \log(5) + \log(30) \ge 6$. □

Theorem 10.2.1 (Lower bound for ψ)**.** For $x \ge 30$, we have $\psi(x) \ge ax - 5 \log x - 1$.

Proof. Use Lemma 10.2.10 and Proposition 10.2.1. □

Proposition 10.2.3 (Upper bound for ψ difference)**.** For $x \ge 30$, we have $\psi(x) - \psi(x/6) \le ax + 5 \log x - 5$.

Proof. Use Lemma 10.2.10 and Proposition **??**. ☐

Sublemma 10.2.1 (Numerical bound for $\psi(x)$ for very small x)**.** For $0 < x \leq 30$, we have $\psi(x) \leq 1.015x$.

Proof. Numerical check (the maximum occurs at $x = 19$). One only needs to check the case when x is a prime power. ☐

Theorem 10.2.2 (Upper bound for ψ)**.** For $x \geq 30$, we have $\psi(x) \leq 6ax/5+(\log(x/5)/\log 6)(5\log x-5)$.

Proof. Iterate Lemma 10.2.3 using Sublemma 10.2.1 . ☐

Sublemma 10.2.2 (Numerical bound for $\psi(x)$ for medium x)**.** For $0 < x \leq 11723$, we have $\psi(x) \leq 1.11x$.

Proof. Verified by brute-force: an $O(N)$ incremental checker confirms $\psi(N) \leq 1.11N$ for every integer $N = 1, \dots, 11723$ via `native_decide`. The sparse checkpoint ladder originally described here is not needed. The real-variable case follows by monotonicity of ψ. ☐

Theorem 10.2.3 (Clean upper bound for ψ)**.** For $x > 0$, we have $\psi(x) \leq 1.11x$.

Proof. Strong induction on x. For $x \leq 11723$ one can use Sublemma 10.2.2. Otherwise, we can use Proposition 10.2.3 and the triangle inequality. ☐

10.3 An inequality of Costa-Pereira

We record here an inequality relating the Chebyshev functions $\psi(x)$ and $\theta(x)$ due to Costa Pereira [7], namely

$$\psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/7}) \leq \psi(x) - \theta(x) \leq \psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}).$$

Sublemma 10.3.1 (Costa-Pereira Sublemma 1.1)**.** For every $x > 0$ we have $\psi(x) = \sum_{k\geqslant 1} \theta(x^{1/k})$.

Proof. This follows directly from the definitions of ψ and θ. ☐

Sublemma 10.3.2 (Costa-Pereira Sublemma 1.2)**.** For every $x > 0$ and n we have $\psi(x^{1/n}) = \sum_{k\geqslant 1} \theta(x^{1/nk})$.

Proof. Follows from Sublemma 10.3.1 and substitution. ☐

Sublemma 10.3.3 (Costa-Pereira Sublemma 1.3)**.** For every $x > 0$ we have

$$\psi(x) = \theta(x) + \psi(x^{1/2}) + \sum_{k\geqslant 1} \theta(x^{1/(2k+1)}).$$

Proof. Follows from Sublemma 10.3.1 and Sublemma 10.3.2. ☐

Sublemma 10.3.4 (Costa-Pereira Sublemma 1.4)**.** For every $x > 0$ we have

$$\psi(x) - \theta(x) = \psi(x^{1/2}) + \sum_{k\geqslant 1} \theta(x^{1/(6k-3)}) + \sum_{k\geqslant 1} \theta(x^{1/(6k-1)}) + \sum_{k\geqslant 1} \theta(x^{1/(6k+1)}).$$

Proof. Follows from Sublemma 10.3.3 and rearranging the sum. ☐

Sublemma 10.3.5 (Costa-Pereira Sublemma 1.5)**.** For every $x > 0$ we have

$$\psi(x^{1/3}) = \sum_{k \geqslant 1} \theta(x^{1/(6k-3)}) + \sum_{k \geqslant 1} \theta(x^{1/(6k)}).$$

Proof. Follows from Sublemma 10.3.2 and substitution. □

Sublemma 10.3.6 (Costa-Pereira Sublemma 1.6)**.** For every $x > 0$ we have

$$\psi(x) - \theta(x) = \psi(x^{1/2}) + \psi(x^{1/3}) + \sum_{k \geqslant 1} \theta(x^{1/(6k-1)}) - \sum_{k \geqslant 1} \theta(x^{1/(6k)}) + \sum_{k \geqslant 1} \theta(x^{1/(6k+1)}).$$

Proof. Follows from Sublemma 10.3.4 and Sublemma 10.3.5. □

Sublemma 10.3.7 (Costa-Pereira Sublemma 1.7)**.** For every $x > 0$ we have

$$\psi(x) - \theta(x) \leqslant \psi(x^{1/2}) + \psi(x^{1/3}) + \sum_{k \geqslant 1} \theta(x^{1/5k}$$

Proof. Follows from Sublemma 10.3.6 and the fact that θ is an increasing function. □

Sublemma 10.3.8 (Costa-Pereira Sublemma 1.8)**.** For every $x > 0$ we have

$$\psi(x) - \theta(x) \geqslant \psi(x^{1/2}) + \psi(x^{1/3}) + \sum_{k \geqslant 1} \theta(x^{1/7k}$$

Proof. Follows from Sublemma 10.3.6 and the fact that θ is an increasing function. □

Theorem 10.3.1 (Costa-Pereira Theorem 1a)**.** For every $x > 0$ we have $\psi(x) - \theta(x) \leqslant \psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5})$.

Proof. Follows from Sublemma 10.3.7 and Sublemma 10.3.2. □

Theorem 10.3.2 (Costa-Pereira Theorem 1b)**.** For every $x > 0$ we have $\psi(x) - \theta(x) \geqslant \psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/7})$.

Proof. Follows from Sublemma 10.3.8 and Sublemma 10.3.2. □

10.4 Converting prime number theorems to prime in short interval theorems

In this section, bounds on E_θ are used to deduce the existence of primes in short intervals. (One could also proceed using E_π, but the bounds are messier and the results slightly weaker.)

Lemma 10.4.1 (Increase in pi iff prime in short interval)**.** There is a prime in $(x, x+h]$ iff $\pi(x+h) > \pi(x)$.

Proof. Both are equivalent to $\sum_{x<p\leq x+h} 1 > 0$. □

Lemma 10.4.2 (Increase in theta iff prime in short interval)**.** There is a prime in $(x, x+h]$ iff $\theta(x+h) > \theta(x)$.

Proof. Both are equivalent to $\sum_{x<p\leq x+h} \log p > 0$. □

Lemma 10.4.3 (Upper bound on Etheta implies prime in short interval)**.** There is a prime in $(x, x+h]$ if $xE_\theta(x) + (x+h)E_\theta(x+h) < h$.

Proof. Lower bound $\theta(x+h) - \theta(x)$ using $\theta(x+h) \geq x + h(1 - E_\theta(x+h))$ and $\theta(x) \leq x(1 + E_\theta(x))$ and apply Lemma 10.4.2. ☐

Lemma 10.4.4 (Numerical bound on Etheta implies prime in short interval)**.** If $E_\theta(x) \leq \varepsilon(x_0)$ for all $x \geq x_0$, and $(2x+h)\varepsilon(x_0) < h$, then there is a prime in $(x, x+h]$.

Proof. Apply Lemma 10.4.3. ☐

Lemma 10.4.5 (Classical bound on Etheta implies prime in short interval)**.** If $E_\theta(x)$ enjoys a classical bound for all $x \geq x_0$, $x \geq \exp(R(2B/C)^2)$ and $(2x+h)A\left(\frac{\log x}{R}\right)^B \exp\left(-C\left(\frac{\log x}{R}\right)^{1/2}\right) < h$, then there is a prime in $(x, x+h]$.

Proof. Apply Lemma 10.4.4 and Lemma 2.0.2. ☐

Lemma 10.4.6 (Prime gap record implies prime in short interval)**.** If there is a prime gap record (g, p), then there is a prime in $(x, x+h]$ whenever $x < p$ and $h \geq g$.

Proof. If p_k is the largest prime less than or equal to x, then $p_{k+1} - p_k < g \leq h$, hence $x < p_{k+1} \leq x + h$, giving the claim. ☐

10.5 The prime number bounds of Rosser and Schoenfeld

In this section we formalize the prime number bounds of Rosser and Schoenfeld [26].

TODO: Add more results and proofs here, and reorganize the blueprint

Theorem 10.5.1 (A medium version of the prime number theorem)**.** $\vartheta(x) = x + O(x/\log^2 x)$.

Proof. This in principle follows by establishing an analogue of Theorem 6.0.1, using mediumPNT in place of weakPNT. ☐

Sublemma 10.5.1 (The Chebyshev function is Stieltjes)**.** The function $\vartheta(x) = \sum_{p \leq x} \log p$ defines a Stieltjes function (monotone and right continuous).

Proof. Trivial ☐

Sublemma 10.5.2 (RS-prime display before (4.13))**.** $\sum_{p \leq x} f(p) = \int_2^x \frac{f(y)}{\log y}\, d\vartheta(y)$.

Proof. This follows from the definition of the Stieltjes integral. ☐

Sublemma 10.5.3 (RS equation (4.13))**.** $\sum_{p \leq x} f(p) = \frac{f(x)\vartheta(x)}{\log x} - \int_2^x \vartheta(y) \frac{d}{dy}\left(\frac{f(y)}{\log y}\right)\, dy$.

Proof. Follows from Sublemma 10.5.2 and integration by parts. ☐

Sublemma 10.5.4 (RS equation (4.14))**.**

$$\sum_{p \leq x} f(p) = \int_2^x \frac{f(y)\, dy}{\log y} + \frac{2f(2)}{\log 2}$$

$$+\frac{f(x)(\vartheta(x) - x)}{\log x} - \int_2^x (\vartheta(y) - y)\frac{d}{dy}\left(\frac{f(y)}{\log y}\right)\, dy.$$

Proof. Follows from Sublemma 10.5.3 and integration by parts. □

Definition 10.5.1 (RS equation (4.16))**.**

$$L_f := \frac{2f(2)}{\log 2} - \int_2^\infty (\vartheta(y) - y)\frac{d}{dy}(\frac{f(y)}{\log y})\ dy.$$

Sublemma 10.5.5 (RS equation (4.15))**.**

$$\sum_{p \le x} f(p) = \int_2^x \frac{f(y)\ dy}{\log y} + L_f$$

$$+\frac{f(x)(\vartheta(x) - x)}{\log x} + \int_x^\infty (\vartheta(y) - y)\frac{d}{dy}(\frac{f(y)}{\log y})\ dy.$$

Proof. Follows from Sublemma 10.5.4 and Definition 10.5.1. □

Sublemma 10.5.6 (RS equation (4.17))**.**

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(y)\ dy}{y \log^2 y}.$$

Proof. Follows from Sublemma 10.5.3 applied to $f(t) = 1$. □

Sublemma 10.5.7 (RS equation (4.18))**.**

$$\sum_{p \le x} \frac{1}{p} = \frac{\vartheta(x)}{x \log x} + \int_2^x \frac{\vartheta(y)(1 + \log y)\ dy}{y^2 \log^2 y}.$$

Proof. Follows from Sublemma 10.5.3 applied to $f(t) = 1/t$. □

Definition 10.5.2 (Meissel-Mertens constant B)**.** $B := \lim_{x\to\infty} \left(\sum_{p\le x} \frac{1}{p} - \log\log x\right)$.

Theorem 10.5.2 (RS equation (4.19) and Mertens' second theorem)**.**

$$\sum_{p \le x} \frac{1}{p} = \log\log x + B + \frac{\vartheta(x) - x}{x \log x}$$

$$-\int_2^x \frac{(\vartheta(y) - y)(1 + \log y)\ dy}{y^2 \log^2 y}.$$

Proof. Follows from Sublemma 10.5.3 applied to $f(t) = 1/t$. One can also use this identity to demonstrate convergence of the limit defining B. □

Definition 10.5.3 (Mertens constant E)**.** $E := \lim_{x\to\infty} \left(\sum_{p\le x} \frac{\log p}{p} - \log x\right)$.

Sublemma 10.5.8 (RS equation (4.19) and Mertens' first theorem)**.**

$$\sum_{p \le x} \frac{\log p}{p} = \log x + E + \frac{\vartheta(x) - x}{x}$$

$$-\int_2^x \frac{(\vartheta(y) - y)\ dy}{y^2}.$$

Proof. Follows from Sublemma 10.5.3 applied to $f(t) = \log t/t$. Convergence will need Theorem 10.5.1. □

Theorem 10.5.3 (RS Theorem 12)**.** We have $\psi(x) < c_0 x$ for all $x > 0$.

Proof. □

10.6 The estimates of Buthe

In this section we collect some results from Buthe's paper [4], which provides explicit estimates on $\psi(x)$, $\theta(x)$, and $\pi(x)$.

TODO: Add more results and proofs here, and reorganize the blueprint

Definition 10.6.1 (Buthe Equation (1.4))**.** $\pi^*(x) = \sum_{k\geq 1} \pi(x^{1/k})/k$.

Theorem 10.6.1 (Buthe Theorem 2(a))**.** If $11 < x \leq 10^{19}$, then $|x - \psi(x)| \leq 0.94\sqrt{x}$.

Proof. □

Theorem 10.6.2 (Buthe Theorem 2(b))**.** If $1423 \leq x \leq 10^{19}$, then $x - \vartheta(x) \leq 1.95\sqrt{x}$.

Proof. □

Theorem 10.6.3 (Buthe Theorem 2(c))**.** If $1 \leq x \leq 10^{19}$, then $x - \vartheta(x) > 0.05\sqrt{x}$.

Proof. □

Theorem 10.6.4 (Buthe Theorem 2(d))**.** If $2 \leq x \leq 10^{19}$, then $|\operatorname{li}(x) - \pi^*(x)| < \frac{\sqrt{x}}{\log x}$.

Proof. □

Theorem 10.6.5 (Buthe Theorem 2(e))**.** If $2 \leq x \leq 10^{19}$, then

$$\operatorname{li}(x) - \pi(x) \leq \frac{\sqrt{x}}{\log(x)}\left(1.95 + \frac{3.9}{\log x} + \frac{19.5}{\log(x)^2}\right).$$

Proof. □

Theorem 10.6.6 (Buthe Theorem 2(f))**.** If $2 \leq x \leq 10^{19}$, then $\operatorname{li}(x) - \pi(x) > 0$.

Proof. □

10.7 Numerical content of BKLNW

Purely numerical calculations from [2]. This is kept in a separate file from the main file to avoid heavy recompilations. Because of this, this file should not import any other files from the PNT+ project, other than further numerical data files.

Sublemma 10.7.1. The entries in Table 14 obey the criterion in Sublemma 10.8.2.

Proof. □

10.8 Tools from BKLNW

In this file we record the results from [2], excluding Appendix A which is treated elsewhere. These results convert an initial estimate on $E_\psi(x)$ (provided by Appendix A) to estimates on $E_\theta(x)$. One first obtains estimates on $E_\theta(x)$ that do not decay in x, and then obtain further estimates that decay like $1/\log^k x$ for some $k = 1, \dots 5$.

10.8.1 Bounding Etheta uniformly

We first focus on getting bounds on $E_\theta(x)$ that do not decay in x. A key input, provided by Appendix A, is a bound on $E_\psi(x)$ of the form

$$E_\psi(x) \leq \varepsilon(b) \quad \text{for } x \geq e^b.$$

We also assume a numerical verification $\theta(x) < x$ for $x \leq x_1$ for some $x_1 \geq e^7$.

Sublemma 10.8.1 (A consequence of Buthe equation (1.7))**.** $\theta(x) < x$ for all $1 \leq x \leq 10^{19}$.

Proof. This follows from Theorem 10.6.3. ☐

Definition 10.8.1 (Default pre-input parameters)**.** We take ε as in Table 8 of [2], and $x_1 = 10^{19}$.

Lemma 10.8.1 (BKLNW Lemma 11a)**.** With the hypotheses as above, we have $\theta(x) \leq (1 + \varepsilon(\log x_1))x)$ for all $x > 0$.

Proof. Follows immediately from the given hypothesis $\theta(x) \leq \psi(x)$, splitting into the cases $x \geq x_1$ and $x < x_1$. ☐

Lemma 10.8.2 (BKLNW Lemma 11b)**.** With the hypotheses as above, we have

$$(1 - \varepsilon(b) - c_0(e^{-b/2} + e^{-2b/3} + e^{-4b/5}))x \leq \theta(x)$$

for all $x \geq e^b$ and $b > 0$, where $c_0 = 1.03883$ is the constant from [26, Theorem 12].

Proof. From Theorem 10.3.1 we have $\psi(x) - \theta(x) \leq \psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5})$. Now apply the hypothesis on $\psi(x)$ and Theorem 10.5.3. ☐

Theorem 10.8.1 (BKLNW Theorem 1a)**.** For any fixed $X_0 \geq 1$, there exists $m_0 > 0$ such that, for all $x \geq X_0$

$$x(1 - m_0) \leq \theta(x).$$

For any fixed $X_1 \geq 1$, there exists $M_0 > 0$ such that, for all $x \geq X_1$

$$\theta(x) \leq x(1 + M_0).$$

For $X_0, X_1 \geq e^{20}$, we have

$$m_0 = \varepsilon(\log X_0) + 1.03883 \left(X_0^{-1/2} + X_0^{-2/3} + X_0^{-4/5}\right)$$

and

$$M_0 = \varepsilon(\log X_1).$$

Proof. Combine Lemmas 10.8.1 with $b = \log X_1$ for the upper bound, and and 10.8.2 with $b = \log X_0$ for the lower bound. ☐

Theorem 10.8.2. One has $x(1-m) \leq \theta(x) \leq x(1+M)$ whenever $x \geq e^b$ and b, M, m obey the condition that $b \geq 20$, $\varepsilon(b) \leq M$, and $\varepsilon(b) + c_0(e^{-b/2} + e^{-2b/3} + e^{-4b/5}) \leq m$.

Proof. ☐

Theorem 10.8.3. The previous theorem holds with (b, M, m) given by the values in [2, Table 14].

Proof. □

Corollary 10.8.1 (BKLNW Corollary 2.1)**.** $\theta(x) \leq (1 + 1.93378 \times 10^{-8})x$.

Proof. We combine together Theorem 10.8.1 and Theorem 10.8.3 with $X_1 = 10^{19}$. □

Definition 10.8.2 (Default input parameters)**.** We take $\alpha = 1.93378 \times 10^{-8}$, so that we have $\theta(x) \leq (1 + \alpha)x$ for all x.

10.8.2 Bounding psi minus theta

In this section we obtain bounds of the shape

$$\psi(x) - \theta(x) \leq a_1 x^{1/2} + a_2 x^{1/3}$$

for all $x \geq x_0$ and various a_1, a_2, x_0.

Definition 10.8.3 (BKLNW Equation 2.4)**.**

$$f(x) := \sum_{k=3}^{\lfloor \log x / \log 2 \rfloor} x^{1/k - 1/3}.$$

Sublemma 10.8.2 (BKLNW Proposition 3, substep 1)**.** Let $x \geq x_0$ and let α be admissible. Then

$$\frac{\psi(x) - \theta(x) - \theta(x^{1/2})}{x^{1/3}} \leq (1 + \alpha) \sum_{k=3}^{\lfloor \frac{\log x}{\log 2} \rfloor} x^{\frac{1}{k} - \frac{1}{3}}.$$

Proof. Bound each $\theta(x^{1/k})$ term by $(1 + \alpha)x^{1/k}$ in Sublemma 10.3.1. □

Sublemma 10.8.3 (BKLNW Proposition 3, substep 2)**.** f decreases on $[2^n, 2^{n+1})$.

Proof. Clear. □

Sublemma 10.8.4 (BKLNW Proposition 3, substep 3)**.** $f(2^n) = 1 + u_n$.

Proof. Clear. □

Sublemma 10.8.5 (BKLNW Proposition 3, substep 4)**.** $u_{n+1} < u_n$ for $n \geq 9$.

Proof. We have

$$u_{n+1} - u_n = \sum_{k=4}^{n} 2^{\frac{n+1}{k} - \frac{n+1}{3}} (1 - 2^{\frac{1}{3} - \frac{1}{k}}) + 2^{1 - \frac{n+1}{3}} = 2^{-\frac{n+1}{3}} \left(2 - \sum_{k=4}^{n} 2^{\frac{n+1}{k}} (2^{\frac{1}{3} - \frac{1}{k}} - 1) \right). \quad (10.1)$$

Define $s(k, n) := 2^{\frac{n+1}{k}} (2^{\frac{1}{3} - \frac{1}{k}} - 1)$. Note that $s(k, n)$ is monotone increasing in n for each fixed $k \geq 4$. By numerical computation (using the trick $x \leq 2^{p/q} \iff x^q \leq 2^p$ to verify decimal lower bounds x), $\sum_{k=4}^{n} s(k, n) \geq \sum_{k=4}^{9} s(k, 9) > 2.12 > 2$. Thus $u_{n+1} - u_n < 0$. □

Sublemma 10.8.6 (BKLNW Proposition 3, substep 5)**.** $f(2^n) > f(2^{n+1})$ for $n \geq 9$.

Proof. This follows from Sublemmas 10.8.4 and 10.8.5. □

Sublemma 10.8.7 (BKLNW Proposition 3, substep 6)**.** $f(x) \leq f(2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1})$ on $[2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1}, \infty)$.

Proof. Follows from Sublemmas 10.8.3 and 10.8.6. $\square$

Sublemma 10.8.8 (BKLNW Proposition 3, substep 7)**.** $f(x) \leq f(x_0)$ for $x \in [x_0, 2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1})$.

Proof. Follows since $f(x)$ decreases on $[2^{\lfloor \frac{\log x_0}{\log 2} \rfloor}, 2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1})$. $\square$

Sublemma 10.8.9 (BKLNW Proposition 3, substep 8)**.** $f(x) \leq \max\Big(f(x_0), f(2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1})\Big)$.

Proof. Combines previous sublemmas. $\square$

Proposition 10.8.1 (BKLNW Proposition 3)**.** Let $x_0 \geq 2^9$. Let $\alpha > 0$ exist such that $\theta(x) \leq (1+\alpha)x$ for $x > 0$. Then for $x \geq x_0$,

$$\sum_{k=3}^{\lfloor \frac{\log x}{\log 2} \rfloor} \theta(x^{1/k}) \leq \eta x^{1/3}, \tag{10.2}$$

where

$$\eta = \eta(x_0) = (1+\alpha)\max\Big(f(x_0), f(2^{\lfloor \frac{\log x_0}{\log 2} \rfloor + 1})\Big) \tag{10.3}$$

with

$$f(x) := \sum_{k=3}^{\lfloor \frac{\log x}{\log 2} \rfloor} x^{\frac{1}{k} - \frac{1}{3}}. \tag{10.4}$$

Proof. Combines previous sublemmas. $\square$

Corollary 10.8.2 (BKLNW Corollary 3.1)**.** Let $b \geq 7$. Assume $x \geq e^b$. Then we have

$$\psi(x) - \theta(x) - \theta(x^{1/2}) \leq \eta x^{1/3},$$

where

$$\eta = (1+\alpha)\max\Big(f(e^b), f(2^{\lfloor \frac{b}{\log 2} \rfloor + 1})\Big) \tag{10.5}$$

Proof. We apply Proposition 10.8.1 with $x_0 = e^b$ where we observe that $x_0 = e^b \geq e^7 > 2^9$. $\square$

Proposition 10.8.2 (BKLNW Proposition 4, part a)**.** If $7 \leq b \leq 2\log x_1$, then we have

$$\theta(x^{1/2}) \leq (1 + \varepsilon(\log x_1))x^{1/2} \quad \text{for } x \geq e^b. \tag{10.6}$$

Proof. Note that in the paper, the inequality in Proposition 4 is strict, but the argument can only show nonstrict inequalities. If $e^b \leq x \leq x_1^2$, then $x^{1/2} \leq x_1$, and thus

$$\theta(x^{1/2}) < x^{1/2} \quad \text{for } e^b \leq x \leq x_1^2.$$

On the other hand, if $x^{1/2} > x_1 = e^{\log x_1}$, then we have by (2.7)

$$\theta(x^{1/2}) \leq \psi(x^{1/2}) \leq (1 + \varepsilon(\log x_1))x^{1/2},$$

since $\log x_1 \geq 7$. The last two inequalities for $\theta(x^{1/2})$ combine to establish (2.8). $\square$

Proposition 10.8.3 (BKLNW Proposition 4, part b)**.** If $b > 2\log x_1$, then we have

$$\theta(x^{1/2}) \leq (1 + \varepsilon(b/2))x^{1/2} \quad \text{for } x \geq e^b.$$

Proof. Note that in the paper, the inequality in Proposition 4 is strict, but the argument can only show nonstrict inequalities. As in the above subcase, we have for $x \geq e^b$

$$\theta(x^{1/2}) \leq \psi(x^{1/2}) \leq (1 + \varepsilon(b/2))x^{1/2},$$

since $x^{1/2} > e^{b/2} > x_1 \geq e^7$. ☐

Definition 10.8.4 (Definition of a1)**.** a_1 is equal to $1 + \varepsilon(\log x_1)$ if $b \leq 2\log x_1$, and equal to $1 + \varepsilon(b/2)$ if $b > 2\log x_1$.

Definition 10.8.5 (Definition of a2)**.** a_2 is defined by

$$a_2 = (1 + \alpha)\max\left(f(e^b), f(2^{\lfloor \frac{b}{\log 2} \rfloor + 1})\right).$$

Theorem 10.8.4 (BKLNW Theorem 5)**.** Let $\alpha > 0$ exist such that

$$\theta(x) \leq (1 + \alpha)x \quad \text{for all } x > 0.$$

Assume for every $b \geq 7$ there exists a positive constant $\varepsilon(b)$ such that

$$\psi(x) - x \leq \varepsilon(b)x \quad \text{for all } x \geq e^b.$$

Assume there exists $x_1 \geq e^7$ such that

$$\theta(x) < x \quad \text{for } x \leq x_1. \tag{10.7}$$

Let $b \geq 7$. Then, for all $x \geq e^b$ we have

$$\psi(x) - \theta(x) < a_1 x^{1/2} + a_2 x^{1/3},$$

where

$$a_1 = \begin{cases} 1 + \varepsilon(\log x_1) & \text{if } b \leq 2\log x_1, \\ 1 + \varepsilon(b/2) & \text{if } b > 2\log x_1, \end{cases}$$

and

$$a_2 = (1 + \alpha)\max\left(f(e^b), f(2^{\lfloor \frac{b}{\log 2} \rfloor + 1})\right).$$

Proof. We have $\psi(x) - \theta(x) = \theta(x^{1/2}) + \sum_{k=3}^{\lfloor \frac{\log x}{\log 2} \rfloor} \theta(x^{1/k})$. For any $b \geq 7$, setting $x_0 = e^b$ in Proposition 4, we bound $\sum_{k=3}^{\lfloor \frac{\log x}{\log 2} \rfloor} \theta(x^{1/k})$ by $\eta x^{1/3}$ as defined in (2.3). We bound $\theta(x^{1/2})$ with Proposition **??** by taking either $a_1 = 1 + \varepsilon(\log x_1)$ for $b \leq 2\log x_1$ or $a_1 = 1 + \varepsilon(b/2)$ for $b > 2\log x_1$. ☐

Corollary 10.8.3 (BKLNW Corollary 5.1)**.** Let $b \geq 7$. Then for all $x \geq e^b$ we have $\psi(x) - \vartheta(x) < a_1 x^{1/2} + a_2 x^{1/3}$, where $a_1 = a_1(b) = 1 + 1.93378 \times 10^{-8}$ if $b \leq 38\log 10$, $1 + \varepsilon(b/2)$ if $b > 38\log 10$, and $a_2 = a_2(b) = (1 + 1.93378 \times 10^{-8})\max\left(f(e^b), f(2^{\lfloor \frac{b}{\log 2} \rfloor + 1})\right)$, where f is defined by (2.4) and values for $\varepsilon(b/2)$ are from Table 8.

Proof. This is Theorem 5 applied to the default inputs in Definition 10.8.2. ☐

10.8.3 Bounding theta(x)-x with a logarithmic decay, I: large x

In this section and the next ones we obtain bounds of the shape

$$x\left(1-\frac{m_k}{\log^k x}\right)\le \theta(x)$$

for all $x \ge X_0$ and

$$\theta(x)\le x\left(1+\frac{M_k}{\log^k x}\right)$$

for all $x \ge X_1$, for various k, m_k, M_k, X_0, X_1, with $k \in \{1, \dots, 5\}$.

For this section we focus on estimates that are useful when x is extremely large, e.g., $x \ge e^{25000}$.

Lemma 10.8.3 (BKLNW Lemma 6)**.** Suppose there exists $c_1, c_2, c_3, c_4 > 0$ such that

$$|\theta(x)-x| \le c_1 x(\log x)^{c_2}\exp(-c_3(\log x)^{\frac12}) \quad \text{for all } x \ge c_4. \tag{10.8}$$

Let $k > 0$ and let $b \ge \max\left(\log c_4, \log\left(\frac{4(c_2+k)^2}{c_3^2}\right)\right)$. Then for all $x \ge e^b$ we have

$$|\theta(x)-x| \le \frac{\mathcal{A}_k(b)x}{(\log x)^k},$$

where

$$\mathcal{A}_k(b) = c_1 \cdot b^{c_2+k}e^{-c_3\sqrt{b}}.$$

Proof. We denote $g(x) = (\log x)^{c_2+k}\exp(-c_3(\log x)^{\frac12})$. By (10.8), $|\theta(x)-x| < \frac{c_1 g(x)x}{(\log x)^k}$ for all $x \ge c_4$. It suffices to bound g: by calculus, $g(x)$ decreases when $x \ge \frac{4(c_2+k)^2}{c_3^2}$. Therefore $|\theta(x)-x| \le \frac{c_1 g(e^b)x}{(\log x)^k}$. Note that $c_1 g(e^b) = \mathcal{A}_k(b)$ and the condition on b follows from the conditions $e^b \ge c_4$ and $e^b \ge \frac{4(c_2+k)^2}{c_3^2}$. ☐

Corollary 10.8.4 (BKLNW Corollary 14.1)**.** Suppose one has an asymptotic bound E_ψ with parameters A, B, C, R, e^{x_0} (which need to satisfy some additional bounds) with $x_0 \ge 1000$. Then E_θ obeys an asymptotic bound with parameters A', B, C, R, e^{x_0}, where

$$A' := A(1+\frac{1}{A}(\frac{R}{x_0})^B\exp(C\sqrt{\frac{x_0}{R}})(a_1(x_0)\exp(\frac{-x_0}{2})+a_2(x_0)\exp(\frac{-2x_0}{3})))$$

and $a_1(x_0), a_2(x_0)$ are as in Corollary 10.8.3.

Proof. We write $\theta(x)-x = \psi(x)-x+\theta(x)-\psi(x)$, apply the triangle inequality, and invoke Corollary 10.8.3 to obtain

$$E_\theta(x) \le A(\frac{\log x}{R})^B\exp(-C(\frac{\log x}{R})^{\frac12}) + a_1(x_0)x^{-\frac12} + a_2(x_0)x^{-\frac23}$$

$$\le A(\frac{\log x}{R})^B\exp(-C(\frac{\log x}{R})^{\frac12})(1+\frac{a_1(x_0)\exp(C\sqrt{\frac{\log x}{R}})}{A\sqrt{x}(\frac{\log x}{R})^B}+\frac{a_2(x_0)\exp(C\sqrt{\frac{\log x}{R}})}{Ax^{\frac23}(\frac{\log x}{R})^B}).$$

The function in brackets decreases for $x \ge e^{x_0}$ with $x_0 \ge 1000$ (assuming reasonable hypotheses on A, B, C, R) and thus we obtain the desired bound with A' as above. ☐

10.8.4 Bounding theta(x)-x with a logarithmic decay, II: medium x

In this section we tackle medium x.
TODO: formalize Lemma 8 and Corollary 8.1

10.8.5 Bounding theta(x)-x with a logarithmic decay, III: small x

In this section we tackle small x.
TODO: formalize (3.17), (3.18), Lemma 9, Corollary 9.1

10.8.6 Bounding theta(x)-x with a logarithmic decay, IV: very small x

In this section we tackle very small x.
TODO: Formalize Lemma 10

10.8.7 Final bound on Etheta(x)

Now we put everything together.
TODO: Section 3.7.1; 3.7.2; 3.7.3; 3.7.4

Theorem 10.8.5 (Theorem 1b)**.** Let k be an integer with $1 \leq k \leq 5$. For any fixed $X_0 > 1$, there exists $m_k > 0$ such that, for all $x \geq X_0$

$$x(1 - \frac{m_k}{(\log x)^k}) \leq \theta(x).$$

For any fixed $X_1 > 1$, there exists $M_k > 0$ such that, for all $x \geq X_1$

$$\theta(x) \leq x(1 + \frac{M_k}{(\log x)^k}).$$

In the case $k = 0$ and $X_0, X_1 \geq e^{20}$, we have

$$m_0 = \varepsilon(\log X_0) + 1.03883 \left(X_0^{-1/2} + X_0^{-2/3} + X_0^{-4/5}\right)$$

and

$$M_0 = \varepsilon(\log X_1).$$

Proof. □

Theorem 10.8.6 (BKLNW Theorem 1b, table form)**.** See [2, Table 15] for values of m_k and M_k, for $k \in \{1, 2, 3, 4, 5\}$.

Proof. □

10.8.8 Computational examples

Now we apply the previous theorem.
TODO: Corollary 11.1, 11.2

10.9 The implications of FKS2

In this file we record the implications in the paper [15]. Roughly speaking, this paper has two components: a "ψ to θ pipeline" that converts estimates on the error $E_\psi(x) = |\psi(x) - x|/x$ in the prime number theorem for the first Chebyshev function ψ to estimates on the error $E_\theta(x) = |\theta(x) - x|/x$ in the prime number theorem for the second Chebyshev function θ; and a "θ to π pipeline" that converts estimates E_θ to estimates on the error $E_\pi(x) = |\pi(x) - \mathrm{Li}(x)|/(x/\log x)$ in the prime number theorem for the prime counting function π. Each pipeline converts "admissible classical bounds" (Definitions 2.0.5 2.0.8, 2.0.9) of one error to admissible classical bounds of the next error in the pipeline.

There are two types of bounds considered here. The first are asymptotic bounds of the form

$$E_\psi(x), E_\theta(x), E_\pi(x) \le A \left(\frac{\log x}{R}\right)^B \exp\left(-C\left(\frac{\log x}{R}\right)^{1/2}\right)$$

for various A, B, C, R and all $x \ge x_0$. The second are numerical bounds of the form

$$E_\psi(x), E_\theta(x), E_\pi(x) \le \varepsilon_{num}(x_0)$$

for all $x \ge x_0$ and certain specific numerical choices of x_0 and $\varepsilon_{num}(x_0)$. One needs to merge these bounds together to obtain the best final results.

10.9.1 Basic estimates on the error bound g

Our asymptotic bounds can be described using a certain function g. Here we define g and record its basic properties.

Definition 10.9.1 (g function, FKS2 (16))**.** For any $a, b, c, x \in \mathbb{R}$ we define $g(a, b, c, x) := x^{-a}(\log x)^b \exp(c(\log x)^{1/2})$.

Sublemma 10.9.1 (FKS2 Sublemma 10-1)**.** We have

$$\frac{d}{dx} g(a, b, c, x) = \left(-a\log(x) + b + \frac{c}{2}\sqrt{\log(x)}\right) x^{-a-1}(\log(x))^{b-1} \exp(c\sqrt{\log(x)}).$$

Proof. This follows from straightforward differentiation. ☐

Sublemma 10.9.2 (FKS2 Sublemma 10-2)**.** $\frac{d}{dx} g(a, b, c, x)$ is negative when $-au^2 + \frac{c}{2}u + b < 0$, where $u = \sqrt{\log(x)}$.

Proof. Clear from previous sublemma. ☐

Lemma 10.9.1 (FKS2 Lemma 10a)**.** If $a > 0$, $c > 0$ and $b < -c^2/16a$, then $g(a, b, c, x)$ decreases with x.

Proof. We apply Lemma 10.9.2. There are no roots when $b < -\frac{c^2}{16a}$, and the derivative is always negative in this case. ☐

Lemma 10.9.2 (FKS2 Lemma 10b)**.** For any $a > 0$, $c > 0$ and $b \ge -c^2/16a$, $g(a, b, c, x)$ decreases with x for $x > \exp((\frac{c}{4a} + \frac{1}{2a}\sqrt{\frac{c^2}{4} + 4ab})^2)$.

Proof. We apply Lemma 10.9.2. If $a > 0$, there are two real roots only if $\frac{c^2}{4} + 4ab \ge 0$ or equivalently $b \ge -\frac{c^2}{16a}$, and the derivative is negative for $u > \frac{\frac{c}{2} + \sqrt{\frac{c^2}{4} + 4ab}}{2a}$. ☐

Lemma 10.9.3 (FKS2 Lemma 10c)**.** If $c > 0$, $g(0, b, c, x)$ decreases with x for $\sqrt{\log x} < -2b/c$.

Proof. We apply Lemma 10.9.2. If $a = 0$, it is negative when $u < \frac{-2b}{c}$. Note: this lemma is mistyped as $\sqrt{\log x} > -2b/c$ in [15]. ☐

Corollary 10.9.1 (FKS2 Corollary 11)**.** If $B \geq 1 + C^2/16R$ then $g(1, 1-B, C/\sqrt{R}, x)$ is decreasing in x.

Proof. This follows from Lemma 10.9.1 applied with $a = 1$, $b = 1 - B$ and $c = C/\sqrt{R}$. ☐

When integrating expressions involving g, the Dawson function naturally appears; and we need to record some basic properties about it.

Definition 10.9.2 (Dawson function, FKS2 (19))**.** The Dawson function $D_+ : \mathbb{R} \to \mathbb{R}$ is defined by the formula $D_+(x) := e^{-x^2} \int_0^x e^{t^2}\ dt$.

Remark 10.9.1 (FKS2 remark after Corollary 11)**.** The Dawson function has a single maximum at $x \approx 0.942$, after which the function is decreasing.

Proof. The Dawson function satisfies the differential equation $F'(x) + 2xF(x) = 1$ from which it follows that the second derivative satisfies $F''(x) = -2F(x) - 2x(-2xF(x) + 1)$, so that at every critical point (where we have $F(x) = \frac{1}{2x}$) we have $F''(x) = -\frac{1}{x}$. It follows that every positive critical value gives a local maximum, hence there is a unique such critical value and the function decreases after it. Numerically one may verify this is near 0.9241 see https://oeis.org/ A133841. ☐

10.9.2 From asymptotic estimates on psi to asymptotic estimates on theta

To get from asymptotic estimates on E_ψ to asymptotic estimates on E_θ, the paper cites results and arguments from the previous paper [2], which is treated elsewhere in this blueprint.

Proposition 10.9.1 (FKS2 Proposition 13)**.** Suppose that A_ψ, B, C, R, x_0 give an admissible bound for E_ψ. If $B > C^2/8R$, then A_θ, B, C, R, x_0 give an admissible bound for E_θ, where

$$A_\theta = A_\psi(1 + \nu_{asymp}(x_0))$$

with

$$\nu_{asymp}(x_0) = \frac{1}{A_\psi}(\frac{R}{\log x_0})^B \exp(C\sqrt{\frac{\log x_0}{R}})(a_1(\log x_0)x_0^{-1/2} + a_2(\log x_0)x_0^{-2/3}).$$

and a_1, a_2 are given by Definitions 10.8.4 and 10.8.5.

Proof. The proof of Corollary 10.8.4 essentially proves the proposition, but requires that $x_0 \geq e^{1000}$ to conclude that the function

$$1+\frac{a_1 \exp(C\sqrt{\frac{\log x}{R}})}{A_\psi\sqrt{x}(\log x/R)^B}+\frac{a_2 \exp(C\sqrt{\frac{\log x}{R}})}{A_\psi x^{2/3}(\log x/R)^B} = 1+\frac{a_1}{A_\psi}g(1/2, -B, C/\sqrt{R}, x)+\frac{a_2}{A_\psi}g(2/3, -B, C/\sqrt{R}, x)$$

is decreasing. By Lemma 10.9.1, since $B > C^2/8R$, the function is actually decreasing for all x. ☐

Corollary 10.9.2 (FKS2 Corollary 14)**.** We have an admissible bound for E_θ with $A = 121.0961$, $B = 3/2$, $C = 2$, $R = 5.5666305$, $x_0 = 2$.

Proof. By Corollary 9.3.23, with $R = 5.5666305$, and using the admissible asymptotic bound for $E_\psi(x)$ with $A_\psi = 121.096$, $B = 3/2$, $C = 2$, for all $x \geq x_0 = e^{30}$, we can obtain $\nu_{asymp}(x_0) \leq 6.3376 \cdot 10^{-7}$, from which one can conclude an admissible asymptotic bound for $E_\theta(x)$ with $A_\theta = 121.0961$, $B = 3/2$, $C = 2$, for all $x \geq x_0 = e^{30}$. Additionally, the minimum value of $\varepsilon_{\theta,asymp}(x)$ for $2 \leq x \leq e^{30}$ is roughly $2.6271\ldots$ at $x = 2$. The results found in [2, Table 13 and 14] give $E_\theta(x) \leq 1 < \varepsilon_{\theta,asymp}(2) \leq \varepsilon_{\theta,asymp}(x)$ for all $2 \leq x \leq e^{30}$. ☐

Remark 10.9.2 (FKS2 Remark 15)**.** If $\log x_0 \geq 1000$ then we have an admissible bound for E_θ with the indicated choice of $A(x_0)$, $B = 3/2$, $C = 2$, and $R = 5.5666305$.

Proof. From [14, Table 6] we have $\nu_{asymp}(x_0) \leq 10^{-200}$. Thus, one easily verifies that the rounding up involved in forming [14, Table 6] exceeds the rounding up also needed to apply this step. Consequently we may use values from A_θ taken from [14, Table 6] directly but this does, in contrast to Corollary 10.9.2, require the assumption $x > x_0$, as per that table. ☐

10.9.3 From asymptotic estimates on theta to asymptotic estimates on pi

To get from asymptotic estimates on E_θ to asymptotic estimates on E_π, one first needs a way to express the latter as an integral of the former.

Sublemma 10.9.3 (FKS2 equation (17))**.** For any $2 \leq x_0 < x$ one has

$$(\pi(x) - \mathrm{Li}(x)) - (\pi(x_0) - \mathrm{Li}(x_0)) = \frac{\theta(x) - x}{\log x} - \frac{\theta(x_0) - x_0}{\log x_0} + \int_{x_0}^{x} \frac{\theta(t) - t}{t \log^2 t} dt.$$

Proof. This follows from Sublemma 10.5.6. ☐

The following definition is only implicitly in FKS2, but will be convenient:

Definition 10.9.3 (Defining an error term)**.** For any $x_0 > 0$, we define

$$\delta(x_0) := |\frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0/\log x_0} - \frac{\theta(x_0) - x_0}{x_0}|.$$

We can now obtain an upper bound on E_π in terms of E_θ:

Sublemma 10.9.4 (FKS2 Equation (30))**.** For any $x \geq x_0 > 0$,

$$|\pi(x) - \mathrm{Li}(x)| \leq \left|\frac{\theta(x) - x}{\log(x)}\right| + \left|\pi(x_0) - \mathrm{Li}(x_0) - \frac{\theta(x_0) - x_0}{\log(x_0)}\right| + \left|\int_{x_0}^{x} \frac{\theta(t) - t}{t(\log(t))^2}\, dt\right|.$$

Proof. This follows from applying the triangle inequality to Sublemma 10.9.3. ☐

Next, we bound the integral appearing in Sublemma 10.9.3.

Lemma 10.9.4 (FKS2 Lemma 12)**.** Suppose that E_θ satisfies an admissible classical bound with parameters A, B, C, R, x_0. Then, for all $x \geq x_0$,

$$\int_{x_0}^{x} \left| \frac{E_\theta(t)}{\log^2 t} dt \right| \leq \frac{2A}{R^B} x m(x_0, x) \exp\left(-C\sqrt{\frac{\log x}{R}}\right) D_+\left(\sqrt{\log x} - \frac{C}{2\sqrt{R}}\right)$$

where

$$m(x_0, x) = \max((\log x_0)^{(2B-3)/2}, (\log x)^{(2B-3)/2}).$$

Proof. NOTE: in order for the proof to work, some lower bounds on x_0 were added to make various limits of integration non-negative.

Since $\varepsilon_{\theta,\mathrm{asymp}}(t)$ provides an admissible bound on $\theta(t)$ for all $t \geq x_0$, we have

$$\int_{x_0}^{x} \left| \frac{\theta(t) - t}{t(\log(t))^2} \right| dt \leq \int_{x_0}^{x} \frac{\varepsilon_{\theta,\mathrm{asymp}}(t)}{(\log(t))^2} = \frac{A_\theta}{R^B} \int_{x_0}^{x} (\log(t))^{B-2} \exp\left(-C\sqrt{\frac{\log(t)}{R}}\right) dt.$$

We perform the substitution $u = \sqrt{\log(t)}$ and note that $u^{2B-3} \leq m(x_0, x)$ as defined in (21). Thus the above is bounded above by

$$\frac{2A_\theta m(x_0, x)}{R^B} \int_{\sqrt{\log(x_0)}}^{\sqrt{\log(x)}} \exp\left(u^2 - \frac{Cu}{\sqrt{R}}\right) du.$$

Then, by completing the square $u^2 - \frac{Cu}{\sqrt{R}} = \left(u - \frac{C}{2\sqrt{R}}\right)^2 - \frac{C^2}{4R}$ and doing the substitution $v = u - \frac{C}{2\sqrt{R}}$, the above becomes

$$\frac{2A_\theta m(x_0, x)}{R^B} \exp\left(-\frac{C^2}{4R}\right) \int_{\sqrt{\log(x_0)} - \frac{C}{2\sqrt{R}}}^{\sqrt{\log(x)} - \frac{C}{2\sqrt{R}}} \exp(v^2)\, dv.$$

Now we have

$$\begin{aligned}\int_{\sqrt{\log(x_0)} - \frac{C}{2\sqrt{R}}}^{\sqrt{\log(x)} - \frac{C}{2\sqrt{R}}} \exp(v^2)\, dv &\leq \int_{0}^{\sqrt{\log(x)} - \frac{C}{2\sqrt{R}}} \exp(v^2)\, dv \\ &= x \exp\left(\frac{C^2}{4R}\right) \exp\left(-C\sqrt{\frac{\log(x)}{R}}\right) D_+\left(\sqrt{\log(x)} - \frac{C}{2\sqrt{R}}\right).\end{aligned}$$

Combining the above completes the proof. □

Definition 10.9.4 (mu asymptotic function, FKS2 (9))**.** For $x_0, x_1 > 0$, we define

$$\mu_{asymp}(x_0, x_1) := \frac{x_0 \log(x_1)}{\epsilon_{\theta,asymp}(x_1) x_1 \log(x_0)} \left| \frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0/\log x_0} - \frac{\theta(x_0) - x_0}{x_0} \right| + \frac{2D_+(\sqrt{\log(x_1)} - \frac{C}{2\sqrt{R}}}{\sqrt{\log x_1}}$$

.

We obtain our final bound for converting bounds on E_θ to bounds on E_π.

Theorem 10.9.1 (FKS2 Theorem 3)**.** If $B \geq \max(3/2, 1 + C^2/16R)$, $x_0 > 0$, and one has an admissible asymptotic bound with parameters A, B, C, x_0 for E_θ, and

$$x_1 \geq \max(x_0, \exp((1 + \frac{C}{2\sqrt{R}}))^2),$$

then

$$E_\pi(x) \leq \epsilon_{\theta,asymp}(x_1)(1+\mu_{asymp}(x_0,x_1))$$

for all $x \geq x_1$. In other words, we have an admissible bound with parameters $(1+\mu_{asymp}(x_0,x_1))A, B, C, x_1$ for E_π.

Proof. The starting point is Sublemma 10.9.4. The assumption ($\varepsilon_{\theta,\mathrm{asymp}}(x)$ provides an admissible bound on $\theta(x)$ for all $x \geq x_0$) to bound $\frac{\theta(x)-x}{\log(x)}$ and Lemma 10.9.4 to bound $\int_{x_0}^x \frac{\theta(t)-t}{t(\log(t))^2}dt$. We obtain

$$|\pi(x) - \mathrm{Li}(x)| \leq |\pi(x_0) - \mathrm{Li}(x_0) - \frac{\theta(x_0)-x_0}{\log(x_0)}| + \frac{x\varepsilon_{\theta,\mathrm{asymp}}(x)}{\log(x)} + \frac{2A_\theta}{R^B} x m(x_0,x)\exp(-C\sqrt{\frac{\log x}{R}})D_+\left(\sqrt{\log x} - \frac{C}{2\sqrt{R}}\right).$$

We recall that $x \geq x_1 \geq x_0$. Note that, by Corollary 10.9.1,

$$\frac{\log(x)}{x\varepsilon_{\theta,\mathrm{asymp}}(x)} = \frac{1}{A_\theta} g(1, 1-B, \frac{C}{\sqrt{R}}, x)$$

is decreasing for all x. Thus,

$$\frac{\log(x)}{x\varepsilon_{\theta,\mathrm{asymp}}(x)} \leq \frac{\log(x_1)}{x_1\varepsilon_{\theta,\mathrm{asymp}}(x_1)}.$$

In addition, we have the simplification

$$\frac{\log(x)}{x\varepsilon_{\theta,\mathrm{asymp}}(x)} \frac{2A_\theta}{R^B} x m(x_0,x) e^{-C\sqrt{\frac{\log x}{R}}} = 2m(x_0,x)(\log(x))^{1-B} = 2(\log(x))^{1-B} \leq 2(\log(x_1))^{1-B},$$

by Definition 2.0.8 and by $m(x_0,x) = (\log(x))^{(2B-3)/2}$, since $B \geq 3/2$. Finally, since $\sqrt{\log(x_1)} - \frac{C}{2\sqrt{R}} > 1$, the Dawson function decreases for all $x \geq x_1$:

$$D_+\left(\sqrt{\log x} - \frac{C}{2\sqrt{R}}\right) \leq D_+\left(\sqrt{\log x_1} - \frac{C}{2\sqrt{R}}\right).$$

We conclude by combining the above:

$$\frac{|\pi(x) - \mathrm{Li}(x)|}{\frac{x\varepsilon_{\theta,\mathrm{asymp}}(x)}{\log(x)}} \leq \frac{\log(x_1)}{x_1\varepsilon_{\theta,\mathrm{asymp}}(x_1)}|\pi(x_0) - \mathrm{Li}(x_0) - \frac{\theta(x_0)-x_0}{\log(x_0)}| + 1 + \frac{2D_+\left(\sqrt{\log x_1} - \frac{C}{2\sqrt{R}}\right)}{\sqrt{\log(x_1)}},$$

from which we deduce the announced bound. □

10.9.4 From numerical estimates on psi to numerical estimates on theta

Here we record a way to convert a numerical bound on E_ψ to a numerical bound on E_θ.

Proposition 10.9.2 (FKS2 Proposition 17)**.** Let $x > x_0 > 2$. If $E_\psi(x) \leq \varepsilon_{\psi,num}(x_0)$, then

$$-\varepsilon_{\theta,num}(x_0) \leq \frac{\theta(x)-x}{x} \leq \varepsilon_{\psi,num}(x_0) \leq \varepsilon_{\theta,num}(x_0)$$

where

$$\varepsilon_{\theta,num}(x_0) = \varepsilon_{\psi,num}(x_0) + 1.00000002(x_0^{-1/2} + x_0^{-2/3} + x_0^{-4/5}) + 0.94(x_0^{-3/4} + x_0^{-5/6} + x_0^{-9/10})$$

Proof. The upper bound is immediate since $\theta(x) \le \psi(x)$ for all x. For the lower bound, we have

$$\frac{\theta(x) - x}{x} = \frac{\psi(x) - x}{x} + \frac{\theta(x) - \psi(x)}{x}.$$

By Theorem 10.3.1, we have

$$\psi(x) - \theta(x) \le \psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}).$$

We use [4, Theorem 2], that for $0 < x < 11$, $\psi(x) < x$, and that $\varepsilon_{\psi,num}(10^{19}) < 2 \cdot 10^{-8}$. In particular when $2 < x < 10^{38}$,

$$\psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}) \le x^{1/2} + x^{1/3} + x^{1/5} + 0.94(x^{1/4} + x^{1/6} + x^{1/10}),$$

when $10^{38} \le x < 10^{54}$,

$$\psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}) \le 1.00000002x^{1/2} + x^{1/3} + x^{1/5} + 0.94(x^{1/6} + x^{1/10}),$$

when $10^{54} \le x < 10^{95}$,

$$\psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}) \le 1.00000002(x^{1/2} + x^{1/3}) + x^{1/5} + 0.94x^{1/10},$$

and finally when $x \ge 10^{95}$,

$$\psi(x^{1/2}) + \psi(x^{1/3}) + \psi(x^{1/5}) \le 1.00000002(x^{1/2} + x^{1/3} + x^{1/5}).$$

The result follows by combining the worst coefficients from all cases and dividing by x. □

10.9.5 From numerical estimates on theta to numerical estimates on pi

Here we record a way to convert a numerical bound on E_θ to a numerical bound on E_π. We first need some preliminary lemmas.

Lemma 10.9.5 (FKS2 Lemma 19)**.** Let $x_1 > x_0 \ge 2$, $N \in \mathbb{N}$, and let $(b_i)_{i=1}^N$ be a finite partition of $[\log x_0, \log x_1]$. Then

$$|\int_{x_0}^{x_1} \frac{\theta(t) - t}{t \log^2 t}\, dt| \le \sum_{i=1}^{N-1} \varepsilon_{\theta,num}(e^{b_i})(\mathrm{Li}(e^{b_{i+1}}) - \mathrm{Li}(e^{b_i}) + \frac{e^{b_i}}{b_i} - \frac{e^{b_{i+1}}}{b_{i+1}}).$$

Proof. We split the integral at each e^{b_i} and apply the bound

$$|\frac{\theta(t) - t}{t}| \le \varepsilon_{\theta,num}(e^{b_i}), \text{ for every } e^{b_i} \le t < e^{b_{i+1}}.$$

Thus,

$$|\int_{x_0}^{x_1} \frac{\theta(t) - t}{t \log^2 t}\, dt| \le \sum_{i=1}^{N-1} \int_{e^{b_i}}^{e^{b_{i+1}}} |\frac{\theta(t) - t}{t \log^2 t}|\, dt \le \sum_{i=1}^{N-1} \varepsilon_{\theta,num}(e^{b_i}) \int_{e^{b_i}}^{e^{b_{i+1}}} \frac{dt}{(\log t)^2}.$$

We conclude by using the identity: for all $2 \le a < b$,

$$\int_a^b \frac{dt}{(\log t)^2} = \mathrm{Li}(b) - \frac{b}{\log b} - (\mathrm{Li}(a) - \frac{a}{\log a}).$$

□

Lemma 10.9.6 (FKS2 Lemma 20a)**.** The function $\mathrm{Li}(x) - \frac{x}{\log x}$ is strictly increasing for $x > 6.58$.

Proof. Differentiate

$$\frac{d}{dx}\left(\mathrm{Li}(x) - \frac{x}{\log(x)}\right) = \frac{1}{\log(x)} + \frac{1-\log(x)}{(\log(x))^2} = \frac{1}{(\log(x))^2}$$

to see that the difference is strictly increasing. Evaluating at $x = 6.58$ and applying the mean value theorem gives the announced result. □

Lemma 10.9.7 (FKS2 Lemma 20b)**.** Assume $x > 6.58$. Then $\mathrm{Li}(x) - \frac{x}{\log x} > \frac{x-6.58}{\log^2 x} > 0$.

Proof. This follows from Lemma 10.9.6 and the mean value theorem. □

Now we can start estimating E_π. We make the following running hypotheses. Let $x_0 > 0$ be chosen such that $\pi(x_0)$ and $\theta(x_0)$ are computable, and let $x_1 \geq \max(x_0, 14)$. Let $\{b_i\}_{i=1}^N$ be a finite partition of $[\log x_0, \log x_1]$, with $b_1 = \log x_0$ and $b_N = \log x_1$, and suppose that $\varepsilon_{\theta,\mathrm{num}}$ gives numerical bounds for $x = \exp(b_i)$, for each $i = 1, \dots, N$.

Sublemma 10.9.5 (FKS2 Theorem 6, substep 1)**.** With the above hypotheses, for all $x \geq x_1$ we have

$$E_\pi(x) \leq \varepsilon_{\theta,num}(x_1) + \frac{\log x}{x}\frac{x_0}{\log x_0}(E_\pi(x_0) + E_\theta(x_0))$$

$$+\frac{\log x}{x}\sum_{i=1}^{N-1}\varepsilon_{\theta,num}(e^{b_i})\left(\mathrm{Li}(e^{b_{i+1}}) - \mathrm{Li}(e^{b_i}) + \frac{e^{b_i}}{b_i} - \frac{e^{b_{i+1}}}{b_{i+1}}\right)$$

$$+\varepsilon_{\theta,num}(x_1)\frac{\log x}{x}\int_{x_1}^{x}\frac{1}{(\log t)^2}\,dt.$$

Proof. This is obtained by combining Sublemma 10.9.4 with the admissibility of $\varepsilon_{\theta,num}$ and Lemma 10.9.5. □

Sublemma 10.9.6 (FKS2 Theorem 6, substep 2)**.** With the above hypotheses, for all $x \geq x_1$ we have

$$\frac{\log x}{x}\int_{x_1}^{x}\frac{dt}{\log^2 t} < \frac{1}{\log x_1 + \log\log x_1 - 1}.$$

Proof. Call the left-hand side $f(x)$. We have

$$f(x) = \frac{\log x}{x}\left(\mathrm{Li}(x) - \frac{x}{\log x} - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1}\right).$$

Using integration by parts, its derivative can be written as

$$f'(x) = -\frac{1}{x\log^2 x} + \frac{2}{x\log^3 x} + \frac{\log x - 1}{x^2}\left(\frac{x_1}{\log^2 x_1} + \frac{2x_1}{\log^3 x_1} - \int_{x_1}^{x}\frac{6}{\log^4 t}dt\right).$$

From which we see that $f'(x_1) = \frac{1}{\log x_1} > 0$, and that $f'(x)$ is eventually negative. Thus there exists a critical point for $f(x)$ to the right of x_1. Moreover, by bounding $\int_{x_1}^{x} \frac{6}{\log^4 t} dt < 6\frac{x - x_1}{\log^4 x_1}$, one finds that $f'(x_1 \log x_1) > 0$ if $x_1 > e$. Now we write $f'(x) = \frac{f_1(x)}{x^2}$ with

$$f_1(x) = \frac{x}{\log x} - (\log x - 1) \int_{x_1}^{x} \frac{1}{\log^2 t} dt.$$

Its derivative is $f_1'(x) = -\frac{1}{x} \int_{x_1}^{x} \frac{1}{\log^2 t} dt$, which is negative for $x > x_1$. Thus $f_1(x)$ decreases and vanishes at most once, giving $f(x)$ at most one critical point, $x_m > x_1$, which is then the maximum of $f(x)$. In other words, x_m satisfies $f_1(x_m) = 0$, i.e. $\mathrm{Li}(x_m) - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1} = -\frac{x_m}{1 - \log x_m}$, which shows that $f(x)$ attains its maximum at $x = x_m$, where

$$f(x_m) = \frac{\log x_m}{x_m} \left(-\frac{x_m}{\log x_m} - \frac{x_m}{1 - \log x_m} \right) = \frac{1}{\log x_m - 1}.$$

Now, because $x_m > x_1 \log x_1$ we obtain the bound

$$f(x) < \frac{1}{\log x_1 + \log(\log x_1) - 1},$$

which gives the announced result. □

Sublemma 10.9.7 (FKS2 Theorem 6, substep 3)**.** With the above hypotheses, for all $x \geq x_1$ we have

$$\frac{\log x}{x} \int_{x_1}^{x} \frac{dt}{\log^2 t} \leq \frac{\log x_2}{x_2} \left(\mathrm{Li}(x_2) - \frac{x_2}{\log x_2} - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1} \right).$$

Proof. Let $f(x)$ be as in the previous sublemma. Notice that by assumption $x_1 \leq x \leq x_2 \leq x_1 \log x_1 < x_m$, so that

$$f(x) \leq f(x_2) = \frac{\log x_2}{x_2} \left(\mathrm{Li}(x_2) - \frac{x_2}{\log x_2} - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1} \right).$$

□

We can merge these sublemmas together after making some definitions.

Definition 10.9.5 (FKS2 equation (11))**.** For $x_1 \leq x_2 \leq x_1 \log x_1$, we define

$$\mu_{num,1}(x_0, x_1, x_2) := \frac{x_0 \log(x_1)}{\epsilon_{\theta,num}(x_1) x_1 \log(x_0)} \left| \frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0 / \log x_0} - \frac{\theta(x_0) - x_0}{x_0} \right| + \frac{\log(x_1)}{\epsilon_{\theta,num}(x_1) x_1} \sum_{i=0}^{N-1} \epsilon_{\theta,num}(e^{b_i}) \Big(\mathrm{Li}(e^{b_{i+1}}) -$$

Definition 10.9.6 (FKS2 equation (12))**.** For $x_2 \geq x_1 \log x_1$, we define

$$\mu_{num,2}(x_0, x_1) := \frac{x_0 \log(x_1)}{\epsilon_{\theta,num}(x_1) x_1 \log(x_0)} \left| \frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0 / \log x_0} - \frac{\theta(x_0) - x_0}{x_0} \right| + \frac{\log(x_1)}{\epsilon_{\theta,num}(x_1) x_1} \sum_{i=0}^{N-1} \epsilon_{\theta,num}(e^{b_i}) \Big(\mathrm{Li}(e^{b_{i+1}}) - \mathrm{L}$$

Definition 10.9.7 (Definition of mu)**.** We define $\mu_{num}(x_0, x_1, x_2)$ to equal $\mu_{num,1}(x_0, x_1, x_2)$ when $x_2 \leq x_1 \log x_1$ and $\mu_{num,2}(x_0, x_1)$ otherwise.

Definition 10.9.8 (FKS2 equation (13))**.** For $x_1 \le x_2$, we define $\epsilon_{\pi,num}(x_0, x_1, x_2) := \epsilon_{\theta,num}(x_1)(1 + \mu_{num}(x_0, x_1, x_2))$.

Remark 10.9.3 (FKS2 Remark 7)**.** If

$$\frac{d}{dx}\frac{\log x}{x}\left(\mathrm{Li}(x) - \frac{x}{\log x} - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1}\right)|_{x_2} \ge 0$$

then $\mu_{num,1}(x_0, x_1, x_2) < \mu_{num,2}(x_0, x_1)$.

Proof. This follows from the definitions of $\mu_{num,1}$ and $\mu_{num,2}$. □

This gives us the final result to obtain numerical bounds for E_π from numerical bounds on E_θ.

Theorem 10.9.2 (FKS2 Theorem 6)**.** Let $x_0 > 0$ be chosen such that $\pi(x_0)$ and $\theta(x_0)$ are computable, and let $x_1 \ge \max(x_0, 14)$. Let $\{b_i\}_{i=1}^N$ be a finite partition of $[\log x_0, \log x_1]$, with $b_1 = \log x_0$ and $b_N = \log x_1$, and suppose that $\varepsilon_{\theta,\mathrm{num}}$ gives computable admissible numerical bounds for $x = \exp(b_i)$, for each $i = 1, \dots, N$. For $x_1 \le x_2 \le x_1 \log x_1$, we define

$$\mu_{num}(x_0, x_1, x_2) = \frac{x_0 \log x_1}{\varepsilon_{\theta,num}(x_0) x_1 \log x_0}\left|\frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0/\log x_0} - \frac{\theta(x_0) - x_0}{x_0}\right|$$

$$+\frac{\log x_1}{\varepsilon_{\theta,num}(x_1) x_1}\sum_{i=1}^{N-1}\varepsilon_{\theta,num}(\exp(b_i))\left(\mathrm{Li}(e^{b_{i+1}}) - \mathrm{Li}(e^{b_i}) + \frac{e^{b_i}}{b_i} - \frac{e^{b_{i+1}}}{b_{i+1}}\right)$$

$$+\frac{\log x_2}{x_2}\left(\mathrm{Li}(x_2) - \frac{x_2}{\log x_2} - \mathrm{Li}(x_1) + \frac{x_1}{\log x_1}\right)$$

and for $x_2 > x_1 \log x_1$, including the case $x_2 = \infty$, we define

$$\mu_{num}(x_0, x_1, x_2) = \frac{x_0 \log x_1}{\varepsilon_{\theta,num}(x_1) x_1 \log x_0}\left|\frac{\pi(x_0) - \mathrm{Li}(x_0)}{x_0/\log x_0} - \frac{\theta(x_0) - x_0}{x_0}\right|$$

$$+\frac{\log x_1}{\varepsilon_{\theta,num}(x_1) x_1}\sum_{i=1}^{N-1}\varepsilon_{\theta,num}(\exp(b_i))\left(\mathrm{Li}(e^{b_{i+1}}) - \mathrm{Li}(e^{b_i}) + \frac{e^{b_i}}{b_i} - \frac{e^{b_{i+1}}}{b_{i+1}}\right)$$

$$+\frac{1}{\log x_1 + \log\log x_1 - 1}.$$

Then, for all $x_1 \le x \le x_2$ we have

$$E_\pi(x) \le \varepsilon_{\pi,num}(x_1, x_2) := \varepsilon_{\theta,num}(x_1)(1 + \mu_{num}(x_0, x_1, x_2)).$$

Proof. This follows by combining the three substeps. □

Corollary 10.9.3 (FKS2 Corollary 8)**.** Let $\{b'_i\}_{i=1}^M$ be a set of finite subdivisions of $[\log(x_1), \infty)$, with $b'_1 = \log(x_1)$ and $b'_M = \infty$. Define

$$\varepsilon_{\pi,num}(x_1) := \max_{1 \le i \le M-1}\varepsilon_{\pi,num}(\exp(b'_i), \exp(b'_{i+1})).$$

Then $E_\pi(x) \le \varepsilon_{\pi,num}(x_1)$ for all $x \ge x_1$.

Proof. This follows directly from Theorem 10.9.2 by taking the supremum over all partitions ending at infinity. □

10.9.6 Putting everything together

By merging together the above tools with various parameter choices, we can obtain some clean corollaries.

Corollary 10.9.4 (FKS2 Corollary 21)**.** Let $B \geq \max(\frac{3}{2}, 1 + \frac{C^2}{16R})$ and $B > C^2/8R$. Let $x_0, x_1 > 0$ with $x_1 \geq \max(x_0, \exp((1 + \frac{C}{2\sqrt{R}})^2))$. If E_ψ satisfies an admissible classical bound with parameters A_ψ, B, C, R, x_0, then E_π satisfies an admissible classical bound with A_π, B, C, R, x_1, where

$$A_\pi = (1 + \nu_{asymp}(x_0))(1 + \mu_{asymp}(x_0, x_1))A_\psi$$

for all $x \geq x_0$, where

$$|E_\theta(x)| \leq \varepsilon_{\theta,asymp}(x) := A(1 + \mu_{asymp}(x_0, x)) \exp(-C\sqrt{\frac{\log x}{R}})$$

where

$$\nu_{asymp}(x_0) = \frac{1}{A_\psi}(\frac{R}{\log x_0})^B \exp(C\sqrt{\frac{\log x_0}{R}})(a_1(\log x_0)x_0^{-1/2} + a_2(\log x_0)x_0^{-2/3})$$

and

$$\mu_{asymp}(x_0, x_1) = \frac{x_0 \log x_1}{\varepsilon_{\theta,asymp}(x_1)x_1 \log x_0}|E_\pi(x_0) - E_\theta(x_0)| + \frac{2D_+(\sqrt{\log x} - \frac{C}{2\sqrt{R}})}{\sqrt{\log x_1}}.$$

Proof. This follows by applying Theorem 10.9.1 with Proposition 10.9.1. The hypothesis $B > C^2/8R$ is not present in original source. □

Corollary 10.9.5 (FKS2 Corollary 22)**.** One has

$$|\pi(x) - \mathrm{Li}(x)| \leq 9.2211x\sqrt{\log x} \exp(-0.8476\sqrt{\log x})$$

for all $x \geq 2$.

Proof. We fix $R = 1$, $x_0 = 2$, $x_1 = e^{100}$, $A_\theta = 9.2211$, $B = 1.5$ and $C = 0.8476$. By Corollary 10.9.2, these are admissible for all $x \geq 2$, so we can apply Theorem 10.9.1 and calculate that

$$\mu_{asymp}(40.78\ldots, e^{20000}) \leq 5.01516 \cdot 10^{-5}. \tag{10.9}$$

This implies that $A_\pi = 121.103$ is admissible for all $x \geq e^{20000}$.

As in the proof of [14, Lemmas 5.2 and 5.3] one may verify that the numerical results obtainable from Theorem 10.9.2, using Corollary 10.9.3, may be interpolated as a step function to give a bound on $E_\pi(x)$ of the shape $\varepsilon_{\pi,asymp}(x)$. In this way we obtain that $A_\pi = 121.107$ is admissible for $x > 2$. Note that the subdivisions we use are essentially the same as used in [14, Lemmas 5.2 and 5.3]. In Table 5 we give a sampling of the relevant values, more of the values of $\varepsilon_{\pi,num}(x_1)$ can be found in Table 4. Far more detailed versions of these tables will be posted online in https://arxiv.org/src/2206.12557v1/anc/PrimeCountingTables.pdf. □

Corollary 10.9.6 (FKS2 Corollary 23)**.** A_π, B, C, x_0 as in [15, Table 6] give an admissible asymptotic bound for E_π with $R = 5.5666305$.

Proof. The bounds of the form $\varepsilon_{\pi,asymp}(x)$ come from selecting a value A for which Corollary **??** provides a better bound at $x = e^{7500}$ and from verifying that the bound in Corollary **??** decreases faster beyond this point. This final verification proceeds by looking at the derivative of the ratio as in Lemma **??**. To verify these still hold for smaller x, we proceed as below. To verify the results for any x in $\log(10^{19}) < \log(x) < 100000$, one simply proceeds as in [14, Lemmas 5.2, 5.3] and interpolates the numerical results of Theorem 10.9.2. For instance, we use the values in Table 4 as a step function and verifies that it provides a tighter bound than we are claiming. Note that our verification uses a more refined collection of values than those provided in Table 4 or the tables posted online in https://arxiv.org/src/2206.12557v1/anc/PrimeCountingTables.pdf. To verify results for $x < 10^{19}$, one compares against the results from Theorem 10.6.1, or one checks directly for particularly small x. □

Corollary 10.9.7 (FKS2 Corollary 24)**.** We have the bounds $E_\pi(x) \leq B(x)$, where $B(x)$ is given by Table 7.

Proof. Same as in Corollary **??**. □

Corollary 10.9.8 (FKS2 Corollary 26)**.** One has

$$|\pi(x) - \mathrm{Li}(x)| \leq 0.4298 \frac{x}{\log x}$$

for all $x \geq 2$.

Proof. We numerically verify that the inequality holds by showing that, for $1 \leq n \leq 25$ and all $x \in [p_n, p_{n+1}]$,

$$\left|\frac{\log(x)}{x}(\pi(x) - \mathrm{Li}(x))\right| \leq \left|\frac{\log(p_n)}{p_n}(\pi(p_n) - \mathrm{Li}(p_{n+1}))\right| \leq 0.4298.$$

For x satisfying $p_{25} = 97 \leq x \leq 10^{19}$, we use Theorems 10.6.5, 10.6.6 and verify

$$\mathcal{E}(x) = \frac{1}{\sqrt{x}}\left(1.95 + \frac{3.9}{\log(x)} + \frac{19.5}{(\log(x))^2}\right) \leq 0.4298.$$

For $x > 10^{19}$, we use Theorem **??** as well as values for $\varepsilon_{\pi,num}(x)$ found in Table 4 to conclude

$$\varepsilon_{\pi,num}(x) \leq 0.4298.$$

□

10.10 Numerical content of eSHP

Purely numerical calculations from [22]. This is kept in a separate file from the main file to avoid heavy recompilations. Because of this, this file should not import any other files from the PNT+ project, other than further numerical data files.

10.11 Prime gap data from eSHP

Numerical results on prime gaps from [22].

Proposition 10.11.1 (Table 8 prime gap record - unit test)**.** For every pair (p_k, g_k) in Table 8 with $p_k < 30$, g_k is the prime gap $p_{k+1} - p_k$, and all prime gaps preceding this gap are less than g_k.

Proof. Direct computation. □

Proposition 10.11.2 (Table 8 prime gap records)**.** For every pair (p_k, g_k) in Table 8, g_k is the prime gap $p_{k+1} - p_k$, and all prime gaps preceding this gap are less than g_k.

Proof. Verified by computer. Unlikely to be formalizable in Lean with current technology, except for the small values of the table. □

Proposition 10.11.3 (Table 8 prime gap records - completeness unit test)**.** Table 8 contains ALL the prime gap records (p_k, g_k) with $p_k \leq 30$.

Proof. Brute force verification. □

Proposition 10.11.4 (Table 8 prime gap records - completeness)**.** Table 8 contains ALL the prime gap records (p_k, g_k) with $p_k \leq 4 \times 10^{18}$.

Proof. Verified by computer. Unlikely to be formalizable in Lean with current technology, except for the small values of the table. □

Proposition 10.11.5 (Maximum prime gap)**.** The maximum prime gap for primes less than or equal to 4×10^{18} is 1476.

Proof. If not, then there would be an entry in Table 8 with $g > 1476$, which can be verified not to be the case. □

Proposition 10.11.6 (Table 9 prime gaps - unit test)**.** For every pair (g, P) in Table 9 with $P < 30$, P is the first prime producing the prime gap g, and all smaller g' (that are even or 1) have a smaller first prime.

Proof. Direct computation. □

Proposition 10.11.7 (Table 9 prime gaps)**.** For every pair (g, P) in Table 9, P is the first prime producing the prime gap g, and all smaller g' (that are even or 1) have a smaller first prime.

Proof. Verified by computer. Unlikely to be formalizable in Lean with current technology, except for the small values of the table. □

Proposition 10.11.8 (Table 9 prime gaps - completeness test)**.** Table 9 contains all first gap records (g, P) with $g < 8$.

Proof. Brute force verification. □

Proposition 10.11.9 (Table 9 prime gaps - completeness)**.** Table 9 contains all first gap records (g, P) with $g < 1346$

Proof. Verified by computer. Unlikely to be formalizable in Lean with current technology, except for the small values of the table. □

Proposition 10.11.10 (Existence of prime gap)**.** Every gap $g < 1346$ that is even or one occurs as a prime gap with first prime at most 3278018069102480227.

Proof. If not, then there would be an entry in Table 8 with $P > 3278018069102480227$, which can be verified not to be the case. ☐

10.12 Dusart's explicit estimates for primes

In this section we record the estimates of Dusart [13] on explicit estimates for sums over primes.

TODO: add more results and proofs here, and reorganize the blueprint

Proposition 10.12.1 (Dusart Proposition 3.2)**.** For $x \geq e^b$ we have $\psi(x) - x| \leq \varepsilon$, where b, ε are given by [13, Table 1].

Proof. ☐

Theorem 10.12.1 (Dusart Theorem 3.3)**.** We have $|\psi(x) - x| < \eta_k x / \log^k x$ for $x \geq 2$ with (k, η) given by the table in [13, Theorem 3.3].

Proof. ☐

Lemma 10.12.1 (Dusart Lemma 4.1)**.** We have $\vartheta(x)$ and $\psi(x) - \vartheta(x)$ for various x given by [13, Table 2], in particular $\vartheta(10^{15}) = 999999965752660.939839805291048\ldots$

Proof. ☐

Theorem 10.12.2 (Dusart Theorem 4.2)**.** We have $|\vartheta(x) - x| < \eta_k x / \log^k x$ for $x \geq x_k$ with (k, η_k, x_k) given by the table in [13, Theorem 4.2].

Proof. ☐

Proposition 10.12.2 (Dusart Proposition 4.3)**.** For $x \geq 121$, we have

$$1 - \frac{4 \log 2}{\log x} \sqrt{x} < \psi(x) - \vartheta(x), \quad \sqrt{\frac{\log^3 x}{x}}\, \vartheta(x^{1/2}).$$

Proof. ☐

Proposition 10.12.3 (Dusart Proposition 4.4)**.** For $x > 0$, we have

$$\psi(x) - \vartheta(x) - \vartheta(\sqrt{x}) < 1.777745 x^{1/3}.$$

Proof. ☐

Corollary 10.12.1 (Dusart Corollary 4.5)**.** For $x > 0$, we have

$$\psi(x) - \vartheta(x) < \left(1 + 1.47 \times 10^{-7}\right) \sqrt{x} + 1.78 x^{1/3}.$$

Proof. ☐

Theorem 10.12.3 (Dusart Theorem 5.1)**.** For $x \geq 4 \times 10^9$, we have

$$\pi(x) = \frac{x}{\log x}\left(1 + \frac{1}{\log x} + \frac{2 \log\log x}{\log^2 x} + O^*\Big(\frac{7.32}{\log^3 x}\Big)\right).$$

Proof. □

Corollary 10.12.2 (Dusart Corollary 5.2 (a))**.** For $x \geq 17$, we have

$$\frac{x}{\log x} \leq \pi(x).$$

Proof. □

Corollary 10.12.3 (Dusart Corollary 5.2 (b))**.** For $x > 1$, we have

$$\pi(x) \leq 1.2551 \frac{x}{\log x}.$$

Proof. □

Corollary 10.12.4 (Dusart Corollary 5.2 (c))**.** For $x \geq 599$, we have

$$\pi(x) \geq \frac{x}{\log x}\Big(1 + \frac{1}{\log x}\Big).$$

Proof. □

Corollary 10.12.5 (Dusart Corollary 5.2 (d))**.** For $x > 1$, we have

$$\pi(x) \leq \frac{x}{\log x}\Big(1 + \frac{1.2762}{\log x}\Big).$$

Proof. □

Corollary 10.12.6 (Dusart Corollary 5.2 (e))**.** For $x \geq 88789$, we have

$$\pi(x) \geq \frac{x}{\log x}\Big(1 + \frac{1}{\log x} + \frac{2}{\log^2 x}\Big).$$

Proof. □

Corollary 10.12.7 (Dusart Corollary 5.2 (f))**.** For $x > 1$, we have

$$\pi(x) \leq \frac{x}{\log x}\Big(1 + \frac{1}{\log x} + \frac{2.53816}{\log^2 x}\Big).$$

Proof. □

Corollary 10.12.8 (Dusart Corollary 5.3 (a))**.** For $x \geq 5393$, we have

$$\frac{x}{\log x - 1} \leq \pi(x)$$

Proof. □

Corollary 10.12.9 (Dusart Corollary 5.3 (b))**.** For $x > e^{1.112}$, we have

$$\pi(x) \le \frac{x}{\log x - 1.112}.$$

Proof. $\square$

Corollary 10.12.10 (Dusart Corollary 5.3 (c))**.** For $x \ge 468049$, we have

$$\frac{x}{\log x - 1 - \frac{1}{\log x}} \le \pi(x).$$

Proof. $\square$

Corollary 10.12.11 (Dusart Corollary 5.3 (d))**.** For $x > 5.6$, we have

$$\pi(x) \le \frac{x}{\log x - 1 - \frac{1.2311}{\log x}}.$$

Proof. $\square$

Sublemma 10.12.1 (Dusart Proposition 5.4, substep 1)**.** For $x \ge 4 \times 10^{18}$, there exists a prime p with

$$x < p \le x\Big(1 + \frac{1}{\log^3 x}\Big).$$

Proof. Use Lemma 10.4.3 and Theorem 10.12.2. $\square$

Sublemma 10.12.2 (Dusart Proposition 5.4, substep 2)**.** For $360653 < x < 4 \times 10^{18}$, there exists a prime p with

$$x < p \le x\Big(1 + \frac{1}{\log^3 x}\Big).$$

Proof. Use Lemma 10.4.6 and Proposition 10.11.2. $\square$

Sublemma 10.12.3 (Dusart Proposition 5.4, substep 3)**.** For $89693 \le x \le 360653$, there exists a prime p with

$$x < p \le x\Big(1 + \frac{1}{\log^3 x}\Big).$$

Proof. This is a computer computation, likely not formalizable within Lean. $\square$

Proposition 10.12.4 (Dusart Proposition 5.4)**.** There exists a constant X_0 (one may take $X_0 = 89693$) with the following property: for every real $x \ge X_0$, there exists a prime p with

$$x < p \le x\Big(1 + \frac{1}{\log^3 x}\Big).$$

Proof. Combine the three substeps. $\square$

Corollary 10.12.12 (Dusart Corollary 5.5)**.** For all $x \ge 468991632$, there exists a prime p such that

$$x < p \le x\Big(1 + \frac{1}{5000 \log^2 x}\Big).$$

Proof. Unfortunately, Proposition 10.12.4 only covers the range $x \geq \exp(5000)$. According to the author, the complete verification of this corollary requires several additional unpublished computations to cover the intermediate range, so this corollary will require significant effort to formalize. □

Theorem 10.12.4 (Dusart Theorem 5.6)**.** We have for $x \geq 2278383$,

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O^*\Big(\frac{0.2}{\log^3 x}\Big).$$

Proof. □

Theorem 10.12.5 (Dusart Theorem 5.7)**.** We have for $x \geq 912560$,

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + E + O^*\Big(\frac{0.3}{\log^2 x}\Big).$$

Proof. □

Theorem 10.12.6 (Dusart Theorem 5.9 (a))**.** We have for $x \geq 2278382$,

$$\prod_{p \leq x} \Big(1 - \frac{1}{p}\Big) = \frac{e^{-\gamma}}{\log x}\Big(1 + O^*\Big(\frac{0.2}{\log^3 x}\Big)\Big).$$

Proof. □

Theorem 10.12.7 (Dusart Theorem 5.9 (b))**.** We have for $x \geq 2278382$,

$$\prod_{p \leq x} \frac{p}{p-1} = e^{\gamma} \log x \Big(1 + O^*\Big(\frac{0.2}{\log^3 x}\Big)\Big).$$

Proof. □

Lemma 10.12.2 (Dusart Lemma 5.10)**.** We have for $k \geq 4$, $p_k \leq k \log p_k$.

Proof. □

Lemma 10.12.3 (Dusart Lemma 5.10)**.** We have for $k \geq 2$, $\log p_k \leq \log k + \log \log k + 1$.

Proof. □

Theorem 10.12.8 (Massias and Robin Theorem B (v))**.** We have for $k \geq 198$,

$$\vartheta(p_k) \leq k \log k + \log \log k - 1 + \frac{\log \log k - 2}{\log k}.$$

Proof. □

Proposition 10.12.5 (Dusart Proposition 5.11)**.** We have for $p_k \geq 10^{11}$,

$$\vartheta(p_k) \geq k \log k + \log \log k - 1 + \frac{\log \log k - 2.050735}{\log k}.$$

Proof. □

Proposition 10.12.6 (Dusart Proposition 5.11)**.** We have for $p_k \geq 10^{15}$,

$$\vartheta(p_k) \geq k \log k + \log\log k - 1 + \frac{\log\log k - 2.04}{\log k}.$$

Proof. □

Proposition 10.12.7 (Dusart Proposition 5.12)**.** We have for $k \geq 781$,

$$\vartheta(p_k) \leq k \log k + \log\log k - 1 + \frac{\log\log k - 2}{\log k} - \frac{0.782}{\log^2 k}.$$

Proof. □

Lemma 10.12.4 (Dusart Lemma 5.14)**.** We have for $k \geq 178974$,

$$p_k \leq k \log k + \log\log k - 1 + \frac{\log\log k - 1.95}{\log k}.$$

Proof. □

Proposition 10.12.8 (Dusart Proposition 5.15)**.** We have for $k \geq 688383$,

$$p_k \leq k \log k + \log\log k - 1 + \frac{\log\log k - 2}{\log k}.$$

Proof. □

Proposition 10.12.9 (Dusart Proposition 5.16)**.** We have for $k \geq 3$,

$$p_k \geq k \log k + \log\log k - 1 + \frac{\log\log k - 2.1}{\log k}.$$

Proof. □

10.13 Summary of results

Here we list some papers that we plan to incorporate into this section in the future, and list some results that have not yet been moved into dedicated paper sections.

References to add:

Dusart [13]

PT: D. J. Platt and T. S. Trudgian, The error term in the prime number theorem, Math. Comp. 90 (2021), no. 328, 871–881.

JY: D. R. Johnston, A. Yang, Some explicit estimates for the error term in the prime number theorem, arXiv:2204.01980.

Theorem 10.13.1 (PT Theorem 1)**.** Let $R = 5.573412$. For each row $\{X, \sigma, A, B, C, \epsilon_0\}$ from [23, Table 1] we have

$$\left|\frac{\psi(x) - x}{x}\right| \leq A \left(\frac{\log x}{R}\right)^B \exp\left(-C\sqrt{\frac{\log x}{R}}\right) \tag{10.10}$$

and

$$|\psi(x) - x| \leq \epsilon_0 x$$

for all $\log x \geq X$.

Proof. □

Corollary 10.13.1 (PT Corollary 1)**.** Let $R = 5.573412$. For each row $\{X, \sigma, A, B, C, \epsilon_0\}$ from [23, Table 1] we have

$$\left|\frac{\theta(x) - x}{x}\right| \leq A_1 \left(\frac{\log x}{R}\right)^B \exp\left(-C\sqrt{\frac{\log x}{R}}\right)$$

where $A_1 = A + 0.1$.

Proof. This follows trivially (and wastefully) from the work of Dusart [**?**, Cor. 4.5] or the authors [23, Cor. 2]. It should also follow from the results of [15]. □

Theorem 10.13.2 (PT Corollary 2)**.** One has

$$|\pi(x) - \mathrm{Li}(x)| \leq 235x(\log x)^{0.52} \exp(-0.8\sqrt{\log x})$$

for all $x \geq \exp(2000)$.

Proof. □

Here we record some results from [31]. TODO: add the Section 3.3 material on the conjecture of Pomerance -

Definition 10.13.1 (Trudgian definition of eps0)**.** One has

$$\epsilon_0(x) = \sqrt{\frac{8}{17\pi}} X^{1/2} e^{-X}, \quad X = \sqrt{(\log x)/R}, \quad R = 6.455.$$

for all $x \geq 2$.

Theorem 10.13.3 (Trudgian Theorem 1 for theta)**.** For $x \geq 149$ one has $|\theta(x) - x| \leq x\epsilon_0(x)$.

Proof. □

Theorem 10.13.4 (Trudgian Theorem 1 for psi)**.** For $x \geq 23$ one has $|\psi(x) - x| \leq x\epsilon_0(x)$.

Proof. □

Lemma 10.13.1 (Trudgian Lemma 1)**.** For $x \geq \exp(35)$ we have

$$|\theta(x) - x| \leq \frac{0.0045x}{\log^2 x}.$$

Proof. □

Theorem 10.13.5 (Trudgian Theorem 2)**.** For $x \geq 229$ one has

$$|\pi(x) - \mathrm{li}(x)| \leq 0.2795 \frac{x}{(\log x)^{3/4}} \exp\left(-\sqrt{\frac{\log x}{6.455}}\right).$$

Proof. □

Theorem 10.13.6 (Trudgian Corollary 2)**.** If $x > 2,898,242$, then there is a prime in the interval
$$\left[x, x\left(1 + \frac{1}{111(\log x)^2}\right)\right].$$

Proof. □

Theorem 10.13.7 (JY Corollary 1.3)**.** One has
$$|\pi(x) - \mathrm{Li}(x)| \leq 9.59x(\log x)^{0.515} \exp(-0.8274\sqrt{\log x})$$
for all $x \geq 2$.

Proof. □

Theorem 10.13.8 (JY Theorem 1.4)**.** One has
$$|\pi(x) - \mathrm{Li}(x)| \leq 0.028x(\log x)^{0.801} \exp(-0.1853 \log^{3/5} x/(\log\log x)^{1/5}))$$
for all $x \geq 2$.

Proof. □

TODO: input other results from JY
The results below are taken from https://tme-emt-wiki-gitlab-io-9d3436.gitlab.io/Art09.html

Theorem 10.13.9 (Schoenfeld 1976)**.** If $x > 2010760$, then there is a prime in the interval
$$\left(x\left(1 - \frac{1}{15697}\right), x\right].$$

Proof. □

Theorem 10.13.10 (Ramaré-Saouter 2003)**.** If $x > 10,726,905,041$, then there is a prime in the interval $(x(1 - 1/28314000), x]$.

Proof. □

Theorem 10.13.11 (Gourdon-Demichel 2004)**.** If $x > \exp(60)$, then there is a prime in the interval
$$\left(x\left(1 - \frac{1}{14500755538}\right), x\right].$$

Proof. □

Theorem 10.13.12 (Prime Gaps 2014)**.** If $x > \exp(60)$, then there is a prime in the interval
$$\left(x\left(1 - \frac{1}{1966196911}\right), x\right].$$

Proof. □

Theorem 10.13.13 (Prime Gaps 2024)**.** If $x > \exp(60)$, then there is a prime in the interval
$$\left(x\left(1 - \frac{1}{76900000000}\right), x\right].$$

Proof. □

Theorem 10.13.14 (Axler 2018 Theorem 1.4(1))**.** If $x \geq 6034256$, then there is a prime in the interval
$$\left(x, x\left(1 + \frac{0.087}{\log^3 x}\right)\right].$$

Proof. □

Theorem 10.13.15 (Axler 2018 Theorem 1.4(2))**.** If $x > 1$, then there is a prime in the interval
$$\left(x, x\left(1 + \frac{198.2}{\log^4 x}\right)\right].$$

Proof. □

Theorem 10.13.16 (Dudek 2014)**.** If $x > \exp(\exp(34.32))$, then there is a prime in the interval
$$\left(x, x + 3x^{2/3}\right].$$

Proof. □

Theorem 10.13.17 (Cully-Hugill 2021)**.** If $x > \exp(\exp(33.99))$, then there is a prime in the interval
$$\left(x, x + 3x^{2/3}\right].$$

Proof. □

Theorem 10.13.18 (RH Prime Interval 2002)**.** Assuming the Riemann Hypothesis, for $x \geq 2$, there is a prime in the interval
$$\left(x - \frac{8}{5}\sqrt{x}\log x, x\right].$$

Proof. □

Theorem 10.13.19 (Dudek 2015 under RH)**.** Assuming the Riemann Hypothesis, for $x \geq 2$, there is a prime in the interval
$$\left(x - \frac{4}{\pi}\sqrt{x}\log x, x\right].$$

Proof. □

Theorem 10.13.20 (Carneiro et al. 2019 under RH)**.** Assuming the Riemann Hypothesis, for $x \geq 4$, there is a prime in the interval
$$\left(x - \frac{22}{25}\sqrt{x}\log x, x\right].$$

Proof. □

Theorem 10.13.21 (Kadiri-Lumley Prime Gaps)**.** [19, Theorem 1.1] If $(\log x_0, m, \delta, T_1, \sigma_0, a, \Delta)$ is a row [19, Table 2], then for all $x \geq x_0$, there is a prime between $x(1 - \Delta^{-1})$ and x.

Proof. □

Theorem 10.13.22 (Ramaré-Saouter 2003 (2))**.** If $x > \exp(53)$, then there is a prime in the interval

$$\left(x\left(1 - \frac{1}{204879661}\right), x\right].$$

Proof. $\square$

Chapter 11

Tertiary explicit estimates

11.1 The least common multiple sequence is not highly abundant for large n

11.1.1 Problem statement and notation

Definition 11.1.1. $\sigma(n)$ is the sum of the divisors of n.

Definition 11.1.2. A positive integer N is called *highly abundant* (HA) if

$$\sigma(N) > \sigma(m)$$

for all positive integers $m < N$, where $\sigma(n)$ denotes the sum of the positive divisors of n.

Informally, a highly abundant number has an unusually large sum of divisors.

Definition 11.1.3. For each integer $n \geq 1$, define

$$L_n := \mathrm{lcm}(1, 2, \ldots, n).$$

We call $(L_n)_{n\geq 1}$ the *least common multiple sequence*.

Clearly L_n has a lot of divisors, and numerical experiments for small n suggests that L_n is often highly abundant. This leads to the following question:

> **Original MathOverflow question.** Is it true that every value in the sequence $L_n = \mathrm{lcm}(1, 2, \ldots, n)$ is highly abundant? Equivalently,
>
> $$\{L_n : n \geq 1\} \subseteq HA?$$

Somewhat surprisingly, the answer is *no*: not every L_n is highly abundant.

It has previously been verified in Lean that L_n is highly aboundant for $n \leq 70$, $81 \leq n \leq 96$, $125 \leq n \leq 148$, $169 \leq n \leq 172$, and not highly abondant for all other $n \leq 10^{10}$. The arguments here establish the non-highly-abundance of L_n for all $n \geq 89683^2$ sufficiently large n, thus completing the determination in Lean of all n for which L_n is highly abundant. This argument was taken from this MathOverflow answer.

11.1.2 A general criterion using three medium primes and three large primes

The key step in the proof is to show that, if one can find six primes $p_1, p_2, p_3, q_1, q_2, q_3$ obeying a certain inequality, then one can find a competitor $M < L_n$ to L_n with $\sigma(M) > \sigma(L_n)$, which will demonstrate that L_n is not highly abundant. More precisely:

Definition 11.1.4. In the next few subsections we assume that $n \geq 1$ and that $p_1, p_2, p_3, q_1, q_2, q_3$ are primes satisfiying

$$\sqrt{n} < p_1 < p_2 < p_3 < q_1 < q_2 < q_3 < n$$

and the key criterion

$$\prod_{i=1}^{3}\Big(1+\frac{1}{q_i}\Big) \leq \left(\prod_{i=1}^{3}\Big(1+\frac{1}{p_i(p_i+1)}\Big)\right)\Big(1+\frac{3}{8n}\Big)\left(1-\frac{4p_1p_2p_3}{q_1q_2q_3}\right). \tag{11.1}$$

NOTE: In the Lean formalization of this argument, we index the primes from 0 to 2 rather than from 1 to 3.

Lemma 11.1.1. We have $4p_1p_2p_3 < q_1q_2q_3$.

Proof. Obvious from the non-negativity of the left-hand side of (11.1). □

11.1.3 Factorisation of L_n and construction of a competitor

Lemma 11.1.2 (Factorisation of L_n)**.** There exists a positive integer L' such that

$$L_n = q_1q_2q_3\,L'$$

and each prime q_i divides L_n exactly once and does not divide L'.

Proof. Since $q_i < n$, the prime q_i divides L_n exactly once (as $q_i^2 > n$). Hence we may write $L_n = q_1q_2q_3L'$ where L' is the quotient obtained by removing these prime factors. By construction, $q_i \nmid L'$ for each i. □

Lemma 11.1.3 (Normalised divisor sum for L_n)**.** Let L' be as in Lemma 11.1.2. Then

$$\frac{\sigma(L_n)}{L_n} = \frac{\sigma(L')}{L'}\prod_{i=1}^{3}\Big(1+\frac{1}{q_i}\Big). \tag{11.2}$$

Proof. Use the multiplicativity of $\sigma(\cdot)$ and the fact that each q_i occurs to the first power in L_n. Then

$$\sigma(L_n) = \sigma(L')\prod_{i=1}^{3}\sigma(q_i) = \sigma(L')\prod_{i=1}^{3}(1+q_i).$$

Dividing by $L_n = L'\prod_{i=1}^{3} q_i$ gives

$$\frac{\sigma(L_n)}{L_n} = \frac{\sigma(L')}{L'}\prod_{i=1}^{3}\frac{1+q_i}{q_i} = \frac{\sigma(L')}{L'}\prod_{i=1}^{3}\Big(1+\frac{1}{q_i}\Big).$$

□

Lemma 11.1.4. There exist integers $m \geq 0$ and r satisfying $0 < r < 4p_1p_2p_3$ and

$$q_1q_2q_3 = 4p_1p_2p_3m + r$$

Proof. This is division with remainder. □

Definition 11.1.5. With m, r as above, define the competitor

$$M := 4p_1p_2p_3mL'.$$

Lemma 11.1.5 (Basic properties of M)**.** With notation as above, we have:

1. $M < L_n$.
2. $$1 < \frac{L_n}{M} = \Big(1 - \frac{r}{q_1q_2q_3}\Big)^{-1} < \Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big)^{-1}.$$

Proof. The first item is by construction of the division algorithm. For the second, note that

$$L_n = q_1q_2q_3L' = (4p_1p_2p_3m + r)L' > 4p_1p_2p_3mL' = M,$$

since $r > 0$. For the third,

$$\frac{L_n}{M} = \frac{4p_1p_2p_3m + r}{4p_1p_2p_3m} = 1 + \frac{r}{4p_1p_2p_3m} = \Big(1 - \frac{r}{4p_1p_2p_3m + r}\Big)^{-1} = \Big(1 - \frac{r}{q_1q_2q_3}\Big)^{-1}.$$

Since $0 < r < 4p_1p_2p_3$ and $4p_1p_2p_3 < q_1q_2q_3$, we have

$$0 < \frac{r}{q_1q_2q_3} < \frac{4p_1p_2p_3}{q_1q_2q_3},$$

so

$$\Big(1 - \frac{r}{q_1q_2q_3}\Big)^{-1} < \Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big)^{-1}.$$

□

11.1.4 A sufficient condition

We give a sufficient condition for $\sigma(M) \geq \sigma(L_n)$.

Lemma 11.1.6 (A sufficient inequality)**.** Suppose

$$\frac{\sigma(M)}{M}\Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big) \;\geq\; \frac{\sigma(L_n)}{L_n}.$$

Then $\sigma(M) \geq \sigma(L_n)$, and so L_n is not highly abundant.

Proof. By Lemma 11.1.5,

$$\frac{L_n}{M} < \Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big)^{-1}.$$

Hence

$$\frac{\sigma(M)}{M} \geq \frac{\sigma(L_n)}{L_n}\Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big)^{-1} > \frac{\sigma(L_n)}{L_n} \cdot \frac{M}{L_n}.$$

Multiplying both sides by M gives

$$\sigma(M) > \sigma(L_n) \cdot \frac{M}{L_n}$$

and hence

$$\sigma(M) \geq \sigma(L_n),$$

since $M/L_n < 1$ and both sides are integers. $\square$

Combining Lemma 11.1.6 with Lemma 11.1.3, we see that it suffices to bound $\sigma(M)/M$ from below in terms of $\sigma(L')/L'$:

Lemma 11.1.7 (Reduction to a lower bound for $\sigma(M)/M$)**.** If

$$\frac{\sigma(M)}{M} \;\geq\; \frac{\sigma(L')}{L'} \left(\prod_{i=1}^{3}\Big(1 + \frac{1}{p_i(p_i+1)}\Big)\right)\Big(1 + \frac{3}{8n}\Big), \tag{11.3}$$

then L_n is not highly abundant.

Proof. Insert (11.3) and (11.2) into the desired inequality and compare with the assumed bound (11.1); this is a straightforward rearrangement. $\square$

11.1.5 Conclusion of the criterion

Lemma 11.1.8 (Lower bound for $\sigma(M)/M$)**.** With notation as above,

$$\frac{\sigma(M)}{M} \geq \frac{\sigma(L')}{L'} \left(\prod_{i=1}^{3}\Big(1 + \frac{1}{p_i(p_i+1)}\Big)\right)\Big(1 + \frac{3}{8n}\Big).$$

Proof. By multiplicativity, we have

$$\frac{\sigma(M)}{M} = \frac{\sigma(L')}{L'} \prod_p \frac{1 + p^{-1} + \cdots + p^{-\nu_p(M)}}{1 + p^{-1} + \cdots + p^{-\nu_p(L')}}.$$

The contribution of $p = p_i$ is

$$\frac{(1 + p_i^{-1} + p_i^{-2})}{1 + p_i^{-1}} = 1 + \frac{1}{p_i(p_i+1)}.$$

The contribution of $p = 2$ is

$$\frac{1 + 2^{-1} + \cdots + 2^{-k-2}}{1 + 2^{-1} + \cdots + 2^{-k}},$$

where k is the largest integer such that $2^k \leq n$. A direct calculation yields

$$\frac{(1 + 2^{-1} + \cdots + 2^{-k-2})}{1 + 2^{-1} + \cdots + 2^{-k}} = \frac{2^{k+3} - 1}{2^{k+3} - 4} = 1 + \frac{3}{2^{k+3} - 4},$$

Finally, since $2^k \leq n < 2^{k+1}$, we have $2^{k+3} < 8n$, so

$$\frac{3}{2^{k+3} - 4} \geq \frac{3}{8n},$$

So the contribution from the prime 2 is at least $1 + 3/(8n)$.

Finally, the contribution of all other primes is at least 1. $\square$

We have thus completed the key step of demonstrating a sufficient criterion to establish that L_n is not highly abundant:

Theorem 11.1.1. Let $n \geq 1$. Suppose that primes $p_1, p_2, p_3, q_1, q_2, q_3$ satisfy

$$\sqrt{n} < p_1 < p_2 < p_3 < q_1 < q_2 < q_3 < n$$

and the key criterion (11.1). Then L_n is not highly abundant.

Proof. By Lemma 11.1.8, the condition (11.3) holds. By Lemma 11.1.7 this implies

$$\frac{\sigma(M)}{M}\Big(1 - \frac{4p_1p_2p_3}{q_1q_2q_3}\Big) \geq \frac{\sigma(L_n)}{L_n}.$$

Applying Lemma 11.1.6, we obtain $\sigma(M) \geq \sigma(L_n)$ with $M < L_n$, so L_n cannot be highly abundant. ☐

Remark 11.1.1. Analogous arguments allow other pairs (c, α) in place of $(4, 3/8)$, such as $(2, 1/4)$, $(6, 17/36)$, $(30, 0.632\ldots)$; or even $(1, 0)$ provided more primes are used on the p-side than the q-side to restore an asymptotic advantage.

11.1.6 Choice of six primes p_i, q_i for large n

To finish the proof we need to locate six primes $p_1, p_2, p_3, q_1, q_2, q_3$ obeying the required inequality. Here we will rely on the prime number theorem of Dusart [13].

Lemma 11.1.9 (Choice of medium primes p_i)**.** Let $n \geq X_0^2$. Set $x := \sqrt{n}$. Then there exist primes p_1, p_2, p_3 with

$$p_i \leq x\Big(1 + \frac{1}{\log^3 x}\Big)^i$$

and $p_1 < p_2 < p_3$. Moreover, $\sqrt{n} < p_1$

Proof. Apply Proposition 10.12.4 successively with $x, x(1+1/\log^3 x), x(1+1/\log^3 x)^2$, keeping track of the resulting primes and bounds. For n large and $x = \sqrt{n}$, we have $\sqrt{n} < p_1$ as soon as the first interval lies strictly above $\sqrt{n}$; this can be enforced by taking n large enough. ☐

Lemma 11.1.10 (Choice of large primes q_i)**.** Let $n \geq X_0^2$. Then there exist primes $q_1 < q_2 < q_3$ with

$$q_{4-i} \geq n\Big(1 + \frac{1}{\log^3 \sqrt{n}}\Big)^{-i}$$

for $i = 1, 2, 3$, and $q_1 < q_2 < q_3 < n$.

Proof. Apply Theorem 10.12.4 with suitable values of x slightly below n, e.g. $x = n(1 + 1/\log^3 \sqrt{n})^{-i}$, again keeping track of the intervals. For n large enough, these intervals lie in $(\sqrt{n}, n)$ and contain primes q_i with the desired ordering. ☐

11.1.7 Bounding the factors in (11.1)

Lemma 11.1.11 (Bounds for the q_i-product)**.** With p_i, q_i as in Lemmas 11.1.9 and 11.1.10, we have

$$\prod_{i=1}^{3}\Big(1+\frac{1}{q_i}\Big) \le \prod_{i=1}^{3}\Big(1+\frac{\big(1+\frac{1}{\log^3\sqrt{n}}\big)^i}{n}\Big). \tag{11.4}$$

Proof. By Lemma 11.1.10, each q_i is at least

$$n\Big(1+\frac{1}{\log^3\sqrt{n}}\Big)^{-j}$$

for suitable indices j, so $1/q_i$ is bounded above by

$$\frac{\big(1+\frac{1}{\log^3\sqrt{n}}\big)^i}{n}$$

after reindexing. Multiplying the three inequalities gives (11.4). □

Lemma 11.1.12 (Bounds for the p_i-product)**.** With p_i as in Lemma 11.1.9, we have for large n

$$\prod_{i=1}^{3}\Big(1+\frac{1}{p_i(p_i+1)}\Big) \ge \prod_{i=1}^{3}\Big(1+\frac{1}{\big(1+\frac{1}{\log^3\sqrt{n}}\big)^{2i}(n+\sqrt{n})}\Big). \tag{11.5}$$

Proof. By Lemma 11.1.9, $p_i \le \sqrt{n}(1+1/\log^3\sqrt{n})^i$. Hence

$$p_i^2 \le n\big(1+\tfrac{1}{\log^3\sqrt{n}}\big)^{2i} \quad \text{and} \quad p_i+1 \le p_i(1+1/\sqrt{n}) \le (1+1/\sqrt{n})p_i.$$

From these one gets

$$p_i(p_i+1) \le \big(1+\tfrac{1}{\log^3\sqrt{n}}\big)^{2i}(n+\sqrt{n}),$$

and hence

$$\frac{1}{p_i(p_i+1)} \ge \frac{1}{\big(1+\frac{1}{\log^3\sqrt{n}}\big)^{2i}(n+\sqrt{n})}.$$

Taking $1+\cdot$ and multiplying over $i=1,2,3$ gives (11.5). □

Lemma 11.1.13 (Lower bound for the product ratio p_i/q_i)**.** With p_i, q_i as in Lemmas 11.1.9 and 11.1.10, we have

$$1-\frac{4p_1p_2p_3}{q_1q_2q_3} \ge 1-\frac{4\big(1+\frac{1}{\log^3\sqrt{n}}\big)^{12}}{n^{3/2}}. \tag{11.6}$$

Proof. We have $p_i \le \sqrt{n}(1+1/\log^3\sqrt{n})^i$, so

$$p_1p_2p_3 \le n^{3/2}\big(1+\tfrac{1}{\log^3\sqrt{n}}\big)^6.$$

Similarly, $q_i \ge n(1+1/\log^3\sqrt{n})^{-i}$, so

$$q_1q_2q_3 \ge n^3\big(1+\tfrac{1}{\log^3\sqrt{n}}\big)^{-6}.$$

Combining,

$$\frac{p_1p_2p_3}{q_1q_2q_3} \le \frac{n^{3/2}\big(1+\frac{1}{\log^3\sqrt{n}}\big)^6}{n^3\big(1+\frac{1}{\log^3\sqrt{n}}\big)^{-6}} = \frac{\big(1+\frac{1}{\log^3\sqrt{n}}\big)^{12}}{n^{3/2}}.$$

This implies (11.6). ☐

11.1.8 Reduction to a small epsilon-inequality

Lemma 11.1.14 (Uniform bounds for large n)**.** For all $n \ge X_0^2 = 89693^2$ we have

$$\frac{1}{\log^3\sqrt{n}} \le 0.000675, \qquad \frac{1}{n^{3/2}} \le \frac{1}{89693}\cdot\frac{1}{n}.$$

and

$$\frac{1}{n+\sqrt{n}} \ge \frac{1}{1+1/89693}\cdot\frac{1}{n}.$$

Proof. This is a straightforward calculus and monotonicity check: the left-hand sides are decreasing in n for $n \ge X_0^2$, and equality (or the claimed upper bound) holds at $n = X_0^2$. One can verify numerically or symbolically. ☐

Lemma 11.1.15 (Polynomial approximation of the inequality)**.** For $0 \le \varepsilon \le 1/89693^2$, we have

$$\prod_{i=1}^{3}(1+1.000675^i\varepsilon) \le \Big(1+3.01\varepsilon+3.01\varepsilon^2+1.01\varepsilon^3\Big),$$

and

$$\prod_{i=1}^{3}\Big(1+\frac{\varepsilon}{1.000675^{2i}}\frac{1}{1+1/89693}\Big)\Big(1+\frac{3}{8}\varepsilon\Big)\Big(1-\frac{4\times 1.000675^{12}}{89693}\varepsilon\Big) \ge 1+3.36683\varepsilon-0.01\varepsilon^2.$$

Proof. Expand each finite product as a polynomial in ε, estimate the coefficients using the bounds in Lemma 11.1.14, and bound the tails by simple inequalities such as

$$(1+C\varepsilon)^k \le 1+kC\varepsilon+...$$

for small ε. All coefficients can be bounded numerically in a rigorous way; this step is a finite computation that can be checked mechanically. ☐

Lemma 11.1.16 (Final polynomial comparison)**.** For $0 \le \varepsilon \le 1/89693^2$, we have

$$1+3.01\varepsilon+3.01\varepsilon^2+1.01\varepsilon^3 \le 1+3.36683\varepsilon-0.01\varepsilon^2.$$

Proof. This is equivalent to

$$3.01\varepsilon+3.01\varepsilon^2+1.01\varepsilon^3 \le 3.36683\varepsilon-0.01\varepsilon^2,$$

or

$$0 \le (3.36683-3.01)\varepsilon-(3.01+0.01)\varepsilon^2-1.01\varepsilon^3.$$

Factor out ε and use that $0 < \varepsilon \le 1/89693^2$ to check that the resulting quadratic in ε is nonnegative on this interval. Again, this is a finite computation that can be verified mechanically. ☐

Proposition 11.1.1 (Verification of (11.1) for large n)**.** For every integer $n \geq X_0^2 = 89693^2 \approx 8.04 \times 10^9$, the primes p_i, q_i constructed in Lemmas 11.1.9 and 11.1.10 satisfy the inequality (11.1).

Proof. Combine Lemma 11.1.11, Lemma 11.1.12, and Lemma 11.1.13. Then apply Lemma 11.1.14 and set $\varepsilon = 1/n$. The products are bounded by the expressions in Lemma 11.1.15, and the inequality in Lemma 11.1.16 then ensures that (11.1) holds. ☐

11.1.9 Conclusion for large n

Theorem 11.1.2 (Non-highly abundant for large n)**.** For every integer $n \geq 89693^2$, the integer L_n is not highly abundant.

Proof. By Proposition 11.1.1, there exist primes $p_1, p_2, p_3, q_1, q_2, q_3$ satisfying the hypotheses of Theorem 11.1.1. Hence L_n is not highly abundant. ☐

11.1.10 Bonus material

The following result is not needed for this application, but is worth recording nevertheless.

Sublemma 11.1.1 (Formula for log of L equals Chebyshev psi)**.** For every n, $\log L_n = \sum_{p \leq n} \lfloor \log n / \log p \rfloor \log p$.

Proof. Compute the number of times p divides L_n and use the fundamental theorem of arithmetic. ☐

Sublemma 11.1.2 (Formula for Chebyshev psi)**.** For every n, $\psi(n) = \sum_{p \leq n} \lfloor \log n / \log p \rfloor \log p$, where ψ is the Chebyshev psi function.

Proof. Compute the number of times p divides L_n and use the fundamental theorem of arithmetic. ☐

Proposition 11.1.2 (Log of L equals Chebyshev psi)**.** For every n, $\log L_n = \psi(n)$, where ψ is the Chebyshev psi function.

Proof. Combine the previous results. ☐

11.2 Erdos problem 392

The proof here is adapted from `https://www.erdosproblems.com/forum/thread/392#post-2696` which in turn is inspired by the arguments in `https://arxiv.org/abs/2503.20170`.

Definition 11.2.1. We work with (approximate) factorizations $a_1 \dots a_t$ of a factorial $n!$.

Definition 11.2.2. The waste of a factorizations $a_1 \dots a_t$ is defined as $\sum_i \log(n/a_i)$.

Definition 11.2.3. The balance of a factorization $a_1 \dots a_t$ at a prime p is defined as the number of times p divides $a_1 \dots a_t$, minus the number of times p divides $n!$.

Lemma 11.2.1. If a factorization has zero total imbalance, then it exactly factors $n!$.

Proof. ☐

Lemma 11.2.2. The waste of a factorization is equal to $t \log n - \log n!$, where t is the number of elements.

Proof. ☐

Definition 11.2.4. The score of a factorization (relative to a cutoff parameter L) is equal to its waste, plus $\log p$ for every surplus prime p, $\log(n/p)$ for every deficit prime above L, $\log L$ for every deficit prime below L and an additional $\log n$ if one is not in total balance.

Lemma 11.2.3. If one is in total balance, then the score is equal to the waste.

Proof. ☐

Sublemma 11.2.1. If there is a prime p in surplus, one can remove it without increasing the score.

Proof. Locate a factor a_i that contains the surplus prime p, then replace a_i with a_i/p. ☐

Sublemma 11.2.2. If there is a prime p in deficit larger than L, one can remove it without increasing the score.

Proof. Add an additional factor of p to the factorization. ☐

Sublemma 11.2.3. If there is a prime p in deficit less than L, one can remove it without increasing the score.

Proof. Without loss of generality we may assume that one is not in the previous two situations, i.e., wlog there are no surplus primes and all primes in deficit are at most L. If all deficit primes multiply to n or less, add that product to the factorization (this increases the waste by at most $\log n$, but we save a $\log n$ from now being in balance). Otherwise, greedily multiply all primes together while staying below n until one cannot do so any further; add this product to the factorization, increasing the waste by at most $\log L$. ☐

Lemma 11.2.4. One can bring any factorization into balance without increasing the score.

Proof. Apply strong induction on the total imbalance of the factorization and use the previous three sublemmas. ☐

Proposition 11.2.1. Starting from any factorization f, one can find a factorization f' in balance whose cardinality is at most $\log n!$ plus the score of f, divided by $\log n$.

Proof. Combine Lemma 11.2.4, Lemma 11.2.3, and Lemma 11.2.2. ☐

Definition 11.2.5. Now let M, L be additional parameters with $n > L^2$; we also need the minor variant $\lfloor n/L \rfloor > \sqrt{n}$.

Definition 11.2.6. We perform an initial factorization by taking the natural numbers between $n - n/M$ (inclusive) and n (exclusive) repeated M times, deleting those elements that are not n/L-smooth (i.e., have a prime factor greater than or equal to n/L).

Sublemma 11.2.4. The number of elements in this initial factorization is at most n.

Proof. ☐

Lemma 11.2.5. The total waste in this initial factorization is at most $n \log \frac{1}{1-1/M}$.

Proof. ☐

Sublemma 11.2.5. A large prime $p \geq n/L$ cannot be in surplus.

Proof. No such prime can be present in the factorization. ☐

Sublemma 11.2.6. A large prime $p \geq n/L$ can be in deficit by at most n/p.

Proof. This is the number of times p can divide $n!$. ☐

Sublemma 11.2.7. A medium prime $\sqrt{n} < p \leq n/L$ can be in surplus by at most M.

Proof. Routine computation using Legendre's formula. ☐

Sublemma 11.2.8. A medium prime $\sqrt{n} < p \leq n/L$ can be in deficit by at most M.

Proof. The number of times p divides $a_1 \dots a_t$ is at least $M\lfloor n/Mp \rfloor \geq n/p - M$ (note that the removal of the non-smooth numbers does not remove any multiples of p). Meanwhile, the number of times p divides $n!$ is at most n/p. ☐

Sublemma 11.2.9. A small prime $p \leq \sqrt{n}$ can be in surplus by at most $M \log n$.

Proof. Routine computation using Legendre's formula, noting that at most $\log n / \log 2$ powers of p divide any given number up to n. ☐

Sublemma 11.2.10. A small prime $L < p \leq \sqrt{n}$ can be in deficit by at most $M \log n$.

Proof. Routine computation using Legendre's formula, noting that at most $\log n / \log 2$ powers of p divide any given number up to n. ☐

Sublemma 11.2.11. A tiny prime $p \leq L$ can be in deficit by at most $M \log n + ML\pi(n)$.

Proof. In addition to the Legendre calculations, one potentially removes factors of the form plq with $l \leq L$ and $q \leq n$ a prime up to M times each, with at most L copies of p removed at each factor. ☐

Proposition 11.2.2. The initial score is bounded by

$$n \log(1-1/M)^{-1} + \sum_{p \leq n/L} M \log n + \sum_{p \leq \sqrt{n}} M \log^2 n / \log 2 + \sum_{n/L < p \leq n} \frac{n}{p} \log \frac{n}{p} + \sum_{p \leq L} (M \log n + ML\pi(n)) \log L.$$

Proof. Combine Lemma 11.2.5, Sublemma 11.2.5, Sublemma 11.2.6, Sublemma 11.2.7, Sublemma 11.2.8, Sublemma 11.2.9, Sublemma 11.2.10, and Sublemma 11.2.11, and combine $\log p$ and $\log(n/p)$ to $\log n$. ☐

Sublemma 11.2.12. If M is sufficiently large depending on ε, then $n \log(1-1/M)^{-1} \leq \varepsilon n$.

Proof. Use the fact that $\log(1-1/M)^{-1}$ goes to zero as $M \to \infty$. ☐

Sublemma 11.2.13. If L is sufficiently large depending on M, ε, and n sufficiently large depending on L, then $\sum_{p \leq n/L} M \log n \leq \varepsilon n$.

Proof. Use the prime number theorem (or the Chebyshev bound). ☐

Sublemma 11.2.14. If n sufficiently large depending on M, ε, then $\sum_{p \leq \sqrt{n}} M \log^2 n / \log 2 \leq \varepsilon n$.

Proof. Crude estimation. $\square$

Lemma 11.2.6.
$$\pi(n) = o(n) \quad \text{as } n \to \infty.$$

Proof. Given $\varepsilon > 0$, choose $a \neq 0$ with $\varphi(a)/a < \varepsilon/2$ (using $\prod_{p \le n}(1 - 1/p) \to 0$). For $n \ge a + 2$,
$$\pi(n) \le \frac{\varphi(a)}{a} \cdot n + \varphi(a) + \pi(a+1) + 1.$$
Since $\varphi(a)/a < \varepsilon/2$, for n large enough the constant terms are absorbed, giving $\pi(n) < \varepsilon n$. $\square$

Sublemma 11.2.15. If n sufficiently large depending on L, ε, then $\sum_{n/L<p\le n} \frac{n}{p} \log \frac{n}{p} \le \varepsilon n$.

Proof. Bound $\frac{n}{p}$ by L and use the prime number theorem (or the Chebyshev bound). $\square$

Sublemma 11.2.16. For all $n \ge 2$, one has
$$\pi(n) \le \sqrt{n} + \frac{2n\log 4}{\log n}.$$

Proof. By Chebyshev's bound, $\prod_{p \le n} p \le 4^n$, so $\sum_{p\le n} \log p \le n \log 4$. The number of primes $p \le \sqrt{n}$ is trivially at most $\sqrt{n}$. For primes $p > \sqrt{n}$, we have $\log p > \frac{1}{2}\log n$, hence
$$\left(\pi(n) - \pi(\sqrt{n})\right) \cdot \tfrac{1}{2}\log n < \sum_{\sqrt{n}<p\le n} \log p \le n \log 4,$$
giving $\pi(n) - \pi(\sqrt{n}) < \frac{2n\log 4}{\log n}$. Adding $\pi(\sqrt{n}) \le \sqrt{n}$ yields the result. $\square$

Sublemma 11.2.17. If n sufficiently large depending on M, L, ε, then $\sum_{p \le L}(M\log n + ML\pi(n))\log L \le \varepsilon n$.

Proof. Use the prime number theorem (or the Chebyshev bound). $\square$

Proposition 11.2.3. The score of the initial factorization can be taken to be $o(n)$.

Proof. Pick M large depending on ε, then L sufficiently large depending on M, ε, then n sufficiently large depending on M, L, ε, so that the bounds in Sublemma 11.2.12, Sublemma 11.2.13, Sublemma 11.2.14, Sublemma 11.2.15, and Sublemma 11.2.17 each contribute at most $(\varepsilon/5)n$. Then use Proposition 11.2.2. $\square$

Theorem 11.2.1. One can find a balanced factorization of $n!$ with cardinality at most $n - n/\log n + o(n/\log n)$.-

Proof. Combine Proposition 11.2.3 with Proposition 11.2.1 and the Stirling approximation. $\square$

Theorem 11.2.2. One can factorize $n!$ into at most $n/2 - n/2\log n + o(n/\log n)$ numbers of size at most n^2.-

Proof. Group the factorization arising in Theorem 11.2.1 into pairs, using Lemma 11.2.1. $\square$

11.3 Numerical verification of Goldbach

We record here a simple way to convert prime in short interval theorems, together with numerical verification of even Goldbach, to numerical verification of odd Goldbach.

Definition 11.3.1 (Even Goldbach conjecture up to a given height)**.** We say that the even Goldbach conjecture is verified up to height H if every even integer between 4 and H is the sum of two primes.

Proposition 11.3.1 (Even Goldbach conjecture - unit test)**.** The even Goldbach conjecture is verified up to height 30.

Proof. This is a simple unit test, which can be verified by hand. □

Definition 11.3.2 (Odd Goldbach conjecture up to a given height)**.** We say that the odd Goldbach conjecture is verified up to height H if every odd integer between 5 and H is the sum of three primes.

Proposition 11.3.2 (Even Goldbach implies odd Goldbach)**.** If the even Goldbach conjecture is verified up to height H, then the odd Goldbach conjecture is verified up to height $H+3$.

Proof. If n is an odd integer between 7 and $H+3$, then $n-3$ is an even integer between 4 and H, so we can write $n-3 = p+q$ for some primes p and q, and hence $n = p+q+3$. □

Proposition 11.3.3 (Even Goldbach plus PNT in short interval implies odd Goldbach)**.** If every interval $(x(1-1/\Delta), x]$ contains a prime for $x \geq x_0$, the even Goldbach conjecture is verified up to height H, and the odd Goldbach conjecture is verified up to height x_0+4, then the odd Goldbach conjecture is verified up to height $(H-4)\Delta+4$.

Proof. If $x \leq x_0+4$ then we are done by hypothesis, so assume $x_0+4 < x \leq H\Delta$. By hypothesis, there is a prime p with $(x-4)(1-1/\Delta) < p \leq x-4$. Then $x-p$ is even, at least 4, and at most $(x-4)/\Delta + 4 \leq H$, so is the sum of two primes, giving the claim. □

Proposition 11.3.4 (Richstein's verification of even Goldbach)**.** [25] The even Goldbach conjecture is verified up to 4×10^{14}.

Proof. Numerical verification. □

Proposition 11.3.5 (Ramaré and Saouter's verification of odd Goldbach)**.** [24, Corollary 1] The odd Goldbach conjecture is verified up to 1.13256×10^{22}.

Proof. Combine Proposition 11.3.4, Proposition 11.3.2, and Theorem 10.13.10. □

Proposition 11.3.6 (e Silva–Herzog–Piranian verification of even Goldbach)**.** [22] The even Goldbach conjecture is verified up to 4×10^{18}.

Proof. Numerical verification. □

Proposition 11.3.7 (Helfgott's verification of odd Goldbach for small n)**.** [17, Appendix C] The odd Goldbach conjecture is verified up to 1.1325×10^{26}.

Proof. Combine Proposition 11.3.6, Proposition 11.3.2, and Theorem 10.13.10. □

The arguments in [17, Appendix C] push the bound further than this, but require unpublished estimates of Ramare. However, similar arguments were established in [19], and we present them here.

Proposition 11.3.8 (e Silva–Herzog–Piranian verification of even Goldbach (extended))**.** [22] The even Goldbach conjecture is verified up to $4 \times 10^{18} + 4$.

Proof. If $N = 4 \times 10^{18}$, use the fact that $211, 313, N - 209, N - 309$ are all prime. ☐

Proposition 11.3.9 (Kadiri–Lumley's verification of odd Goldbach for small n)**.** [19, Corollary 1.2] The odd Goldbach conjecture is verified up to $1966196911 \times 4 \times 10^{18}$.

Proof. Combine Proposition 11.3.8, Proposition 11.3.2, and Theorem 10.13.21 with $x_0 = e^{60}$ and $\Delta = 1966090061$. (Actually, if one is only allowed to use the previous results listed here, one first needs to use the values $x_0 = e^{59}$ and $\Delta = 1946282821$ to cover an intermediate range of values.) ☐

11.4 Ramanujan's inequality

Use of prime number theorems to establish Ramanujan's inequality

$$\pi(x)^2 < \frac{ex}{\log x}\pi\Big(\frac{x}{e}\Big)$$

for sufficiently large x, following [12].

Sublemma 11.4.1 (Criterion for Ramanujan's inequality, substep 1)**.** Let $M_a \in \mathbb{R}$ and suppose that for $x > x_a$ we have

$$\pi(x) < x \sum_{k=0}^{4} \frac{k!}{\log^{k+1} x} + \frac{M_a x}{\log^6 x}.$$

Then for $x > x_a$ we have

$$\pi^2(x) < x^2 \Big\{ \frac{1}{\log^2 x} + \frac{2}{\log^3 x} + \frac{5}{\log^4 x} + \frac{16}{\log^5 x} + \frac{64}{\log^6 x} + \frac{\epsilon_{M_a}(x)}{\log^7 x} \Big\} \tag{11.7}$$

where

$$\epsilon_{M_a}(x) = 72 + 2M_a + \frac{2M_a + 132}{\log x} + \frac{4M_a + 288}{\log^2 x} + \frac{12M_a + 576}{\log^3 x} + \frac{48M_a}{\log^4 x} + \frac{M_a^2}{\log^5 x}.$$

(cf. [12, Lemma 2.1])

Proof. Direct calculation ☐

Sublemma 11.4.2 (Criterion for Ramanujan's inequality, substep 2)**.** Let $m_a \in \mathbb{R}$ and suppose that for $x > x_a$ we have

$$\pi(x) > x \sum_{k=0}^{4} \frac{k!}{\log^{k+1} x} + \frac{m_a x}{\log^6 x}.$$

Then for $x > ex_a$ we have

$$\frac{ex}{\log x}\pi\Big(\frac{x}{e}\Big) > x^2\Big\{\frac{1}{\log^2 x} + \frac{2}{\log^3 x} + \frac{5}{\log^4 x} + \frac{16}{\log^5 x} + \frac{65}{\log^6 x} + \frac{\epsilon'_{m_a}(x)}{\log^7 x}\Big\},$$

where

$$\epsilon'_{m_a}(x) = 206 + m_a + \frac{364}{\log x} + \frac{381}{\log^2 x} + \frac{238}{\log^3 x} + \frac{97}{\log^4 x} + \frac{30}{\log^5 x} + \frac{8}{\log^6 x}.$$

(cf. [12, Lemma 2.1])

Proof. We have

$$\frac{ex}{\log x}\pi\Big(\frac{x}{e}\Big) > \frac{x^2}{\log x}\Big(\sum_{k=0}^{4}\frac{k!}{(\log x - 1)^{k+1}}\Big) + \frac{m_a x}{(\log x - 1)^6}$$

Substituting

$$\begin{aligned}\frac{1}{(\log x - 1)^{k+1}} &= \frac{1}{\log^{k+1} x}\Big(1 + \frac{1}{\log x} + \frac{1}{\log^2 x} + \frac{1}{\log^3 x} + \cdots\Big)^{k+1} \\ &> \frac{1}{\log^{k+1} x}\Big(1 + \frac{1}{\log x} + \cdots + \frac{1}{\log^{5-k} x}\Big)^{k+1}\end{aligned}$$

we obtain the claim. □

Proposition 11.4.1 (Criterion for Ramanujan's inequality)**.** [12, Lemma 2.1] Let $m_a, M_a \in \mathbb{R}$ and suppose that for $x > x_a$ we have

$$x\sum_{k=0}^{4}\frac{k!}{\log^{k+1} x} + \frac{m_a x}{\log^6 x} < \pi(x)$$

and for $x > ex_a$ one has

$$\pi(x) < x\sum_{k=0}^{4}\frac{k!}{\log^{k+1} x} + \frac{M_a x}{\log^6 x}.$$

Then Ramanujan's inequality is true for $x > x_0$ if

$$x_0 \geq ex_a$$

and

$$\epsilon_{M_a}(x_0) - \epsilon'_{m_a}(x_0) < \log x.$$

Proof. Combine the previous two sublemmas. □

Lemma 11.4.1 (Integral identity for pi - Li)**.** If $x \geq 2$, then

$$\pi(x) - \mathrm{Li}(x) = \frac{\theta(x) - x}{\log x} + \frac{2}{\log 2} + \int_2^x \frac{\theta(t) - t}{t\log^2 t}\,dt.$$

Proof. Follows from Sublemma 10.5.6 and the fundamental theorem of calculus. □

Sublemma 11.4.3 (Upper bound for pi)**.** Suppose that for $x \geq 2$ we have $|\theta(x) - x| \log^5 x \leq xa(x)$. Then

$$\pi(x) \leq \frac{x}{\log x} + a(x)\frac{x}{\log^6 x} + \int_2^x \frac{dt}{\log^2 t} + \int_2^x \frac{a(t)}{\log^7 t}\, dt.$$

(cf. [23, §5])

Proof. Follows from Lemma 11.4.1 and the triangle inequality. □

Sublemma 11.4.4 (Lower bound for pi)**.** Suppose that for $x \geq 2$ we have $|\theta(x) - x| \log^5 x \leq xa(x)$. Then

$$\pi(x) \geq \frac{x}{\log x} - a(x)\frac{x}{\log^6 x} + \int_2^x \frac{dt}{\log^2 t} - \int_2^x \frac{a(t)}{\log^7 t}\, dt.$$

(cf. [23, §5])

Proof. Follows from Lemma 11.4.1 and the triangle inequality. □

Lemma 11.4.2 (Bound for integral of an inverse power of log)**.** For $x \geq 2$ we have

$$\int_2^x \frac{dt}{\log^7 t} < \frac{x}{\log^7 x} + 7\Big(\frac{\sqrt{x}}{\log^8 2} + \frac{2^8 x}{\log^8 x}\Big).$$

(cf. [12, Section 2.3])

Proof. Integrate by parts to write the left-hand side as $\frac{x}{\log^7 x} - \frac{2}{\log^7 2} + 7\int_2^x \frac{t}{\log^8 t}dt$. Discard the middle term. For the final term, split between $\int_2^{\sqrt{x}}$ and $\int_{\sqrt{x}}^x$. For the first, use the bound $\int_2^{\sqrt{x}} \frac{t}{\log^8 t}dt < \int_2^{\sqrt{x}} \frac{t}{\log^8 2}dt$, and for the second, use the bound $\int_{\sqrt{x}}^x \frac{t}{\log^8 t}dt < \int_{\sqrt{x}}^x \frac{t}{\log^8 x}dt$. □

Sublemma 11.4.5 (Error estimate for theta, range 1)**.** For $2 \leq x < 599$ we have

$$E_\theta(x) \leq 1 - \frac{\log 2}{3}.$$

(cf. [23, (18)])

Proof. For $x \in [2, 3)$ we have $\theta(x) = \log 2$, so $E_\theta(x) = 1 - \log 2/x < 1 - \log 2/3$ since $x < 3$. For $x \in [3, 599)$ we use the LeanCert ChebyshevTheta engine: `checkAllThetaRelErrorReal 3 599 (768/1000) 20` via `native_decide` gives $|\theta(x) - x| \leq 0.768x$, hence $E_\theta(x) \leq 0.768 \leq 1 - \log 2/3$. □

Sublemma 11.4.6 (Error estimate for theta, range 2)**.** For $599 < x \leq \exp(58)$ we have

$$E_\theta(x) \leq \frac{\log^2 x}{8\pi\sqrt{x}}.$$

(cf. [23, (18)])

Proof. This is [23, Lemma 6]. □

Sublemma 11.4.7 (Error estimate for theta, range 3)**.** For $\exp(58) < x < \exp(1169)$ we have

$$E_\theta(x) \leq \sqrt{\frac{8}{17\pi}}\left(\frac{\log x}{6.455}\right)^{\frac{1}{4}} \exp\left(-\sqrt{\frac{\log x}{6.455}}\right).$$

(cf. [23, (18)])

Proof. This follows from Theorem 10.13.3. ☐

Sublemma 11.4.8 (Error estimate for theta, range 4)**.** For $\exp(1169) \le x < \exp(2000)$ we have

$$E_\theta(x) \le 462.0 \left(\frac{\log x}{5.573412}\right)^{1.52} \exp\left(-1.89\sqrt{\frac{\log x}{5.573412}}\right).$$

(cf. [23, (18)])

Proof. This follows from Corollary 10.13.1. ☐

Sublemma 11.4.9 (Error estimate for theta, range 5)**.** For $\exp(2000) \le x < \exp(3000)$ we have

$$E_\theta(x) \le 411.5 \left(\frac{\log x}{5.573412}\right)^{1.52} \exp\left(-1.89\sqrt{\frac{\log x}{5.573412}}\right).$$

(cf. [23, (18)])

Proof. This follows from Corollary 10.13.1. ☐

Sublemma 11.4.10 (Error estimate for theta, range 6)**.** For $x > \exp(3000)$ we have

$$E_\theta(x) \le 379.7 \left(\frac{\log x}{5.573412}\right)^{1.52} \exp\left(-1.89\sqrt{\frac{\log x}{5.573412}}\right).$$

(cf. [23, (18)])

Proof. This follows from Corollary 10.13.1. ☐

Proposition 11.4.2 (Equation (18) of Platt-Trudgian)**.** For $x \ge 2$ we have

$$E_\theta(x)(\log x)^5 \le a(x).$$

Proof. This follows from the previous five sublemmas. ☐

Lemma 11.4.3 (Monotonicity of a(x))**.** The function $a(x)$ is nonincreasing for $x \ge x_a$.

Proof. Follows from Lemma 2.0.1. ☐

Lemma 11.4.4 (Specific upper bound on pi)**.** For $x > ex_a$,

$$\pi(x) < x \sum_{k=0}^{4} \frac{k!}{\log^{k+1} x} + \frac{M_a x}{\log^6 x}.$$

.

Proof. This follows from the previous lemmas and calculations, including Lemma 11.4.2. ☐

Lemma 11.4.5 (Specific lower bound on pi)**.** For $x > x_a$,

$$\pi(x) > x \sum_{k=0}^{4} \frac{k!}{\log^{k+1} x} + \frac{m_a x}{\log^6 x}.$$

.

Proof. This follows from the previous lemmas and calculations, including Lemma 11.4.2. ☐

Lemma 11.4.6 (Bound for M - m)**.** We have $\epsilon_{M_a} - \epsilon'_{m_a} < \log(ex_a)$.

Proof. This is a direct calculation. An AI verification can be found at https://chatgpt.com/share/69a64f96-b1cc-800e-8f85-850168d23094 ☐

Theorem 11.4.1 (Ramanujan's inequality)**.** For $x \geq ex_a$ we have $\pi(x)^2 < \frac{ex}{\log x}\pi\left(\frac{x}{e}\right)$.

Proof. [23, Theorem 2] This follows from the previous lemmas and calculations, including the criterion for Ramanujan's inequality. ☐

Chapter 12

Iwaniec-Kowalski

12.1 Blueprint for Iwaniec-Kowalski Chapter 1

Here we collect facts from Chapter 1 that are not already in Mathlib. We will try to upstream as much as possible.

Definition 12.1.1. Additive arithmetic function: satisfies $f(mn) = f(m)+f(n)$ for coprime m, n.

Definition 12.1.2. Completely additive arithmetic function: satisfies $f(mn) = f(m)+f(n)$ for all $m, n \neq 0$.

Theorem 12.1.1. A completely additive function is additive.

Proof. $\square$

Theorem 12.1.2. If a and b are coprime, then any divisor d of ab can be uniquely expressed as a product of a divisor of a and a divisor of b.

Proof. Let a and b be coprime natural numbers, and let d be a divisor of ab. Since a and b are coprime, we can use the fact that the divisors of ab correspond to pairs of divisors of a and b. Specifically, we can express d as $d = d_a \cdot d_b$, where d_a divides a and d_b divides b. The uniqueness of this decomposition follows from the coprimality of a and b, which ensures that the divisors of a and b do not share any common factors. Therefore, there is a one-to-one correspondence between the divisors of ab and the pairs of divisors of a and b, which guarantees the uniqueness of the decomposition. $\square$

Theorem 12.1.3. If f is a multiplicative arithmetic function, then for coprime, nonzero a and b, we have that $\sum_{d|ab} f(d) = (\sum_{d|a} f(d)) \cdot (\sum_{d|b} f(d))$.

Proof. Since f is multiplicative, we can express the sum over divisors of ab in terms of the sums over divisors of a and b. The key idea is to use the fact that the divisors of ab can be expressed as products of divisors of a and divisors of b, due to the coprimality condition. Specifically, each divisor d of ab can be uniquely written as $d = d_a * d_b$, where d_a divides a and d_b divides b. Therefore, we can rewrite the sum as a double sum over the divisors of a and b, which factorizes into the product of the two separate sums. $\square$

Theorem 12.1.4. If g is a multiplicative arithmetic function, then for any $n \neq 0$, $\sum_{d|n} \mu(d) \cdot g(d) = \prod_{p|n}(1 - g(p))$.

Proof. Multiply out and collect terms. □

Theorem 12.1.5. The Dirichlet convolution of ζ with itself is τ (the divisor count function).

Proof. By definition of ζ, we have $\zeta(n) = 1$ for all $n \geq 1$. Thus, the Dirichlet convolution $(\zeta * \zeta)(n)$ counts the number of ways to write n as a product of two positive integers, which is exactly the number of divisors of n, i.e., $\tau(n)$. □

Theorem 12.1.6. The L-series of τ equals the square of the Riemann zeta function for $\Re(s) > 1$.

Proof. From the previous theorem, we have that the Dirichlet convolution of ζ with itself is τ. Taking L-series on both sides, we get $L(\tau, s) = L(\zeta, s) \cdot L(\zeta, s)$. Since $L(\zeta, s)$ is the Riemann zeta function $\zeta(s)$, we conclude that $L(\tau, s) = \zeta(s)^2$ for $\Re(s) > 1$. □

Definition 12.1.3. d_k is the k-fold divisor function: the number of ways to write n as an ordered product of k natural numbers. Equivalently, the Dirichlet convolution of ζ with itself k times.

Theorem 12.1.7. d_0 is the multiplicative identity (indicator at 1).

Proof. By definition, d_k is the k-fold Dirichlet convolution of ζ. When $k = 0$, this corresponds to the empty convolution, which is defined to be the multiplicative identity in the algebra of arithmetic functions. The multiplicative identity is the function that takes the value 1 at $n = 1$ and 0 elsewhere, which can be expressed as ζ^0. □

Theorem 12.1.8. d_1 is ζ.

Proof. By definition, d_k is the k-fold Dirichlet convolution of ζ. When $k = 1$, this means we are taking the convolution of ζ with itself once, which simply gives us ζ. Therefore, $d_1 = \zeta^1 = \zeta$. □

Theorem 12.1.9. d_2 is the classical divisor count function τ.

Proof. By definition, d_k is the k-fold Dirichlet convolution of ζ. When $k = 2$, this means we are taking the convolution of ζ with itself twice, which gives us $\zeta * \zeta$. From the earlier theorem, we know that $\zeta * \zeta = \tau$, where τ is the divisor count function. Therefore, $d_2 = \zeta^2 = \tau$. □

Theorem 12.1.10. Recurrence: $d_{k+1} = d_k * \zeta$.

Proof. By definition, d_k is the k-fold Dirichlet convolution of ζ. Therefore, d_{k+1} is the $(k + 1)$-fold convolution of ζ, which can be expressed as the convolution of d_k (the k-fold convolution) with ζ. Thus, we have $d_{k+1} = d_k * \zeta$. □

Theorem 12.1.11. The L-series for d_k is summable for $\Re(s) > 1$.

Proof. Since d_k is defined as the k-fold Dirichlet convolution of ζ, and we know that the L-series of ζ converges for $\Re(s) > 1$, it follows that the L-series of d_k also converges for $\Re(s) > 1$. This is because the convolution of functions with convergent L-series will also have a convergent L-series in the same region. Therefore, we can conclude that $L(d_k, s)$ is summable for $\Re(s) > 1$. □

Theorem 12.1.12. The L-series of d_k equals $\zeta(s)^k$ for $\Re(s) > 1$.

Proof. From the definition of d_k as the k-fold Dirichlet convolution of ζ, we can express d_k as ζ^k. The L-series of a Dirichlet convolution corresponds to the product of the L-series of the individual functions. Since $L(\zeta, s)$ is the Riemann zeta function $\zeta(s)$, it follows that $L(d_k, s) = L(\zeta^k, s) = (L(\zeta, s))^k = \zeta(s)^k$ for $\Re(s) > 1$ where the series converges. □

Theorem 12.1.13. d_k is multiplicative for all k.

Proof. The function d_k is defined as the k-fold Dirichlet convolution of ζ. Since ζ is a multiplicative function, and the Dirichlet convolution of multiplicative functions is also multiplicative, it follows that d_k is multiplicative for all k. This can be shown by induction on k, using the fact that the convolution of a multiplicative function with another multiplicative function remains multiplicative. □

Theorem 12.1.14. Explicit formula: $d_k(p^a) = (a + k - 1).choose(k - 1)$ for prime p and $k \geq 1$.

Proof. The function d_k counts the number of ways to write a natural number as an ordered product of k natural numbers. For a prime power p^a, the number of ways to factor it into k factors corresponds to the number of non-negative integer solutions to the equation $x_1 + x_2 + ... + x_k = a$, where each x_i represents the exponent of p in the factorization of the corresponding factor. This is a classic combinatorial problem, and the number of solutions is given by the formula $(a + k - 1).choose(k - 1)$, which counts the ways to distribute a indistinguishable items (the prime factors) into k distinguishable boxes (the factors). □

Theorem 12.1.15. (1.25) in Iwaniec-Kowalski: a formula for d_k for all n.

Proof. The function d_k is multiplicative, so to compute $d_k(n)$ for a general natural number n, we can factor n into its prime power decomposition: $n = p_1^{a_1} p_2^{a_2} ... p_m^{a_m}$. Since d_k is multiplicative, we have:

$$d_k(n) = d_k(p_1^{a_1}) \cdot d_k(p_2^{a_2}) \cdot ... \cdot d_k(p_m^{a_m})$$

Using the explicit formula for prime powers from the previous theorem, we can substitute to get:

$$d_k(n) = \prod_{i=1}^{m} (a_i + k - 1).choose(k - 1)$$

This gives us a complete formula for $d_k(n)$ in terms of the prime factorization of n. □

Definition 12.1.4. Divisor power sum with complex exponent.

Theorem 12.1.16. For natural exponents, σ^R agrees with σ.

Proof. The function σ^R is defined as the sum of the s-th powers of the divisors of n. When s is a natural number k, this definition coincides with the classical divisor power sum function $\sigma_k(n)$, which also sums the k-th powers of the divisors of n. Therefore, for natural exponents, we have $\sigma_k^R(n) = \sigma_k(n)$ when we view $\sigma_k(n)$ as a complex number. This can be shown by directly comparing the definitions and noting that both functions sum over the same set of divisors with the same exponentiation. □

Definition 12.1.5. Arithmetic function with complex parameter ν. Evaluates as $n \mapsto n^\nu$ for $n \neq 0$ and 0 at $n = 0$.

Theorem 12.1.17. $\sigma^R(\nu) = \zeta * \text{pow}^R(\nu)$, where ζ is the constant function 1.

Proof. The function $\sigma^R(\nu)$ is defined as the sum of the ν-th powers of the divisors of n. The function $\text{pow}^R(\nu)$ is defined as $n \mapsto n^\nu$ for $n \neq 0$ and 0 for $n = 0$. The Dirichlet convolution of ζ (the constant function 1) and $\text{pow}^R(\nu)$ is exactly $\sigma^R(\nu)$, since for each divisor d of n, we have $(\zeta * \text{pow}^R(\nu))(n) = \sum_{d|n} 1 \cdot d^\nu = \sigma^R(\nu)(n)$. Thus, we have $\sigma^R(\nu) = \zeta * \text{pow}^R(\nu)$. $\square$

Theorem 12.1.18. $L(\text{pow}^R(\nu), s) = \zeta(s - \nu)$ for $\Re(s - \nu) > 1$.

Proof. The function $\text{pow}^R(\nu)$ is defined as $n \mapsto n^\nu$ for $n \neq 0$ and 0 for $n = 0$. The L-series of $\text{pow}^R(\nu)$ at s is given by the sum $\sum_{n=1}^{\infty} n^{\nu-s}$. This series converges to the Riemann zeta function $\zeta(s - \nu)$ for $\Re(s - \nu) > 1$, since the zeta function is defined as $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $\Re(s) > 1$. Therefore, we have $L(\text{pow}^R(\nu), s) = \zeta(s - \nu)$ under the condition that $\Re(s - \nu) > 1$. $\square$

Theorem 12.1.19. The abscissa of absolute convergence of $L(\text{pow}^R(\nu), s)$ is at most $\Re(\nu)+1$.

Proof. We apply **??** which states that if there exists a constant C such that $\|f(n)\| \leq C \cdot n^r$ for all n sufficiently large, then the abscissa of absolute convergence of $L(f, s)$ is at most $r + 1$. In our case, we can take $f(n) = n^\nu$ and observe that $\|n^\nu\| = n^{\Re(\nu)}$. Thus, we can choose $C = 1$ and $r = \Re(\nu)$, which gives us the desired result that the abscissa of absolute convergence of $L(\text{pow}^R(\nu), s)$ is at most $\Re(\nu) + 1$. $\square$

Theorem 12.1.20. $\zeta(s)\zeta(s - \nu) = \sum_{n=1}^{\infty} \sigma_\nu(n) n^{-s}$ for $\Re(s) > 1$ and $\Re(s - \nu) > 1$.

Proof. The divisor power sum function σ_ν is the Dirichlet convolution of the constant function 1 (i.e., ζ) and the power function $n \mapsto n^\nu$. The L-series of a Dirichlet convolution is the product of the L-series of the individual functions. Since $L(1, s) = \zeta(s)$ and $L(n \mapsto n^\nu, s) = \zeta(s-\nu)$, we have $L(\sigma_\nu, s) = \zeta(s) \cdot \zeta(s-\nu)$ for $\Re(s) > 1$ and $\Re(s-\nu) > 1$. $\square$

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